Hydrogen in Residential Heating: A Merit-Order Assessment for the EU, the United Kingdom and Switzerland under Net-Zero by 2050

Abdurahman Alsulaiman[1,*] Ali Alajwad[2,3]

[1]Oxford Institute for Energy Studies, Oxford, UK
[2]RENEW Lab, Johns Hopkins University, Baltimore, MD, USA
[3]Cornell University, Ithaca, NY, USA

[*]*Corresponding author.*

Draft v20, June 2026

Abstract

This paper provides a merit-order assessment of green hydrogen's role in European decarbonisation, across the EU-27, the United Kingdom and Switzerland, 29 countries over 2025 to 2050, built on a residential heat-demand reconstruction resolved to the third level of the Nomenclature of Territorial Units for Statistics (NUTS3) and assessed at the country level. The context is the buildings sector, which accounts for approximately 40 per cent of final energy consumption across the European Union and Switzerland and is therefore central to the net-zero-by-2050 agenda [1, 2]. Heat-pump electrification leads the field [3, 4], yet the role of green hydrogen in heating remains contested, and is usually settled by assuming a deployment share instead of deriving one [5, 6]. We reconstructed residential heat demand bottom-up from building footprints and archetype intensities and carried it through a 200-draw Monte Carlo over four scenarios and a least-cost optimisation.

Hydrogen was priced as a system operator prices it, on short-run marginal cost in dispatch, which bounds the heating share it could hold by where it wins the winter-peak merit order. That bound is what the dispatch tests deliver; the share each scenario realises is a policy target we set, and it did not move when the cost comparison did. Three tests carry the assessment, cost competitiveness on a levelised basis, merit-order dispatch, and investment viability or capital recovery (the missing-money test). The third is decisive rather than incidental. No hydrogen peaker recovers its capital from market rent in any country or scenario we run, at about 9 per cent capacity-weighted across the markets that build. The paper then traces the electrified heat load into the power system and assesses hydrogen as a dispatchable backup generator.

On the levelised-cost test the outcome settles, and it settles on the cost of holding the molecule between seasons. Geology sets that cost for two of the seven surviving hydrogen leads; the retail tariff on the rival carrier sets it for the other five. At 2050 medians of €118.5 for the best heat pump, €122.8 for the hydrogen boiler and €132.7 for the gas boiler per MWh of useful heat, with each carrier charged its own last-mile network and with hydrogen charged the seasonal store that its winter-weighted load shape requires, the best heat pump is the cheaper of the two clean options in 22 of 29 countries and the hydrogen boiler in seven, at a median country gap of €15.4/MWh in the heat pump's favour (mean €19.7, and €1.4 weighted by each country's 2025 heat demand). Five of those seven are salt-cavern markets, including the four widest. Space heating draws about three-quarters of its energy between October and March while electrolytic production is far flatter, so the mismatch must be stored, and hydrogen's base-load case concentrates where the geology makes storing it cheap.

That is the same condition that decides the peaking arena, so the base-load and dispatch tests now agree on one physical quantity, the cost of holding the molecule between seasons, which is also why their agreement corroborates less than two

separate confirmations would. The two assumptions still most generous to hydrogen both run the same way. By 2050 the median study-area household pays roughly €209/MWh for taxed retail electricity against roughly €50/MWh for hydrogen delivered and untaxed, so equalising the fiscal treatment would widen the heat pump's margin, and repricing the molecule as blue at the €81/MWh of our own blue route, which is the provenance the one dedicated assessment of heating hydrogen concludes it would have, adds a fuel wedge of €31/MWh, wider than the widest surviving hydrogen lead. The count turns on the hydrogen price path and not the policy scenario, standing at 13, 22, 26 and 29 of 29 for the heat pump as that path runs from rapid decline to stranded.

The dispatch test then bounds hydrogen across three merit-order arenas, each pricing it on short-run marginal cost in a different market: the building winter-peak heating stack, the district-heating stack, and the power-sector peaker that supplies the electrified load. Under the most favourable scenario (H2 Push, and all three counts below are on that scenario, charging both carriers alike) hydrogen wins the building winter-peak dispatch in 14 of 29 countries; undercuts gas combined-heat-and-power in the district-heating stack in seven; and displaces the carbon-priced gas peaker in the power sector that the electrified load runs on in seven of 29, all of them salt-cavern markets, on the even-handed series that holds both turbines at the same efficiency. Letting the two efficiencies diverge as each technology improves, which hands hydrogen the better machine, raises that count to 15, but the additional eight sit within €8/MWh-e of parity while the seven win decisively, so seven is the count we carry. Where it wins, the dispatch role is secured by operating cost and the carbon price on the fossil competitor, more than by technology or cheap hydrogen alone; high achievement in one arena does not carry to another.

Even where hydrogen wins the dispatch competition, a capital-recovery hurdle remains, and it is the binding one. The peaker runs too few hours to recover its capital from market rent, the classic missing-money problem, so any deliberate build requires a capacity payment. A technology-neutral payment does not deliver one. Across the five markets that build, the hydrogen peaker's standing payment exceeds the gas peaker's in four of them, covering 96 per cent of the winning capacity, so a neutral auction procures unabated gas. Selecting hydrogen instead requires conditioning the payment on emissions, worth about €172/tCO_2 of displaced gas generation, or €239 if the gas rent is floored at zero. Because electricity is priced on operating cost, hydrogen keeps its dispatch role; what it lacks is the investment case. The power role is the clearest case. The firm peaking fleet that the electrified heat load calls for is about 278 GW against a study-area system peak of about 889 GW, roughly 31 per cent of peak capacity at about 139 full-load hours a year, a capacity contribution. Those power-sector figures rest on one realisation of a single synthetic weather year, which is the largest simplification here, so they size a role and are not an adequacy assessment.

Infrastructure costing places this in context. The cost-optimal decarbonisation pathway is led by district heat at about 53 per cent of 2050 supply and heat pumps at about 45 per cent, with residual fossil at about 1.4 per cent; its hydrogen share of about a third of one per cent is bounded by a ceiling that is zero in 27 of the 29 markets and so adds no independent evidence on hydrogen. That pathway carries the carbon price of the second emissions trading system for buildings and road transport (ETS2) inside the cost of every fossil option. Tightening the emissions cap from a 75 to a 100 per cent reduction moves the present-value system cost by 0.05 per cent, so the cost of the ambition is small.

The peaking verdict is robust to the structural choices it once rested on. The power-sector result comes from the reduced-form dispatch and is cross-checked against an explicit unit commitment, which reproduces it to within €1.1 bn with its constraints off and €1.3 bn with them on, the near-zero 2050 retrofit is foresight-invariant under a myopic rolling-horizon test, and it holds under an endogenous treatment of Switzerland with electricity-price feedback, each of which strengthens the conclusion without changing it. The least-cost reading is nonetheless a property of the cost-optimal frontier.

On the peaking and capital-recovery evidence, then, hydrogen's direct heating niche is small. Its widest count, 20 of 29 in district heating, is a ceiling on a residential gas tariff rather than a competitive score, and falls to 7 at a wholesale gas price, which is what a district-heat operator would pay. The larger potential is indirect, as a candidate clean peaker, but only under a payment conditioned on emissions. A neutral one buys gas.

Keywords: hydrogen; residential heating; heat pumps; decarbonisation; merit order; missing money; net zero; European Union; Switzerland.

JEL codes: Q41, Q48, Q54, H23

Contents

List of Figures

Figures numbered A.1, B.1, C.1 and so on are in the appendices; plain numbers are in the main body.

List of Tables

Tables numbered A.1, B.1, C.1 and so on are in the appendices; plain numbers are in the main body.

1 Introduction

Which technology should decarbonise Europe's homes, and where, is now a commitment of capital that cannot easily be reversed. Between now and 2050 hundreds of billions of euros of heating infrastructure will be replaced, and the choice between electrifying the heat and reusing the gas network through hydrogen decides how that money is spent. The stakes are large because the sector is large. Buildings account for approximately 40 per cent of final energy consumption in the European Union, the greater part of it burned for space heating and hot water, so the sector sits at the centre of climate policy from the European Green Deal through to the Energy Performance of Buildings Directive [1, 2].

What makes the choice hard is the form the heat takes. It is delivered through hundreds of millions of individual dwellings, each with its own vintage, fabric, climate and occupant, and replaced only slowly as boilers fail and buildings change hands [7, 8]. Residential heat must therefore be abated one household at a time, against real constraints of cost, disruption and capital turnover [9, 10], and it is the heterogeneity of the stock that makes the technology choice consequential, because a route that is cheap in one dwelling can be costly in the next.

European policy has converged on heat pumps as the principal route. The *Fit for 55* package, the revised Energy Performance of Buildings Directive and the REPowerEU plan all treat electrification as the default for buildings, and the case is strong on the merits [11, 12]. A heat pump delivers a coefficient of performance (COP) of two to four, so it undercuts any combustion appliance on delivered energy, and it decarbonises automatically as the grid does [3, 4]. Deployment is nonetheless slowed by frictions that fall unevenly across the stock, among them the cost of retrofitting poorly insulated dwellings, distribution-grid reinforcement in dense urban areas, a COP that degrades in the coldest weather and in the oldest fabric, split incentives between landlords and tenants, and uneven public acceptance [13, 14]. These frictions do not overturn electrification's cost case; they create pockets, poorly insulated buildings, grid-constrained zones and regions of dense industrial heat demand, where the upfront cost or engineering burden of electrification is locally steep, and into which a hydrogen alternative has been proposed.

Hydrogen is the alternative most often advanced, and the most contested. Its appeal is institutional as much as technical. A low-carbon gas could in principle reuse the existing distribution network and the appliance habit of a gas-heated home, sidestepping the fabric upgrades and grid reinforcement that electrification demands, and serving precisely the hard-to-electrify segment that heat pumps struggle to reach [15, 16]. REPowerEU raised the EU's 2030 hydrogen ambition to 20 Mt of renewable and imported supply, with buildings named among the candidate demand centres [11, 17]. The energy-security shock that followed the 2022 gas crisis sharpened the appetite for a domestic, storable, network-compatible fuel [18, 19].

A substantial sceptical literature has answered that hydrogen for space heating is thermodynamically wasteful and expensive, that it competes for scarce electrolytic supply with industry and transport, where substitutes are fewer, and that the network incumbency argument understates the cost of conversion and the value of the heat a pump extracts from the air [5, 6]. This dispute matters in practice. It shapes whether gas distribution networks are written down or repurposed, where renovation subsidy is directed, and how the heating infrastructure replaced before 2050 is committed. Switzerland, ringed by states with active hydrogen strategies and without domestic gas resources of its own, faces a sharper version of the same question, namely whether imported low-carbon hydrogen has any role in a national heat system that it would otherwise electrify [20, 21].

Despite the stakes, the quantitative basis for resolving the dispute is thinner than the policy debate implies, and the literature can be mapped onto three recurrent limitations, each exemplified by influential work. The first limitation is one of spatial resolution. The continental scenario assessments that frame the heating debate operate at national or whole-system resolution, averaging over exactly the spatial heterogeneity, in building stock, climate, demand density and storage geology, that determines where, if anywhere, hydrogen can compete. The pan-European race between hydrogen and heat pumps in Billerbeck et al. [22] and the whole-system electrification accounts of Hoseinpoori et al. [23] and Brown et al. [24] resolve the technology contest finely but the geography coarsely, while the spatially explicit studies that do exist remain single-country, as in the Spanish optimisation of Maestre et al. [25] and the German techno-economic comparison of Huckebrink and Bertsch [26]. The cross-country pattern of where hydrogen's economics diverge from the average is thus left unmapped.

The second limitation concerns how hydrogen's heating share enters the analysis. Studies typically fix that share by assumption, whether as a policy target, a scenario lever or a feasibility ceiling, and then evaluate the consequences, a procedure visible in the scenario-driven shares of Billerbeck et al. [22] and the technology-portfolio framing of Weidner and Guillén-Gosálbez [27]. That procedure cannot answer the prior question of whether a hydrogen heating market would form at all, because it imposes the very outcome it ought to derive.

Pricing hydrogen on its levelised cost over a year, as most such comparisons do, compounds the problem by misstating how a dispatchable fuel competes. A system operator does not run the cheapest technology on average, but the one with the lowest short-run marginal cost at each hour, so a fuel that loses the annual cost contest may still win, or lose, the few hundred hours of the winter peak on which dispatchable capacity actually earns its keep [28, 29]. The third limitation is a severed link between end uses.

The electrification that displaces fossil heat does not make the heat-supply problem disappear; it relocates part of it into the power sector, which must then meet a cold-snap heating load on the same scarce winter evenings. That interaction has largely been treated apart, the buildings-side studies of the heating choice and the power-side studies of system adequacy addressing it separately. The heat-peak work of Zeyen et al. [30] and the Swiss winter-deficit analysis of Mellot et al. [31] take the two halves on their own, not as one merit order. The updated REPowerEU context, the accelerated hydrogen targets and the introduction of a second emissions trading system for buildings and road transport (ETS2) further postdate much of the published modelling.

These three limitations define the questions this paper sets out to answer. The core question is European and country-level: under what conditions, on economic and environmental cost grounds, is decarbonised hydrogen competitive in the buildings sector across the EU-27, the United Kingdom and Switzerland under a net-zero by 2050 case, and where those conditions hold. It resolves into three research questions, each closing one of the limitations above, restated as the three gaps of Section 2.8, and each answered in full by the analysis that follows.

RQ1 (EU, UK and Switzerland, country-level) takes up the first two gaps together, testing the hydrogen-versus-heat-pump trade-off on a consistent operating-cost basis and deriving hydrogen's heating share from where the fuel can win the cold-snap peak instead of imposing it as a target [5, 32]. RQ2 (Switzerland, national case) closes the third gap for Switzerland as a high-cost, hydro-rich, cavern-poor system reliant on imported hydrogen [20, 33]. RQ3 (cross-sector synergy, scoped to the power sector) names the buildings-to-power coupling that the severed link implies, the one synergy this paper carries through in full [24, 30]. The three questions are stated in full, with the scope boundary that separates what this paper models from what it leaves to subsequent work, in Section 1.1.

Three contributions. This paper resolves the three limitations just set out, the missing spatial resolution, the imposed and not derived heating share, and the severed heating-to-power link, within a single, internally consistent framework, and in doing so makes three contributions.

1. *Methodological.* We build a bottom-up techno-economic model of residential heating across the EU-27, the United Kingdom and Switzerland, 29 countries, over 2025 to 2050. Useful heat demand is reconstructed at NUTS3 resolution from EUBUCCO building footprints and TABULA heat intensities, validated against the Hotmaps regional database, and propagated through eight heating technologies and four multi-lever scenarios by a 200-draw Monte Carlo that carries parametric uncertainty end to end, alongside an independent least-cost optimisation [34, 35]. The NUTS3 reconstruction earns its place. It grounds the demand and its climate-and-density gradient in observed building stock, not a top-down national split, the step that makes the country aggregates trustworthy. The economic verdict is then formed at the country level, where hydrogen's prospects diverge from a continental average because the conditions its case turns on, salt-cavern storage geology, the carbon-

priced gas it must undercut and dense district-heating loads, differ markedly from one country to the next, a divergence that a single continental average dilutes to invisibility. This matters more for hydrogen than for heat pumps, because a heat pump is broadly available wherever the grid reaches and its cost varies smoothly with climate and fabric, so an average represents it tolerably.

2. *Reframing.* Rather than fixing a heating share by policy assumption, we *derive* it through cost and dispatch tests that are run separately. On the full annual cost of delivered heat, with each carrier charged its own equipment capital, its own last-mile network and the seasonal storage its own load shape requires, the best heat pump is cheaper than a gas boiler in 24 of the 29 countries and cheaper than a hydrogen boiler in 22, so most of the stock electrifies. Two of the seven exceptions are geological and five turn on the retail tariff the heat pump pays. Electrified heat is then an electricity load, which moves part of the heating problem onto the power sector and its scarcest winter evenings. That displaced load is priced the way a system operator actually prices electricity, by short-run marginal cost at each hour. Hydrogen enters at both the first and the last of those steps, as a base-load competitor on annual cost in the countries where retail electricity is dearest, and again as a candidate peaker competing on operating cost against the carbon-taxed gas plant. Hydrogen is represented along its full supply chain, from production route and import corridor through seasonal storage to the distribution network, with a geographically bounded storage adder of roughly €50/MWh in countries with salt-cavern endowment and roughly €100/MWh where caverns are absent.

3. *Heating-to-power bridge.* We carry the analysis across the heating-to-power bridge that the electrification pathway creates, applying the same dispatch logic in three linked arenas, the building winter peak, the district-heating stack, and the power sector on which the electrified heat load now depends. It is this bridge that makes the power-sector merit order a heating question at all, and it is the synergy named in RQ3.

Three findings follow, set out with their uncertainty bounds in Sections 4 and 5 and previewed here.

First, the base-load and dispatch tests converge instead of cutting against each other. On the full annual cost of delivered heat in 2050, with each carrier charged its own last-mile network and with hydrogen charged the seasonal store its winter-weighted load shape requires, the EU median is roughly €118.5/MWh for the best heat pump, €122.8/MWh for a hydrogen boiler and €132.7/MWh for a gas boiler. The best heat pump is the cheaper of the two clean options in 22 of 29 countries at a median country gap of €15.4/MWh, and hydrogen holds the remaining seven, five of them salt-cavern markets.

Priced instead on the short-run marginal cost a system operator actually uses, hydrogen undercuts the carbon-taxed fossil competitor at the peak hour in a cavern-endowed minority, and the count rises with policy ambition across the four scenarios. Charging both carriers alike, so that hydrogen pays the same last-mile adder its base-load comparison uses and the district-heat operator's gas is bought at the wholesale hub price, it wins at the building winter peak in 0 countries under Current Policies, 2 under Stated Policies, 7 under Net Zero and 14 under H2 Push; in the district-heating stack in 0, 0, 7 and 7 countries respectively; and in the power sector in 0, 0, 7 and 15 on the evolving-efficiency turbine series, or 0, 0, 7 and 7 on the even-handed series that holds both turbines at 0.40. On the asymmetric accounting an earlier draft carried, which charges hydrogen no last-mile network and prices the district-heat gas unit at the residential retail tariff, the first two arenas read 0, 4, 9 and 16 and 0, 5, 11 and 20.

These counts measure where hydrogen would actually be dispatched, not where it is deployed by policy or where it breaks even on capital. The power-sector count of 15 is bimodal and should be read with care, the decisive operating-cost advantage holding in only the seven salt-cavern markets and the remaining eight sitting at a marginal €0 to 8/MWh-e (Section 5.1). Both tests therefore turn on one physical endowment, which is also why their agreement corroborates less than two separate confirmations would.

What lead hydrogen keeps in those markets is a fiscal result before a technological one, since by 2050 the median European household pays roughly €209/MWh for taxed retail electricity against roughly €50/MWh for hydrogen delivered and untaxed, and equalising the two treatments would widen the heat pump's margin further.

Second, the electrified load opens a power-sector niche that no market would build unaided. Once heat pumps supply roughly half of useful heat, residential heating becomes about 9 per cent of 2050 electricity demand and a driver of the system's winter peak, calling for a firm peaking fleet of about 278 GW against a study-area system peak of about 889 GW, roughly 31 per cent of peak capacity, which is a capacity role and not an energy one. Both figures are computed on a single synthetic weather year, so they size that role and do not assess adequacy.

The same merit-order test inside the power sector puts the hydrogen turbine level with the carbon-priced gas peaker at the cross-country median, about €238 against €239 per megawatt-hour of electricity, and decisively below it in the cavern markets. Yet the peaker runs too few hours to recover its capital from market rent, the classic missing-money problem, recovering about 9 per cent across the cavern markets on a capacity-weighted basis, so even the dispatched role would require a capacity payment or equivalent support. An independent least-cost optimisation, which minimises total system cost over the whole technology portfolio with no hour-by-hour dispatch, reaches a 2050 mix of about 53 per cent district heat and 45 per cent heat pumps, with residual fossil at about 1.4 per cent. Its hydrogen share of about a third of one per cent is bounded by a per-country ceiling that is zero in 27 markets, so that program corroborates the district-heat ordering and not the hydrogen verdict.

Third, policy sets the emissions outcome. Across four scenarios sharing a common ≈644 $MtCO_2$ 2025 baseline, 2050 residential-heating emissions range from 352 to 10 $MtCO_2$. Read together, the base-load, dispatch and investment tests converge on one condition. Hydrogen holds base load in seven countries, five of which have the salt caverns that make seasonal storage cheap, wins the cold-snap dispatch only in the cavern endowment itself, and nowhere recovers a peaker's capital from market rent, including there. The three tests therefore identify the narrow and policy-supported conditions under which a hydrogen heating market could exist.

The remainder of the paper is organised as follows. Section 1.1 states the research questions and fixes the scope. Section 2 reviews the literature on buildings decarbonisation, heat pumps, hydrogen for heat, merit order and system value, spatial bottom-up methods, and the Switzerland context, and from it derives the research gap. Section 3 sets out the data sources, the demand trajectory and the four-scenario lever design. Section 4 reports demand, the technology mix, levelised costs, emissions and the base-load hydrogen gap, and Section 5 carries the analysis across the heating-to-power bridge, from the merit order to the firm-capacity requirement and the peaker's capital recovery. Section 6 interprets the results and Section 7 concludes with the policy implications. Four appendices carry the rest, and the first of them is substantial rather than supplementary. Appendix A holds the bottom-up demand build and its validation, the merit-order, missing-money and emissions formulations, the Monte Carlo machinery, the hourly dispatch and unit commitment, the hydrogen supply and last-mile network costs, the district-heating stack, the infrastructure bill and the Switzerland case; a reader following the method will spend as much time there as in Section 3. Appendix B presents the least-cost optimisation and the implied-carbon-price finding, Appendix C tests robustness, and Appendix D states the principal limitations.

1.1 Research Questions and Scope

This study is organised around three research questions, each posed as the answer to one of the three gaps identified in Section 2.8: the missing NUTS3-resolved demand basis for a continental assessment in the updated REPowerEU context (gap one), the absence of a hydrogen-versus-heat-pump comparison on a consistent operating-cost basis that bounds hydrogen's share by where it can win and caps each scenario at or below that bound (gap two), and the severed link between the heating choice and the power sector that electrified heat runs on, of which the import-reliant Swiss case is the sharpest instance (gap three).

RQ1 closes gaps one and two together at the EU scale; RQ2 takes up gap three for Switzerland as a full national case in its own right; and RQ3 names the buildings-to-power coupling that the severed link implies. Each question is carried in full so that the reader can hold the analysis against the question it sets out to settle. RQ1 and RQ2 are answered in full by the analysis that follows; RQ3 is scoped to the buildings-to-power synergy, which this paper prices on the same operating-cost basis as the rest of the analysis, while the cross-sector synergies with industry and transport are treated qualitatively and deferred to future work, the precise boundary being set out in the RQ3 statement and the scope subsection below.

RQ1 (EU, UK and Switzerland, country-level). Under what conditions is decarbonised hydrogen cost-competitive for building heat by 2050 under a net-zero 2050 requirement, where do those conditions hold, and how do the resulting pathways compare on economic and environmental cost bases? RQ1 takes up the first two gaps together. It addresses hydrogen's heating role at NUTS3 resolution across the EU-27, the United Kingdom and Switzerland, and tests the hydrogen-versus-heat-pump trade-off on a consistent operating-cost basis, the trade-off the critical literature finds decisive but parameter-sensitive [5, 32]. It asks where hydrogen is deployed, on what cost basis, and in competition with which alternative, and it answers by deriving hydrogen's heating share from where the fuel can win the cold-snap peak in a merit-order dispatch instead of imposing it as an exogenous target, reading levelised cost of heat, carbon abatement and infrastructure cost off the same modelled pathway.

RQ2 (Switzerland, national case). How can Switzerland make use of the EU's adoption of hydrogen-based technologies to contribute to its own net-zero 2050 aim? RQ2 closes the third gap for Switzerland, a high-cost, hydro-rich, cavern-poor system reliant on imported hydrogen [20, 33], carried through the same framework as a full national case, neither subordinate to the EU-wide assessment nor split off into a separate study, because its distinctive position, a near-decarbonised grid, no domestic salt-cavern storage [36], a reliance on imported hydrogen whose delivered price is the principal lever, and a winter-electricity adequacy gap that any firm-capacity role would have to address [31], warrants investigation on its own terms and on the same operating-cost basis as the rest of the analysis. We examine the sensitivity of the optimal Swiss buildings pathway to the import price across the central, rapid and stranded supply trajectories, and ask under what conditions, if any, hydrogen earns a place in Swiss residential heat.

RQ3 (cross-sector synergy, scoped to the power sector). What synergies can be established between hydrogen in the buildings sector and other sectors? RQ3 names the buildings-to-power coupling that the severed link of the third gap implies. Buildings do not decarbonise in isolation. The electrified heat load reshapes power demand, and any hydrogen used for heat draws on the same production, storage and transport system that serves industry and transport. This paper answers in full the one synergy that follows most directly from its own results, the coupling between buildings-sector hydrogen and the power sector, carrying the electrified heat load into the electricity merit order and assessing hydrogen's role in the firm-capacity stack on the same operating-cost basis (Section 5), the role the system-value literature locates in scarce peak hours, not in annual energy [24, 30]. The wider synergies with industry and transport are treated qualitatively in the Discussion (Section 6); a full cross-sector optimisation lies beyond the present scope and is left to subsequent work.

1.2 Research Scope and Methodological Boundaries

The following scope statement defines the geographic, temporal, and technological boundaries within which RQ1, RQ2, and RQ3 are addressed. The analysis covers residential space heating and hot water demand across 29 countries (the EU-27, the United Kingdom, and Switzerland) at NUTS3 regional resolution. The modelling period spans 2025 to 2050 at milestone years (2025, 2030, 2040, 2050). Three building types are distinguished, single-family homes, multi-family dwellings (including apartment blocks), and other residential buildings, reflecting differences in technology feasibility and retrofit economics. Eight heating technologies compete in each region: gas boiler, oil boiler, biomass boiler, resistance heater, air-source heat pump, ground-source heat pump, district heating, and hydrogen boiler. To this building-level competition the study adds two further layers that test hydrogen beyond the boiler comparison: the district-heating supply stack and the power sector on which electrified heat depends. The power layer commits the firm fleet with an explicit unit-commitment optimisation (ramp, minimum stable output, reserve and start-up constraints).

Three boundaries delimit the study. First, it treats the residential sub-sector only. Commercial and public buildings, industrial process heat, and end uses other than space heating and hot water are excluded. Second, it stops at the meter and the wholesale market for the buildings question itself, with network reinforcement entering through an explicit infrastructure-cost account, short of a nodal grid model. Third, and following directly from RQ3 above, sector coupling beyond the buildings-to-power link is not modelled endogenously. Synergies between hydrogen in buildings and its use in industry or transport are discussed but not co-optimised, and are flagged throughout as a direction for future work. This last boundary is a deliberate design choice, not an omission. The buildings-to-power link is the synergy that follows directly from the paper's own dispatch results and can be priced on the same operating-cost basis. A full industry-and-transport co-optimisation would require a separate model and is held over and not treated superficially here. Within these bounds the paper aims to price hydrogen for heat the way a system operator actually would, and to let its role be decided by that test, not assumed.

2 Literature Review

This review is organised by theme. It moves from the scale and policy framing of buildings decarbonisation (Section 2.1), through the two rival heating pathways of electrification (Section 2.2) and hydrogen (Section 2.3) and the hydrogen supply chain that any heating role would draw on (Section 2.4), to the merit-order and system-value literature that this paper extends (Section 2.5), the spatial and bottom-up methods on which its demand reconstruction rests (Section 2.6), and finally the policy and Switzerland context (Section 2.7). Each theme closes by feeding the gap statement in Section 2.8.

Together, these themes establish the urgency of heating decarbonisation and set out the technical and economic evidence that hydrogen's competitive role, if any, is severely constrained. This paper puts that evidence to the test across 29 countries and three merit-order arenas, and the test does not return a uniform confirmation. Once each carrier is charged its own equipment capital and its own last-mile network, the base-load cost comparison divides geographically instead of running one way across Europe (Section 4), and the case against a spontaneous hydrogen heating market rests instead on capital recovery in dispatch and on the asymmetric fiscal treatment of the two carriers.

Three policy instruments recur throughout the themes below and are defined here before first use. The extension of the European Union Emissions Trading System to buildings and road transport (ETS2) prices the fossil incumbent directly from 2027; the REPowerEU framework, the European Union's response to the 2022 energy crisis, reframed efficiency and electrification as instruments of security of supply; and the Nomenclature of Territorial Units for Statistics at its third tier (NUTS3), the sub-national grain at which this paper reconstructs demand, is the resolution at which heating need and stock heterogeneity become visible. Together these form the policy and spatial scaffold on which the review and the analysis rest.

2.1 Buildings-Sector Decarbonisation: Scale, Drivers and the Policy Frame

The buildings sector accounts for over one-third of European energy-related emissions and is therefore the decisive test of mid-century climate feasibility; its decarbonisation must address three interconnected challenges, namely the structural constraint of slow stock turnover, the heterogeneity of building types and climates across Europe, and the policy instruments (ETS2, the Energy Performance of Buildings Directive (EPBD), and REPowerEU) designed to drive the transition. It is a first-order contributor to global and European energy use and greenhouse gas emissions, and space heating is its largest single end use [37, 38]. In the European Union, buildings account for approximately 36% of energy-related CO_2 emissions [2], a share large enough that the recent synthesis literature treats the sector as decisive for whether mid-century climate targets are attainable at all [7, 39].

The challenge is structural and not merely technical, which reframes decarbonisation as a problem of allocation, not of selecting a single winner. The stock turns over slowly, most of the 2050 building stock already stands, and embodied as well as operational emissions bind the achievable trajectory [40, 41]. A planetary-boundaries assessment of European heating options finds that no single carrier is unambiguously dominant across every environmental dimension, which frames decarbonisation as a problem of allocating the right technology to the right context instead of selecting a universal winner [27].

Decarbonisation strategies for the sector fall into three broad categories that the literature treats as complementary and not substitutable: demand reduction through envelope efficiency, fuel switching to low-carbon carriers, and structural change in district energy systems [7, 42]. The pace required is steep. Comparative scenario work for Europe shows that the speed of envelope renovation and heating electrification needed to reach successive climate goals rises sharply with ambition, and that delay raises the eventual cost [43, 44].

The critical barrier to scaling these strategies is the slow rate of deep renovation, and it is as much economic and behavioural as technical. The European stock renovates at roughly 1% per year, far below the pace implied by net-zero pathways, and deep retrofits remain a small fraction of that flow [45, 46]. Split incentives between landlords and tenants, household budget constraints that push retrofits into partial and staged sequences, and weak information all slow uptake [9, 10]. Decision-support and household-choice studies emphasise that the renovation decision and the heating-technology decision are coupled, because envelope quality conditions the performance of a heat pump and therefore its economics [13, 47].

The European policy frame has tightened around this diagnosis. The European Green Deal, the 2018 long-term strategy, and the climate-neutrality objective set the direction [2, 48]; the recast EPBD of 2024 then converted it into binding instruments, including minimum energy performance standards and a trajectory towards zero-emission buildings [12, 44]. The 2022 energy crisis and the REPowerEU response accelerated the agenda, reframing efficiency and electrification as instruments of security of supply as well as decarbonisation [11, 17]. The official reference projections now carry these policies through to mid-century [49]. The empirical literature on implementation, however, is cautious. It documents wide divergence in transposition and renovation outcomes across member states, and historical analyses of EU-28 residential heat confirm that fuel mixes and consumption levels differ markedly between countries and over time [44, 50]. This heterogeneity, recurring across drivers, stock, and policy, is the first reason a spatially explicit treatment is warranted.

2.2 Heat Pumps, Electrification and District Heating

Heat pumps are the leading near-term pathway for decarbonising building heat, combining high thermodynamic efficiency with a carbon intensity that falls automatically as the grid decarbonises [3, 51]. The reported coefficient of performance places a well-installed unit several times ahead of any combustion boiler in delivered-heat terms, and life-cycle comparisons confirm that the advantage holds in present and projected grids [52, 53]. Air-source systems are increasingly cost-competitive with gas boilers in new build and moderate-renovation contexts, while ground-source systems offer superior seasonal performance at higher capital cost [23, 54].

The central qualification in the literature concerns cold-climate and poorly insulated stock, where the coefficient of performance falls as outdoor temperature drops and the design heat load rises; field and review evidence nonetheless shows that modern units retain workable performance at low temperatures, and that the binding constraint is more often envelope quality and emitter sizing than the device itself [13, 55]. Whole-system and optimal-control studies add that smart and hybrid operation can materially raise the value of electrified heat and ease its peak demand [23, 56]. Adoption, however, is mediated by policy and consumer preference as much as by economics, and several studies show that awareness, trust, and the perceived disruption of installation shape willingness to switch [57, 58].

District heating is the complementary structural pathway, and the literature treats it as the natural home for waste and ambient heat that individual systems cannot capture [42, 59]. The fourth-generation paradigm lowers network temperatures so that large heat pumps, industrial excess heat, and other low-grade sources can supply the stack economically, displacing combustion at the margin [59, 60]. The Heat Roadmap Europe (HRE) programme mapped the spatial economics of distribution and showed that dense urban areas can be served at competitive cost by networks fed from such sources [60, 61]. Cost-allocation and carbon-accounting studies of cogeneration emphasise that the merit order inside a district system, and the emissions credited to it, depend sensitively on how heat and power are jointly priced, a point this paper inherits when it ranks the district-heat stack [62, 63].

Thermal storage and sector coupling further raise the value of electrified district heat by allowing it to absorb cheap off-peak power [64, 65]. Across both the individual and networked routes, the recurring finding is that electrification, suitably staged and stored, is the cost-effective backbone of decarbonised heat; the open question the present paper addresses is what, if anything, is left for a combustible carrier, both at the cold-snap peak that electrified supply finds hardest to serve and on the annual cost of delivered heat once each carrier is charged its own capital and its own network.

2.3 Hydrogen in the Buildings Sector: The Case, the Critique and the Costs

Hydrogen's potential role in building heat is among the more contested questions in the energy-transition literature, and the disagreement runs deeper than terminology. Proponents emphasise infrastructure compatibility, fuel flexibility and the avoidance of deep electrical reinforcement, arguing that repurposed gas networks could deliver hydrogen to existing boiler stock with limited disruption [15, 16]. Technically, hydrogen boilers operate on principles close to those of natural-gas boilers and can in principle use retrofitted distribution pipework, subject to material compatibility and safety limits, while fuel cells offer a higher-efficiency route that supplies heat and power together where simultaneous demand is high [66, 67]. Blending studies map the intermediate option of admixing hydrogen into the gas grid, but identify binding technical and regulatory ceilings well below full substitution [68, 69]. Public-perception research finds that the lower behavioural disruption of a like-for-like boiler swap is part of the appeal, even as it cautions that acceptance is conditional on cost and safety assurances [14, 70].

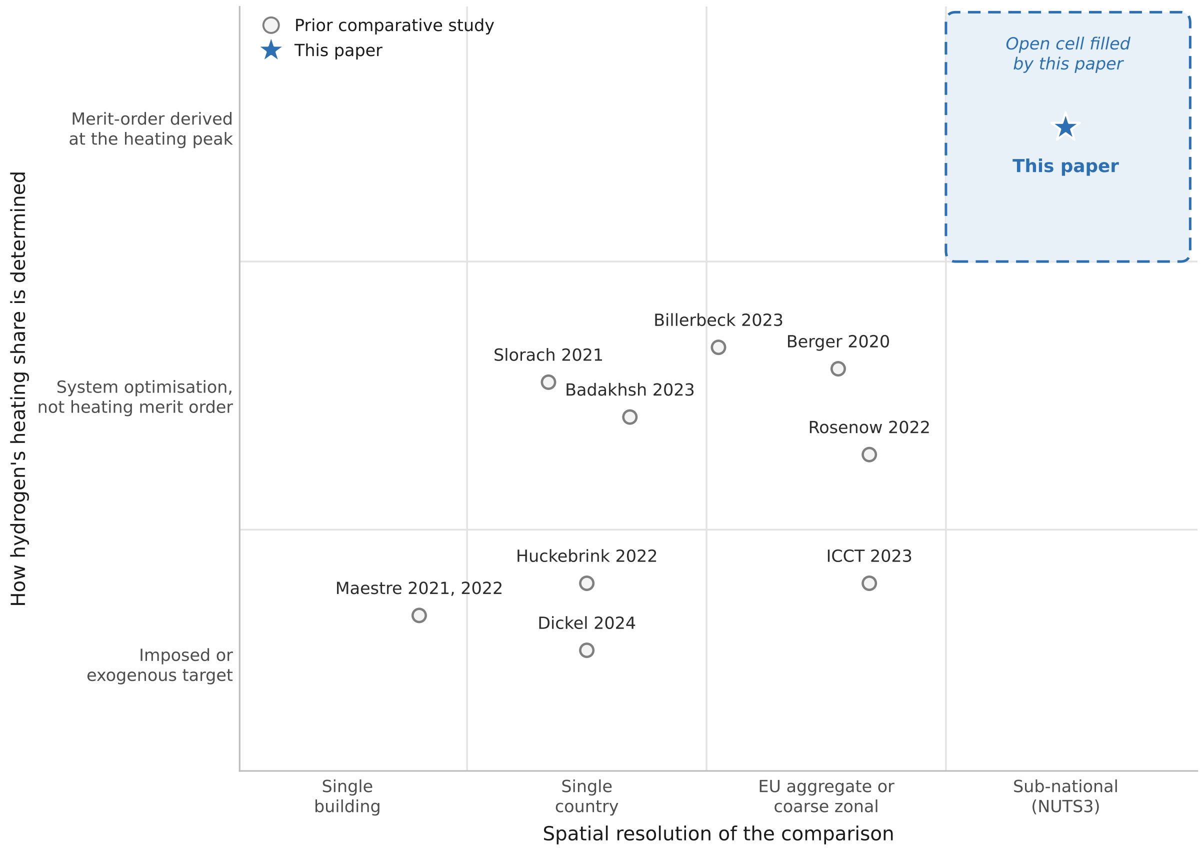
Yet a larger and increasingly convergent critical literature disputes this claim on thermodynamic and cost grounds, and its strands form a single causal chain from which it concludes that hydrogen is implausible for mass-market space heating. The root cause is thermodynamic. Hydrogen produced by electrolysis and burned in a boiler delivers only a small fraction of the useful heat that the same electricity would provide through a heat pump, so hydrogen heating multiplies the renewable generation and electrolyser capacity required [5, 6]. The first consequence of that efficiency multiplier is cost; evidence reviews and independent assessments consistently show that, across the large majority of analysed cases, hydrogen is more expensive than heat pumps or district heating, and its plausible niche is narrow. A meta-review of 54 independent studies finds none supporting a widespread role for hydrogen in heating [5, 71, 72].

The International Council on Clean Transportation (ICCT) assessment of EU households in 2050 and the Oxford Institute for Energy Studies (OIES) analysis of the German heating sector both reach this conclusion, the former across the climates it examines and the latter about a wholesale hydrogen conversion [32, 73]. The second consequence is that the result, while consistent in direction, is brittle in magnitude. Scenario and sensitivity studies show that hydrogen's apparent role is acutely sensitive to input assumptions on green-hydrogen cost, carbon pricing, and infrastructure, so that small parameter changes swing it between marginal and material [74, 75]. The third consequence follows from the first two; once cost and brittleness are imposed within an explicit system model, hydrogen is confined to a residual role, as the model-based race between hydrogen and heat pumps for space and water heating finds, with heat pumps dominating the cost-effective frontier under central assumptions and hydrogen relegated to residual and peaking duty [22].

The dedicated modelling literature on hydrogen in buildings remains limited in scope and scale, covering mostly single countries or technologies, and it divides into two strands that share a common boundary: single-country techno-economic studies, and dispatch- and system-focused work. The first strand, the single-country techno-economic studies, of Spain [25, 76], Italy [77], Germany [26], Ireland [78] and a sub-Saharan context [79], together with national planning assessments [80], each confirm that hydrogen's economic case depends on green-hydrogen cost trajectories, carbon pricing, and infrastructure availability, but none extends the comparison to the full European context. The second strand, the dispatch- and system-focused work, reaches the same conclusion from the other direction. Berger et al. [81] conducted a linear-programming assessment of power-to-gas in cross-sector decarbonisation at coarse spatial resolution, and Slorach and Stamford [82] assessed net-zero heating for the United Kingdom to 2050; both evaluate hydrogen within wider systems but do not isolate its merit-order position against heat pumps in the heating stack across diverse climates. Each strand is thus confined to one country, one technology, or one building, and the two together leave the comparative European question open.

The analytical gap the present paper addresses is the absence of a study that prices hydrogen on short-run marginal cost in a full merit-order dispatch across the entire diversity of European climate and building stock, deriving its heating share endogenously from where it wins the peak instead of imposing it as a scenario target. This is the contribution that distinguishes the present analysis from the literature reviewed above, and it motivates the supply, merit-order and spatial literatures reviewed next. Because the analysis also charges each carrier the equipment capital implied by the peak load it must serve, and the last-mile network its own delivery requires, its base-load comparison departs from the convergent finding above in a defined group of countries, a divergence taken up in Section 4 and traced there to the fiscal treatment of retail electricity, not to the cost of hydrogen. Figure 1 positions the comparative studies cited here on these two axes and shows the resulting gap directly. The cell combining sub-national resolution with a merit-order derived heating share is empty, and it is the cell this paper fills.

Figure 1: **The comparative hydrogen-versus-heat-pump literature.** Placed by spatial resolution and by whether the hydrogen heating share is imposed or derived.



Source: Authors' contribution.

2.4 Production, Supply, Storage and Transport of Hydrogen

The production cost of green hydrogen is the foundation of any heating claim, because any heating role for the carrier draws on its cost and availability, so the production and supply literature bounds the question. The global reviews establish the baseline. Electrolytic (green) hydrogen is presently more expensive than the unabated fossil routes it must displace, but its cost is projected to fall steeply with electrolyser learning and cheap renewable electricity [66, 83]. Observatory and calculator data for Europe place the levelised cost of production and its sensitivity to electricity price, utilisation, and capital cost on a common footing [84, 85]. Techno-economic studies of European and national supply, several using explicit optimisation or Monte Carlo treatment of uncertainty, broadly agree that green hydrogen reaches the range of tens of euros per megawatt-hour by mid-century in favourable locations, while remaining costlier where renewable resources are poor [86, 87]. Blue hydrogen from natural gas with carbon capture and pink hydrogen from nuclear power are characterised as transitional or niche options whose competitiveness depends on gas prices and capture performance, and which the European cost literature does not find cheapest at mid-century [88, 89].

The delivered cost of hydrogen is dominated by seasonal storage and transport, both geographically concentrated. Production cost is only a busbar figure (the cost at the production gate, before storage and transport). What a heating comparison turns on is the delivered, stored cost at the point of combustion. Heating demand is acutely seasonal and the cold-snap peak is precisely when a dispatchable hydrogen unit would run, so the cost of holding hydrogen from summer production to winter use, and of moving it to where the stock sits, can exceed the production cost itself; both layers are therefore as spatially variable as production, and both must be resolved at the same country grain. The technical-potential mapping of salt caverns shows that low-cost seasonal storage is geographically concentrated in a minority of European countries, which the present paper translates into a country-specific storage adder [36].

Network repurposing and import costs vary by country and by delivery method. Network studies of the European Hydrogen Backbone and gas-for-climate pathways quantify the cost of repurposing gas pipelines and building dedicated hydrogen corridors, and of per-property conversion at the distribution edge [90, 91]. Trade and import analyses, including the OIES import-route assessments and the IRENA trade work, characterise the delivered cost of hydrogen reaching European markets by pipeline and as ammonia, and the diseconomies of small-scale shipping [92, 93]. Optimisation studies of national supply chains and seasonal power-to-hydrogen storage show that transport technology choice and the value of inter-seasonal storage are themselves model outputs sensitive to geography [94, 95]. The implication this paper carries forward is that the delivered, stored cost of hydrogen at the point of combustion, not the busbar production cost, is what a heating or peaking comparison must use, and that this cost is irreducibly spatial. The further implication is that hydrogen's heating role can only be assessed through the merit-order dispatch framework that allocates resources to the unit running at short-run marginal cost. This framework, reviewed next, is the paper's analytical core.

2.5 Merit Order, Capacity Value and the Missing-Money Problem

The analytical core of this paper is drawn from the electricity-market literature on short-run marginal cost, capacity value, and missing money, which it applies to heat. The foundational result is that competitive dispatch prices energy at the short-run marginal cost of the marginal unit, so that capital, being sunk, does not set the spot price; a peaking plant that runs few hours must therefore recover its fixed costs from scarcity rent and not from energy margin [28]. When that rent is insufficient, the plant faces the classic missing-money problem [28] and will not be built without a capacity mechanism [96].

The carbon-pass-through literature [97] establishes the complementary mechanism that this paper finds recurring. An emissions price is passed into the marginal cost of fossil generation and thereby into the market price, so that it is the carbon charge on the fossil competitor, more than any fall in the price of the clean alternative, that reorders the stack. Studies of technology value in decarbonising power systems show that the system worth of a firm, dispatchable resource lies in the scarce hours it covers, not in the energy it supplies, which motivates separating a capacity role from an energy role [24, 98].

A direct line of work asks where hydrogen and related power-to-X options sit in this picture, and its findings frame the present analysis. The same conclusion, that hydrogen turbines and electrolytic flexibility are valuable chiefly for covering rare residual-load peaks and bridging low-renewable spells, a capacity rather than an energy contribution, recurs across model scopes. For highly renewable Europe, the sector-coupled optimisation of Brown et al. [24] and the heat-demand-peak study of Zeyen et al. [30] both place hydrogen in this residual role; for Great Britain, the integrated value-chain optimisation of Quarton and Samsatli [29] reaches the same finding at the national scale; and at the global scale, the fully renewable transition pathways of Bogdanov et al. [99] and Victoria et al. [100] likewise confine dispatchable hydrogen to bridging the residual peak. That the result holds from the national through the continental to the global scope is what makes it robust to local conditions. The complementary peak-mitigation and climatological-adequacy literature shows that the depth of the cold, low-wind event (the Dunkelflaute) governs how much firm capacity the system needs, and that this need is robust but bounded [30, 31].

Heat-specific applications of the same logic, including studies of peat displacement and thermal storage in district systems, confirm that dispatchable units at the top of a heat stack run for limited hours and so face the same recovery problem [63, 64]. To operationalise this framework, the paper draws its cost and performance parameters from standard technology databases, European electricity-market reviews and grid-intensity indicators, and carbon-price trajectories from ETS2 modelling, with the sources and assumptions detailed in Section 3. The European market reviews and grid-intensity indicators that anchor present-day electricity prices and carbon intensities are by now a settled empirical base for such comparisons [101, 102]. This paper's contribution is to carry the merit-order, capacity-value and missing-money triad, established for power, into the heating peak and the district-heat stack, and then back into the power sector that electrified heat runs on, applying one consistent operating-cost test across all three arenas.

2.6 Spatial and Bottom-Up Methods for Building-Stock Modelling

Regional heterogeneity is a first-order determinant of both heating demand and technology feasibility, so the method by which demand is built matters as much as the technologies compared. Climate zones, vintage distributions, urbanisation and national regulation all vary substantially across Europe, and aggregate national models can obscure the spatial targeting that effective policy requires [60, 103]. The bottom-up tradition reconstructs demand from physical stock characteristics, where the top-down tradition infers it from national totals, and reviews of the approach set out its strengths and its data demands [104, 105]. Three data infrastructures underpin the present reconstruction. The Hotmaps project provides an open NUTS3 heat-demand dataset and toolbox derived from stock and climate data, which this paper uses as an independent validation reference [103, 106]. The EUBUCCO database assembles harmonised footprints and attributes for more than 200 million individual European buildings, supplying the geometric basis for a footprint-up demand estimate, and is complemented by emerging global building datasets [34, 107]. The TABULA typologies translate stock into energy by attaching per-class, per-cohort space-heating and hot-water intensities across European countries, with national brochures providing the country detail [35, 108].

Census, geographic, and observatory sources complete the reconstruction and its benchmarking. Dwelling counts and building-type splits come from the European census and national equivalents, with the NUTS reference geographies aligning the spatial units [109, 110]. The EU Building Stock Observatory and the Odyssee-MURE profiles provide independent national totals and intensities for multi-source reconciliation, alongside the peer-reviewed historical series for EU-28 residential heat [111]. Heating demand is driven by climate through heating degree days, whose European datasets and projected changes the present analysis uses for the climate multiplier and warming sensitivity [112]. Recent work disaggregating energy consumption and emissions to municipal and sub-national resolution for individual countries demonstrates both the feasibility and the value of high-resolution stock modelling, while remaining national in coverage [113]. On the network side, sub-national European hydrogen work does exist and is continental in scope [114, 115]; what it returns is a network plan rather than a per-market balance sheet for a single plant.

On the choice and dispatch side, the technology-share step in this paper draws on discrete-choice theory [116, 117], the least-cost layer on linear-optimisation generators and reviews of bottom-up energy-system models [118, 119], the multi-criteria reconciliation on compromise-programming methods [120, 121], the demand decomposition on the LMDI index method [122], and the uncertainty treatment on reviews of uncertainty assessment and time-series aggregation in energy-system optimisation [123, 124]. What is missing from this otherwise rich methodological literature is a study that combines a footprint-up demand reconstruction at NUTS3 resolution with an explicit merit-order test of hydrogen across the full European country set; the methods exist in parts but have not been assembled for this question.

2.7 Policy, Carbon Pricing and the Switzerland Context

The policy and carbon-pricing literature supplies the levers that drive this paper's scenarios, and the recurring mechanism it documents is that pricing the fossil incumbent, more than subsidising the alternative, reorders the heating stack. ETS2 is the central instrument. Modelling of its price trajectory spans a wide range depending on complementary policies, and the official framework and independent projections place it at the heart of buildings decarbonisation after 2027 [125, 126]. National measures increasingly act on the same margin. Boiler phase-out schedules, building-energy acts, and future-homes standards remove the fossil option directly, complementing the price signal [127, 128]. The security-of-supply literature shows how the 2022 crisis recast gas demand reduction and electrification as strategic as well as climate objectives, and how gas-network access and household exposure vary across Europe [18, 129]. The price and consumption series that calibrate the scenarios are drawn from Eurostat and the national regulators [130, 131].

Switzerland occupies a distinctive position that the limited literature treats as a case apart: a non-EU member deeply integrated with EU energy markets, surrounded by countries with ambitious hydrogen strategies, and with substantial hydropower and nuclear generation that could support domestic low-carbon hydrogen [20, 21]. The Energy Perspectives 2050+ and the national and cantonal strategies set the decarbonisation pathway, and the 2024 national hydrogen strategy and climate legislation frame the hydrogen question specifically [132, 133]. Country-specific studies of Swiss heat decarbonisation, building-stock thermal performance and the energy-performance gap establish that the Swiss stock is comparatively well insulated but heavily reliant on fossil heating, and that electrification and district heating dominate the cost-effective options [134, 135].

The winter-electricity-deficit literature is particularly relevant, since it identifies the cold-season adequacy gap that any Swiss firm-capacity or hydrogen-peaking role would have to address, and the domestic tariff data bound the comparison [31, 136]. The Swiss case in this literature, however, is examined largely on its own terms. The lacuna is specific. No prior study has examined whether Switzerland, positioned at the intersection of EU hydrogen ambitions and its own net-zero commitment, can justify hydrogen imports for residential heating given its existing low-carbon grid and competing district-heat expansion. The question of how imported decarbonised hydrogen, priced and stored, would compete in Swiss building heat under a net-zero-by-2050 constraint, which motivates Research Question 2, has not been addressed.

2.8 Research Gap

The foregoing review reveals three gaps that this paper addresses, each emerging from a theme above. First, the buildings, hydrogen, and spatial literatures together establish that heating demand and technology feasibility are irreducibly heterogeneous across

Europe, yet no continental, quantitative assessment of hydrogen's role in EU buildings decarbonisation has been built on a bottom-up heat-demand reconstruction resolved to NUTS3, in the updated REPowerEU policy context. Second, the hydrogen-in-buildings and merit-order literatures show that hydrogen's apparent role is highly sensitive to how it is priced and to local conditions, yet no prior study has systematically evaluated the hydrogen-versus-heat-pump trade-off on a consistent operating-cost basis across the full diversity of European building stock and climate, deriving hydrogen's share from where it can win the peak instead of imposing it.

Third, Switzerland's specific strategic position, surrounded by hydrogen-ambitious EU members, with substantial hydroelectric and nuclear capacity, has not been modelled in a comparative context. Specifically, whether imported decarbonised hydrogen can compete with electrification in Swiss residential heating under net-zero constraints remains unaddressed, and it is this that motivates Research Question 2. This paper addresses all three gaps through the integrated modelling framework described in Section 3.

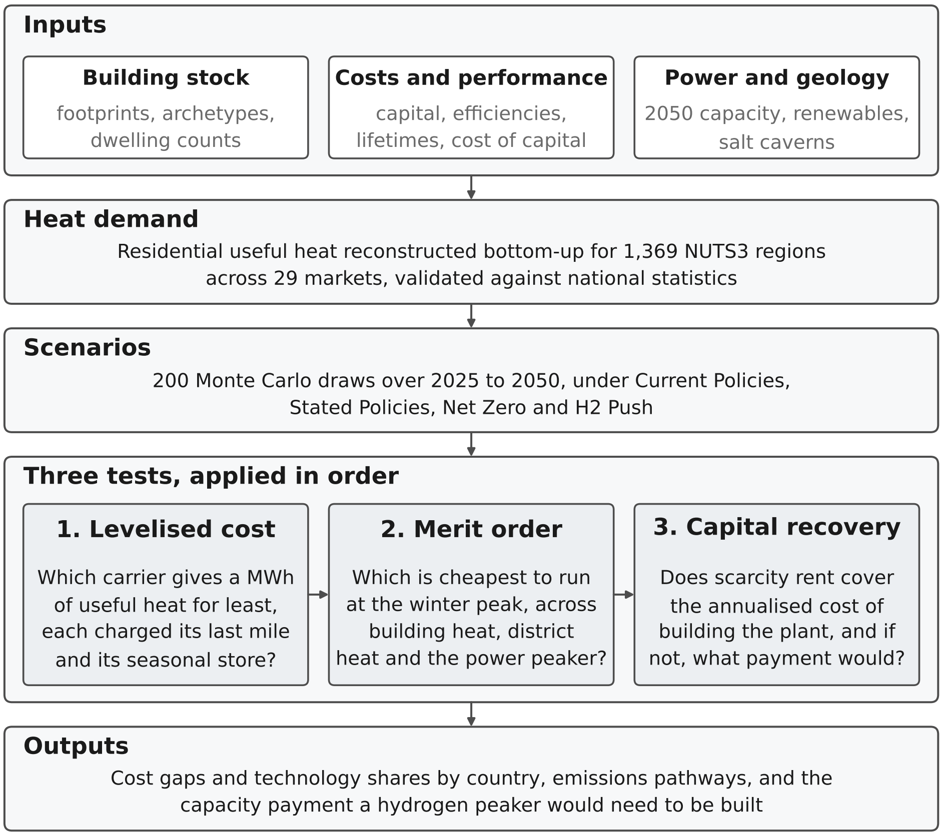
3 Data and Methodology

3.1 Model Overview

The model is built demand-first; a NUTS3-level estimate of useful heat demand is established first, building-type shares are then attached to it, and technology shares are applied last to split that demand across heating technologies for each region and year. The useful-heat surface is drawn from the Hotmaps regional database [103], building-type shares from the Eurostat Census 2021 [109] and the United Kingdom Census 2021 accommodation tables [110].

Demand is reconstructed along two independent strands that are reconciled against each other (Figure 2). A *top-down* baseline takes NUTS3 heat demand from the Hotmaps regional database and attaches census-derived building-type shares; an independent *bottom-up* build reconstructs residential heat demand from open building-footprint data (EUBUCCO) and a harmonised building typology (TABULA). The two agree to $-0.8\,\%$ across the EU-27, the United Kingdom and Switzerland and to within $\pm 25\,\%$ in 28 of 29 countries (Appendix A.1, Figure A.1). The headline results reported here use the *bottom-up* demand basis, with the Hotmaps top-down baseline retained as the independent reconciliation benchmark. The single reconciled demand surface then propagates through the shared technology, cost and policy layers and through the scenario Monte Carlo machinery, so the choice of demand basis does not propagate a different downstream treatment; the merit-order and missing-money tests that carry the paper's hydrogen findings (Appendix A.2) sit at the end of that common chain. The header band of Figure 2 sets out the three input layers that feed the pipeline and the policy frame they run inside.

Figure 2: Model architecture. Inputs, the demand reconstruction and the scenario draws feed three tests applied in order, and a technology that fails an earlier test is not rescued by a later one. The first two are cost comparisons at different time scales, annual and peak-hour; the third asks whether the plant that wins the peak can be financed at all. The subsections below give the data sources and parameter ranges the boxes summarise.



Source: Authors' contribution.

3.2 Data Sources

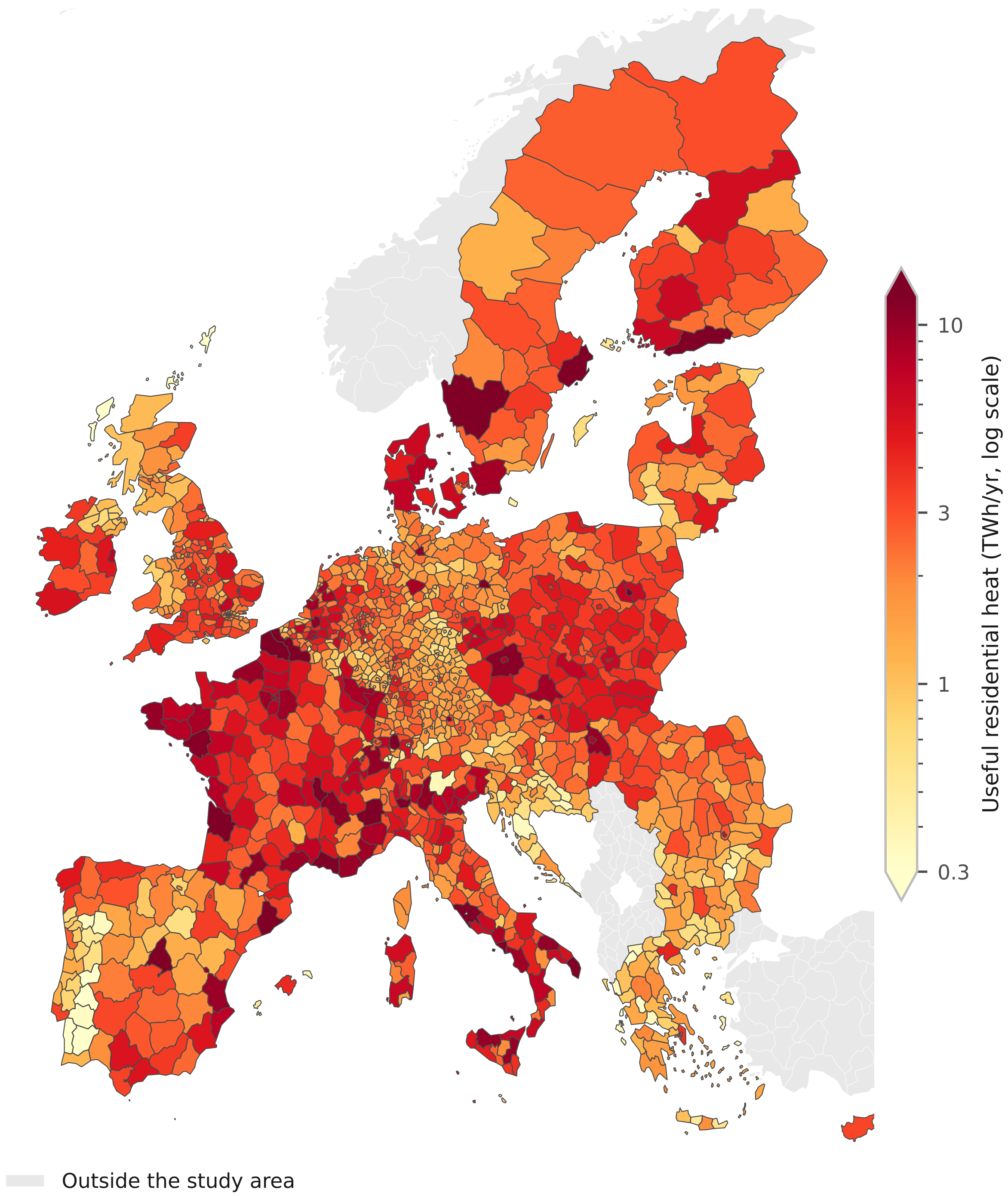
3.2.1 Heat Demand Baseline

NUTS3-level useful heat demand for space heating and hot water is taken from the Hotmaps regional demand dataset [103], which estimates demand from building-stock characteristics and climate data for the EU member states. The reference year is approximately 2015. Coverage extends to 1,369 NUTS3 regions across the EU-27, Switzerland and the United Kingdom, totalling 3,863 TWh yr^{-1} of residential heat demand. The spatial gradient in demand is steep and is driven jointly by climate severity and population density, as Figure 3 shows.

3.2.2 Building Stock

Building-type shares at NUTS3 level for the EU-27 and Switzerland are derived from the Eurostat Census 2021 microdata table CENS_21DWBNO_R3 [109], which reports

Figure 3: Useful residential heat demand by NUTS3 region (bottom-up build). The colour scale is logarithmic and runs from 0.3 to 12 TWh/yr, decade-friendly bounds close to the 2nd and 98th percentiles of the 1,359 regions the map draws, which are 0.36 and 10.2; the arrowheads mark the 1.0 per cent that fall below the range and the 1.4 per cent above it. The darkest regions are metropolitan. Rome, Nord, Naples, Turin, Milan, Stockholm, Paris, Madrid, Warsaw, Barcelona and Berlin lead the distribution, and within France the eight darkest are Lille, Paris, the Lens and Arras belt, Bordeaux, Marseille, Lyon, Nantes and Grenoble.



Source: Authors' contribution.

dwelling counts by building type and NUTS3 region. Three model building types are defined: single-family homes (SFH, corresponding to Eurostat codes RES1 and RES2), multi-family and high-rise buildings (MFH, code RES_GE3), and other residential (the remaining codes). For the United Kingdom, national building-type shares are derived from the ONS Census 2021 TS044 dataset (accommodation type by lower-tier local authority) [110] and applied uniformly to the UK NUTS3 regions.

In the top-down census split, single-family homes account for 769 TWh (20 per cent) of the 3,863 $\mathrm{TWh\,yr^{-1}}$ baseline, multi-family or high-rise buildings for 594 TWh (15 per cent), and a large 'other/mixed' residual for the remainder (2,501 TWh, 65 per cent). That residual is a known artefact of the Hotmaps stock disaggregation, which over-counts non-residential records; it is addressed in the Limitations section and is one motivation for the bottom-up build, whose class composition is markedly cleaner (Appendix A.1).

3.2.3 NUTS3 Reference Geography

NUTS3 regional boundaries and typology codes (urban or rural, coastal, mountain) are sourced from the Eurostat GISCO service [137] at 1:1 million scale, coordinate reference system EPSG:4326.

The detailed treatment of bottom-up validation of the residential building stock is given in Appendix A.1.

3.3 Heat Demand Trajectory

Useful heat demand is projected forward from a base year $t_0 = 2025$ with a per-country, multi-driver ratio identity, and the historical 2015 $\rightarrow$ 2025 change is separately attributed across the same drivers with a Logarithmic Mean Divisia Index (LMDI; 122) decomposition. The vintage-matched backcast set out below is a third and separate construction, and it scales by dwelling stock alone. The LMDI is an exact decomposition that isolates the independent contribution of each driver to a scalar outcome; we adopt it because it is factor-neutral and path-independent, leaves no unexplained residual term, and assigns each driver a unique weight without the ad hoc parameterisation that arithmetic-mean Divisia, Fisher-ideal or regression-based alternatives require [122]. The base-year demand D_0 is the NUTS3 region/building-type baseline of Section 3.2 (the Hotmaps estimate in the top-down chain; the EUBUCCO and TABULA snapshot in the bottom-up specification), treated as the demand level in t_0. The model operates at the NUTS3 region and building-type level and aggregates to the country before the trajectory is applied. National useful heat demand then evolves as the 2025 baseline scaled by three demographic and efficiency drivers: population growth and household-size decline (occupancy), both of which add demand, set against envelope-renovation intensity, which removes it. The formal decomposition is

$$D_c(t) = D_c^0 \, \underbrace{\frac{\mathrm{Pop}_{c,t}}{\mathrm{Pop}_{c,25}}}_{\text{population}} \; \underbrace{\frac{(\mathrm{Dw/Pop})_{c,t}}{(\mathrm{Dw/Pop})_{c,25}}}_{\text{occupancy}} \; \underbrace{\left(1 - r_{c,s}\right)^{t-t_0}}_{\text{envelope}}, \qquad t \geq t_0, \; t_0 = 2025, \tag{1}$$

where $D_c^0 = \sum_{r \in c,\, b} D_0(r, b)$ is the national base-year demand, $\mathrm{Pop}_{c,t}$ is projected population (the Eurostat EUROPOP2023 baseline for the EU-27, the ONS 2022-based projection for the UK, and the BFS Referenzszenario A-00-2025 for Switzerland), $(\mathrm{Dw/Pop})_{c,t}$ is dwellings per capita (the inverse of average household size, linearly extrapolated from the observed 2015 to 2023 Eurostat `ilc_lvph01` trend), and $r_{c,s}$ is the per-country, per-scenario *envelope-design* intensity decline rate, so the closing factor $\left(1 - r_{c,s}\right)^{t-t_0}$ is an exponential decline in the building envelope's design heat demand at the annual rate $r_{c,s}$. Average dwelling size (m^2/Dw) is held flat over the projection window, since national series drift by less than 2 per cent per decade.

The occupancy term captures the increased average heat demand per capita when household size shrinks and dwellings are occupied by fewer residents; it assumes the built envelope (m^2/Dw) and the heated floor area per m^2 remain constant, so a rising dwellings-per-capita ratio raises aggregate demand by adding dwellings to heat, not by enlarging each dwelling. Moving the anchor from 2015 to t_0 keeps the demand path consistent with the technology-share, efficiency and turnover ramps, which all originate in 2025, and ensures that the base-year demand equals the baseline estimate, not that estimate net of a decade of assumed efficiency gains. The LMDI form makes the demand reduction an *emergent* quantity, the net of population growth and household-size decline (both of which add demand) against envelope renovation (which removes it), rather than an exogenously imposed 2050 target.

The parameter that varies by scenario is the envelope-design intensity decline rate $r_{c,s}$, read per country and scenario from the scenario parameter register accompanying the supplementary material and sampled $r_{c,s} \sim \mathcal{N}(\mu_{c,s}, \sigma_{c,s})$ (truncated to 0 to 5 per cent/yr) in the Monte Carlo. The central values are anchored on each country's current envelope-relevant renovation pace (EU stock-weighted ≈0.55 to 0.7 per cent/yr; 45, 138). **Stated Policies** holds the announced-policy pace; **Current Policies** multiplies it by 0.70 (current-measures-only, so renovation slows); **Net Zero** multiplies it by 2.73 (the EU-average 1.5 to 1.9 per cent/yr implied by a 3 per cent/yr renovation rate at ≈50 per cent envelope depth, which is the obligation the Energy Efficiency Directive places on public buildings and roughly the deep-renovation pace advocacy work argues the stock needs, above the Renovation Wave's own commitment to double the rate to about 2 per cent/yr); and **H2 Push** multiplies it by 1.64 (an intermediate pace, since hydrogen substitution does not alter the envelope). Because the same multiplier is applied to every country, the scenarios scale each country's own baseline *proportionally* instead of converging all countries to a common rate, so the cross-country renovation-pace heterogeneity of

Stated Policies is preserved in every scenario (Appendix D).

Vintage-matched reconciliation. Although the bottom-up build is used for all reported results without calibration to Hotmaps, the vintage-matched reconciliation provides an independent 2015 cross-check that the underlying methodology is sound, not a step that adjusts the baseline. In the bottom-up specification D_0 describes the present-day stock (EUBUCCO ≈2020 to 2023), and the Hotmaps validation benchmark is dated 2015, so the two are separated by roughly a decade of stock and efficiency change and cannot be compared directly. We therefore backcast the national base-year demand $D_c^0 = \sum_{r\in c,\,b} D_0(r,b)$ to 2015 so that the two independent estimates can be tested for convergence once vintage is accounted for. The backcast is deliberately not the forward identity of Eq. (1) run backwards. It scales the snapshot by dwelling-stock growth alone:

$$D_c(2015) = D_c^0 \left(1 + g_c\right)^{-10}, \tag{2}$$

where g_c is the country's net annual dwelling-stock growth rate, running from 0.15 per cent a year in Portugal to 2.5 in Malta on a 0.90 median, sourced from Eurostat (households, census, populations) and cross-checked with national statistical offices (Destatis, INSEE, ISTAT, INE, CBS, SCB, GUS, ONS/MHCLG, CSO, KSH, INS, NSI, CYSTAT, CSB, STATEC, SURS, NSO Malta and BFS; the UK and Switzerland from ONS and BFS respectively). Across the 29 countries this compounds to an aggregate factor of 0.924. The quantity being matched is a stock vintage and not a demand pathway, and the stock series is the one input the two estimates share, which is why the backcast carries no population, occupancy or envelope term. It follows that the vintage-matched deviation sits below the raw one by construction and is the more conservative of the two, not a correction to it.

The intensity factor r_c is the country's *envelope-relevant* renovation rate, derived from the renovation depth bands of the European Commission's renovation-activity study [45] with envelope-saving weights $r_c = 0.50\,\mathrm{deep}_c + 0.25\,\mathrm{medium}_c + 0.05\,\mathrm{light}_c$ (EU stock-weighted $\sim 0.55\,\%\,\mathrm{yr}^{-1}$). The weights are the fraction of each depth band's renovations that delivers a lasting cut in envelope-design demand, set from the distribution of renovation depths in that study [45]; deep retrofits are credited at half, medium at a quarter and light at a twentieth, with the full derivation given in the supplementary material. The envelope rate is used because the full metered Eurostat intensity decline would credit conversion-efficiency improvement (boilers, heat pumps) and behavioural prebound response, neither of which changes the *design* demand the model represents. The vintage-matched deviation against the Hotmaps benchmark H_c^{2015} is

$$\Delta_c^{\mathrm{VM}} = \frac{D_c(2015) - H_c^{2015}}{H_c^{2015}}. \tag{3}$$

Across the EU-27+UK+CH aggregate, the four LMDI drivers contribute approximately +1.7 % (population), +4.6 % (occupancy, that is, shrinking household size), 0 % (dwelling size) and −6.3 % (envelope retrofit) to the change 2015 → 2025 in TWh terms. The two stock-side drivers sum to +6.3 %, matched almost exactly by the envelope-retrofit term, so the net change across the EU-27+UK+CH is only +0.06 %, a near-coincidence of this particular decade. The EU's actual energy renovation rate has remained close to its long-run $\sim 0.55\,\%\ \mathrm{yr}^{-1}$ envelope-relevant baseline through 2015 to 2023, without accelerating towards either the Renovation Wave's doubling commitment or the $3\,\%\ \mathrm{yr}^{-1}$ pace used as the Net Zero anchor [138]; population and household decomposition added stock at a similar pace, leaving the net flat.

The matched-vintage deviation is $\Delta^{\mathrm{VM}} = -8.3\,\%$ across the EU-27+UK+CH, within the ±15 % acceptance band, so the model passes its independent 2015 validation against Hotmaps without any Hotmaps-targeted calibration. Per-country deviations range from −32.3 % (Luxembourg, where fast population growth sits on a low stock) to +20.1 % (Slovenia, followed by Latvia at +18.0 and Bulgaria at +16.9, where the comfort-regime deflator is already applied but residual stock factors remain). On the raw and not the vintage-matched basis the range is −21.1 % (the United Kingdom) to +35.6 % (Malta). The per-country acceptance bands of the build are set on the raw deviation and are not recomputed on Δ^{VM}. An independent Eurostat-derived 2015 reference (climate-corrected to the WMO 1991 to 2020 normal) is reported alongside; it captures *final* metered energy and lands at ∼ 54 % of the Hotmaps useful demand. The prebound and performance-gap effect runs in that direction, at a median of about −11 per cent in metered stock studies [139], so it accounts for part of the gap and not for all of it.

The lower Eurostat anchor. That gap is large enough to ask what the results would look like on the lower anchor, so we re-run the chain on it. Every ranking survives. The heat pump is still cheapest in 22 of 29 markets, the same seven parity markets appear, the salt-cavern power-peak wins are still seven, peaker capital recovery is still 9 per cent, and per-country emissions stay monotone in ambition.

Every extensive quantity moves roughly in proportion to the anchor itself. The 2050 emissions fan falls from 352 (Current Policies), 231 (Stated Policies), 123 (H2 Push) and 10 (Net Zero) to 192, 126, 67 and 5 $\mathrm{MtCO_2}$, and the firm peaker fleet falls from 278 to about 151 GW and the cumulative peaker shortfall from €357 to about €194 bn. Each of those quantities is derived where it is first reported, the emissions fan in Section 4.4 and the fleet and the shortfall in Section 5; they stand here only as the scale the anchor sets. The anchor therefore sets the scale of every level in this paper and none of its orderings, which is the sense in which the choice of anchor is conservative for the comparison at issue.

Three-way benchmark. A three-way benchmark places the bottom-up build against two further independent sources. Against Hotmaps the aggregate gap is −0.8 per cent, against the EU Building Stock Observatory −11.9 per cent, and against Odyssee-MURE +52.9 per cent, the last because Odyssee reports final metered energy and so carries the same prebound gap as the Eurostat reference. The build therefore sits between the two useful-heat sources and well above the metered one, which is where a useful-heat reconstruction should sit.
The detailed treatment of the technology mix and its underlying techno-economic assumptions is given in Appendix A.9 and Appendix A.11, while the fuel-price assumptions and the hydrogen supply-cost and distribution-infrastructure build are set out in Appendix A.12 and Appendix A.13 respectively.

3.4 Scenario Definitions

Four multi-lever scenarios are defined to bracket the plausible range of European buildings decarbonisation trajectories to 2050 (Table 1).

Table 1: Scenario definitions, lever settings and outcomes. The three 2050 share rows are Monte Carlo means and not hard caps. Each draw takes the tabulated value as the centre of a normal, with a standard deviation of 5 percentage points on the heat-pump share, 4 on the fossil share and 3 each on the district-heat and hydrogen shares, censored at its admissible interval rather than resampled. A realised median can therefore sit a little above or below its tabulated setting. Hydrogen's drawn value is applied as a hard per-draw ceiling after the cost reallocation, but the ceiling binds that draw rather than the tabulated mean, so the realised medians sit just above their settings in all four scenarios, at 0.5, 5.5, 3.5 and 8.5 per cent against 0, 5, 3 and 8.

Lever	Current Policies	Stated Policies	Net Zero	H2 Push
Envelope rate (vs announced)	×0.70	×1.00	×2.73	×1.64
Carbon price path	LOW	CENTRAL	HIGH	HIGH
Hydrogen price path	STRANDED	CENTRAL	CENTRAL	RAPID
Grid-decarbonisation speed	×1.15 (slower)	×1.00	×0.70 (faster)	×1.00
Fossil share 2050 (ambition)	40%	27%	5%	15%
H_2 ceiling 2050 (set $\leq$ derived bound)	0%	5%	3%	8%
Demand change 2050 (outcome)	−5%	−9%	−31%	−18%

Source: Authors' contribution.

The four scenarios span the plausible range of buildings decarbonisation to 2050. **Current Policies** freezes policy at today's implemented measures. Renovation slows, the carbon price stays low, hydrogen markets strand, and the grid decarbonises 15 per

cent slower than announced, the implementation-gap bookend. **Stated Policies** delivers announced policy (EPBD bans, ETS2 central, NECP renovation), broadly consistent with the IEA World Energy Outlook trajectory and the European Commission's 2018 long-term strategy [48]. **Net Zero** represents climate-neutrality-consistent ambition in line with the IEA Net Zero scenario [140]: Renovation-Wave-pace envelope upgrades, a high carbon price, and a fast-decarbonising grid. **H2 Push** is the deliberately hydrogen-favourable world envisaged in REPowerEU [11], in which hydrogen prices fall rapidly along the rapid delivered-hydrogen path [93], testing how large the buildings niche can become once the fuel-cost barrier is removed. Unlike earlier exogenous-share designs, the hydrogen share in every scenario is capped at the ceiling *derived* from the merit-order dispatch analysis (Section 4.6). Hydrogen wins the share its peak-slice economics support, and each scenario's ceiling is then set at or below that bound. All levers are sampled stochastically around their central values in the Monte Carlo.
The detailed treatment of peak dispatch, peaking investment and network capital is given in Appendix A.2, and that of the uncertainty quantification in Appendix A.3.

4 Results

One finding orders this section. Cost and policy set the decarbonisation pathway, and technology availability does not constrain it. Demand is sticky, so the transition is supply-side; the base-load cost ordering settles in the heat pump's favour once each carrier is charged the storage its own load shape requires; and policy ambition separates the scenario outcomes. Hydrogen is the test case for that claim. It is the cheaper base-load option in only seven markets, five of them with the geology that makes seasonal storage cheap; it fails the investment test wherever it wins the peaking dispatch; and the niche that survives needs policy support, so it does not drive a spontaneous transition at any scale. The 200-sample Monte Carlo simulations across the four multi-lever scenarios (Current Policies, Stated Policies, Net Zero and H2 Push; Section 3) are reported in that logical order.

The first four subsections establish the spine of demand, technology mix, cost ordering and emissions. The remaining subsections then put hydrogen through three explicit tests that together answer the first research question (RQ1; Section 1.1), whose two facets, base-load competitiveness and peaking competitiveness, are distinguished below.

Test 1, the building-level base-load levelised cost of heat (Section 4.5), asks in how many countries a hydrogen boiler undercuts the best heat pump in 2050. *Test 2*, the peaking dispatch (Section 4.6), reprices hydrogen as the dispatchable source that serves only the cold-snap peak, at the building, grid and district-heating scales. *Test 3*, the investment-feasibility check nested within it (Section 4.7 and Appendix A.7), asks whether a peaker that wins the dispatch can also recover its capital from market rents. Appendix A.15 then compares the network-infrastructure bill of each pathway, and Appendix A.8 applies the cost model to Switzerland as a standalone case bearing on the second research question (RQ2).

The buildings-to-power coupling named by RQ3 is carried in the dedicated power-sector treatment (Section 5) and enters here only through the peaking tests.

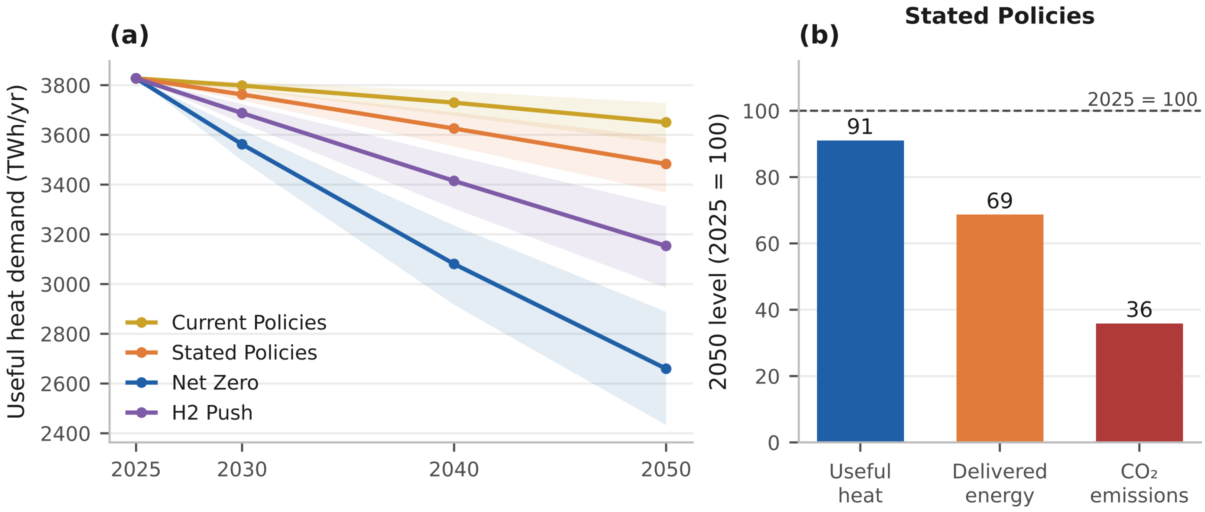
4.1 Useful Heat Demand Trajectory across the 29 Markets

Useful heat demand across the 29-country study area barely falls to 2050, because envelope renovation only just outpaces the growth of the building stock. Demand therefore falls far less than emissions do over the period, and far less than the policy ambition of the deeper scenarios might suggest. In the 2025 base year it is estimated at approximately 3,827 TWh per year, consistent with the Hotmaps regional baseline [106]. The three closely-spaced baseline totals that appear in this paper are on different bases. The 3,827 TWh here is our bottom-up 2025 reconstruction, the 3,858 TWh in the validation table (Table A.1) is the Hotmaps 2015 all-class benchmark we validate against, and the 3,863 TWh quoted in the methodology is the raw Hotmaps total before the all-class alignment.

The forward trajectory is a per-country LMDI multi-driver projection over population, occupancy and envelope-design intensity (Section 3). Under the Stated Policies scenario, in which announced renovation policy is delivered and envelope renovation proceeds near its current EU pace (approximately 0.7 per cent per year; [138]), median demand declines only modestly, to 3,762 TWh by 2030, 3,626 TWh by 2040 and 3,483 TWh by 2050, a cumulative reduction of 9 per cent relative to 2025 (Figure 4). The 80 per cent Monte Carlo interval at 2050 spans 3,367 to 3,589 TWh. Because the per-country envelope-renovation rates are sampled with a shared EU-policy correlation ($\rho = 0.5$; Appendix A.3) rather than independently, this 29-country aggregate band reflects the limited cross-country diversification of a common policy environment, and is about 2.2 times wider than a naive independent-draw band (Appendix C, Table C.3).

The four scenarios span a wide demand range by 2050, but a narrow one relative to their emissions spread. Current Policies, with renovation slowed to its current-measures-only pace, reduces demand by just 5 per cent, to 3,651 TWh. Net Zero achieves the deepest reduction, to 2,660 TWh (a fall of 31 per cent), through Renovation-Wave-pace envelope upgrades, with envelope intensity falling approximately 1.9 per cent per year in the central draw. H2 Push reaches 3,154 TWh (a fall of 18 per cent). Hydrogen substitution changes the supply mix while leaving the building envelope alone, so its useful-demand reduction is intermediate. Only the Net Zero pathway reaches the IEA Net Zero Emissions range for European building heat demand (a fall of 30 to 55 per cent by 2050; [140]); the others sit below it, reflecting the deliberate assumption that the Renovation Wave is not realised in full.

Figure 4: Sticky useful heat demand and the supply-side decarbonisation. Panel (a) carries all four scenarios, with the shaded band the 10th to 90th Monte Carlo percentiles. Panel (b) is Stated Policies alone. It indexes 2050 useful heat, delivered energy and emissions to their 2025 levels in that one scenario, so its three bars are not study-wide figures and the numbers on them do not apply to the other three pathways.



Source: Authors' contribution.

4.2 Technology Transition Pathways

The scenario lever, more than cross-country variation in the building stock, is what moves the technology mix. Every scenario starts from the same observed 2025 mix (heat pumps 8 per cent of useful heat, gas 44 per cent, oil 14 per cent, biomass 17 per cent, district heat 11 per cent and resistance heating 6 per cent, taken per country from Eurostat and national statistics on a per-country useful-heat basis) and interpolates towards scenario-specific 2050 targets, with a cost-responsive reallocation among the non-fossil options and policy constraints (boiler bans and heat-pump mandates) overlaid. The fossil share is governed by the scenario's policy-ambition lever, not by cost alone, and the hydrogen share is targeted on the exogenous adoption setting tested against the dispatch screen of Section 4.6. All 2050 mixes reported below are Monte Carlo medians, renormalised.

4.2.1 Current Policies

With policy frozen at today's measures, heat pumps reach 29 per cent of useful heat by 2050 and fossil boilers retain 40 per cent, the largest residual fossil share of any scenario. District heat stays near 18 per cent, biomass near 11 per cent, and hydrogen is negligible at half a per cent. Under a low carbon price and a stranded hydrogen market, neither the carbon penalty on gas nor the hydrogen price opens any niche.

4.2.2 Stated Policies

Delivering announced policies lifts heat pumps to 37 per cent by 2050, of which 18 per cent is reached already by 2030, and pushes fossil down to 27 per cent. Gas falls from 44 per cent of useful heat in 2025 to 38 per cent by 2030 and continues to decline as replacement bans (effective 2028 to 2035 across most member states under the EPBD 2024 recast; [12]) and ETS2 carbon costs bite. District heat reaches 19 per cent, biomass 11 per cent, and hydrogen 5 per cent, close to its merit-order ceiling.

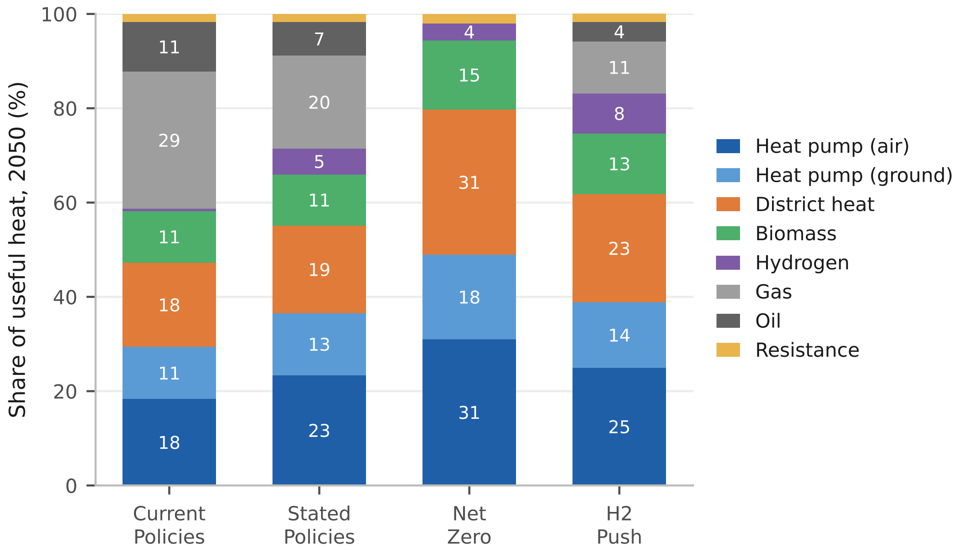
4.2.3 Net Zero

The climate-neutrality pathway combines deep renovation, a high carbon price and a fast-decarbonising grid. Heat pumps reach 49 per cent, district heat expands markedly to 31 per cent, fossil is squeezed to zero at the median, biomass holds 15 per cent, and hydrogen stays a peaking sliver, 3 per cent at the scenario ceiling and 3.6 at the realised median. The expansion of district heating, and not individual electrification alone, is a defining feature of the deep pathway, with direct infrastructure consequences (Appendix A.15).

4.2.4 H2 Push

Even in a deliberately hydrogen-favourable world, in which the rapid price decline brings delivered hydrogen to a 2050 range of about €22 to 39/MWh across the 29 markets, against €39 to 66 on the central path, hydrogen reaches only 8 per cent of useful heat at the scenario ceiling and 8.5 at the realised median. That ceiling is one we set, and it sits below the 9.3 per cent the merit order of Section 4.6 would support. Heat pumps still lead at 39 per cent, district heat at 23 per cent and fossil at 15 per cent. The scenario shows what cheap hydrogen *can* do for buildings. It grows the niche from the half to 5 per cent of the other scenarios to 8 per cent, and it does not change the architecture of the transition.

Figure 5: 2050 heating technology mix across the four scenarios (EU-27+UK+CH).



Source: Authors' contribution.

Figure 5 sets the four 2050 mixes side by side. Cross-country heterogeneity within any one of them is material but secondary to the scenario lever. Within a scenario, national 2050 heat-pump shares vary by approximately 4 to 8 points between the tenth and ninetieth percentiles, and by up to 10 percentage points. They are highest in the northern markets with detached stock and clean grids, led by the Netherlands, Denmark, Finland, Poland and the United Kingdom, and lowest in the warm south, where Malta, Cyprus and Greece trail the field because the heating load itself is small. The scenario lever, by contrast, moves the 29-country share by 20 points, from 29 to 49 per cent. Eastern European countries with delayed boiler-ban timelines retain higher residual gas and biomass shares in 2030.

4.2.5 What the Cost-Optimal Mix Says About These Shares

The four scenarios above set technology shares by policy lever. A least-cost linear program run over the same demand and the same technology portfolio sets them by minimising system cost, and is reported in full in Appendix B. Read against the scenario pathways it is informative on one comparison and silent on another.

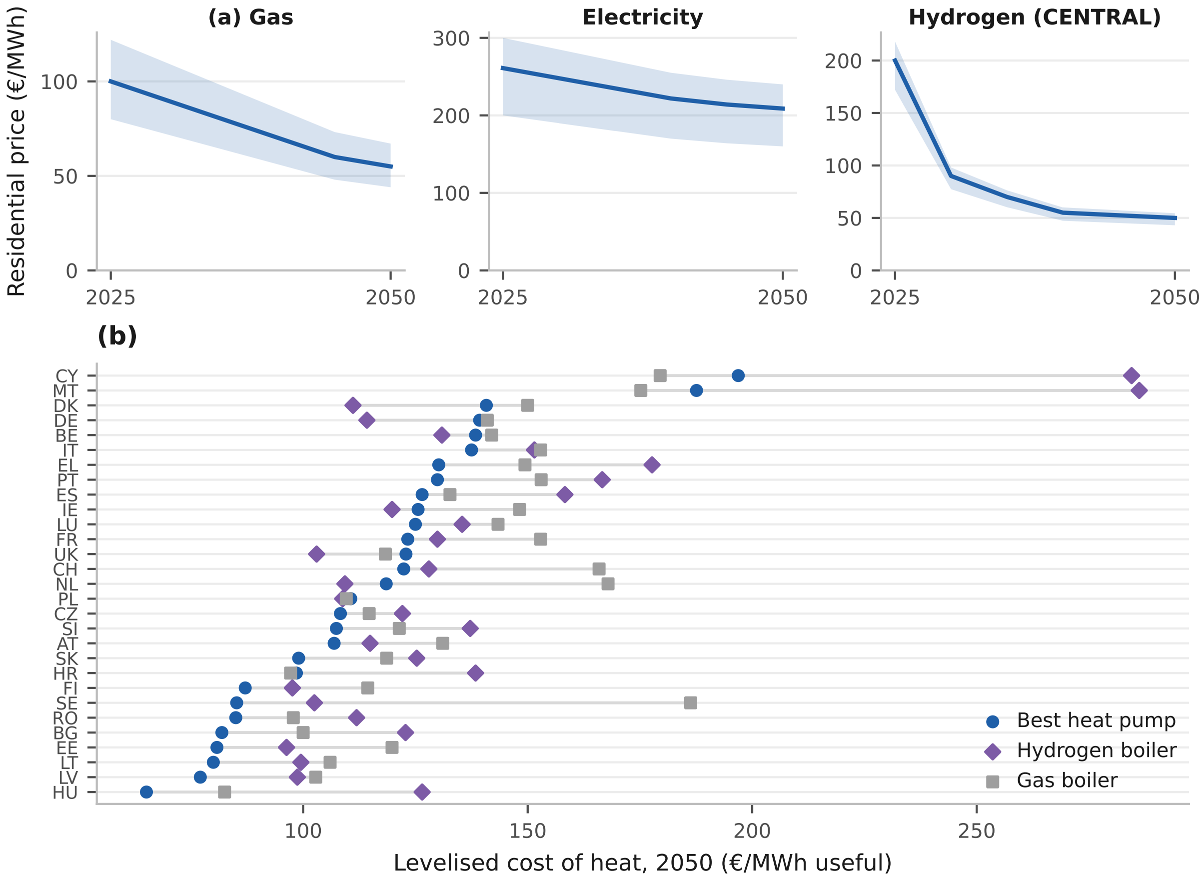
It is silent on hydrogen, and the silence has to be stated plainly because the comparison would otherwise look like corroboration. The cost-optimal 2050 mix carries hydrogen at about a third of one per cent, against 3.6 per cent in the Net Zero pathway. But the program is bounded by a per-country hydrogen ceiling that is zero in 27 of the 29 markets, so in most of the study area it is not permitted to select any hydrogen whatever the cost. Its near-zero share is therefore a property of that bound and carries no independent evidence on the direct-heating verdict, which rests on the base-load, dispatch and investment tests of this section alone.

It disagrees on which clean option carries the residual. The cost-optimal mix reaches about 53 per cent district heat and 45 per cent heat pumps and retires biomass altogether, where the Net Zero pathway holds district heat at 31 per cent, heat pumps at 49 and biomass at 15. The gap is the biomass share. Biomass is available to the optimisation, at 17 per cent of the 2025 base mix, and the optimisation drops it to zero because district heat delivers the same heat more cheaply wherever the network reaches, while the scenario pathways carry a biomass share forward as a policy assumption. The divergence therefore sits between district heat and biomass and not between the molecules and the machines, which is the comparison RQ1 turns on.

4.3 Levelised Cost of Heat Comparison

The cost ordering among heating technologies resolves by 2050, and it resolves against combustion. Throughout this paper the *best available heat pump* means, for each country, whichever of the air-source and ground-source units is the cheaper there, so the comparison is against the better of the two rather than an average of them. At the 29-country median the order runs district heat at €96.5/MWh, the best available heat pump at €118.5/MWh, the hydrogen boiler at €122.8/MWh and the gas boiler at €132.7/MWh, which places the individual heat pump second of the four and ahead of both molecules, and the country-level picture behind that median runs the same way outside a small group (Section 4.5). We establish that ordering here from the aggregate levelised costs and the fuel prices that drive them (Figure 6); the decomposition of those costs into their components, which shows what sets the ordering, is carried in Appendix A.4 (Figure A.3). Figure 6 sets out the aggregate picture. Panel a gives the input-price trajectories that drive most of the inter-country variation in heating costs (residential gas, electricity and delivered-hydrogen paths, as the 29-country median with an interquartile band; sources are the Eurostat household energy price series for H1 2025 and the OIES ET32 central hydrogen path), and panel b the resulting cost ordering (the 2050 levelised cost of the best available heat pump, the hydrogen boiler and the gas boiler in every country).

Figure 6: Residential fuel-price trajectories and the 2050 cost ordering. Panel (a) is the three small panels along the top: median residential gas, electricity and delivered hydrogen prices across the 29 countries from 2025 to 2050, with the band the interquartile range across countries and the hydrogen path the OIES ET32 central trajectory. Panel (b) is the strip below: the 2050 levelised cost of the best available heat pump, the hydrogen boiler and the gas boiler in each country, one row per country and ordered by the heat-pump cost.



Source: Authors' contribution.

In 2025 the air-source heat pump and the gas boiler stand at parity across the study area, at €155.8/MWh against €156.0/MWh at the median, with the heat pump ahead outright in 16 of the 29 countries. Two forces offset each other here. The high residential gas prices of the post-2022 period (the Eurostat H1 2025 consumption-weighted EU aggregate is €114.3/MWh including taxes, against a 29-country median of €100/MWh in the per-country series the model actually charges), combined with heat-pump seasonal coefficients of performance of 2.8 to 4.5, favour the heat pump; the equipment capital it must recover, charged against a machine sized to peak heat demand and so not reduced by that seasonal efficiency, works the other way. The coefficient band sits at the modern variable-speed end of the field evidence, not the older two-to-three fleet average, so a more conservative fleet-average coefficient would move the 2025 comparison against the heat pump (Appendix A.11).

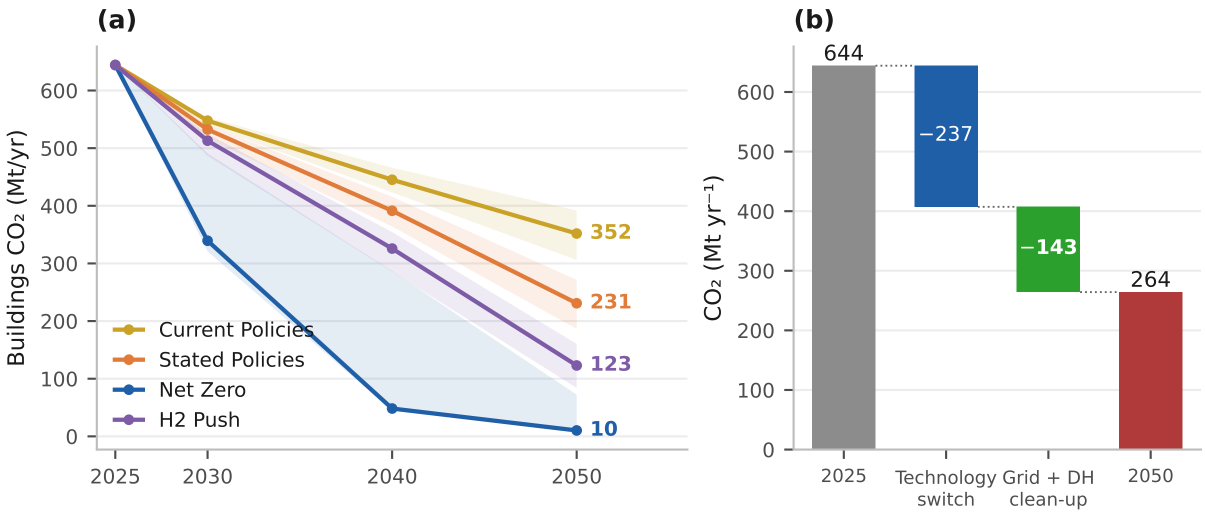
By 2050 the heat pump pulls clear of gas in 24 of the 29 countries, at a median €118.5/MWh for the best available heat pump per country and the same figure for the air-source unit alone, against €132.7/MWh for gas; the five exceptions are Cyprus, Croatia, Malta, Poland and the United Kingdom. That particular count is carbon-sensitive, because the gas boiler pays ETS2 and the heat pump does not. The heat pump leads gas in 19 of 29 countries on the low carbon path, 24 on the central path and all 29 on the high path. The policy implication has moved with the accounting. Equipment capital is a first-order term in the heat-pump cost, at 27.7 per cent of its 2050 stack and the largest capital share of the three technologies (Appendix A.4), so upfront cost is a genuine cost disadvantage to be closed and not a financing friction, and it stands alongside installation capacity and regulatory inertia as a barrier to adoption.

ETS2 widens the gas-versus-heat-pump gap further from 2027. Each €50/tCO_2 of carbon price adds about €11/MWh to the gas-boiler levelised cost of heat (LCOH), so the central €122/tCO_2 path adds about €27/MWh, against a median €0.7/MWh across the 29 countries on the heat-pump LCOH at 2030. That charge is the increment of electricity's carbon cost above its 2025 level, and it falls to zero at the median from 2040 and to zero in every market by 2050 as the grid decarbonises. The carbon price therefore leaves the 2050 hydrogen-versus-heat-pump comparison unmoved, not because neither technology pays it but because by then neither pays any increment, so the country counts reported below are identical under the low, central and high carbon paths and it is the hydrogen price path that moves them. Under the central trajectory the hydrogen boiler shows an LCOH of €299.3/MWh in 2025, falling to €122.8/MWh by 2050. On that same central path the best available heat pump undercuts it in 22 of the 29 countries by 2050, a key finding for RQ1 that Section 4.5 takes up in full.
The detailed treatment of the anatomy of the cost ordering is given in Appendix A.4.

4.4 CO_2 Emissions Trajectory

The inventory behind this subsection, its emission factors and the country grid-intensity paths it applies, is set out in Appendix A.10.

Figure 7: The buildings emissions fan and its Stated Policies attribution. Panel (a) carries all four scenarios from 2025 to 2050, with the shaded band the 10th to 90th Monte Carlo percentiles and the 2050 median labelled on each. Panel (b) decomposes the Stated Policies fall alone into the part the technology switch delivers and the part grid and district-heat decarbonisation deliver. Both panels report the same quantity but on different bases, so their endpoints differ. Panel (a) plots the Monte Carlo median, while panel (b) runs on a single central draw, which shares the 644 $MtCO_2$ 2025 level but reaches 264 $MtCO_2$ in 2050 against the ensemble median's 231. The decomposition is taken entirely on that second basis, so the two bars sum to its own fall and not to the ensemble median's.



Source: Authors' contribution.

Where demand is sticky, emissions are not. The outcomes diverge sharply across scenarios (Figure 7, Table 2). The four pathways fan out from a common baseline of approximately 644 $MtCO_2$ in 2025 to 352 (Current Policies), 231 (Stated Policies), 123 (H2 Push) and 10 (Net Zero) $MtCO_2$ by 2050, reductions of 45, 64, 81 and 98 per cent respectively. The 10th to 90th percentile Monte Carlo ranges at 2050 are 306 to 392 (Current Policies), 186 to 272 (Stated Policies), 84 to 160 (H2 Push) and 9 to 72 $MtCO_2$ (Net Zero). The medians are separated by 121, 108 and 113 $MtCO_2$, each of them larger than any single band is wide, so the scenario ranking is robust to parameter uncertainty (Table 2). The ranking claim rests on that median separation and not on the bands themselves, which are indicative at this sample size because the deciles have not converged at $N = 200$ (Appendix A.3). *Ambition, not technology availability, separates the outcomes.* The same technology set is available in every scenario, and the spread is driven by the policy levers, namely fossil-phaseout ambition, renovation depth, the carbon price, and grid-decarbonisation speed. The 2050 gap between Current Policies and Net Zero, of roughly 342 $MtCO_2$ per year, is the emissions value of closing the implementation gap in residential heat.

Table 2: Buildings-sector CO_2 emissions by scenario and year.

Scenario	2030		2040		2050		2050 range
	$MtCO_2$	Reduction	$MtCO_2$	Reduction	$MtCO_2$	Reduction	[10th to 90th]
Current Policies	548	−15%	445	−31%	352	−45%	[306 to 392]
Stated Policies	532	−17%	391	−39%	231	−64%	[186 to 272]
H2 Push	513	−20%	326	−49%	123	−81%	[84 to 160]
Net Zero	339	−47%	48	−93%	10	−98%	[9 to 72]

Emission factors: IPCC AR6 (combustion); EMBER 2024 (electricity actuals); declining paths to 2050 calibrated to European decarbonisation targets [141], which are scenario-consistent trajectories rather than published country projections (Appendix A.10). All four scenarios share the common observed 2025 baseline of approximately 644 $MtCO_2$, and every reduction in the table is computed against it. The figure is not scenario-specific, since each scenario starts from the same observed 2025 mix, and every aggregation order returns it. The Net Zero ensemble median, the sum of its per-country medians and the median of its per-draw sums all give 644.2 Mt, within a 200-draw spread of less than one megatonne.

Source: Authors' contribution.

The detailed treatment of decomposition is given in Appendix A.5.

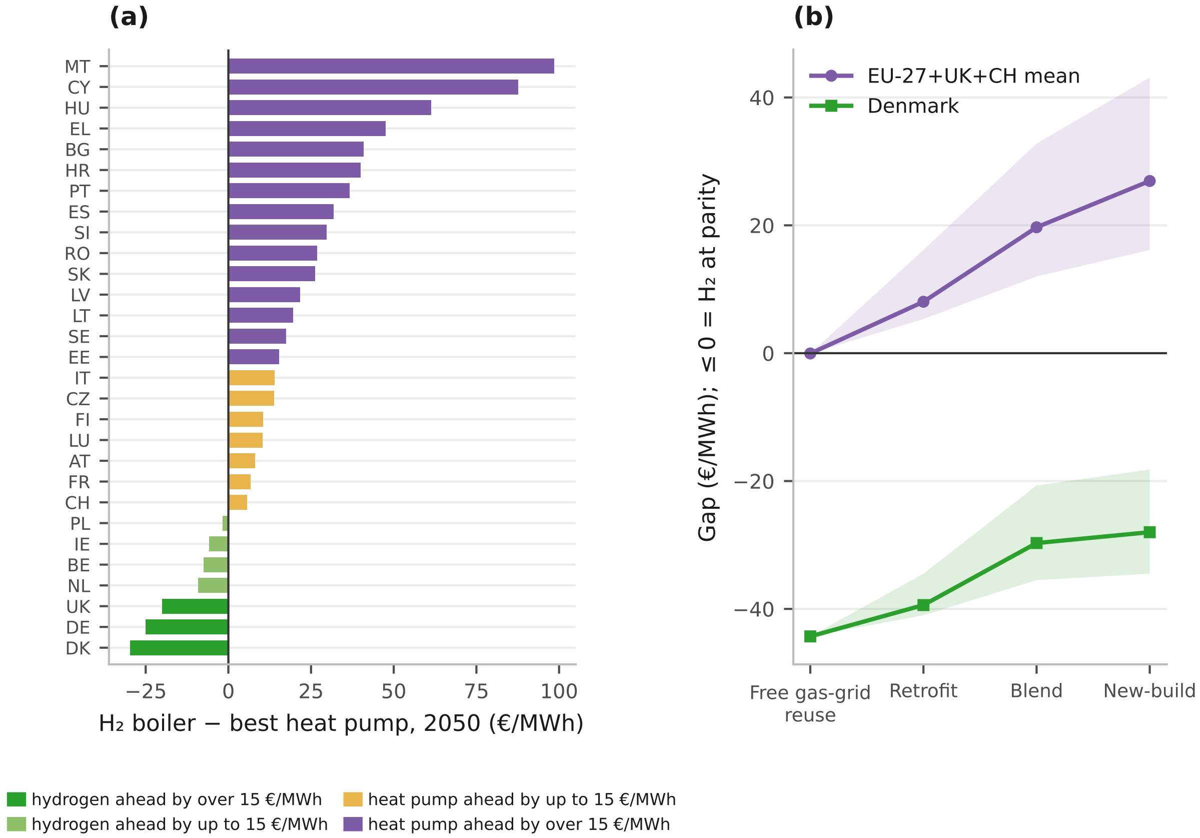
4.5 Test 1: Base-Load Competitiveness in EU Buildings (RQ1)

This is *Test 1*, the base-load facet of RQ1: whether hydrogen can compete at the building level across the 29 countries, priced as a base-load supplier of annual heat in the levelised-cost framework of Section 4.3. The operational question is specific: in which countries, if any, does a hydrogen boiler undercut the best available heat pump in 2050, and by what margin elsewhere? We answer it by ranking every country by hydrogen's 2050 cost gap to the best heat pump, and that ranking is the base-load answer to RQ1; the complementary peaking facet (Test 2) follows in Section 4.6.

Figure 8 presents the ranking under the model's reference assumptions, namely the central hydrogen trajectory with each carrier charged for its own last-mile network, a partially converted distribution system for hydrogen and low-voltage reinforcement for the heat-pump winter peak, and with hydrogen charged the seasonal store that its own winter-weighted load shape requires (Appendix A.13); the sensitivity of the ranking to the network treatment is taken up in Appendix A.6. Its headline is that the best available heat pump is cheaper in 22 of the 29 countries and the hydrogen boiler in the remaining seven, five of which are salt-cavern markets. How much of that concentration the storage rate itself carries is set out below, and the answer is less than half of it.

Two features of the network treatment move that count and belong beside it. The first is an asymmetry running against the heat pump. The reference case charges electric heating for distribution reinforcement but not for the firm generating capacity the same winter evenings call on. Charged pro rata to heating's share of power demand that capacity adds about €1/MWh and leaves the count at 22; charged in full to the heat-pump load, which is an upper bound no one would defend, it adds €11 to 14/MWh and the count falls to 17 and then 15 of 29 (Appendix D). The second is the reinforcement level itself. At the top of its plausible band, with hydrogen paying nothing for its own network, the heat pump leads in only 10 of 29; charging hydrogen the network it is charged by default reproduces the headline 22 at that same bound (Appendix C.7). The count is therefore robust to the reinforcement level and sensitive to whether both carriers are charged for their own last mile, which is the symmetry this paper is built on.

Figure 8: The per-country hydrogen-boiler gap, and who pays for delivery. Seven of the 29 countries sit at or below the line in panel (a), Denmark widest. Panel (a) is drawn on the coverage-weighted blending assumption, the third point in panel (b) and the central case throughout this paper; the count below the line moves from 12 of 29 under free gas-grid reuse to 3 of 29 if every hydrogen customer needs new-build distribution (Appendix A.6).



Source: Authors' contribution.

The ranking in Figure 8a resolves RQ1 at country level, and it is a ranking, not a single threshold. Under the central hydrogen trajectory, with the per-country delivered-hydrogen cost set by the bottom-up supply model of Appendix A.13 and with the seasonal store charged on the winter-weighted fraction of the hydrogen heating load, the best heat pump is cheaper than the hydrogen boiler in 22 of the 29 countries in 2050 and dearer in the other 7. The 29-country median gap, hydrogen minus the best heat pump, is +€15.4/MWh and the mean +€19.7/MWh; weighted by heat demand it narrows to +€1.4/MWh, because the large north-western markets are the ones that sit on hydrogen's side of the line.

The distribution is right-skewed (skewness +0.86), so the mean is pulled upward by a tail of countries in which hydrogen loses heavily. The heat pump's margin is widest in Malta (+€99/MWh), Cyprus (+€88), Hungary (+€61), Greece (+€48) and Bulgaria (+€41). Hydrogen retains a lead in seven markets, Denmark (−€30/MWh), Germany (−€25), the United Kingdom (−€20), the Netherlands (−€9), Belgium (−€8), Ireland (−€6) and Poland (−€2). Five of them, including the four widest, are salt-cavern markets; Belgium and Ireland survive on very high retail electricity prices while paying the higher non-cavern storage rate, and only Poland's margin is narrower.

That concentration is the structural finding of Test 1. Space heating draws about three-quarters of its annual energy between October and March, and electrolytic production is far flatter, so a hydrogen heating load creates a seasonal imbalance that has to be stored. Storing it costs about €1.5/kg where salt caverns are available and about €3.0/kg where they are not (Appendix A.13), which is worth €12.1/MWh of useful heat in cavern-endowed Germany against €28.4/MWh in cavern-free Spain. Hydrogen's baseload case therefore concentrates where the geology makes seasonal storage cheap, which is the same condition that decides the peaking arena outright.

How much of that concentration the geology actually carries needs stating separately, because the storage charge sets how wide each margin is without selecting the set. Priced at the punitive non-cavern rate everywhere, Denmark still leads by €17.6/MWh, Germany by €13.0 and the United Kingdom by €7.8, while Belgium and Ireland pay that rate already and hold on €7.5 and €5.8. Only the Netherlands and Poland change hands. Running the counterfactual the other way, charging the cheap cavern rate to all 29, moves the heat-pump count to 19 of 29 against 22 in the baseline and 24 with no cavern anywhere, so the whole geological binary is worth three markets below the baseline 22 and two above it. Cavern endowment decides two of the seven holds. What decides the other five is the retail electricity tariff on the rival carrier, which is a fiscal choice and not a physical endowment, and that is the larger cause of the exception. The two country-specific prices that earlier drafts read as the drivers of the ordering now act as modifiers within it:

1. **Cheap domestic green hydrogen.** Countries with strong wind and low capital costs (Ireland, Denmark, the Baltics and Finland) produce green hydrogen below the 29-country median (a supply multiplier of 0.77 to 0.87; Appendix C.6), which narrows the gap against the heat pump; warm, high-WACC and island economies (Greece, Iberia, Cyprus and Malta) cannot, and sit at the far end of the heat-pump side.

2. **Expensive heat pumps.** A cold climate and high residential electricity prices raise heat-pump operating cost, narrowing the gap from the other side. This is why Germany and Belgium, despite only median-cost hydrogen, are among the seven, and why the warm islands, with very cheap heat-pump heat, sit at the opposite extreme.

Neither price condition is on its own decisive, and neither is the geology. Ireland has the cheapest delivered hydrogen in the study area and still holds a lead of only €6/MWh, while the widest leads belong to the countries in which cheap molecules, dear taxed electricity and cavern storage coincide.

One qualification governs how much weight the seven remaining hydrogen leads can bear. The comparison sets an untaxed molecule against taxed retail electricity. By 2050 the median study-area household pays roughly €209/MWh for delivered electricity inclusive of levies, taxes and network charges, against roughly €50/MWh for hydrogen delivered and untaxed, so what survives of hydrogen's base-load lead is a difference in fiscal treatment as much as a difference in technology. Equalising the levy treatment of the two carriers would remove most of it, and would therefore widen the heat pump's margin, not narrow it. The asymmetry now runs with the headline result, and its bearing has moved accordingly. It no longer qualifies a hydrogen lead across half the study area; it explains why any hydrogen lead remains at all.

A second objection bears on the molecule's assumed provenance, and it points the same way. Our central trajectory prices green hydrogen at about €50/MWh delivered to the gate. The one dedicated assessment of hydrogen for buildings concludes that any hydrogen actually serving heat would be blue, not green, at approximately €81/MWh, because the volumes heating requires at the times it requires them are not the volumes a green electrolysis fleet is being built to supply [73]. That is a wedge of €31/MWh of fuel, wider than the widest of the seven surviving hydrogen leads, set against a median country gap of €15.4/MWh that already runs against hydrogen. Repricing the molecule on that basis moves every country further towards the heat pump, and how far depends on how a single German estimate is read. Read as a European price, a flat €81/MWh at every gate leaves the heat pump cheaper in all 29 and closes every one of the seven leads. Read as a green-to-blue ratio, so that each country keeps its relative position, it leaves the heat pump cheaper in 28 and Denmark alone still holding. The base-load verdict against hydrogen therefore survives the provenance objection on either reading, and the flat reading is the one both this paper and the submission quote.

The two objections are independent and both run with the headline and not against it. The base-load ranking of Figure 8a is therefore a conservative statement of the heat pump's advantage, since the two assumptions that remain most generous to hydrogen, its fiscal treatment and its assumed provenance, would each widen that advantage if they were corrected. The two objections do not carry equally to the peaking result, and the difference matters. The fiscal one does not touch it, since both turbines price at wholesale and neither pays a household levy. The provenance one does.

Priced flat at the €81/MWh of the model's own blue route [88], which is the provenance the one dedicated assessment of heating hydrogen concludes it would have [73], the hydrogen turbine reaches about €289/MWh-e against the gas peaker's €256, and the cavern win reverses in all seven markets; read instead as a green-to-blue ratio the seven hold, with the advantage falling from €72 to 89 down to €30 to 57/MWh-e. The capital-recovery result is invariant either way, because a winner's rent is the scarcity premium times its run hours and carries no fuel term. So the peaking win is materially exposed to hydrogen's provenance and may not survive it. The missing-money finding survives it. Hydrogen deployment in buildings before 2050 therefore turns on targeted support justified by co-benefits beyond delivered cost, namely gas-grid asset utilisation, energy-security diversification, and the avoidance of costly retrofit in hard-to-treat building segments. The detailed treatment of hydrogen distribution-infrastructure cost is given in Appendix A.6.

4.6 Test 2: Peaking Dispatch (Building and Grid Scale), the Merit-Order Niche

Test 1 priced hydrogen as a base-load supplier and found the best heat pump ahead in 22 markets and hydrogen ahead in seven, five of them cavern-endowed. The base-load frame is, however, only one facet of RQ1. Hydrogen's advocates implicitly price it as a *peaking* producer, the dispatchable source that serves only the cold-snap peak, when air-source heat-pump coefficients of performance collapse and the power system is stressed. This is *Test 2*, the peaking facet of RQ1. We reprice hydrogen as a peaker and ask where its marginal operating cost wins. Test 2 identifies which countries' hydrogen wins the peaking dispatch; but Test 3, nested within it (Section 4.7 and Appendix A.7), asks whether that peaker can then recover its capital from market rents, so the two tests are not independent and must be read together to bound hydrogen's role.

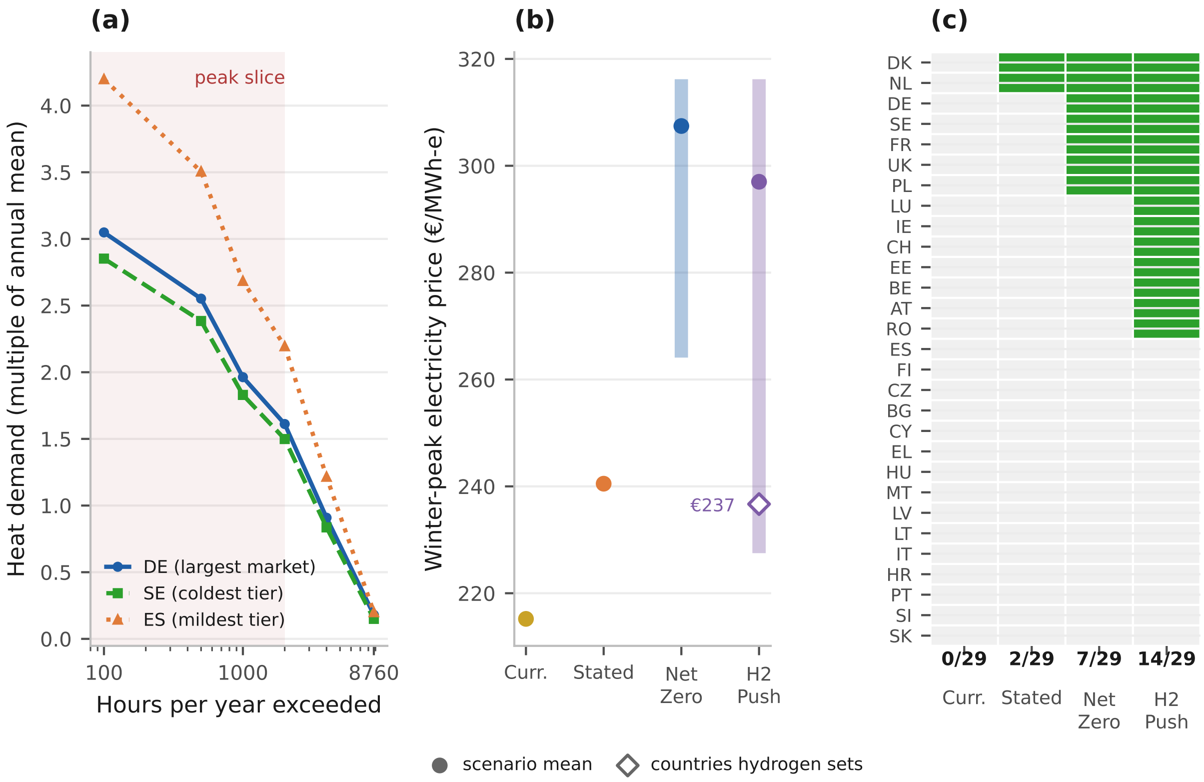
The peaking frame does not replace the base-load answer; it is the second of the two facets the opening distinguished, and the integrated answer combines them at the end of this section. We run the peaking test as a single nested argument in three parts: the peak at the building and grid scale (immediately below), the peak inside district heating (Appendix A.14), and the investment-feasibility check (Test 3, Section 4.7 and Appendix A.7), into which the hour-resolved verification of the cold snap is folded. That last part asks whether a peaker that wins the dispatch can also be financed, and its negative answer bounds the peaking niche on its own terms, independently of what the base-load comparison found.

We therefore model the winter peak explicitly. The top slice of each country's heat load-duration curve, the energy above the 2,000-hour demand level and approximately

12 to 18 per cent of annual heat, must be met by a quick-start dispatchable source. Three candidates compete on marginal operating cost: a gas peaker (fuel plus ETS2 carbon), a hydrogen peaker (delivered fuel plus a seasonal-storage adder of approximately €50/MWh-heat where salt caverns exist and approximately €100/MWh where they do not, our own assumptions, consistent in order with the bulk-storage rates of Lord et al. [142], Papadias and Ahluwalia [143] and Talukdar et al. [144], applied to the cavern geology of Caglayan et al. [36]), and the cold-snap heat pump (running at approximately 0.6 of its seasonal coefficient of performance against a scarcity electricity price).

The scarcity electricity price is *derived* from the power sector's own merit order through Eq. (7) (Appendix A.2). The marginal winter unit, a gas OCGT or an H_2-ready turbine at wholesale fuel prices, carbon-priced, plus a scarcity premium π^s, sets it per country and scenario. This endogenous price averages €215/MWh under Current Policies, €240 under Stated Policies and €307 under Net Zero. The high carbon price loads the marginal gas unit and makes the cold snap *dearer* in high-ambition worlds. Under H2 Push, hydrogen turbines become the price-setting unit in cavern-endowed countries, so cheap hydrogen partially cannibalises its own heating niche by lowering the very peak price that hydrogen-for-heat must beat, to an average of €237 across the units it sets.

Figure 9: The winter heat peak, in load, price and dispatch. Panel (a) is the heat load duration curve for three representative countries, as a multiple of each one's annual mean, with the shaded band the peak slice the peaker is sized against. Panel (b) gives the modelled cold-snap electricity price by scenario, the vertical bar spanning the 29 countries and the dot their mean. The open diamond on H2 Push is the mean over the seven countries in which the hydrogen turbine is the marginal unit at the winter peak, which is the €237 the text quotes. Panel (c) is a grid of which countries hydrogen wins on the building winter peak in each scenario, one row per country and shaded where it wins, with the win count out of 29 under each column, on the symmetric accounting basis that charges hydrogen its last-mile network.



Source: Authors' contribution.

Hydrogen's derived heating share is then the peak slice it actually wins, and the result bounds hydrogen tightly. Panel c of Figure 9 carries the country grid behind the counts that follow, on the symmetric basis that charges hydrogen the same last-mile adder its base-load comparison uses. Hydrogen is the cheapest peaking option in **0 of 29** countries under Current Policies, **2** under Stated Policies (NL and DK), **7** under Net Zero, and **14** under H2 Push, led in each case by the cavern-endowed north-west European cluster. Leaving that adder off, as an earlier draft did, gives 0, 4, 9 and 16. The ordering is instructive. It is the carbon price on the competing gas peaker that opens the niche, since the low-carbon Current and Stated worlds admit few or none, and cheap hydrogen layered on a high carbon price then widens it, so H2 Push, carrying both levers, wins the most. Weighted by demand, the winning peak slices correspond to a 29-country hydrogen share of 0, 0.6, 7.5 and 9.0 per cent of useful heat across the four scenarios, against 0, 5.1, 7.9 and 9.3 on the earlier draft's basis. Those are the merit order's own derived shares, and the scenario ceilings of Section 4.2 sit at or below them, so the ceiling is bounded by the dispatch and is not the same quantity.

The two facets of RQ1 can now be read together, and they agree because they measure the same thing, the cost of holding a molecule from the season in which it is cheap to make to the season in which it is wanted. Geology sets that cost in both, which is why the base-load exceptions and the widest peaking margins fall in the same cavern-endowed north-west.

The integrated answer to RQ1 is that hydrogen does not drive a spontaneous building-heat transition, and the reason is now single, not compound. In most of the study area hydrogen loses the base-load comparison because it must pay for the seasonal store that its own load shape creates, and cheap storage is what most of its exceptions have in common; inside the cavern endowment it can also win the cold-snap dispatch, yet the peaker that would serve that dispatch cannot recover its capital from market rents, as Test 3 now shows. Hydrogen's role in heat remains conditional on storage geology and on policy support, not on delivered cost alone.

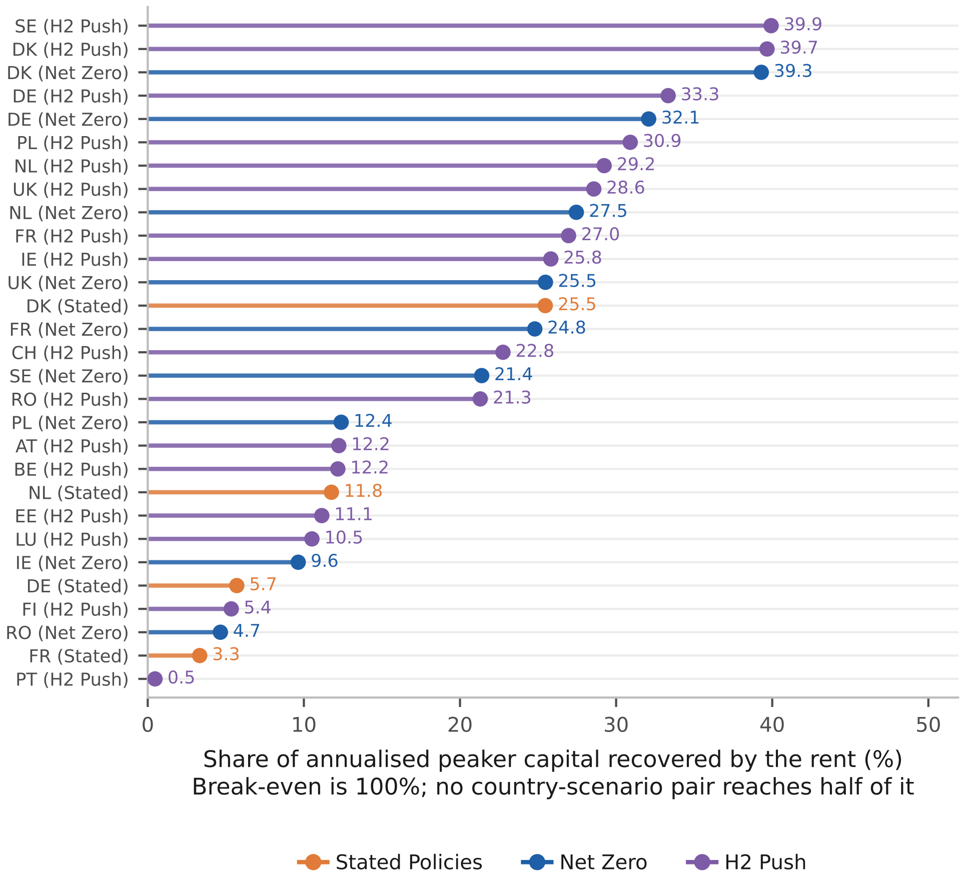
4.7 Test 3: Investment Feasibility, the Missing-Money Caveat

Winning the dispatch is necessary but not sufficient. A private investor must also recover the peaker's capital from the few hours it runs. Reinterpreting the Test 2 peakers through a capital-recovery lens, *no hydrogen peaker is privately financeable from market rents, in any country and in any scenario*, even where its operating cost wins the cold-snap dispatch. Over its roughly 500 running hours of the heat-side peak slice, the inframarginal rent it earns (the clearing price minus its own marginal cost) falls far short of its annualised capital requirement of about €52 to 73/kW-yr.

In the power arena, whose fleet, running hours and cumulative ledger are derived in Section 5 and quoted here only for the comparison, the capital-recovery condition of Eq. (9) (Appendix A.2) gives Denmark as the best case at about 40 per cent, most winners recover well under 20, and the capacity-weighted average across the five cavern markets that build one is about 9 per cent. The heat-side peaker recovers about 20 per cent, aggregated across the 29 winning country-scenario combinations, which 16 countries generate across three scenarios. It recovers the more of the two for a reason that is arithmetic and not economic. The heat-side slice runs about 519 hours against the power fleet's 139, so the same scarcity margin is earned over nearly four times as many hours. Neither recovers its capital. Figure 10 ranks all 29 heat-side winners on the share of capital their rent returns, and the full per-country and per-scenario table is reported in the supplementary material. Not one reaches half of break-even. The best pair, Sweden and Denmark under H2 Push, stops just under 40 per cent, and 12 of the 29 sit below 20.

This is the classic *missing-money* problem of low-utilisation peaking plant [28], and it is reported here as a caveat to the operating-cost result. It does not stop hydrogen setting the price and taking the role, but it means a deliberate hydrogen-peaker build implies a standing capacity payment. It is this investment-side shortfall that bounds the peaking niche of Test 2, and it does so independently of the base-load comparison, since a niche that cannot be financed from market rent does not widen on delivered cost alone. Crediting the same power peaker with a district-heating heat offtake (a combined power-and-heat mode) lifts its cumulative position by only about €8 to 16 billion, 2 to 4 per cent of the roughly €357 billion shortfall. Running only about 139 hours a year, it has little heat to sell, so a fast-start power peaker is a poor heat-led cogeneration plant. Hydrogen's genuine cogeneration value lies in the separate, heat-led district-heating case (Appendix A.14), where the unit runs an order of magnitude more hours.

Figure 10: **Capital recovery of the heat-side hydrogen peaker.** Every country-scenario combination that wins the cold-snap dispatch. Recovery is the inframarginal rent over the annualised capital requirement, so full recovery is 100 per cent. The axis is scaled to the values the model produces, whose maximum is 39.9, and break-even therefore sits off it to the right.



Source: Authors' contribution.

The hour-resolved dispatch (Appendix A.7), the heat-led district-heating merit order (Appendix A.14), the network-infrastructure bill of each pathway (Appendix A.15) and the Switzerland import case for RQ2 (Appendix A.8) are treated in full in the supplementary methodology.

5 Hydrogen in the Power Sector: From Electrified Heat to the Merit Order

The heating merit-order analysis of Section 4.6 located hydrogen inside the heating stack; on operating cost, hydrogen undercuts the carbon-priced fossil competitor at the building and district-heat peak in the high-carbon scenarios, while on base-load levelised cost it leads the heat pump in the cold, high-electricity-price north-west and trails it across the warm south and east. This section asks a distinct question about the power system that carries the electrified heat. As heating electrifies, a rising share of heat demand becomes electricity demand, and the decarbonised grid that serves it still requires firm capacity for the cold-snap peak. We therefore ask, on the same operating-cost basis a system operator actually uses, where hydrogen sits in the *power* merit order, and how large a role that position implies.

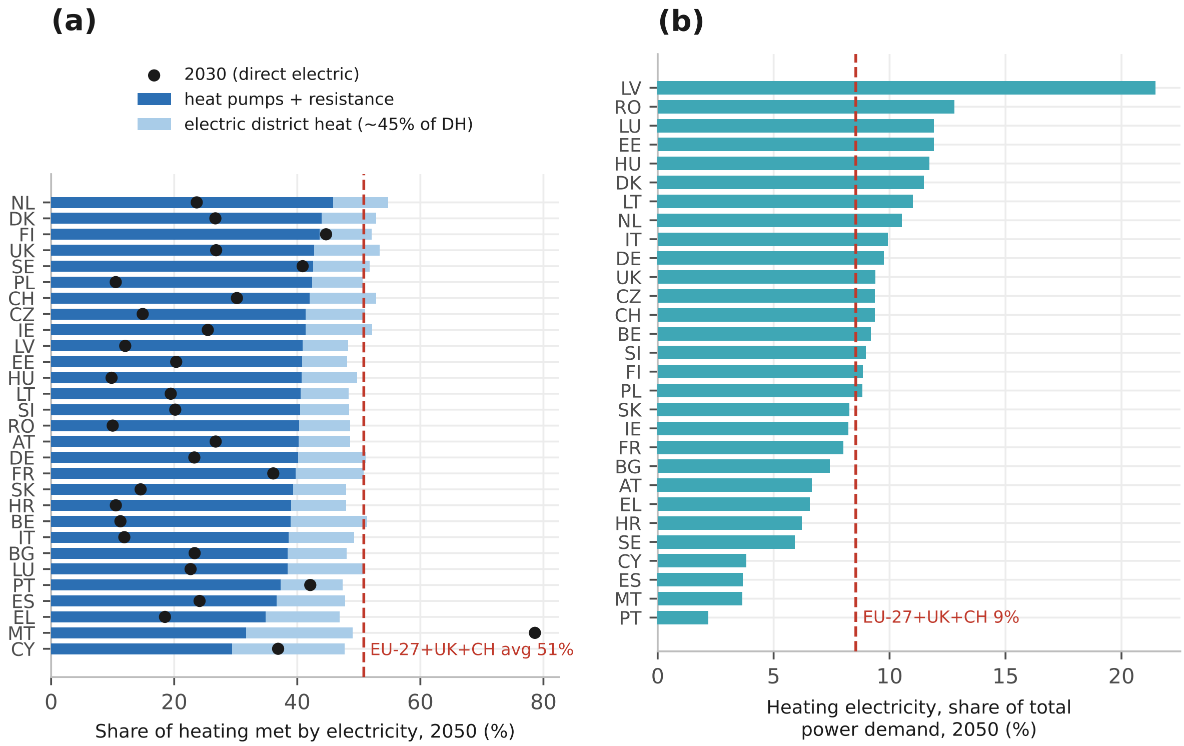
This question is the buildings-to-power dimension of the third research question (RQ3), the synergies between hydrogen in buildings and the wider energy system, and it is the one cross-sector coupling this paper takes up. The scope is deliberately bounded to that coupling. The further synergies with industry and transport are out of scope and are deferred to subsequent work, as set out in Section 1.1. Throughout, electricity is priced as wholesale markets price it, at the short-run marginal cost of the marginal generator, because capital is a sunk cost that does not set the spot price; the investment question is returned to, as a caveat, in Section 5.3.

The dispatchable layer is resolved by a reduced-form merit order, which sets the fleet size and the running hours quoted throughout, and is cross-checked against an explicit unit-commitment optimisation of the firm dispatchable fleet (firm CCGT against open-cycle gas or hydrogen peakers, with minimum stable output, ramp and minimum up and down times, a 5 per cent operating reserve, start-up costs and storage state of charge; Appendix A.2). With the commitment constraints disabled the two agree to within 0.33 per cent on dispatched energy and 0.40 per cent on peak capacity in 28 of the 29 countries; Romania is the exception, and Appendix A.7 sets out why. The exercise tests where the heating result lands in the power stack and bounds the implied role.

Electrified heating represents a material but bounded share of 2050 EU power demand. Across all 29 markets, heating electricity is about 9 per cent of total power demand (Figure 11b), large enough to shape the load but not to dominate it. This is the scaled load that determines where hydrogen competes. The cold-snap peak of that demand block is the segment the power system must cover with firm dispatchable capacity, and it is there that hydrogen enters the merit order.

The component shares behind that 9 per cent follow from the degree of electrification. By 2050 heating is heavily electrified. Heat pumps and resistance heating meet about 41 per cent of useful heat directly under H2 Push, rising to about 51 per cent once the electric portion of district heat (large heat pumps), taken as its share of total useful heat, without being added in full, is included, and about 51 per cent directly under Net Zero (Figure 11a). Converting those shares to electricity through the temperature-dependent coefficient of performance and adding the result to a national baseline load grown to 2050 yields the 9 per cent share of total power demand above.

Figure 11: The bridge from electrified heating to 2050 electricity demand, under H2 Push, all 29 markets. The red dashed line in each panel is the study-area average. Panel (a) counts district heat at its 0.45 electrification factor, which is why its 51 per cent exceeds the 41 per cent direct-electrification average of Table 3.



Source: Authors' contribution.

The same picture, country by country and across the decade, is set out in Table 3. The heat-weighted EU-27+UK+CH share of heat met directly by electricity climbs from about 23 per cent in 2030 to about 31 per cent in 2040 and about 41 per cent in 2050, and once the electric share of district heat is added the 2050 figure rises to about 51 per cent.

Table 3: Share of residential heat met directly by electricity (heat pumps and resistance heating), per country, 2030/2040/2050, under H2 Push, with the heat-weighted EU-27+UK+CH average in the final row.

Country	2030	2040	2050
NL	24	29	46
FI	45	46	44
DK	27	35	44
SE	41	44	43
UK	27	29	43
PL	10	31	42
CH	30	34	42
IE	25	28	41
HU	10	29	41
LT	19	34	41
LV	12	33	41
CZ	15	32	41
EE	20	36	41
SI	20	33	40
RO	10	31	40
FR	36	36	40
DE	23	28	40
AT	27	32	40
BE	11	27	39
HR	11	30	39
IT	12	28	39
SK	15	30	39
LU	23	27	38
BG	23	37	38
ES	24	32	37
PT	42	41	37
EL	18	28	35
MT	79	52	32
CY	37	31	29
EU-27+UK+CH (heat-weighted)	23	31	41

Source: Authors' contribution, scenario Monte Carlo medians.

5.1 Where Hydrogen Sits in the Power Merit Order

Hydrogen competes for the peaker role on short-run marginal cost, the fuel-plus-carbon operating cost that a system operator dispatches on, with no capital charge. Under H2 Push, which pairs the high carbon path with the rapid hydrogen cost decline, hydrogen undercuts the gas peaker on operating cost in **seven of 29 countries** (Figures 12 and 13), exactly the salt-cavern set, a point we return to below because the winning set being identical to the classification means the seven is conditional on it rather than independent of it, implying it is dispatched ahead of gas at the cold-snap peak in those markets. That count holds both turbines at the same 0.40 efficiency, which is the even-handed comparison and the one we carry throughout.

On the *evolving-efficiency* series, in which the gas turbine reaches 0.42 by 2050 and the hydrogen turbine 0.48 under H2 Push, so the hydrogen unit is deliberately given the better machine, the count rises to 15. We report both because they answer different questions, the first what a plausible 2050 fleet would dispatch and the second whether hydrogen wins on fuel cost alone; the seven that survive the even-handed test are the same seven that win decisively on the looser one. This finding is consistent with the system-value literature on complementary firm and flexible low-carbon resources [98, 145], which locates hydrogen's contribution at the dispatchable margin, and not in bulk energy. The count is contingent on the 2050 carbon-price path, which is the dominant lever on the ranking (Appendix C). A lower carbon price narrows the gas peaker's operating-cost penalty and would reduce the count, a higher one would raise it.

The advantage is, however, sharply bimodal and concentrated where seasonal storage is cheap. In the seven salt-cavern markets, the United Kingdom, Denmark, the Netherlands, Romania, Poland, Germany and France, hydrogen undercuts the gas peaker by a decisive €85 to 99/MWh-e on this evolving-efficiency series, and by €72 to 89/MWh-e on the equal-efficiency series the headline count uses; in the remaining eight of the 15 the margin is only €0 to 8/MWh-e, within the noise of the cost estimate. The operative result is therefore better read as about seven markets with a material operating-cost advantage, all of them cavern-endowed, and a further handful at the margin. It is not a uniform 15 (Figures 13 and 14).

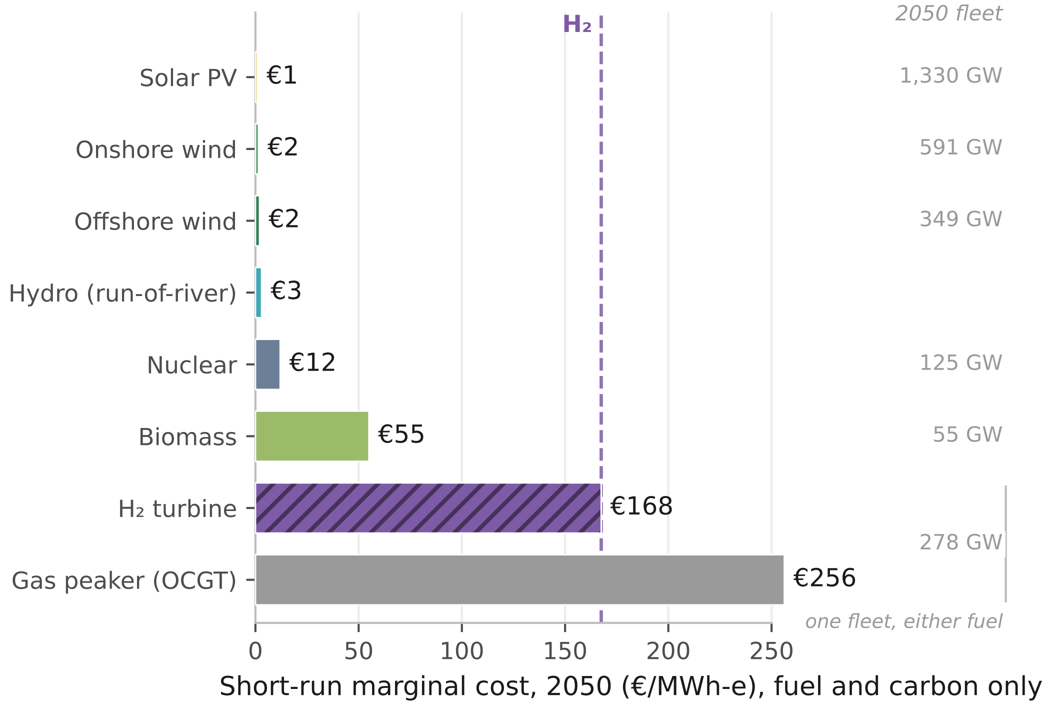
The mechanism is the comparison of two operating costs. At the cold-snap peak the price is set by the dearest dispatched generator, and the candidate firm units rank by short-run marginal cost:

$$c^{\text{gas}}_{\text{SRMC}} = \frac{p^{\text{gas}} + \varphi\, e_g}{\eta_g}, \qquad c^{\text{H}_2}_{\text{SRMC}} = \frac{p^{\text{H}_2}}{\eta_h} + \sigma, \tag{4}$$

where each short-run marginal cost is expressed in EUR/MWh-e of electricity sent out: p^{gas} and p^{H_2} are the delivered fuel prices (EUR/MWh of fuel), φ is the carbon price (EUR/tCO$_2$), e_g the gas emission factor (tCO$_2$/MWh of fuel) and η_g, η_h the dimensionless electric efficiencies of the gas and hydrogen turbines. The first term of each expression is the fuel-and-carbon operating cost; both exclude capital, which is sunk and does not set the spot price (the capital recovery of the hydrogen unit is the separate test of Section 5.3). The seasonal-storage adder σ is the annualised cost of holding and cycling hydrogen in seasonal store; it is purely a storage-and-cycling charge and excludes transmission losses. It is approximately €50/MWh-heat where caverns exist and approximately €100/MWh-heat where they do not, equivalently €45 and €90 per MWh of hydrogen at the 0.9 boiler efficiency used throughout, at rates consistent in order with the bulk-storage literature [142, 143, 144] and applied on the cavern geology of Caglayan et al. [36]. Equation (4) needs it per MWh of electricity sent out, so the implementation divides by the turbine efficiency, giving about €112 and €225/MWh-e at $\eta_h = 0.40$; it is the largest single driver of the hydrogen unit's cost gap, and its derivation is set out under seasonal storage in Appendix A.13.

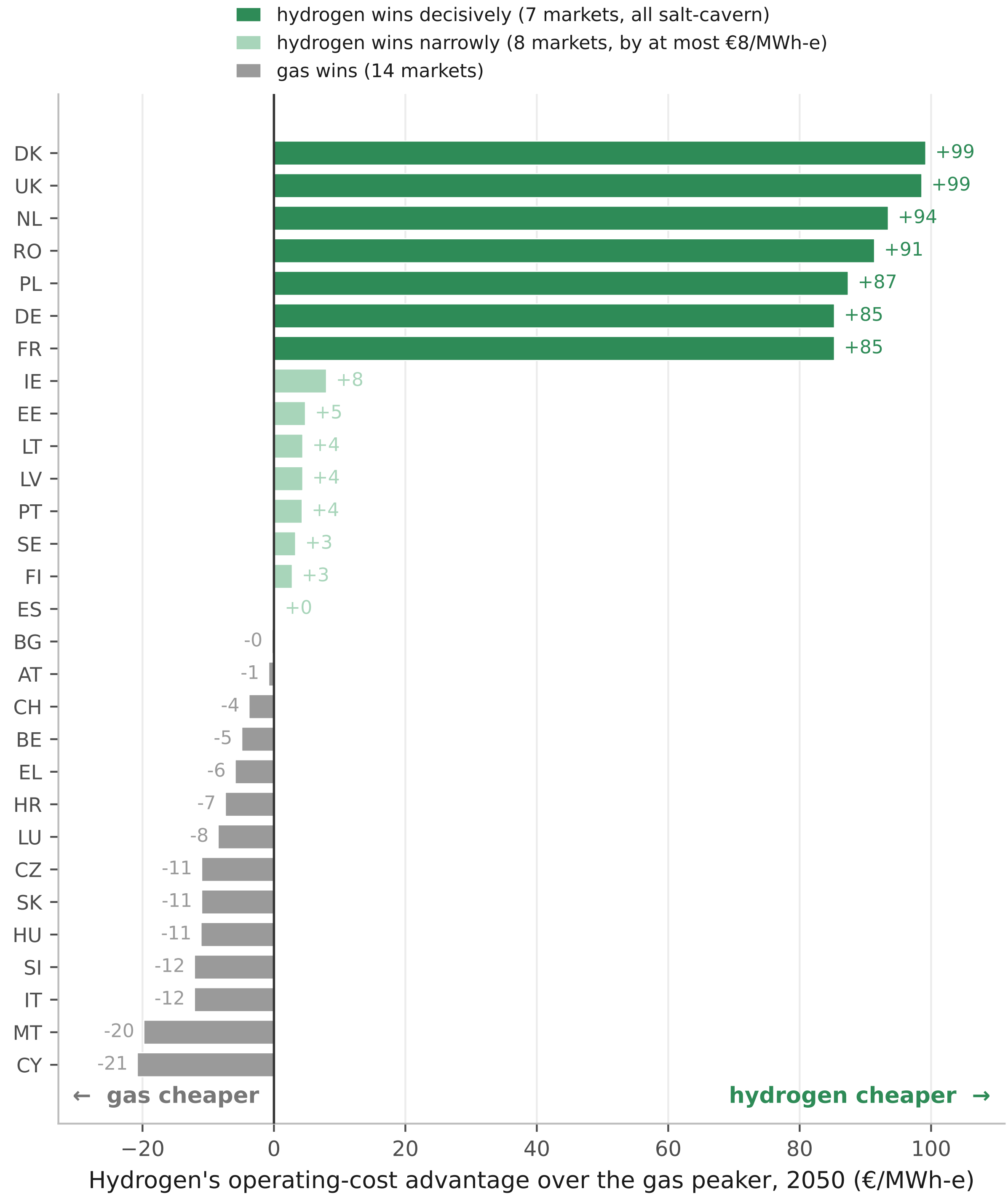
Ordered by Equation (4) against the low-carbon base of the 2050 fleet, the merit order runs from near-zero variable renewables, through nuclear (about €12/MWh) and biomass (about €55/MWh), up to the dispatchable tip. There the hydrogen turbine is bimodal across countries, so a single average describes it poorly. On the evolving-efficiency series used for capital recovery, which gives the hydrogen turbine the better machine, at the cross-country median it sits at about €238/MWh-e against a carbon-priced gas peaker at about €239/MWh-e, which is a tie. Held at equal efficiency, the series that sets the win counts, the same medians are about €286 against €256/MWh-e, so hydrogen sits clearly above gas in the typical market. The cross-country mean of about €221/MWh-e, some €18/MWh-e below gas, is pulled down by the cavern tail and should not be read as the typical market. On that same evolving-efficiency series, in the seven cavern markets the hydrogen turbine falls to about €141 to 155/MWh-e, decisively below gas; held at equal efficiency the equivalent range is €168 to 184. Eight of the remaining 22 sit within €8/MWh-e of it, inside the noise of the estimate, and the other 14 sit above it (Figure 12).

Figure 12: **2050 power merit order in Denmark, under H2 Push.** A salt-cavern market. Short-run marginal cost by technology, on the *equal-efficiency* turbine series that holds both machines at the same conversion efficiency. That series prices the gas peaker at €256/MWh-e and the hydrogen turbine at €168 here; the evolving-efficiency series puts these same two Danish machines at €241 and €142, because it moves the gas machine's efficiency and fuel price as well as the hydrogen turbine's. The €239 and €221 quoted in the paragraph above are cross-country statistics, the median gas cost and the mean hydrogen cost, and are not Denmark's. Costs are Denmark's; the capacity column is the study-area 2050 fleet, and the two dispatchable rows carry the same firm fleet because they are alternative fuels for it. Across the 22 non-cavern markets the hydrogen turbine sits about €36/MWh-e above the gas peaker, so the median country places hydrogen above gas.



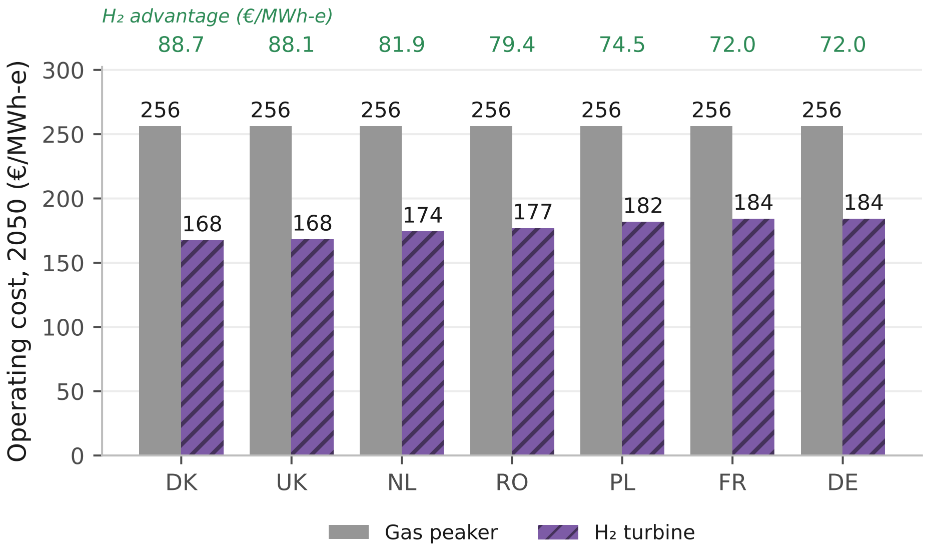
Source: Authors' contribution.

Figure 13: **Hydrogen's operating-cost advantage over the gas peaker, 2050.** By country, under H2 Push, on short-run marginal cost with capital excluded, on the *evolving-efficiency* turbine series. Hydrogen clears zero in 15 markets, but the shading separates two cases the count runs together. Seven markets, all salt-cavern, lead by €85 to 99/MWh-e and carry the result reported in the text. Eight more clear zero by at most €8/MWh-e, which is inside any reasonable read of model noise, and they should not be counted with the seven.



Source: Authors' contribution.

Figure 14: **Where hydrogen is decisively cheaper.** The seven salt-cavern markets, on the *equal-efficiency* turbine series that holds both machines at the same conversion efficiency. This is the series the headline seven is reported on, so its margins, €72 to 89/MWh-e, are narrower than the €85 to 99/MWh-e that Figure 13 shows for the same seven markets on the evolving-efficiency series, where the hydrogen turbine is credited with the better machine.



Source: Authors' contribution.

The 2050 snapshot is the end point of a trajectory. Table 4 traces the two operating costs from 2025. Hydrogen's falls steeply as delivered hydrogen cheapens and the turbine improves, while the carbon-priced gas peaker rises, so the two cross only late and only under H2 Push.

The same scenario monotonicity seen at the heating level recurs here. Table 5 collects the count of countries in which hydrogen wins across the three arenas and the four scenarios. In the power merit order hydrogen wins nowhere under Current or Stated Policies, but in seven countries under Net Zero and 15 under H2 Push, precisely the high-carbon worlds. The same monotonicity appears at the building winter peak (0, 2, 7, 14) and inside district heating (0, 0, 7, 7), both on the symmetric basis, and on the asymmetric one at (0, 4, 9, 16) and (0, 5, 11, 20). The mechanism is the carbon pass-through that raises the gas peaker's marginal cost [97], sharpened by the extension of carbon pricing to buildings and road transport under the second Emissions Trading System, ETS2 [125, 146]; the position of hydrogen in the stack is set as much by the fortunes of its fossil rival as by its own cost.

Table 4: **Mean peaker operating cost, by scenario and year.** Across the EU-27, the United Kingdom and Switzerland; short-run marginal cost in EUR/MWh-e, fuel and carbon only, no capital. Hydrogen's cost falls as delivered hydrogen cheapens and the turbine improves; the carbon-priced gas peaker rises with the carbon path (about 70 to 350 EUR/tCO_2 in the two high-carbon scenarios). The two cross only late and only under H2 Push (221 versus 239 in 2050 on this MEAN basis; on the median the same year is a tie at 238 against 239, and the AE submission reports the median because this distribution is bimodal and the mean is dragged by the salt-cavern tail, so the sign of the crossing depends on which statistic is quoted); in the lower-carbon scenarios gas stays cheaper throughout. The EU-27+UK+CH mean understates the salt-cavern markets, where the hydrogen cost is far lower (Figure 13).

Scenario	Peaker	2025	2030	2040	2050
Current Policies	Gas	114	121	141	143
	H_2	667	605	487	416
Stated Policies	Gas	127	147	165	167
	H_2	667	417	316	281
Net Zero	Gas	135	186	224	239
	H_2	667	417	316	281
H2 Push	Gas	135	186	224	239
	H_2	667	393	267	221

Source: Authors' contribution.

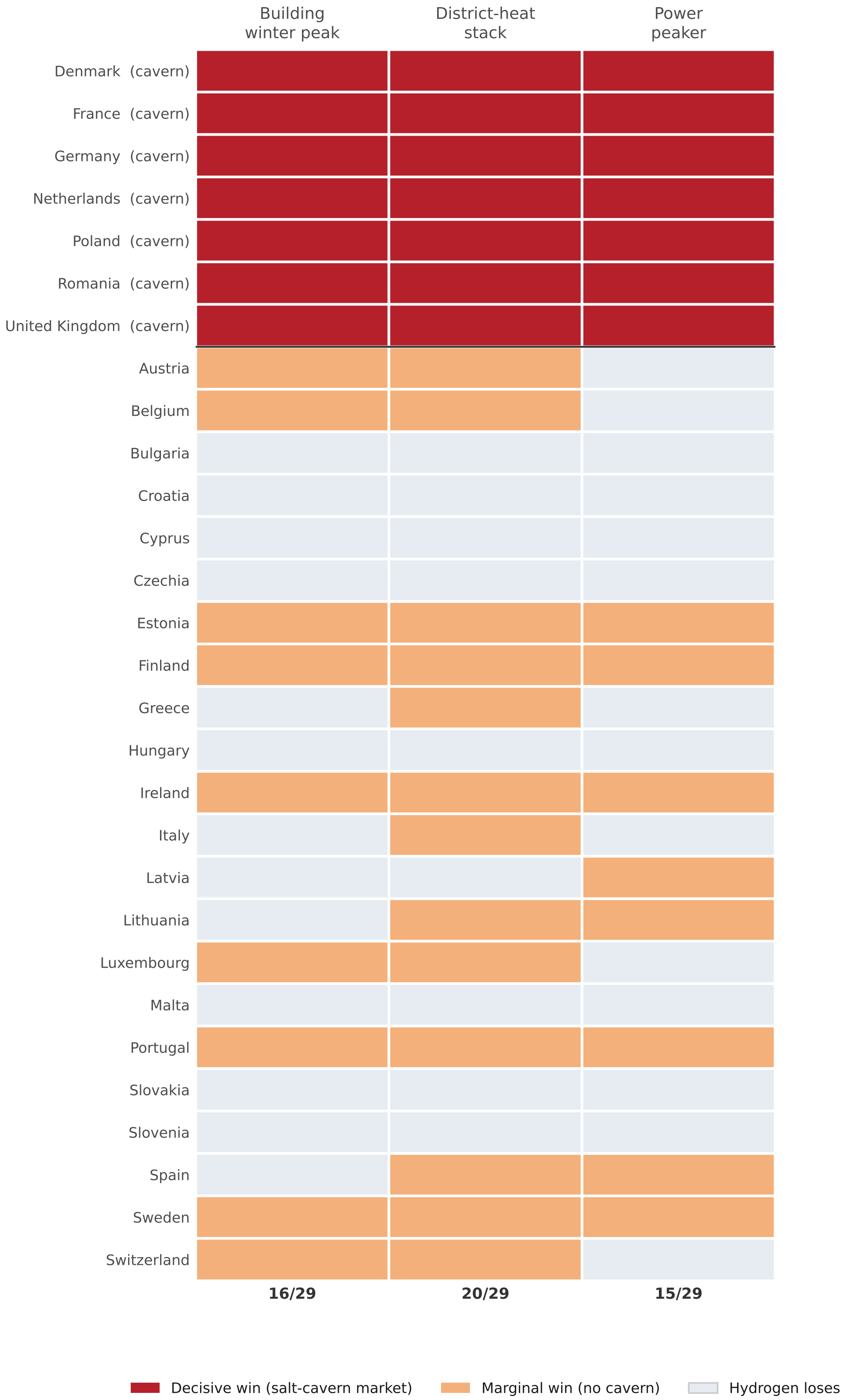
Table 5: Operating-cost wins do not translate into capital self-financing. The power-sector row is on the evolving-efficiency turbine series; the even-handed series, holding both turbines at 0.40, gives seven wins under both Net Zero and H2 Push, exactly the salt-cavern markets.

	Current	Stated	Net Zero	H2 Push
Panel A. Operating-cost wins (merit-order dispatch; count of 29)				
Symmetric basis: hydrogen pays its own last-mile adder, gas enters district heat at the wholesale hub price. Headline.				
Building winter peak (vs gas peaker)	0	2	7	14
District-heat peak (vs gas CHP)	0	0	7	7
Asymmetric basis: hydrogen pays no last-mile network, gas is priced at the residential retail tariff.				
Building winter peak	0	4	9	16
District-heat peak	0	5	11	20
Neither charge enters the power arena, so these serve both.				
Power peaker, evolving turbine efficiency	0	0	7	15
Power peaker, both turbines held at 0.40	0	0	7	7
Panel B. Capital recovery on market rent (count of 29)				
Recovers peaker capital from market rent	0	0	0	0

Cumulatively to 2050 the H2 Push hydrogen peaker loses about €357 bn against about €142 bn for an equivalent gas peaker, so neither fuel recovers its capital in any country (Section 5.3). District-heating wins in Panel A are countries in which hydrogen undercuts gas combined heat and power (CHP) on cost; gas CHP nonetheless remains the marginal source that sets the district-heat peak in 20 of 29 countries (hydrogen in 9), because hydrogen is dispatched below it. Costs follow the Danish Energy Agency Technology Catalogue [147] with the scenario carbon and hydrogen price paths (Section 3).

The win counts of Table 5 aggregate a heterogeneous country picture, and that heterogeneity is the paper's first contribution. Figure 15 resolves it, mapping for each of the 29 countries whether hydrogen wins or loses in each of the three arenas under H2 Push, and distinguishing the decisive salt-cavern wins from the marginal non-cavern ones. The pattern is bimodal. The seven cavern-endowed markets (Germany, the Netherlands, France, Denmark, the United Kingdom, Poland and Romania) win across all three arenas and carry the firm-capacity tip, while the remaining wins are marginal and, as the missing-money result will show, unfunded. So although hydrogen wins on operating cost in 14, 7 and 15 of 29 countries across the three arenas, only about seven of those wins are decisive.

Figure 15: Where hydrogen wins, by country and arena (H2 Push, 2050). The power-peaker column is the evolving-efficiency turbine series, which gives 15 wins; the equal-efficiency series that the headline count uses gives seven, all salt-cavern. Shading separates decisive wins, in the cavern markets, from marginal ones.



Source: Authors' contribution.

5.2 The Role This Implies, a Capacity Role Measured in Peak Hours

How much of the power system can hydrogen take? The answer divides in two, and the division *is* the result. The firm peaking fleet, sized to a loss-of-load standard in the hourly dispatch of Appendix A.7, is about 278 GW across the EU-27, the United Kingdom and Switzerland, about 31 per cent of the 889 GW system peak. Both figures, and every power-sector number that follows from them, are computed on *one realisation of a single synthetic weather year* and not on reanalysis or metered weather. That is the largest simplification in this paper, it is set out in full in Appendix D, and a reader who wants a system adequacy number should not take one from here. Four further qualifications belong with those two figures, and two of them bound the 278 GW itself. The fleet is sized on a completely inflexible electrified heat load. Making 10 per cent of the heat-pump load shiftable within three hours leaves it at about 277 GW, and at the most generous corner we sweep, 30 per cent shiftable across thirteen hours, three energy-conserving smoothing operators span 265.6 to 248.0 GW, a reduction of 4.3 to 10.7 per cent.

And the single synthetic cold-and-still event that sizes it is assumed to last six days; sweeping that length from two days to ten moves the fleet sized on the inflexible load from 253.7 to 302.0 GW, so 278 GW carries a band of roughly ±9 per cent from that assumption alone, wider than the flexibility correction. Both make 278 GW, and the €357 bn shortfall that scales with it, upper bounds. Neither is a point estimate. The 889 GW sums 29 national peaks that do not coincide, so it bounds the requirement from above in an interconnected system, which is what the copper-plate arm of Appendix A.7 tests. The 278 GW is an increment on a firm combined-cycle fleet of about 195 GW that the dispatch holds fixed and runs ahead of the peaker, so total firm dispatchable capacity is about 473 GW, or 53 per cent of that peak; what follows concerns the peaking increment, which is the part hydrogen competes for. Where hydrogen undercuts gas it can serve that backup tip; aggregating the firm capacity of the 15 winning countries,

$$\Phi_{\text{cap}} = \frac{\sum_{c\,:\,c_c^{\text{H}_2} < c_c^{\text{gas}}} K_c^{\text{pk}}}{\sum_c D_c^{\text{pk}}} \approx \frac{233\text{ GW}}{889\text{ GW}} \approx 26\%, \qquad \Phi_{\text{en}} = \frac{E^{\text{pk}}}{D^{\text{tot}}} \approx \frac{28\text{ TWh}}{4{,}513\text{ TWh}} \approx 0.63\%, \tag{5}$$

where, summing over countries c, the subset $c : c_c^{\text{H}_2} < c_c^{\text{gas}}$ in the numerator is the 15 countries in which hydrogen wins the dispatch on operating cost; K_c^{pk} is country c's firm peaker capacity (GW), D_c^{pk} its annual peak electricity demand (GW), E^{pk} the EU peaker energy (TWh) and D^{tot} total annual power demand (TWh).

One convention decides that 15, and we have not settled it. The looser short-run cost series that carries the capital-recovery accounting prices the molecule at 0.85 of its delivered price on the reasoning that a turbine sited on the hydrogen backbone avoids the last-mile margin, while our own cost layer treats it as though it does not. Removing the discount takes the count from 15 to 8, the seven cavern-endowed markets plus Ireland, and with it Φ_{cap} from 26.3 to 23.3 per cent and Φ_{en} from 0.63 to 0.36 per cent. The 15 and the shares above should therefore be read as the top of an 8 to 15 range on a convention we do not resolve, and the seven-market equal-efficiency count that carries the headline is unaffected, because it prices both turbines on the same basis.

The 4,513 TWh denominator rests on an assumption we state here. Non-heat national electricity demand is set at its 2023 level, 2,926 TWh across the 29 markets on ENTSO-E and Ember reporting, and grown by a smooth factor reaching 1.40 by 2050 (1.00, 1.10 and 1.25 at 2025, 2030 and 2040) to stand for the electrification of road transport and industry. That is an assumed path, equivalent to about 1.35 per cent a year compounding, which no model produces, and a single factor is applied across countries because the comparison at issue is between two firm technologies inside a single demand envelope; demand futures are not being compared. Heat-pump electricity from the heating model is added on top, at 417 TWh in 2050, giving the 4,513 TWh total. Both shares scale inversely with the growth factor; a flat baseline would raise this winners' energy share to about 0.85 per cent, and the whole-fleet share to about 1.2 per cent and leave the capacity share, set by the winter peak rather than by annual energy, materially unchanged.

Φ_{cap} is thus the share of system peak capacity that hydrogen can serve and Φ_{en} the share of generated energy it supplies. Hydrogen can take about 26 per cent of study-area peak *capacity* through the winning countries, which is some 84 per cent of the 278 GW peaker fleet, yet supply about 0.63 per cent of generated *energy*, because the peaker runs only about 139 full-load hours a year. Counting the whole fleet, and not the winners alone, gives the 31 per cent and 0.86 per cent quoted elsewhere in this paper; the difference between the two pairs is the 14 non-winning markets, of which 9 hold some peaker capacity. The two figures describe the same plant from the two sides of the system, capacity and energy. Hydrogen's electricity role is accordingly a capacity role, and its energy contribution is negligible. It can serve as the clean firm backstop for the electrified heat load while remaining a negligible share of generated kilowatt-hours. The reading echoes the wider finding that the system value of a clean firm resource resides in the scarce hours it covers, which its annual output does not measure [29, 98].

5.3 The Investment Caveat

Setting the price and taking the role on operating cost does not, on its own, finance the plant. The missing-money test of Section 4.7 applies unchanged to the power peaker. The inframarginal rent recovers only about 9 per cent of the power peaker's annualised capital by 2050, capacity-weighted across the five cavern markets that build a hydrogen peaker, which average 80 running hours against 139 for the fleet as a whole (Figure 16).

Two recovery percentages appear. Figure 16 plots the study-area aggregate, which reaches 14.6 per cent by 2050, and the 9 per cent quoted here weights only the five cavern markets that build a hydrogen peaker, by the capacity each builds, and Germany's 88 GW at 48 running hours dominates that weighting.

The aggregate is still rising at 2050. It turns positive in 2037 and climbs monotonically to the horizon, so it bounds nothing beyond it and we make no claim about where it settles.

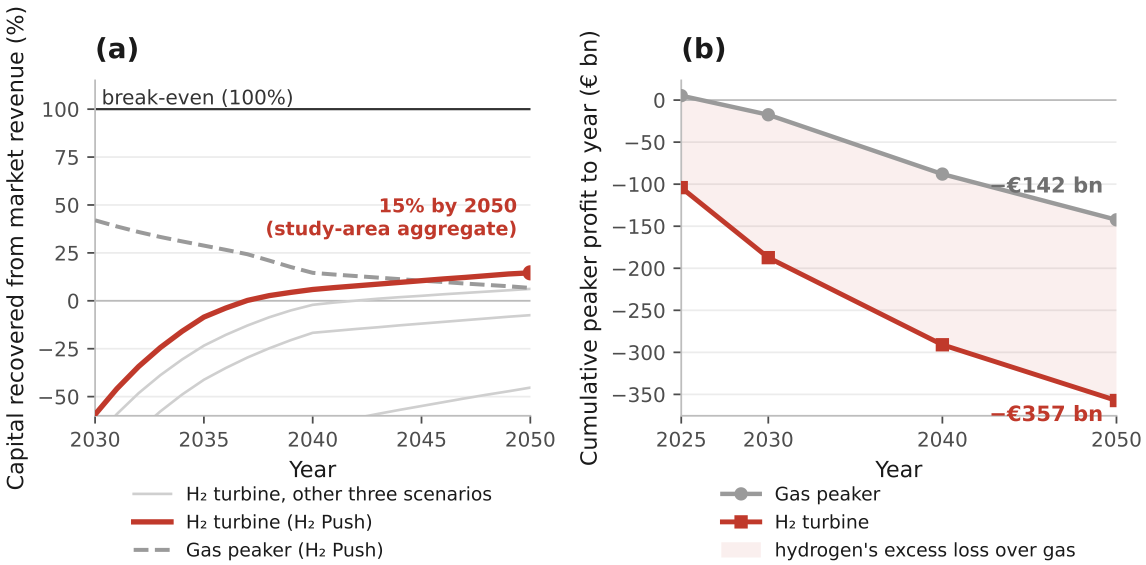
Cumulatively to 2050 the H2 Push hydrogen peaker loses €357.3 bn against €142.4 bn for an equivalent gas peaker, on the reduced-form dispatch that also sets the fleet size and the running hours. The unit-commitment layer runs the same economics on its own dispatch and lands close to them, at losses of €358.4 bn and €143.2 bn with its constraints switched off and €356.0 bn and €142.3 bn with them on, because the constraints raise peaker run-hours and the plant earns in more hours as well as burning more fuel. That spread is the effect of the commitment constraints, reported in Appendix A.7.

Because electricity is priced on operating cost, this shortfall does not prevent hydrogen from setting the price or winning the dispatch in the 15 countries; it means that a deliberate hydrogen-peaker build implies a standing capacity payment or equivalent support, the classic missing-money outcome of low-utilisation peaking plant [28, 148]. The verdict is economic, not one about reliability. The modelled system remains adequate, with a loss of load averaging about two hours a year (Appendix A.7), and the difficulty is paying for the firm capacity, not keeping the system adequate. The dominant driver of the gap is the seasonal hydrogen-storage adder σ in Equation (4), which the delivered fuel price does not set; even at hydrogen's physical production floor the peaker's run-hours set the ceiling on what market rent can recover, and its fuel cost does not.

Two qualifications belong with that verdict, and both cut against reading it as a technology result. The first is that the shortfall is a property of the market design we impose, and the machine itself does not cause it. Rent per kW is the scarcity price times the hours the plant runs, so with the scarcity premium fixed at €60/MWh the recovery percentage follows from run-hours alone by Eq. (9) (Appendix A.2). Germany runs 48 hours, which at €60/MWh is €2.9/kW-yr against €53.4/kW-yr of annualised capital, or 5.4 per cent. The model also enforces a three-hour loss-of-load standard and prices unserved energy at a value of lost load of €8,000/MWh, but pays that price to no generator. Crediting the peaker for those three hours at the model's own value of lost load adds about €24/kW-yr and lifts capacity-weighted recovery from 8.7 to about 51 per cent, with Denmark moving from 40 to about 87. The missing-money finding is therefore a statement about an energy-only market with an administrative price cap, set out as such in Appendix D.4. It does not say the plant cannot be paid for.

The second is that a capacity payment large enough to build firm capacity would not, on these numbers, build *hydrogen*. The hydrogen peaker's rent shortfall is a standing payment of €31 to 63/kW-yr across the five markets that build, against €41 to 52/kW-yr for a gas peaker in the same markets. The gas figure is the smaller of the two in four of those five, covering about 96 per cent of the winning capacity, because the hydrogen-ready machine costs about €600/kW against about €450/kW and that capital gap outweighs its smaller rent shortfall. A technology-neutral mechanism procures unabated gas, which is what the emissions limits in Article 22(4) of the Electricity Regulation are meant to prevent and, at 550 gCO_2/kWh and 350 $kgCO_2$/kWe-yr, do not [149, 150]; the 2024 British four-year-ahead auction is a mechanism doing exactly this, procuring 27.3 GW of unabated gas out of 43.1 GW at £60/kW-yr [151]. Selecting hydrogen instead requires the payment to be conditioned on emissions, and the implied conditioning premium is about €172/tCO_2 on top of the carbon price the gas plant already pays, rising to roughly €239/tCO_2 if the gas peaker's negative rent is floored at zero on the ground that no operator would hold that position. The direction is unchanged either way, and the lower figure is the conservative end.

Figure 16: The power peaker is competitive on operating cost yet unfunded, under H2 Push. Panel (a) is the share of the hydrogen turbine's annualised capital its market rent returns, against the same ratio for a gas peaker, with the three other scenarios drawn faint for context. Its 15 per cent is the study-area aggregate, the sum of rent over the sum of annualised capital across the countries the dispatch builds a peaker in. The 9 per cent quoted in the text is a different statistic on the same run, the capacity-weighted mean of the per-country ratios across the five cavern markets that build, in which Germany's 88 GW at 48 full-load hours dominates the weighting while Denmark's 40 per cent recovery carries only 8 GW. Panel (b) is the cumulative profit of each fleet to the year shown, and the shaded wedge between them is what hydrogen loses over and above gas.



Source: Authors' contribution.

5.4 Hydrogen Supply and Network Infrastructure

Two infrastructure questions bound the picture without changing it. On *supply*, the delivered hydrogen cost that enters Equation (4) is dominated by domestic green hydrogen.

Across the 29 countries the cheapest 2050 route is domestic electrolysis in 23 and pipeline import in six, at a delivered cost of about €37 to €64/MWh (about €1.24 to €2.14/kg), with shipped ammonia and blue or pink hydrogen more expensive [92, 152]. A hydrogen economy for peaking needs only electrolysers, renewable supply and storage, and so avoids the critical-mineral and lithium dependence of battery backup; its principal cost is the seasonal storage that Equation (4) already prices, which we treat as a cited trajectory rather than a built network, and leave, together with ammonia and other carriers, to further work.

On *networks*, each pathway carries a distinct bill (cumulative 2025 to 2050, central estimate, all 29 markets; Appendix A.15 and Figure A.11): about €249 bn under Current Policies, €325 bn under Stated Policies, €506 bn under Net Zero, and €412 bn under H2 Push. In every pathway the bill is dominated by district-heat expansion and grid reinforcement, which a hydrogen network does not drive. H2 Push is the cautionary case. It pays about €74 bn for a hydrogen distribution network *and* still about €89 bn of grid reinforcement, because hydrogen serves only the peak while heat pumps continue to set the distribution-level load. A hydrogen role genuinely confined to firm peaking, as the merit order implies, argues for reusing the existing gas transmission backbone and salt caverns for seasonal balancing [36, 90], and not for building a parallel hydrogen distribution grid to dwellings. This is the buildings-to-power synergy in its sharpest form. Whatever role hydrogen holds sits at the firm-capacity end of the power system, and Section 5.3 has already shown that the plant holding it recovers about 9 per cent of its capital from energy-only rents, so the role depends on a standing payment the market does not currently provide. The synergy is therefore a claim about where hydrogen belongs if it is built, and not a claim that building it pays.

6 Discussion

6.1 Interpreting the Results in Light of the Research Questions

This paper set out to answer three questions. Two are empirical and are taken up here, namely under what conditions decarbonised hydrogen is cost-competitive with heat pumps for residential heating across the EU-27, the United Kingdom and Switzerland (RQ1), and how Switzerland can draw on the wider European hydrogen build-out, including imports, to serve its own net-zero buildings strategy (RQ2). The evidence yields negative conclusions for both questions, though for reasons that differ from those the framing of the questions anticipated. Hydrogen heating does not drive a building-heat transition on its own economics (RQ1), and it offers limited strategic value for Switzerland's net-zero buildings strategy despite that country's access to European supply (RQ2). Hydrogen reaches between under 1 and about 8 per cent of useful heat at the modelled end-date, and only under scenarios with substantial carbon pricing (Figure 5). That ceiling is set by the peaking niche hydrogen can win, by the investment economics of the plant that would serve it, and by a base-load comparison that now points the same way.

Once equipment capital is charged against machines sized to peak heat demand, once each carrier pays for its own last-mile network, and once hydrogen pays for the seasonal store that its winter-weighted load shape creates, the best available heat pump undercuts the hydrogen boiler in 22 of 29 countries by 2050, at a median gap of +€15.4/MWh (Figure 6). Five of the seven exceptions are salt-cavern markets. Space heating draws about three-quarters of its annual energy in the October-to-March half of the year while electrolytic production is far flatter, so a hydrogen heating load has to be stored across the seasons, and hydrogen's base-load case concentrates where the geology makes that store cheap. This is the same endowment that decides the three peaking arenas, the building winter peak, the district-heating stack and the power-sector peaker.

The arenas therefore close the same door for one physical reason, and their agreement is worth less as corroboration than it may appear. All three price the hydrogen side through the same storage parameter in the same functional form, so their country rankings are correlated by construction, and finding that they line up is closer to a property of the model than to three independent confirmations. What they do establish is scope, since an operating-cost win in one arena does not carry to another and none clears the capital-recovery hurdle. No hydrogen peaker recovers its capital from market rent in any country or any scenario, on the assumed €60/MWh scarcity premium and with no capacity payment, and what base-load lead survives in the seven rests on a fiscal asymmetry between an untaxed molecule and taxed retail electricity, not on a technology advantage.

The finding qualifies the framing embedded in REPowerEU [11] and in several national hydrogen strategies, which implicitly position hydrogen as a near-term substitute for natural gas in buildings. The analysis suggests instead that the transition will be heat-pump-led, and that any material role for hydrogen in buildings before 2050 would have to rest on targeted policy support justified by co-benefits. Where the delivered-cost comparison still favours hydrogen it does so on a tax-inclusive electricity price set against an untaxed molecule, so a strategy built on that comparison rests on a policy choice a future government can reverse, in the only countries where the comparison survives at all.

6.2 Comparison with Prior Literature on Heating

The results align with, but qualify, a maturing body of comparative work. Berger et al. [81] found that heat pumps dominate gas boilers on lifecycle cost across most of Europe by 2030 under moderate carbon pricing. The present analysis supports that finding on the 2050 horizon, where the best available heat pump undercuts the gas boiler in 24 of

29 countries on the central carbon path, and qualifies it in the near term, where the two stand at parity at the study median in 2025 and the heat pump is ahead in 16 of 29. That 2050 count is the one result here that moves with the carbon price, running from 19 of 29 on the low path to all 29 on the high path, precisely because the gas boiler pays ETS2 and the heat pump does not. The qualification traces to equipment capital, which is charged here against a machine sized to peak heat demand and so does not shrink with the seasonal coefficient of performance. Figure 6(b) ranks the 2050 levelised cost of heat by country and technology under the central carbon and hydrogen trajectories, plotting the best available heat pump, the hydrogen boiler and the gas boiler for each.

Huckebrink and Bertsch [26] concluded that heat pumps outperform hydrogen boilers for German residential heating, attributing the result to the coefficient-of-performance advantage. The present model still does not reproduce that ordering for Germany, and the discrepancy is worth setting out. Reading the German entry of that ranking, the best available heat pump reaches €139.3/MWh by 2050 and the hydrogen boiler comes in about €25/MWh below it, which places Germany among the seven markets in which hydrogen still leads on base load. Three things put it there, and unusually cheap German hydrogen is not among them. The first is that equipment capital is charged against a machine sized to peak heat demand, which withdraws an implicit credit that a high seasonal coefficient of performance would otherwise confer on the heat pump's capital charge.

The second, and the larger, is the German treatment of retail electricity, in which levies, taxes and network charges lift the household tariff far above the wholesale cost of the kilowatt-hour, while hydrogen is costed delivered and untaxed and carries no equivalent burden. The third is geological. Germany has salt caverns, so it holds its seasonal hydrogen imbalance at about €1.5/kg, worth €12.1/MWh of useful heat, against €28.4/MWh in a cavern-free market such as Spain, and it is that endowment that keeps Germany on hydrogen's side of a line that most of the study area now sits the other side of. The German case is therefore a statement about relative fiscal treatment and storage geology. It makes no claim that the coefficient-of-performance mechanism has ceased to operate. That mechanism is intact, and a levy-neutral comparison would restore much of the ordering Huckebrink and Bertsch [26] reports.

The cost-stack decomposition of Figure A.3, set out in full in Appendix A.4, bears on that literature in three ways. It corroborates the emphasis on the running-cost penalty of combustion heating, since fuel is the largest single component of all three stacks. It locates hydrogen's disadvantage in the logistics of the molecule and not in the cost of producing it, since delivery and seasonal storage take more than a quarter of the hydrogen stack. And it does not support treating the heat pump as a capital-dominated technology set against fuel-dominated rivals, because fuel is the largest item in the heat-pump stack too, at 54.2 per cent, even though the heat pump carries the heaviest capital share of the three.

The scenario comparison of Billerbeck et al. [22], whose model-based 'race' between hydrogen and heat pumps for space and water heating finds heat pumps winning across most configurations, and the techno-economic caveats catalogued by Badakhsh and Bhagavathy [153] for green hydrogen in buildings, reach the same verdict on the aggregate outcome. The results here agree on that outcome and locate it more precisely, since the ceiling on hydrogen arises from the cost of holding a winter-weighted load across the seasons, which acts on the base-load comparison, the peaking dispatch and capital recovery alike, and which is lower in the seven markets whose salt caverns can hold that store cheaply.

The emissions ranking is consistent with the life-cycle literature. Slorach and Stamford [82] showed that hydrogen boilers carry higher lifecycle emissions than heat pumps even on low-carbon hydrogen, owing to upstream production and distribution losses; the planetary-boundaries assessment of Weidner and Guillén-Gosálbez [27] reaches a comparable verdict across a wider set of impact categories. The emissions results here corroborate the ordering. Net Zero achieves the lowest near-term CO_2 trajectory, at 339 $MtCO_2$ by 2030, against 513 $MtCO_2$ for H2 Push, 532 for Stated Policies and 548 for Current Policies, because deep renovation combined with grid decarbonisation drives heat-pump emissions down faster than hydrogen production chains can themselves decarbonise.

At the spatial margin, Maestre et al. [25] found that hydrogen may be competitive in specific high-density urban building segments in Spain yet is unlikely to achieve mass-market adoption on economic grounds; the country-level results here place Spain firmly on the heat-pump side of the base-load comparison, because it lacks the salt caverns that halve the seasonal-storage charge, so any Spanish hydrogen case rests on the dense urban segments Maestre and colleagues identify and not on the national cost ordering. The contribution relative to this literature is threefold: a residential heat-demand reconstruction built bottom-up at NUTS3 resolution, a merit-order test that prices hydrogen on short-run marginal cost in a dispatch, not on levelised cost alone, and an investment test that asks whether the peaker can recover its capital, which the levelised comparisons do not pose.

6.3 Switzerland's Low Hydrogen Demand for Heating, a Consequence of Absent Cavern Storage

The Swiss case answers RQ2 directly, and every arena now answers it the same way. On base-load levelised cost Switzerland sits on the heat pump's side of the line, at a 2050 heat-pump band of €122.4 to 131.1/MWh against €128.0/MWh for the hydrogen boiler, so the best Swiss heat pump is about €5.6/MWh the cheaper. The reason is the same one that orders the study area as a whole. Switzerland has no salt caverns, so a Swiss hydrogen heating load is stored at the €3.0/kg rate, not the €1.5/kg cavern rate, and that charge is what holds expensive, heavily taxed Swiss retail electricity ahead of an imported molecule. The case against direct hydrogen heating there is reinforced by the other two arenas and by the country's position in the supply chain.

Hydrogen wins no Swiss peak in any of the four scenarios once the Swiss reservoir fleet is credited as firm winter capacity, the Swiss hydrogen peaker costing €135 to 236/MWh-heat against a heat-pump peak of €62 to 90/MWh-heat. Without that credit the uncoupled 29-country comparison prices the Swiss peak at €155/MWh-heat and hydrogen does win it under H2 Push, which is why Switzerland is one of the 16 in that row; the credit removes the win, and the endogenous arm is the better-specified test for a country whose reservoir headroom the uncoupled series does not see. No Swiss peaker recovers its capital on either arm; and Switzerland is cavern-poor and import-dependent, so every molecule burned in a Swiss home is one bought on a competitive European market. Switzerland illustrates a more general principle alongside this. Countries that combine a clean electricity grid with high building-fabric standards weaken the *emissions* case for hydrogen in buildings relative to the EU-27+UK+CH average, so the question that matters is why direct hydrogen heating is a poor strategic bet there in the first place. France, whose grid intensity of 80 gCO_2/kWh in 2025 falls to 8 gCO_2/kWh by 2050, and Sweden, where it falls from 45 to 5 gCO_2/kWh [101, 141], share this characteristic. For these countries the policy priority is to accelerate heat-pump installation and the supporting grid reinforcement.

6.4 What Switzerland Can Draw from the European Build-Out

The finding that direct hydrogen heating is a poor strategic bet for Switzerland does not exhaust RQ2, since it leaves open whether Switzerland can draw on the European hydrogen build-out for ends other than heat. Two qualifications close the loop in that direction. First, the low domestic demand for hydrogen heat is a strength rather than a vulnerability for an import-dependent, cavern-poor system. A heat-pump-led buildings sector spares Switzerland from competing for scarce imported molecules to warm its homes, and frees its access to European supply for uses where the molecule carries a higher value.

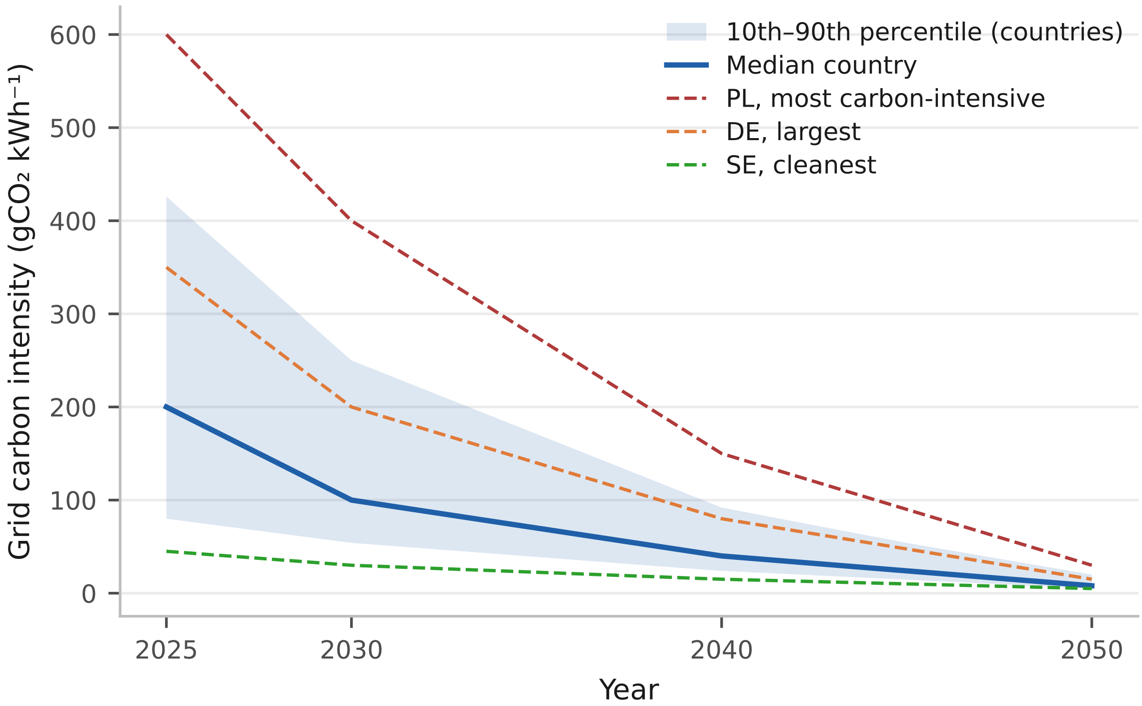
Second, those higher-value uses lie outside buildings. Swiss access to the emerging European hydrogen network may create value as a consumer for hard-to-abate industry, parts of heavy transport, and a firm-capacity role in the power system against the country's winter-electricity adequacy gap, the very sectors in which hydrogen substitutes for costly or absent alternatives and not for a mature heat pump. The strategic answer to RQ2 is therefore that its leverage over the European build-out is best realised away from the building stock, where the same access that does little for residential heating may yet underwrite decarbonisation in the harder sectors; the buildings-to-power link is taken up under RQ3 below, and the wider industrial and transport uses are left to future work.

6.5 Coal-Grid Markets and the Trajectory Argument

The harder case is the set of Eastern European countries whose grids are still carbon-intensive today, and here the argument turns on the trajectory of grid carbon intensity rather than on its present level. In Poland, at 600 gCO_2/kWh in 2025 [101], heat pumps are already cost-competitive on levelised cost yet carry a higher near-term operating carbon footprint than in the clean-grid countries. That near-term penalty is transitional. The EEA projections have these grids decarbonising substantially by 2040, at which point heat-pump carbon performance converges with that of Western Europe [141]. The policy implication follows directly from this convergence. Eastern European countries should prioritise the gas-to-heat-pump switch now, while accelerating grid decarbonisation in parallel, instead of treating present grid carbon intensity as a reason to defer electrification, since the carbon penalty is temporary and the infrastructure choices made now are durable.

Figure 17 provides the supporting evidence for this convergence, tracing the grid-intensity trajectories of the 29 modelled countries from 2025 to 2050 with the most carbon-intensive, the largest and the cleanest grids marked. Because heat-pump operating emissions scale directly with grid intensity, the narrowing of the inter-country band from a 2025 spread of roughly 600 to 45 gCO_2/kWh to about a quarter of that width by 2040, and to an order of magnitude smaller by 2050, is what removes the near-term carbon gap on which the deferral argument rests, and quantifies the penalty that an Eastern-European heat pump carries over a Western-European one in any given year as the vertical distance between the trajectories.

Figure 17: Electricity grid carbon-intensity trajectories, 2025 to 2050, on the exogenous paths of Section A.10. The figure has no scenario dimension. It shows the median country, the tenth-to-ninetieth percentile band across the 29, and Poland, Germany and Sweden as the most carbon-intensive, the largest and the cleanest grids. These are inputs to the model and not results of it.



Source: Authors' contribution.

6.6 Policy Implication I: The Structural Role of ETS2 Carbon Pricing

The RQ1 finding carries two policy implications that the remainder of the discussion draws out; the electricity merit-order and peaking results that inform them are the substance of RQ3 and are developed in Section 5, so the present subsections read them as implications of the heating result. The first implication concerns carbon pricing. The second emissions trading system, ETS2, which extends carbon pricing to buildings and road transport from 2027, does more to move the pathway than any other policy lever in the pathway, and its incidence on the cost stack is quantified in Appendix A.4 (Figure A.3). The central claim of this subsection is that carbon pricing accelerates the transition instead of determining its direction. Heat pumps already stand at parity with gas boilers on levelised cost before ETS2 takes effect, at the study median in 2025 and ahead outright in 16 of 29 countries, so the carbon price acts as an accelerant on an ordering that is close to balanced without it.

The differential ETS2 burden is what tips the remaining markets. Each €50/tCO_2 of carbon price adds about €11/MWh of useful heat to the gas boiler, so the central path carries a 2030 median of about €27/MWh against a median €0.7/MWh on the heat pump. It closes the residual economic case for retaining gas boilers where they are still ahead, and raises the opportunity cost of delayed electrification everywhere else. The pass-through of the carbon price to the consumer is a material source of uncertainty. The model samples pass-through between 75 and 100 per cent, a range consistent with the empirical estimates of Fabra and Reguant [97]; incomplete pass-through, for instance where governments cap energy bills, would blunt the price signal and slow the transition.

This carbon-pass-through mechanism, set out above, recurs at the building peak, the district-heat stack and the power peak alike, and the country counts across the three arenas bear it out. Table 5 reports, for each of the four scenarios (Current Policies, Stated Policies, Net Zero and H2 Push), the count of the 29 countries in which hydrogen undercuts its fossil rival on operating cost in three arenas, the building winter peak, the district-heating peak and the power-sector peak. No row falls as carbon-pricing ambition rises, every row starts at zero under Current Policies, and the two building-scale rows reach their widest count under H2 Push, so no scenario without a carbon-price wedge on the fossil competitor opens the niche at all. The even-handed power row does not rise past seven from Net Zero on, because the carbon wedge that opens the niche is already at its full width there.

6.7 Policy Implication II: Infrastructure and Stranded-Asset Risk in Buildings

The second implication concerns infrastructure. If the Stated Policies or Net Zero pathways materialise, the gas distribution grids that serve residential buildings face steeply declining utilisation from the 2030s, which raises the risk of a tariff spiral. Falling volumes force fixed network costs onto a shrinking customer base through higher unit charges, the higher charges accelerate disconnections, and the disconnections raise charges further. This dynamic has been modelled for the United Kingdom by Competition and Markets Authority [96] and bears directly on Germany's municipal heat-planning (*Wärmeplanung*) mandate under the 2024 Building Energy Act.

Figure A.11 (Appendix A.15) quantifies the scale of the exposure. Under H2 Push the pathway builds a hydrogen distribution network and still needs more electricity reinforcement than that network costs, because hydrogen serves only the peak sliver while heat pumps carry the bulk of the load, so hydrogen does not, in aggregate, substitute for the network investment that electrification demands. The H2 Push pathway only partly forecloses the tariff spiral, since repurposing the gas grid to carry hydrogen preserves utilisation for the 9.3 per cent peak niche that the merit order supports under that scenario (Section 4.6), and the missing-money result implies that even that niche would require a capacity payment to be financed privately.

That 9.3 per cent is a merit-order and financing ceiling that the base-load comparison no longer disputes, since delivered cost now favours hydrogen only in a similarly narrow set, and the dispatch awards it a role in the seven cavern-endowed markets and in no others. The infrastructure-bill comparison therefore sharpens the stranded-asset tension instead of resolving it. The trade-off between stranding gas assets and building out hydrogen infrastructure prematurely is not squarely resolved in current EU or national strategy documents, and it warrants explicit policy analysis, the more so because the pipeline-blending route that is sometimes invoked as a bridge is itself bounded by the technical and regulatory blend limits reviewed by Erdener et al. [68] and by the distribution-grid material constraints assessed by Giehl et al. [69].

6.8 Cross-Sector Synergies and the Scope of RQ3

Beyond the carbon-pricing and infrastructure implications above, a third question follows. RQ3 asks what synergies hold between hydrogen in buildings and its use in other sectors. This paper answers the buildings-to-power coupling in full, on the same operating-cost basis as the rest of the analysis (Section 5), and defers the wider synergies with industry and transport to subsequent work, for the reasons set out in Section 1.1. On the deferred part, the present finding speaks in one direction only. As the Swiss case above already shows, hydrogen ranks low among the competing uses for the molecule, because low-temperature space heating measures it against a mature heat pump while hard-to-abate industry and parts of heavy transport measure it against costly or absent alternatives [24, 74]. The plausible spillover therefore runs from those uses back to buildings, where scaling electrolysis and seasonal storage for industry and power may lower the cost of the narrow building niche identified in Section 4.6 without widening it. A full treatment, in which buildings, industry, transport and power compete for a common hydrogen supply under a single cost-minimising frame, is left to a subsequent paper.

6.9 Limitations

Three limitations bear on the discussion above, ahead of the fuller treatment in Appendix D. First, while the power layer now commits the firm fleet with an explicit unit-commitment dispatch, the framework still does not embed a dynamic capacity-expansion model. The mass deployment of heat pumps raises electricity demand materially; our own bridge puts study-area heat-pump electricity at 417 TWh in 2050 (Section 5), which calls for grid reinforcement and flexibility measures that the dispatch coupling here quantifies but does not co-optimise. Second, the demand reconstruction operates at NUTS3 resolution, the finer spatial unit for which harmonised European data are available, with three building-type categories, single-family, multi-family including apartments, and other residential, trading intra-regional heterogeneity, in particular the dense historic urban cores that might support a hydrogen niche, for cross-national comparability across the 29 countries. Third, and now the most consequential of the three, the non-fuel cost components of the hydrogen boiler, its capital and operating costs, are less well established than those of mature technologies and carry wider parameter uncertainty than the heuristic ranges applied here convey.

That uncertainty was close to a rounding term while fuel accounted for almost the whole of the delivered cost. It no longer is, since more than two-fifths of the hydrogen boiler's 2050 cost now sits in capital, delivery and seasonal storage, the components with the thinnest evidence base, and it is that cost which decides the country-level base-load ranking. To those three limitations should be added the fiscal asymmetry set out above, which runs in hydrogen's favour and so bounds the result only from the side that would widen the heat pump's margin, and which Appendix D takes up in full. That appendix ranks the simplifications by consequence and leads with a different four, the synthetic weather year behind every power-sector number, the two network charges the symmetry does not cover, the same fiscal asymmetry, and the energy-only market behind the recovery figures. The three named here bear on the discussion above; the four named there bear on the results, and a reader weighing what the paper can carry should read both lists.

7 Conclusion

7.1 Summary of Key Findings

This section answers RQ1 and RQ2 as stated in Section 1.1; RQ3 is answered in Section 5. Each figure below is derived where it is first reported.

On RQ1 the verdict runs one way across every arena, and the reason is the same in each. Against the fossil incumbent it is settled. The best heat pump is cheaper than a gas boiler in 24 of the 29 markets by 2050, a count that runs from 19 to all 29 across the carbon-price paths because only the gas boiler pays that price. Against hydrogen the base-load contest settles too, with each carrier charged its own equipment capital, its own last-mile network and the seasonal storage its own load shape requires. The best heat pump leads in 22 of 29, at 2050 medians of €118.5/MWh against €122.8/MWh and a median country gap of €15.4/MWh in its favour, and hydrogen holds the remaining seven, five of them salt-cavern markets.

That concentration is the paper's central structural result, and its cause divides. Heating draws about three-quarters of its energy between October and March while electrolytic production is far flatter, so the molecule must be carried between seasons, at about €1.5/kg where salt caverns exist and €3.0/kg where they do not. Hydrogen's base-load case concentrates where that store is cheap, which is the same condition that decides the peaking arena, so the two tests turn on one physical quantity. That shared parameter is also why their agreement is a property of the model as much as an independent confirmation. Geology does not carry the whole of the concentration. Charging every country the cavern rate leaves the heat pump ahead in 19 of 29 and the non-cavern rate leaves it ahead in 24, so the geological binary is worth five markets and decides two of the seven exceptions. The other five turn on the retail tariff the heat pump pays.

What hydrogen keeps in those seven is not a technology result. It follows from the fiscal treatment of the two carriers. Pricing the molecule as blue hydrogen at the €81/MWh of our own blue route [88], the provenance one dedicated assessment concludes heating hydrogen would have [73], removes the lead in all seven; equalising the fiscal treatment would widen the heat pump's margin in the same direction, though the levy-equalised comparison that would size the effect is one we did not run (Appendix D). The base-load count moves with the hydrogen price path and not with the policy scenario, from 13 markets on the fastest cost decline to all 29 if hydrogen costs stall, because neither appliance pays a carbon price.

Where hydrogen's dispatch case is strongest it fails the investment test. Repriced as a winter peaker it wins on operating cost, under the most carbon-ambitious H2 Push scenario, in seven markets on the even-handed efficiency series and 15 on the evolving one, and those wins are decisive in only the seven salt-cavern markets of the north-west cluster that carries the firm-capacity tip, the remainder marginal (Section 5.2). In no modelled country-scenario does the inframarginal rent recover the peaker's capital. That is an economic signal and not a reliability one, since the system stays adequate at a modelled loss of load averaging about two hours a year. The case against hydrogen for heat therefore rests on the base-load and investment arenas together, and on the same endowment in each.

Three extensions test these results without altering them. An explicit unit-commitment cross-check of the firm fleet and an endogenous Swiss integration with electricity-price feedback both bear on the peaking verdict, and a myopic rolling-horizon test bears on the cost-optimal retrofit share.

On RQ2 the Swiss base-load and peaking comparisons both run against hydrogen. The best Swiss heat pump ends €5.6/MWh below the hydrogen boiler on the uncoupled arm we quote as the conservative one, and €9.6/MWh below it on the endogenous arm. Switzerland has short pipeline distances to the major European producers and dear retail electricity, but no salt caverns, so its heating load would be stored at the higher rate, and that charge holds the ordering. The peaking comparison points the same way with far more room, since hydrogen wins no Swiss winter peak in any scenario once the Swiss reservoir fleet is credited as firm winter capacity. Switzerland accordingly makes best use of the European build-out by directing it, as federal strategy already does, to industry and mobility, and meeting residential heat through electrification supported by its hydropower flexibility.

Common to both questions is an emissions divergence of about 342 $MtCO_2$/yr between drift and net zero that the four scenarios do not close. From a common ≈644 $MtCO_2$ 2025 baseline they reach 352 (Current Policies), 231 (Stated Policies), 123 (H2 Push) and 10 (Net Zero) $MtCO_2$ by 2050. Because the same technologies are available in every scenario, that gap measures ambition.

The same divergence appears on the cost side. Each pathway carries a distinct network-infrastructure bill cumulative to 2050, €249 bn under Current Policies, €325 bn under Stated Policies, €412 bn under H2 Push and €506 bn under Net Zero, dominated in the deep pathway by district-heating expansion at €356 bn against €119 bn of electricity reinforcement. The hydrogen-push variant spends €74 bn on a dedicated hydrogen distribution network and still requires €89 bn of reinforcement, so hydrogen investment does not displace grid expansion and the two networks must co-develop even in the hydrogen-heavy case.

7.2 Contribution

The contribution of this paper is threefold.

First, it provides what is, to our knowledge, the first merit-order assessment of hydrogen's role in residential heating across the EU-27, the United Kingdom and Switzerland (29 countries, 2025 to 2050), built on a bottom-up heat-demand reconstruction resolved to NUTS3, in which the heating share of hydrogen is bounded above by where it can win the cold-snap peak, not assumed. Building demand is reconstructed bottom-up from EUBUCCO footprints and TABULA intensities, validated against Hotmaps, and carried through a 200-draw Monte Carlo over four scenarios and an independent least-cost optimisation.

Second, it prices hydrogen the way a system operator does, on short-run marginal cost in a dispatch, and tests it in three linked arenas, the building winter peak, the district-heating stack and the power-sector peak. Testing all three is what makes the exercise informative, because all three converge on a single condition. Hydrogen can be the cheaper appliance on base-load cost, and it can win the winter-peak dispatch, and in both tests the condition that decides it is the cost of holding a winter-weighted load in store from one season to the next. Cheap salt-cavern storage is what makes that condition affordable, so the same endowment is decisive in each. They are not the same seven countries in each.

Five of the seven base-load holds are cavern markets, while the seven peaking wins are exactly the cavern set, and the two lists differ in Belgium and Ireland, which hold base load on very high retail tariffs, and in France and Romania, which have caverns yet lose the base-load comparison. Even in its most favourable niche a hydrogen peaker cannot recover its capital from market rent, because the scarcity hours on which it earns are too few. That is the missing-money problem. That convergence across arenas, shown across 29 countries and four scenarios, is what distinguishes this approach from single-arena assessments, which return whichever verdict the arena they chose happens to carry and cannot tell whether the arenas share a mechanism.

Third, the analysis isolates a finding of wider import. In the dispatch arenas it is the carbon price on the fossil competitor, more than the price of hydrogen, that opens what niche hydrogen has, while the base-load arena, in which neither clean option pays carbon at all, turns on the cost of seasonal storage first and on how the two carriers are taxed at the meter second. On the least-cost side the ambition is cheap without being free. The present-value system cost moves by 0.05 per cent between a 75 and a 100 per cent reduction cap, so decarbonising scope-1 residential heat is close to costless at the system level, though the cap does bind in 4 of 29 countries by 2050 (Cyprus, Croatia, Poland and the United Kingdom) at a shadow price of up to €81/tCO_2, and in 7 countries at some point on the path, which is where an implicit carbon price beyond ETS2 would have to be found. That reading holds under both perfect foresight and the myopic rolling-horizon test (Appendix B).

7.3 Policy Implications

Five policy implications follow from these findings.

1. Policy should make heat-pump and district-heat deployment the primary near-term priority. Against the fossil incumbent the case is already made in 2025 price data, heat pumps being cheaper than a gas boiler in 16 of 29 countries then and in 24 by 2050, and ETS2 widens that margin. The constraints are partly economic and partly not. Equipment capital is now the largest single fixed item in a heat pump's cost, at about 28 per cent of its 2050 levelised cost, which is a heavier capital burden than either the gas or the hydrogen boiler carries, and reinforcing low- and medium-voltage networks for the winter heat-pump peak adds between €1.1 and 38.2/MWh depending on how much of that cost the load is asked to bear. Installer capacity, access to upfront capital and regulatory inertia sit on top of those. Policy is therefore best directed at skilled-installer training, subsidised financing (for example through the REPowerEU home-renovation programme), streamlined planning permission for residential heat pumps, and at bringing down the delivered capital and network cost to which the levelised comparison is most sensitive.

2. Policy should allow ETS2 to pass through to end users, conditional on empirical measurement of the household switching response, to preserve the switching incentive. This paper quantifies the carbon-cost differential between gas and heat-pump heating (an adder of €11/MWh per €50/tCO_2 against a median €0.7/MWh by 2030) but does not model the behavioural elasticity of the consumer response to it, a gap that future work should close; the implication that follows is therefore a logical one conditioned on that response being non-trivial. On that condition, because energy-bill caps blunt the differential signal, measures that shield residential consumers from ETS2 in the aggregate, such as broad energy-bill caps and untargeted rebates, would slow the transition. Targeted social protection for fuel-poor households is appropriate and necessary, but it should be designed to preserve the switching incentive, not to suppress the carbon price across the board.

3. Policy should require a capacity-payment design and a co-benefit justification for hydrogen in buildings, and should not rest the decision on the headline cost comparison in either direction. Where hydrogen still comes out cheaper than a heat pump on delivered annual cost, in those seven markets, the margin comes from the levies and network charges loaded onto retail electricity and not onto the molecule, so it is an artefact of tax policy that tax policy can remove, and it would survive neither an equalisation of the two treatments nor a shift of heating hydrogen onto a blue supply route. Where hydrogen looks strongest in dispatch, in the cavern-endowed countries at a high carbon price, no private investor recovers the peaker's capital from market rents alone.

Any deliberate hydrogen-for-heat deployment would therefore require explicit justification on grounds beyond the headline comparison, such as co-benefits in particular geographies or network configurations (for example regions where major district-heating expansion is infeasible), and should be paired with a standing capacity payment or equivalent support to signal the absence of revenue-market recovery. These co-benefits are not quantified in the present model and are noted as qualitative considerations and not modelled results. The corollary on the tax side is that the levy structure on residential electricity is itself a heating-technology policy, since reforming it would move the base-load ranking in a large group of countries. Policy should make these distinctions explicit, not present hydrogen as a general-purpose decarbonisation solution for buildings.

4. Policy should give Eastern European countries tailored transition support. Countries with coal-heavy electricity grids and constrained fiscal capacity, such as Poland, the Czech Republic, Romania and Bulgaria [101, 141], face the most demanding transition, since heat pumps are cost-competitive but carry a higher near-term grid carbon intensity, and boiler bans risk imposing disproportionate adjustment costs. This is also the region where the cost evidence most strongly favours electrification, the heat pump's 2050 margin over the hydrogen boiler being widest across the warm south and east, at €99/MWh in Malta, €88/MWh in Cyprus, €61/MWh in Hungary, €48/MWh in Greece and €41/MWh in Bulgaria. Bulgaria, the Czech Republic and Hungary have no salt-cavern endowment to make a seasonal hydrogen store cheap, so in those markets there is no base-load case for a hydrogen alternative to hedge against, and support can be directed squarely at the grid and the installation base.

Poland and Romania are the exceptions within the group and should be treated separately. Both hold caverns, and Poland is one of the seven markets in which the hydrogen boiler still leads on delivered annual cost, though by only €2/MWh, a margin narrower than the fiscal wedge between the two carriers. Romania, despite the same geology, sits €27/MWh on the heat pump's side, which shows that cavern access is not on its own sufficient to give hydrogen the base-load case; Belgium and Ireland, which hold base load without caverns, show that it is not strictly necessary either. Neither case justifies a hydrogen heating programme, but both warrant the levy-incidence reform of recommendation 3 before the base-load comparison in those countries is treated as settled. Just Transition Fund resources should be linked explicitly to both grid decarbonisation and building electrification, so that the two imperatives become complementary.

5. Policy should recognise that Switzerland's case for hydrogen-import infrastructure for residential buildings is weak, and weak on delivered cost as well. The delivered-cost margin set out above is narrow, so it should not carry the argument alone, and the grounds that bear on an irreversible national commitment all point the same way.

Equalising the fiscal treatment of taxed Swiss retail electricity and an untaxed imported molecule would widen the heat pump's margin. Hydrogen wins no Swiss winter peak in any of the four scenarios on the reservoir-credited arm we read the Swiss verdict from, on the figures given above, so an import programme would not answer the winter adequacy problem that electrified heat creates; and the same imported molecules have higher-value uses in industry and mobility, where substitutes are fewer. Federal hydrogen strategy is therefore better directed towards industrial and mobility applications, as the current strategy already does [154], while residential resources are better directed towards heat-pump installation programmes, grid flexibility, and seasonal thermal storage. This is offered as a direction following from the base-load, peaking and capital-recovery results together and from competing uses, and not as a modelled policy appraisal. The analysis also leaves out modelled Swiss policy options, local infeasibility factors and other constraints.

7.4 Directions for Future Research

The study points to four productive directions for future work.

First, the least-cost optimisation layer currently fixes demand at the Stated Policies trajectory and optimises the supply mix; a joint demand-and-supply optimisation that co-optimises renovation depth with technology choice would close the remaining gap between the simulation and optimisation frameworks. On the supply side, the explicit unit-commitment dispatch introduced here (Appendix A.7) now commits the firm fleet against ramp, minimum-stable-output, reserve and start-up constraints, but it still relies on calibrated availability profiles and a causal storage heuristic, not on co-optimised storage and demand response. Endogenising storage sizing and demand-side flexibility within the commitment model, and propagating the capacity stack through the four policy scenarios, not a single central trajectory, would further tighten the peaking and missing-money bounds.

Second, city-level spatial analysis within NUTS3 regions would capture intra-regional heterogeneity that matters for the hydrogen case. Dense urban cores with historic multi-family stock have feasibility profiles quite different from those of peri-urban detached housing, and the aggregate NUTS3 treatment masks this variation.

Third, the unit-commitment hourly dispatch quantifies one side of the sector-coupling problem, namely the firm peaking capacity that electrified heat requires for winter adequacy ($\approx$278 GW across the EU-27+UK+CH centrally, about 31 per cent of the study-area system peak, with a wider band across the interconnection and Dunkelflaute-depth sweep) and the inability of energy-market revenue to finance it. A fully co-optimised treatment, in which the increase in EU electricity demand from building electrification by 2050 is balanced against endogenous transmission, storage and flexibility investment, would convert this adequacy signal into explicit grid-investment and capacity-market requirements. This is also the natural setting in which to take up the wider part of RQ3, the cross-sector synergies between buildings and industry or transport that the present paper has deferred. A hydrogen turbine that cannot recover its capital from heating-peak rent alone may yet be financeable when its capacity value is shared across sectors, and quantifying that shared value is the central open question the merit-order verdict leaves behind.

Fourth, within RQ2 a dedicated city-level analysis of Zurich, Geneva and Basel would provide the granularity to distinguish hydrogen niches in the historic, dense urban stock that the NUTS3 model treats as an aggregate, sharpening the RQ2 answer for exactly the hard-to-treat districts where any residual case for hydrogen would have to be made. This is a refinement of the existing scope rather than an extension of it.

References

[1] International Energy Agency, "Buildings," IEA, Tech. Rep., 2022. [Online]. Available: https://www.iea.org/energy-system/buildings

[2] European Commission, "The European Green Deal," European Commission, Tech. Rep., 2020. [Online]. Available: https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en

[3] International Energy Agency, "The future of heat pumps," IEA, Paris, Tech. Rep., 2022. [Online]. Available: https://www.iea.org/reports/the-future-of-heat-pumps

[4] M. H. Abbasi *et al.*, "Heat transition in the European building sector: Overview of the heat decarbonisation practices through heat pump technology," *Sustainable Energy Technologies and Assessments*, 2021.

[5] J. Rosenow, "Is heating homes with hydrogen all but a pipe dream? An evidence review," *Joule*, vol. 6, no. 10, pp. 2225–2228, 2022.

[6] D. Cebon, "Hydrogen for heating: A comparison with heat pumps," 2022. [Online]. Available: https://h2sciencecoalition.com/blog/hydrogen-for-heating-a-comparison-with-heat-pumps-part-1/

[7] I. L. Azevedo, M. G. Morgan, and M. C. Moura, "The challenges of deep decarbonization in the buildings sector," *Joule*, vol. 6, no. 4, pp. 696–708, 2022.

[8] S. Chatterjee, G. Molnár, B. Kiss, D. Topál, and D. Ürge-Vorsatz, "Navigating the transition: Modelling the path for net-zero European building sector," *Renewable and Sustainable Energy Reviews*, 2024. [Online]. Available: https://doi.org/10.1016/j.rser.2024.114827

[9] I. Maia, D. Harringer, and L. Kranzl, "Household budget restrictions as reason for staged retrofits: A case study in Spain," *Energy Policy*, 2024. [Online]. Available: https://doi.org/10.1016/j.enpol.2024.114047

[10] M. Arbulu, X. Oregi, L. E. Igiñiz, and R. J. Hernández-Minguillón, "Barriers and challenges of the assessment framework of the Commission Recommendation (EU) 2019/786 on building renovation by European RTD projects," *Energy and Buildings*, 2022. [Online]. Available: https://doi.org/10.1016/j.enbuild.2022.112267

[11] European Commission, "REPowerEU plan," European Commission, Brussels, Communication from the Commission COM(2022) 230 final, May 2022. [Online]. Available: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52022DC0230

[12] European Parliament and Council, "Directive 2024/1275 on the energy performance of buildings (recast)," Official Journal of the European Union, 2024. [Online]. Available: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=OJ:L_202401275

[13] J. Lingard, "Residential retrofit in the UK: The optimum retrofit measures necessary for effective heat pump use," *Building Services Engineering Research and Technology*, 2020. [Online]. Available: https://doi.org/10.1177/0143624420975707

[14] J. A. Gordon, N. Balta-Ozkan, and S. A. Nabavi, "Divergent consumer preferences and visions for cooking and heating technologies in the United Kingdom: Make our homes clean, safe, warm and smart!" *Energy Research & Social Science*, 2023. [Online]. Available: https://doi.org/10.1016/j.erss.2023.103204

[15] P. E. Dodds *et al.*, "Hydrogen and fuel cell technologies for heating: A review," *International Journal of Hydrogen Energy*, 2015.

[16] D. Bothe and M. Janssen, "The value of hydrogen in the heating market," Frontier Economics, for FNB Gas, Tech. Rep., 2021, accessed 2026-08-01. [Online]. Available: https://www.frontier-economics.com/media/byojuyut/the-value-of-hydrogen-in-the-heating-market.pdf

[17] European Commission, "Implementing the REPowerEU action plan: Investment needs, hydrogen accelerator and achieving the bio-methane targets," European Commission Staff Working Document, Tech. Rep., 2022. [Online]. Available: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=SWD:2022:230:FIN

[18] A. Goldthau and R. Youngs, "The EU Energy Crisis and a New Geopolitics of Climate Transition," *JCMS Journal of Common Market Studies*, 2023. [Online]. Available: https://doi.org/10.1111/jcms.13539

[19] B. Zakeri, K. Paulavets, L. Barreto-Gomez, L. G. Echeverri, S. Pachauri, B. Boza-Kiss *et al.*, "Pandemic, War, and Global Energy Transitions," *Energies*, 2022. [Online]. Available: https://doi.org/10.3390/en15176114

[20] P. Gabrielli, F. Charbonnier, A. Guidolin, and M. Mazzotti, "Enabling low-carbon hydrogen supply chains through use of biomass and carbon capture and storage: A Swiss case study," *Applied Energy*, 2020.

[21] Swiss Federal Office of Energy, "Thesen zur künftigen bedeutung von wasserstoff in der schweizer energieversorgung," BfE, Tech. Rep., 2023. [Online]. Available: https://www.bfe.admin.ch/bfe/en/home/research-and-cleantech/research-programmes/hydrogen.html

[22] A. Billerbeck, C. P. Kiefer, J. Winkler, C. Bernath, F. Sensfuß, L. Kranzl *et al.*, "The race between hydrogen and heat pumps for space and water heating: A model-based scenario analysis," *Energy Conversion and Management*, 2023. [Online]. Available: https://doi.org/10.1016/j.enconman.2023.117850

[23] P. Hoseinpoori *et al.*, "A whole-system approach for quantifying the value of smart electrification for decarbonising heating in buildings," *Energy Conversion and Management*, 2022.

[24] T. Brown, D. Schlachtberger, A. Kies, S. Schramm, and M. Greiner, "Synergies of sector coupling and transmission reinforcement in a cost-optimised, highly renewable European energy system," *Energy*, 2018. [Online]. Available: https://doi.org/10.1016/j.energy.2018.06.222

[25] V. M. Maestre, A. Ortiz, and I. Ortiz, "Transition to a low-carbon buildings stock: Techno-economic and spatial optimization of renewables-hydrogen strategies in Spain," *Journal of Energy Storage*, 2022.

[26] D. Huckebrink and V. Bertsch, "Decarbonising the residential heating sector: A techno-economic assessment of selected technologies," *Energy*, 2022.

[27] T. Weidner and G. Guillén-Gosálbez, "Planetary boundaries assessment of deep decarbonisation options for building heating in the European Union," *Energy Conversion and Management*, 2023. [Online]. Available: https://doi.org/10.1016/j.enconman.2022.116602

[28] P. L. Joskow, "Capacity payments in imperfect electricity markets: Need and design," *Utilities Policy*, vol. 16, no. 3, pp. 159–170, 2008.

[29] C. J. Quarton and S. Samsatli, "The value of hydrogen and carbon capture, storage and utilisation in decarbonising energy: Insights from integrated value chain optimisation," *Applied Energy*, 2019. [Online]. Available: https://doi.org/10.1016/j.apenergy.2019.113936

[30] E. Zeyen, V. Hagenmeyer, and T. Brown, "Mitigating heat demand peaks in buildings in a highly renewable European energy system," *Energy*, 2021. [Online]. Available: https://doi.org/10.1016/j.energy.2021.120784

[31] A. Mellot, C. Moretti, T. Tröndle, and A. Patt, "Mitigating future winter electricity deficits: A case study from Switzerland," *Energy Conversion and Management*, 2024. [Online]. Available: https://doi.org/10.1016/j.enconman.2024.118426

[32] C. Baldino, J. O'Malley, S. Searle, and A. Christensen, "Hydrogen for heating? decarbonization options for households in the European Union in 2050,"

International Council on Clean Transportation (ICCT), Tech. Rep., Mar. 2021. [Online]. Available: https://theicct.org/publication/hydrogen-for-heating-decarbonization-options-for-households-in-the-european-union-in-2050/

[33] International Energy Agency, "Switzerland energy policy review 2023," IEA, Paris, Tech. Rep., 2023. [Online]. Available: https://www.iea.org/reports/switzerland-2023

[34] N. Milojevic-Dupont, F. Wagner, F. Nachtigall, J. Hu, G. B. Brüser, M. Zumwald, F. Biljecki, N. Heeren, L. H. Kaack, P.-P. Pichler, and F. Creutzig, "EUBUCCO v0.1: European building stock characteristics in a common and open database for 200+ million individual buildings," *Scientific Data*, vol. 10, no. 1, p. 147, 2023. [Online]. Available: https://www.nature.com/articles/s41597-023-02040-2

[35] T. Loga, B. Stein, and N. Diefenbach, "TABULA building typologies in 20 European countries — Making energy-related features of residential building stocks comparable," *Energy and Buildings*, vol. 132, pp. 4–12, 2016.

[36] D. G. Caglayan, N. Weber, H. U. Heinrichs, J. Linssen, M. Robinius, P. A. Kukla, and D. Stolten, "Technical potential of salt caverns for hydrogen storage in Europe," *International Journal of Hydrogen Energy*, vol. 45, no. 11, pp. 6793–6805, 2020.

[37] Intergovernmental Panel on Climate Change, "Global warming of 1.5°C," IPCC, Tech. Rep., 2018. [Online]. Available: https://www.ipcc.ch/sr15/

[38] I. Smith, *The Buildings Sector as a Contributor to Climate Change.* Springer, 2018.

[39] K. Calvin, D. Dasgupta, G. Krinner, A. Mukherji, P. Thorne, C. H. Trisos *et al.*, "Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team, H. Lee and J. Romero (eds.)]," *Intergovernmental Panel on Climate Change (IPCC), Geneva, Switzerland*, 2023. [Online]. Available: https://doi.org/10.59327/ipcc/ar6-9789291691647

[40] M. Röck, M. R. M. Saade, M. Balouktsi, F. N. Rasmussen, H. Birgisdóttir, R. Frischknecht *et al.*, "Embodied GHG emissions of buildings – The hidden challenge for effective climate change mitigation," *Applied Energy*, 2019. [Online]. Available: https://doi.org/10.1016/j.apenergy.2019.114107

[41] F. Pomponi, J. Hart, J. H. Arehart, and B. D'Amico, "Buildings as a Global Carbon Sink? A Reality Check on Feasibility Limits," *One Earth*, 2020. [Online]. Available: https://doi.org/10.1016/j.oneear.2020.07.018

[42] B. V. Mathiesen, N. Bertelsen, N. Schneider, L. Sánchez-García, S. Paardekooper, J. Z. Thellufsen *et al.*, “Towards a decarbonised heating and cooling sector in Europe: Unlocking the potential of energy efficiency and district energy,” *VBN Forskningsportal (Aalborg Universitet)*, 2019.

[43] M. Victoria, E. Zeyen, and T. Brown, “Speed of technological transformations required in Europe to achieve different climate goals,” *Joule*, 2022. [Online]. Available: https://doi.org/10.1016/j.joule.2022.04.016

[44] C. Maduta, D. D’Agostino, S. Tsemekidi-Tzeiranaki, L. Castellazzi, G. Melica, and P. Bertoldi, “Towards climate neutrality within the European Union: Assessment of the Energy Performance of Buildings Directive implementation in Member States,” *Energy and Buildings*, 2023. [Online]. Available: https://doi.org/10.1016/j.enbuild.2023.113716

[45] A. Esser, A. Dunne, T. Meeusen, S. Quaschning, D. Wegge, A. Hermelink, S. Schimschar, M. Offermann, A. John, M. Reiser, A. Pohl, and J. Grözinger, “Comprehensive study of building energy renovation activities and the uptake of nearly zero-energy buildings in the EU: Final report,” Ipsos Belgium and Navigant, for the European Commission, Directorate-General for Energy, Brussels, Tech. Rep., Nov. 2019. [Online]. Available: https://energy.ec.europa.eu/system/files/2019-12/1.final_report_0.pdf

[46] F. Filippidou and J. P. Jiménez Navarro, “Achieving the cost-effective energy transformation of Europe’s buildings,” Joint Research Centre, European Commission, Tech. Rep. EUR 29906 EN, JRC117739, 2019.

[47] I. D. Mel, F. Bierkens, X. Liu, M. Leach, M. Chitnis, L. Liu *et al.*, “A decision-support framework for residential heating decarbonisation policymaking,” *Energy*, 2023. [Online]. Available: https://doi.org/10.1016/j.energy.2023.126651

[48] European Commission, “A Clean Planet for All: A European Strategic Long-Term Vision for a Prosperous, Modern, Competitive and Climate Neutral Economy,” European Commission, Brussels, Communication from the Commission COM(2018) 773 final, Nov. 2018. [Online]. Available: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52018DC0773

[49] ——, “Eu reference scenario 2024: Energy, transport and ghg emissions — trends to 2050,” Directorate-General for Energy, Tech. Rep., 2025. [Online]. Available: https://energy.ec.europa.eu/data-and-analysis/energy-modelling/eu-reference-scenario-2024_en

[50] N. Bertelsen and B. V. Mathiesen, "EU-28 residential heat supply and consumption: Historical development and status," *Energies*, vol. 13, no. 8, p. 1894, 2020. [Online]. Available: https://www.mdpi.com/1996-1073/13/8/1894

[51] Heat Pumping Technologies, "Heat pumps at the heart of global decarbonisation: Insights from the IEA World Energy Outlook 2025," IEA Heat Pumping Technologies Programme, Tech. Rep., 2026. [Online]. Available: https://heatpumpingtechnologies.org/articles/heat-pumps-at-the-heart-of-global-decarbonisation-insights-from-the-iea-world-energy-outlook-2025/

[52] J. Famiglietti, T. Toppi, D. Bonalumi, and M. Motta, "Heat pumps for space heating and domestic hot water production in residential buildings, an environmental comparison in a present and future scenario," *Energy Conversion and Management*, 2022. [Online]. Available: https://doi.org/10.1016/j.enconman.2022.116527

[53] S. Dong *et al.*, "Carbon reduction analysis of electric heat pumps in carbon neutrality in China," *Sustainable Cities and Society*, 2023.

[54] H. Kim and L. Junghans, "Economic feasibility of achieving net-zero emission building (NZEB) by applying solar and geothermal energy sources to heat pump systems," *Journal of Cleaner Production*, 2023.

[55] D. Gibb, J. Rosenow, R. Lowes, and N. Hewitt, "Coming in from the cold: Heat pump efficiency at low temperatures," *Joule*, 2023. [Online]. Available: https://doi.org/10.1016/j.joule.2023.08.005

[56] J. Vivian and N. Mazzi, "An algorithm for the optimal management of air-source heat pumps and PV systems," *Journal of Physics Conference Series*, 2019. [Online]. Available: https://doi.org/10.1088/1742-6596/1343/1/012069

[57] S. Kokoni and M. Leach, "Policy mechanisms to support heat pump deployment: A UK case study based on techno-economic modelling," *Renewable and Sustainable Energy Transition*, 2021. [Online]. Available: https://doi.org/10.1016/j.rset.2021.100009

[58] M. Corbett, E. Rhodes, A. Pardy, and Z. Long, "Pumping up adoption: The role of policy awareness in explaining willingness to adopt heat pumps in Canada," *Energy Research & Social Science*, 2023. [Online]. Available: https://doi.org/10.1016/j.erss.2022.102926

[59] H. Lund, P. A. Østergaard, M. Chang, S. Werner, S. Svendsen, P. Sorknæs *et al.*, "The status of 4th generation district heating: Research and results," *Energy*, 2018. [Online]. Available: https://doi.org/10.1016/j.energy.2018.08.206

[60] B. Möller, E. Wiechers, U. Persson, L. Grundahl, R. S. Lund, and B. V. Mathiesen, "Heat Roadmap Europe: Towards EU-Wide, local heat supply strategies," *Energy*, 2019. [Online]. Available: https://doi.org/10.1016/j.energy.2019.04.098

[61] U. Persson, E. Wiechers, B. Möller, and S. Werner, "Heat Roadmap Europe: Heat distribution costs," *Energy*, 2019. [Online]. Available: https://doi.org/10.1016/j.energy.2019.03.189

[62] H. Dorotić, T. Pukšec, D. R. Schneider, and N. Duić, "Evaluation of district heating with regard to individual systems – Importance of carbon and cost allocation in cogeneration units," *Energy*, 2021. [Online]. Available: https://doi.org/10.1016/j.energy.2021.119905

[63] N. Javanshir, S. Syri, A. Teräsvirta, and V. Olkkonen, "Abandoning peat in a city district heat system with wind power, heat pumps, and heat storage," *Energy Reports*, 2022. [Online]. Available: https://doi.org/10.1016/j.egyr.2022.02.064

[64] D. Enescu, G. Chicco, R. Porumb, and G. Seritan, "Thermal Energy Storage for Grid Applications: Current Status and Emerging Trends," *Energies*, 2020. [Online]. Available: https://doi.org/10.3390/en13020340

[65] A. Vecchi, K. Knobloch, T. Liang, H. Kildahl, A. Sciacovelli, K. Engelbrecht *et al.*, "Carnot Battery development: A review on system performance, applications and commercial state-of-the-art," *Journal of Energy Storage*, 2022. [Online]. Available: https://doi.org/10.1016/j.est.2022.105782

[66] International Energy Agency, "Global hydrogen review 2022," IEA, Paris, Tech. Rep., 2022. [Online]. Available: https://www.iea.org/reports/global-hydrogen-review-2022

[67] R. Scita, P. P. Raimondi, and M. Noussan, "Green hydrogen: The holy grail of decarbonisation? an analysis of the technical and geopolitical implications of the future hydrogen economy," *SSRN*, 2020.

[68] B. C. Erdener, B. Sergi, O. J. G. Fernandez, A. L. Chueca, K. A. Pambour, C. Brancucci *et al.*, "A review of technical and regulatory limits for hydrogen blending in natural gas pipelines," *International Journal of Hydrogen Energy*, 2022. [Online]. Available: https://doi.org/10.1016/j.ijhydene.2022.10.254

[69] J. Giehl, J. Hollnagel, and J. Müller-Kirchenbauer, "Assessment of using hydrogen in gas distribution grids," *International Journal of Hydrogen Energy*, 2023.

[70] G. Thomas, N. Pidgeon, and K. Henwood, "Hydrogen, a less disruptive pathway for domestic heat? Exploratory findings from public perceptions

research," *Cleaner Production Letters*, 2023. [Online]. Available: https://doi.org/10.1016/j.clpl.2023.100047

[71] J. Rosenow, "A meta-review of 54 studies on hydrogen heating," *Cell Reports Sustainability*, vol. 1, no. 1, p. 100010, 2024.

[72] J. Rosenow, R. Lowes, O. Broad, G. Hawker, J. Wu, M. Qadrdan *et al.*, "The pathway to net zero heating in the UK," *Open Research Exeter (University of Exeter)*, 2020. [Online]. Available: https://doi.org/10.5286/ukerc.edc.000941

[73] R. Dickel, "Decarbonising Germany's heating sector," Oxford Institute for Energy Studies, Tech. Rep. ET29, 2024. [Online]. Available: https://www.oxfordenergy.org/publications/decarbonising-germanys-heating-sector/

[74] C. J. Quarton, O. Tlili, L. Welder, C. Mansilla, H. Blanco, H. Heinrichs *et al.*, "The curious case of the conflicting roles of hydrogen in global energy scenarios," *Sustainable Energy & Fuels*, 2019. [Online]. Available: https://doi.org/10.1039/c9se00833k

[75] O. A. Marzouk, "Expectations for the Role of Hydrogen and Its Derivatives in Different Sectors through Analysis of the Four Energy Scenarios: IEA-STEPS, IEA-NZE, IRENA-PES, and IRENA-1.5°C," *Energies*, 2024. [Online]. Available: https://doi.org/10.3390/en17030646

[76] V. M. Maestre, A. Ortiz, and I. Ortiz, "Challenges and prospects of renewable hydrogen-based strategies for full decarbonization of stationary power applications," *Renewable and Sustainable Energy Reviews*, 2021.

[77] R. F. D. Masi, V. Festa, D. Penchini, S. Ruggiero, F. Tariello, G. P. Vanoli *et al.*, "State of art of hydrogen utilization for building sector and set-up with preliminary experimental results of 1 kWel solid oxide fuel cell installed in a nearly zero energy house," *Energy*, 2024. [Online]. Available: https://doi.org/10.1016/j.energy.2024.131810

[78] M. Genovese *et al.*, "Power-to-hydrogen and hydrogen-to-X energy systems for the industry of the future in europe," *International Journal of Hydrogen Energy*, 2023.

[79] B. Modu *et al.*, "A systematic review of hybrid renewable energy systems with hydrogen storage: Sizing, optimization, and energy management strategy," *International Journal of Hydrogen Energy*, 2023.

[80] Y. Jung, Y. Heo, H. Cho, Y. T. Kang, Y. Kim, and H. Lee, "A plan to build a net zero energy building in hydrogen and electricity-based energy

scenario in South Korea," *Journal of Cleaner Production*, 2023. [Online]. Available: https://doi.org/10.1016/j.jclepro.2023.136537

[81] M. Berger, D. Radu, R. Fonteneau *et al.*, "The role of power-to-gas and carbon capture technologies in cross-sector decarbonization strategies," *Electric Power Systems Research*, vol. 180, 2020.

[82] P. C. Slorach and L. Stamford, "Net zero in the heating sector: Technological options and environmental sustainability from now to 2050," *Energy Conversion and Management*, 2021. [Online]. Available: https://doi.org/10.1016/j.enconman.2021.113838

[83] International Energy Agency, "Global hydrogen review 2025," IEA, Paris, Tech. Rep., 2025. [Online]. Available: https://www.iea.org/reports/global-hydrogen-review-2025

[84] European Hydrogen Observatory, "Cost of hydrogen production," European Hydrogen Observatory / Clean Hydrogen JU, Tech. Rep., 2024. [Online]. Available: https://observatory.clean-hydrogen.europa.eu/hydrogen-landscape/production-trade-and-cost/cost-hydrogen-production

[85] ——, "Levelised cost of hydrogen (LCOH) calculator manual," European Hydrogen Observatory / Clean Hydrogen JU, Tech. Rep., 2025. [Online]. Available: https://observatory.clean-hydrogen.europa.eu/sites/default/files/2025-05/Manual%20-%20LCOH%20calculator_Update%202025_v3_1.pdf

[86] L. Sens, Y. Piguel, U. Neuling, S. Timmerberg, K. Wilbrand, and M. Kaltschmitt, "Cost minimized hydrogen from solar and wind – Production and supply in the European catchment area," *Energy Conversion and Management*, 2022. [Online]. Available: https://doi.org/10.1016/j.enconman.2022.115742

[87] M. T. Zun and B. McLellan, "Cost Projection of Global Green Hydrogen Production Scenarios," *Hydrogen*, 2023. [Online]. Available: https://doi.org/10.3390/hydrogen4040055

[88] IEAGHG, "Low-carbon hydrogen from natural gas: Global roadmap," IEA Greenhouse Gas R&D Programme, Tech. Rep. 2022-07, 2022.

[89] I. Staffell, D. Scamman, A. V. Abad, P. Balcombe, P. E. Dodds, P. Ekins *et al.*, "The role of hydrogen and fuel cells in the global energy system," *Energy & Environmental Science*, 2018. [Online]. Available: https://doi.org/10.1039/c8ee01157e

[90] Gas for Climate, "European hydrogen backbone: How a dedicated hydrogen infrastructure can be created," Guidehouse, Tech. Rep., 2020.

[91] European Hydrogen Backbone, "European hydrogen backbone: Country-specific developments and import corridors," Gas for Climate / Guidehouse, Tech. Rep., 2023. [Online]. Available: https://ehb.eu/page/country-specific-developments

[92] V. Lenivova and A. Hove, "Potential development of renewable hydrogen imports to European markets until 2030," Oxford Institute for Energy Studies, Tech. Rep. ET08, 2022. [Online]. Available: https://www.oxfordenergy.org/publications/potential-development-of-renewable-hydrogen-imports-to-european-markets-until-2030/

[93] R. Rikabi, "Green hydrogen imports into Europe: An assessment of potential sources," Oxford Institute for Energy Studies, Tech. Rep. ET32, 2024. [Online]. Available: https://www.oxfordenergy.org/publications/green-hydrogen-imports-into-europe-an-assessment-of-potential-sources/

[94] A. Lahnaoui, C. Wulf, and D. Dalmazzone, "Optimization of Hydrogen Cost and Transport Technology in France and Germany for Various Production and Demand Scenarios," *Energies*, 2021. [Online]. Available: https://doi.org/10.3390/en14030744

[95] I. Petkov and P. Gabrielli, "Power-to-hydrogen as seasonal energy storage: an uncertainty analysis for optimal design of low-carbon multi-energy systems," *Applied Energy*, 2020. [Online]. Available: https://doi.org/10.1016/j.apenergy.2020.115197

[96] Competition and Markets Authority, "Energy licence modification appeals 2021: Final determination," Competition and Markets Authority, London, Tech. Rep., Oct. 2021. [Online]. Available: https://www.gov.uk/cma-cases/energy-licence-modification-appeals-2021

[97] N. Fabra and M. Reguant, "Pass-through of emissions costs in electricity markets," *American Economic Review*, vol. 104, no. 9, pp. 2872–2899, 2014. [Online]. Available: https://www.aeaweb.org/articles?id=10.1257/aer.104.9.2872

[98] J. E. Bistline and G. J. Blanford, "Value of technology in the U.S. electric power sector: Impacts of full portfolios and technological change on the costs of meeting decarbonization goals," *Energy Economics*, vol. 86, p. 104694, 2020.

[99] D. Bogdanov, J. Farfan, K. Sadovskaia, A. Aghahosseini, M. Child, A. Gulagi *et al.*, "Radical transformation pathway towards sustainable electricity via evolutionary steps," *Nature Communications*, 2019. [Online]. Available: https://doi.org/10.1038/s41467-019-08855-1

[100] M. Victoria, N. M. Haegel, I. M. Peters, R. Sinton, A. Jäger-Waldau, C. del Cañizo *et al.*, "Solar photovoltaics is ready to power a sustainable future," *Joule*, 2021. [Online]. Available: https://doi.org/10.1016/j.joule.2021.03.005

[101] D. Jones *et al.*, "European electricity review 2024," Ember, London, Tech. Rep., Feb. 2024. [Online]. Available: https://ember-energy.org/latest-insights/european-electricity-review-2024/

[102] European Environment Agency, "GHG emission intensity of electricity generation in europe, indicator ENER 038," EEA, Tech. Rep., 2024. [Online]. Available: https://www.eea.europa.eu/data-and-maps/indicators/overview-of-the-electricity-production-3/assessment

[103] Hotmaps Project, "Hotmaps regional heat demand dataset," Hotmaps, Tech. Rep., 2020. [Online]. Available: https://gitlab.com/hotmaps/regional_demand

[104] I. Sue Wing, "The synthesis of bottom-up and top-down approaches to climate policy modeling: Electric power technology detail in a social accounting framework," *Energy Economics*, vol. 30, no. 2, pp. 547–573, 2008.

[105] M. G. Prina, G. Manzolini, D. Moser, B. Nastasi, and W. Sparber, "Classification and challenges of bottom-up energy system models - A review," *Renewable and Sustainable Energy Reviews*, 2020. [Online]. Available: https://doi.org/10.1016/j.rser.2020.109917

[106] Hotmaps Project, "Hotmaps — the open source mapping and planning tool for heating and cooling," EU Horizon 2020 project, grant agreement No. 723677, 2019, open building-stock and heat-demand dataset and toolbox. [Online]. Available: https://www.hotmaps-project.eu/

[107] X. X. Zhu, S. Chen, F. Zhang, Y. Shi, and Y. Wang, "GlobalBuildingAtlas: An open global and complete dataset of building polygons, heights and LoD1 3D models," *Earth System Science Data*, vol. 17, pp. 6647–6668, 2025, dataset: mediaTUM ID 1782307, https://mediatum.ub.tum.de/1782307. [Online]. Available: https://essd.copernicus.org/articles/17/6647/2025/

[108] W. Cyx, N. Renders, M. Van Holm, and S. Verbeke, "IEE TABULA — typology approach for building stock energy assessment, belgian scientific report," VITO — Flemish Institute for Technological Research, Tech. Rep. 2011/TEM/R/091763, August 2011. [Online]. Available: https://episcope.eu/fileadmin/tabula/public/docs/scientific/BE_TABULA_ScientificReport_VITO.pdf

[109] Eurostat, "CENS_21DWBNO_R3: Dwellings by number of occupants and type of building, NUTS3," Eurostat, Tech. Rep., 2021. [Online]. Available: https://ec.europa.eu/eurostat/databrowser/view/CENS_21DWBNO_R3

[110] Office for National Statistics, "TS044: Accommodation type, Census 2021," ONS, Tech. Rep., 2021. [Online]. Available: https://www.ons.gov.uk/datasets/TS044/editions/2021

[111] European Commission, DG ENER, "EU building stock observatory database," DG ENER, managed by CINEA via LIFE Clean Energy Transition subprogramme, Tech. Rep., 2025. [Online]. Available: https://building-stock-observatory.energy.ec.europa.eu/

[112] Eurostat, "Cooling and heating degree days by country — annual data," European Commission, Eurostat (source: JRC AGRI4CAST), Tech. Rep., 2025, dataset code nrg_chdd_a; base temperature 15°C. [Online]. Available: https://ec.europa.eu/eurostat/databrowser/view/nrg_chdd_a/default/table

[113] S. Patil, N. Pflugradt, J. M. Weinand, J. P. Kropp, and D. Stolten, "Disaggregated Municipal Energy Consumption and Emissions in End-use Sectors in Germany and Spain for 2022," *Scientific Data*, 2025. [Online]. Available: https://doi.org/10.1038/s41597-025-05938-1

[114] F. Neumann, E. Zeyen, M. Victoria, and T. Brown, "The potential role of a hydrogen network in Europe," *Joule*, vol. 7, no. 8, pp. 1793–1817, 2023.

[115] I. Kountouris, R. Bramstoft, T. Madsen, J. Gea-Bermúdez, M. Münster, and D. Keles, "A unified European hydrogen infrastructure planning to support the rapid scale-up of hydrogen production," *Nature Communications*, 2024. [Online]. Available: https://doi.org/10.1038/s41467-024-49867-w

[116] D. McFadden, "Conditional logit analysis of qualitative choice behavior," in *Frontiers in Econometrics*, P. Zarembka, Ed. New York: Academic Press, 1974, pp. 105–142.

[117] K. E. Train, *Discrete Choice Methods with Simulation*, 2nd ed. Cambridge University Press, 2009.

[118] U. Krien, P. Schönfeldt, J. Launer, S. Hilpert, C. Kaldemeyer, and G. Pleßmann, "oemof.solph–A model generator for linear and mixed-integer linear optimisation of energy systems," *Software Impacts*, 2020. [Online]. Available: https://doi.org/10.1016/j.simpa.2020.100028

[119] Á. A. Juan, P. Keenan, R. Martí, S. McGarraghy, J. Panadero, P. Carroll *et al.*, “A review of the role of heuristics in stochastic optimisation: from metaheuristics to learnheuristics,” *Annals of Operations Research*, 2021. [Online]. Available: https://doi.org/10.1007/s10479-021-04142-9

[120] P.-L. Yu, “A class of solutions for group decision problems,” *Management Science*, vol. 19, no. 8, pp. 936–946, 1973.

[121] M. Zeleny, “Compromise programming,” in *Multiple Criteria Decision Making*, J. L. Cochrane and M. Zeleny, Eds. Columbia, SC: University of South Carolina Press, 1973, pp. 262–301.

[122] B. W. Ang, “The LMDI approach to decomposition analysis: a practical guide,” *Energy Policy*, vol. 33, no. 7, pp. 867–871, 2005.

[123] X. Yue, S. Pye, J. DeCarolis, F. G. Li, F. Rogan, and B. Ó Gallachóir, “A review of approaches to uncertainty assessment in energy system optimization models,” *Energy Strategy Reviews*, vol. 21, pp. 204–217, 2018.

[124] M. Hoffmann, L. Kotzur, D. Stolten, and M. Robinius, “A Review on Time Series Aggregation Methods for Energy System Models,” *Energies*, 2020. [Online]. Available: https://doi.org/10.3390/en13030641

[125] European Commission, “EU ETS2: Emissions trading for buildings and road transport,” European Commission DG CLIMA, Tech. Rep., 2024. [Online]. Available: https://climate.ec.europa.eu/eu-action/eu-emissions-trading-system-eu-ets/ets2-buildings-road-transport-and-additional-sectors_en

[126] C. Günther, M. Pahle, K. Govorukha, S. Osorio, and T. Fotiou, “Carbon prices on the rise? Shedding light on the emerging second EU emissions trading system (EU ETS 2),” *Climate Policy*, 2025. [Online]. Available: https://www.tandfonline.com/doi/full/10.1080/14693062.2025.2485196

[127] European Heat Pump Association, “Boiler ban tracker: National phase-out schedules for fossil fuel heating,” EHPA, Tech. Rep., November 2025. [Online]. Available: https://www.ehpa.org/policy/boiler-ban-tracker/

[128] German Federal Government, “Gebäudeenergiegesetz (GEG): Building energy act 2024 amendment,” Bundesministerium für Wirtschaft und Klimaschutz, Tech. Rep., 2024. [Online]. Available: https://www.bmwk.de/Redaktion/DE/Artikel/Energie/gebaeudeenergiegesetz.html

[129] N. Helwig, “EU Strategic Autonomy after the Russian Invasion of Ukraine: Europe’s Capacity to Act in Times of War,” *JCMS Journal of Common Market Studies*, 2023. [Online]. Available: https://doi.org/10.1111/jcms.13527

[130] Eurostat, “Energy consumption in households – natural gas share by country, nrg_d_hhq,” Eurostat, Tech. Rep., 2023. [Online]. Available: https://ec.europa.eu/eurostat/databrowser/view/nrg_d_hhq

[131] ——, “Gas and electricity prices for household consumers (nrg_pc_202, nrg_pc_204), h1 2025,” Online data set, 2025, accessed 2026-05-26. [Online]. Available: https://ec.europa.eu/eurostat/web/energy/database

[132] Swiss Federal Office of Energy and Prognos AG, “Energy perspectives 2050+ (EP2050+),” SFOE, Bern, Tech. Rep., 2023. [Online]. Available: https://www.bfe.admin.ch/bfe/en/home/policy/energy-perspectives-2050-plus.html

[133] Conference of Cantonal Energy Directors (EnDK), “Energy-strategic guidelines of the Conference of Cantonal Energy Directors (EnDK) 2050+,” EnDK, Bern, Tech. Rep., 2022. [Online]. Available: https://www.endk.ch/de/endk-2050-plus

[134] K. Narula, J. Chambers, K. N. Streicher, and M. Patel, “Strategies for decarbonising the Swiss heating system,” *Energy*, 2018. [Online]. Available: https://doi.org/10.1016/j.energy.2018.12.082

[135] K. N. Streicher, P. Padey, D. Parra, M. Bürer, and M. Patel, “Assessment of the current thermal performance level of the Swiss residential building stock: Statistical analysis of energy performance certificates,” *Energy and Buildings*, 2018. [Online]. Available: https://doi.org/10.1016/j.enbuild.2018.08.032

[136] ElCom, “Electricity tariffs in switzerland,” Swiss Federal Electricity Commission tariff data, 2025. [Online]. Available: https://www.elcom.admin.ch/

[137] Eurostat GISCO, “NUTS reference shapefiles 2021,” Eurostat, Tech. Rep., 2021. [Online]. Available: https://gisco-services.ec.europa.eu/distribution/v2/nuts/

[138] Buildings Performance Institute Europe, “EU buildings climate tracker,” BPIE, Brussels, Tech. Rep., 2024. [Online]. Available: https://www.bpie.eu/

[139] S. Cozza, J. Chambers, and M. Patel, “Measuring the thermal energy performance gap of labelled residential buildings in Switzerland,” *Energy Policy*, 2019. [Online]. Available: https://doi.org/10.1016/j.enpol.2019.111085

[140] International Energy Agency, “Net Zero Roadmap: A Global Pathway to Keep the 1.5 °C Goal in Reach, 2023 Update,” International Energy Agency, Paris, Tech.

Rep., Sep. 2023. [Online]. Available: https://www.iea.org/reports/net-zero-roadmap-a-global-pathway-to-keep-the-15-c-goal-in-reach

[141] European Environment Agency, "Trends and projections in europe 2023," European Environment Agency, Copenhagen, EEA Report 07/2023, Oct. 2023. [Online]. Available: https://www.eea.europa.eu/en/analysis/publications/trends-and-projections-in-europe-2023

[142] A. S. Lord, P. H. Kobos, and D. J. Borns, "Geologic storage of hydrogen: Scaling up to meet city transportation demands," *International Journal of Hydrogen Energy*, vol. 39, no. 28, pp. 15 570–15 582, 2014.

[143] D. D. Papadias and R. K. Ahluwalia, "Bulk storage of hydrogen," *International Journal of Hydrogen Energy*, vol. 46, no. 70, pp. 34 527–34 541, 2021.

[144] M. Talukdar, P. Blum, N. Heinemann, and J. Miocic, "Techno-economic analysis of underground hydrogen storage in Europe," *iScience*, vol. 27, no. 1, p. 108771, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2589004223028481

[145] J. Lotze, M. Moser, G. Savvidis, D. Keles, and V. Hagenmeyer, "The complementary role of energy system and power grid models: An analysis of the European energy transformation from a holistic perspective," *Energy Conversion and Management*, 2024. [Online]. Available: https://doi.org/10.1016/j.enconman.2024.118761

[146] BloombergNEF, "EU ETS II pricing scenarios," BloombergNEF, Tech. Rep., 2025. [Online]. Available: https://about.bnef.com/insights/clean-energy/eu-ets-ii-pricing-scenarios/

[147] Danish Energy Agency, "Technology data for energy plants for electricity and district heating generation," Danish Energy Agency, Copenhagen, Tech. Rep., 2023. [Online]. Available: https://ens.dk/en/our-services/projections-and-models/technology-data

[148] D. Newbery, "Missing money and missing markets: Reliability, capacity auctions and interconnectors," *Energy Policy*, vol. 94, pp. 401–410, 2016.

[149] European Union, "Regulation (EU) 2019/943 of the European Parliament and of the Council of 5 June 2019 on the internal market for electricity (recast)," Official Journal of the European Union L 158, 14.6.2019, pp. 54–124, 2019. [Online]. Available: https://eur-lex.europa.eu/eli/reg/2019/943/oj

[150] European Union Agency for the Cooperation of Energy Regulators, "Security of EU electricity supply: Report on capacity mechanisms," ACER, Ljubljana, Tech. Rep., 2023. [Online]. Available: https://www.acer.europa.eu/

[151] Department for Energy Security and Net Zero, "2024 four year ahead capacity auction (T-4) delivery year 2028/29: Auction monitor report," UK Government, London, Tech. Rep., 2025. [Online]. Available: https://www.gov.uk/government/publications/capacity-market-auction-auction-monitor-report-for-t-4-auction-for-2028-to-2029

[152] L. Bühler and H. Scharf, "Green hydrogen in the European Union: A large-scale assessment of the supply potential and economic viability," *Applied Energy*, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0306261925013170

[153] A. Badakhsh and S. M. Bhagavathy, "Caveats of green hydrogen for decarbonisation of heating in buildings," *Applied Energy*, 2023. [Online]. Available: https://doi.org/10.1016/j.apenergy.2023.122152

[154] Swiss Federal Office of Energy, "Hydrogen Strategy for Switzerland," Swiss Federal Office of Energy (SFOE), DETEC, Bern, Tech. Rep., Dec. 2024. [Online]. Available: https://www.bfe.admin.ch/bfe/en/home/supply/hydrogen-and-power-to-x.html

[155] N. Milojevic-Dupont, F. Wagner, F. Nachtigall, J. Hu, G. B. Brüser, M. Zumwald, F. Biljecki, N. Heeren, L. H. Kaack, P.-P. Pichler, and F. Creutzig, "EUBUCCO v0.2," Zenodo data release, 2023, extends v0.1 to the United Kingdom and Norway.

[156] European Heat Pump Association, "European heat pump market and statistics report 2023/2024," EHPA, Tech. Rep., 2024. [Online]. Available: https://www.ehpa.org/

[157] Northern Gas Networks, Equinor, and Cadent, "H21 North of England," Northern Gas Networks, Tech. Rep., 2018.

[158] International Renewable Energy Agency, "Heat pumps: Technology brief," IRENA, Abu Dhabi, Tech. Rep., 2022. [Online]. Available: https://www.irena.org/publications/2022/Nov/Heat-pumps

[159] Joint Research Centre, "Techno-economic projections until 2050 for smaller heating and cooling technologies," European Commission Joint Research Centre, Tech. Rep., 2023. [Online]. Available: https://publications.jrc.ec.europa.eu/repository/bitstream/JRC109034/kjna28861enn.pdf

[160] ——, “Long-term (2050) projections of techno-economic performance of low-carbon technologies,” European Commission Joint Research Centre, Tech. Rep., 2023. [Online]. Available: https://publications.jrc.ec.europa.eu/repository/bitstream/JRC109006/kjna28859enn.pdf

[161] International Renewable Energy Agency, “Heat pump costs and markets,” IRENA, Abu Dhabi, Tech. Rep., 2022. [Online]. Available: https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2022/Nov/IRENA_Heat_Pumps_Costs_Markets_2022.pdf

[162] European Heat Pump Association, “Heat pumps: Key figures and market data,” EHPA, Tech. Rep., 2023. [Online]. Available: https://www.ehpa.org/market-data/

[163] IRENA, “Global hydrogen trade to meet the 1.5°c climate goal: Part i,” International Renewable Energy Agency, Tech. Rep., 2022.

[164] A. Alsulaiman, “Renewable hydrogen import routes into the EU,” Oxford Institute for Energy Studies, Tech. Rep. ET24, 2023. [Online]. Available: https://www.oxfordenergy.org/publications/renewable-hydrogen-import-routes-into-the-eu/

[165] Gas for Climate, “Gas decarbonisation pathways 2020–2050 (optimised gas scenario),” Guidehouse, Tech. Rep., 2020. [Online]. Available: https://gasforclimate2050.eu/

[166] Hydrogen Science Coalition, “Principles for the use of hydrogen,” 2023. [Online]. Available: https://h2sciencecoalition.com/principles/

[167] Eurostat, “Disaggregated final energy consumption in households (nrg_d_hhq),” Online data set, space-heating end-use by fuel, 2024, accessed 2026-05-26. [Online]. Available: https://ec.europa.eu/eurostat/databrowser/product/page/NRG_D_HHQ

[168] Odyssee-Mure project, “Energy efficiency trends and policies – country profiles,” Online database, 2024. [Online]. Available: https://www.odyssee-mure.eu/publications/efficiency-trends-policies-profiles/

[169] J. Spinoni, J. V. Vogt, P. Barbosa, A. Dosio, N. McCormick, A. Bigano, and H.-M. Füssel, “Changes of heating and cooling degree-days in Europe from 1981 to 2100,” *International Journal of Climatology*, vol. 38, no. S1, pp. e191–e208, 2018.

[170] I. Staffell and S. Pfenninger, “Using bias-corrected reanalysis to simulate current and future wind power output,” *Energy*, vol. 114, pp. 1224–1239, 2016.

[171] S. Pfenninger and I. Staffell, "Long-term patterns of European PV output using 30 years of validated hourly reanalysis and satellite data," *Energy*, vol. 114, pp. 1251–1265, 2016.

[172] M. Kittel, A. Roth, and W.-P. Schill, "Long-duration electricity storage needs for coping with Dunkelflaute events in Europe," *Nature Communications*, vol. 17, p. 4210, 2026. [Online]. Available: https://www.nature.com/articles/s41467-026-72681-5

A Supplementary Methodology and Detailed Results

A.1 Bottom-up Validation of the Residential Building Stock

The top-down heat-demand baseline (Section 3.2) rests on the Hotmaps regional dataset combined with census building-type shares. Because that baseline is itself a modelled product, we constructed an independent *bottom-up* estimate of residential heat demand and use it as a validation cross-check. The bottom-up build is a single pipeline parameterised per country, so that adding a country is a data exercise.

Method. For each country, every building footprint in the EUBUCCO v0.2 open building dataset [155] is classified. The published description of the database covers release v0.1 [34]; we use v0.2, which is what extends coverage to the United Kingdom, one of the 29 markets. Each footprint is classified into one of four types, namely single-family house (SFH), low-rise and high-rise multi-family (low-rise and high-rise), and non-residential, using sequential rules on footprint area, floor count and the native EUBUCCO type string. Heated floor area is the footprint polygon area multiplied by the EUBUCCO floor count and a 0.85 useable-area fraction. Each building is then assigned a space-heating intensity from the TABULA European building typology [35], indexed by building class and construction-vintage cohort and multiplied by a heating-degree-day climate correction and a national retrofit-state blend, with a domestic-hot-water term added. The resulting per-building annual demand is summed to a national residential total and reconciled against the Hotmaps baseline. Countries without a national TABULA dataset use a climate-corrected neighbour as a proxy (Luxembourg via Belgium [108]; Finland via Sweden), with the correction taken from the ratio of Eurostat heating-degree days [112].

Coverage. Across the 29 country builds, non-residential buildings dominate the raw building *count*, between roughly half and two-thirds of all footprints (garages, saunas and agricultural outbuildings in the open data); because they carry zero heated floor area, they are excluded from the residential total. Single-family homes carry 58% of residential useful heat and multi-family buildings 42%, with the single-family share ranging from ≈80% in Ireland to under 10% in Malta. The per-vintage intensity gradient that TABULA encodes is steep. Across countries, the median space-heating intensity falls by roughly a factor of four from the pre-1945 cohort to post-2020 construction, reflecting successive thermal building regulations.

Validation result. Table A.1 and Figure A.1 compare the bottom-up estimate with the Hotmaps baseline for all 29 countries. On the raw 2025 snapshot, 19 of the 29 countries fall within the ±15 % 'consistent' tier and 28 within the ±25 % acceptance band (only Malta, the smallest stock, lies outside), while the EU-27+UK+CH aggregate deviates by just −0.8 %. Three countries illustrate the spread. Luxembourg (−20.2 %) sits near the lower edge of the band, Finland (−11.8 %) and France (+11.4 %) inside the tighter tier on opposite sides of the benchmark. The bottom-up build is reported without any calibration to the benchmark. This level of agreement, achieved from entirely independent data (building footprints and a physical typology, set against the Hotmaps top-down accounting), supports the use of the baseline for the scenario analysis.

A count discards the size of a miss and an aggregate hides it, because the country deviations carry opposite signs and cancel. On the raw basis the mean absolute deviation across the 29 countries is 11.6 %, the demand-weighted mean 9.8 %, the root-mean-square deviation 14.4 percentage points and the median absolute deviation 8.5 %, the largest single miss being 35.6 % at Malta. On the vintage-matched basis the same four read 12.4, 10.7, 15.2 and 9.6 %. The −0.8 % aggregate is therefore below a tenth of the median country deviation, which is the quantitative form of the reading we take throughout: the study-area aggregate is the supported figure, and the country-level verdicts are indicative.

The table's final column reports the vintage-matched deviation, which backcasts each 2025 snapshot to 2015 along the country's historical demand trend before comparison. It shifts every country downward (the EU-27+UK+CH aggregate moves to −8.3 %) and is reported so that the model and the 2015-vintage benchmark are compared on a consistent year. The remaining gaps are discussed in the Limitations section. In particular, EUBUCCO construction-year coverage varies sharply between countries (France approximately 72 %, Luxembourg and Finland approximately 0 to 2 %), so for low-coverage countries the result is driven by a stock-weighted fallback intensity in place of a true per-vintage calculation. Because the aggregate band is tight but the country tails are not (Luxembourg reaches about −32 % on the vintage-matched basis, and Bulgaria and Romania about +21 to +23 % on the raw basis), the country-level verdicts for those tail countries carry materially more uncertainty than the EU-27+UK+CH aggregate, and should be read as indicative.

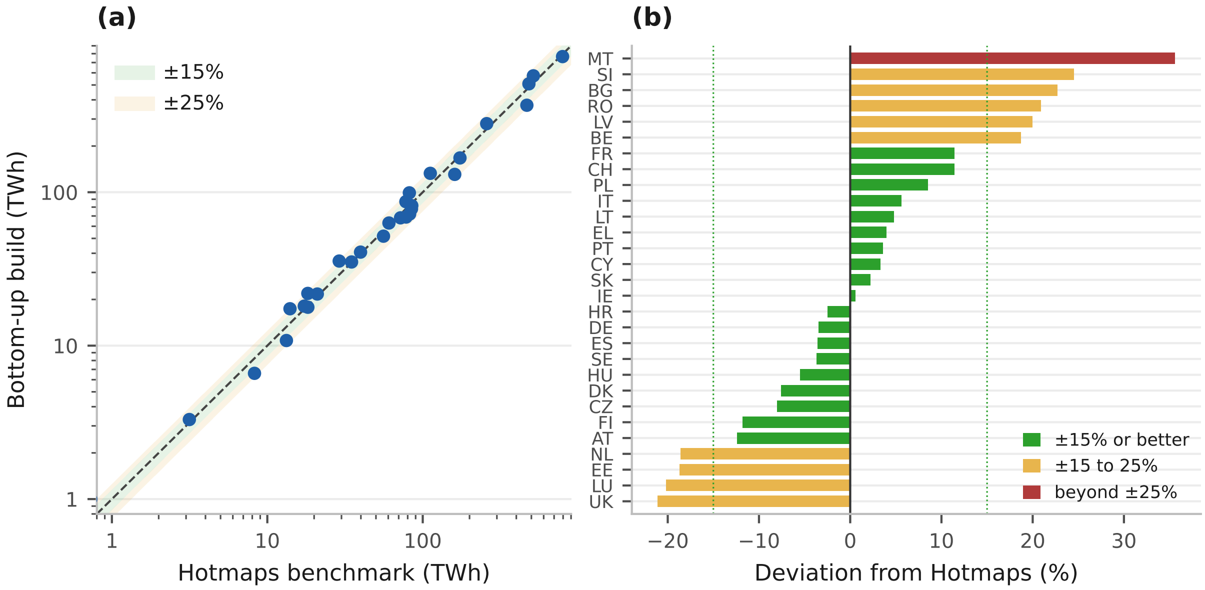
Table A.1: Bottom-up heat demand vs Hotmaps 2015, all 29 countries.

Country	EUBUCCO buildings	Bottom-up ($\mathrm{TWh\,yr^{-1}}$)	Hotmaps 2015 ($\mathrm{TWh\,yr^{-1}}$)	Raw deviation (%)	Vintage-matched deviation (%)
United Kingdom	24,861,926	368.9	467.7	−21.1	−27.9
Luxembourg	186,171	6.6	8.3	−20.2	−32.3
Estonia	1,141,841	10.8	13.3	−18.7	−26.1
Netherlands	11,562,667	130.7	160.7	−18.6	−25.6
Austria	4,876,313	72.1	82.4	−12.4	−22.3
Finland	6,633,364	68.9	78.1	−11.8	−20.9
Czech Republic	5,764,786	77.9	84.6	−8.0	−17.9
Denmark	6,152,463	51.7	55.9	−7.6	−15.0
Hungary	5,933,779	68.1	72.0	−5.5	−8.2
Sweden	7,121,754	81.8	85.0	−3.7	−15.1
Spain	17,937,712	167.3	173.6	−3.6	−8.3
Germany	67,146,981	765.8	793.7	−3.5	−9.6
Croatia	2,908,621	17.8	18.2	−2.5	−11.1
Ireland	3,580,975	35.1	34.8	0.6	−7.0
Slovakia	3,993,065	40.7	39.8	2.2	−6.5
Cyprus	890,693	3.3	3.1	3.3	−4.8
Portugal	6,373,919	21.7	21.0	3.6	2.1
Greece	5,971,312	63.0	60.6	4.0	1.3
Lithuania	2,525,315	18.1	17.3	4.8	−5.1
Italy	25,577,441	508.9	482.0	5.6	1.0
Poland	21,993,388	279.9	257.9	8.5	−5.6
Switzerland	3,000,960	86.8	77.9	11.4	0.4
France	52,610,604	573.8	515.1	11.4	1.8
Belgium	9,170,389	132.8	111.8	18.7	7.5
Latvia	1,368,162	21.9	18.2	20.0	18.0
Romania	12,645,080	99.1	82.0	20.9	12.7
Bulgaria	4,087,643	35.6	29.0	22.7	16.9
Slovenia	1,430,391	17.4	14.0	24.5	20.1
Malta	143,277	1.0	0.7	35.6	9.6
EU+UK+CH	317,590,992	3827.5	3858.8	−0.8	−8.3

Bottom-up estimates from EUBUCCO v0.2 building footprints and TABULA per-vintage intensities; reported as-is, without calibration. Hotmaps figures are the all-class NUTS3 totals for 2015. A positive deviation means the bottom-up build is above the benchmark. Of the 29 countries, 19 fall within ±15 % and 28 within ±25 % on the raw deviation (only Malta, the smallest stock, exceeds the band), and the EU+UK+CH aggregate deviates by −0.8 %; the vintage-matched column shifts each country down by the 2015 to 2025 demand trend and is reported so the comparison is vintage-consistent. Three illustrative rows (Luxembourg, France, Finland) are discussed in the text.

Source: Authors' contribution.

Figure A.1: Validation of the bottom-up build against the Hotmaps baseline. Panel (a) plots each country's bottom-up demand against its Hotmaps benchmark on log axes, with the dashed line parity and the shaded bands ±15 and ±25 per cent. Panel (b) gives the same deviation per country as a sorted bar chart, coloured green inside ±15 per cent, amber to ±25 and red beyond.



Source: Authors' contribution.

A.2 Peak Dispatch, Peaking Investment, and Network Capital

Levelised costs compare technologies as base-load suppliers of annual heat. Hydrogen's most favourable case, however, is as a *peaking* producer serving only the cold-snap slice of the load-duration curve, the energy above the 2000-hour demand level, ≈12 to 18% of annual heat (Figure 9a), when air-source heat-pump performance collapses and the power system is stressed. We therefore model the winter peak explicitly as a merit-order dispatch among the three quick-start options. Their marginal costs per MWh of useful heat are

$$c_c^{\text{gas}} = \frac{p_c^{\text{gas}} + \varphi_y\, e_{\text{g}}}{\eta_{\text{g}}}, \qquad c_c^{\text{H}_2} = \frac{p_c^{\text{H}_2}}{\eta_{\text{h}}} + \sigma_c, \qquad c_c^{\text{HP}} = \frac{\pi_c^{\text{peak}}}{\text{COP}_c^{\text{cold}}}, \tag{6}$$

where φ_y is the carbon price, e_{g} the gas emission factor (0.202 tCO_2/MWh), η_{g}=0.92 and η_{h}=0.90 the boiler efficiencies, σ_c the seasonal-storage adder (≈€50/MWh-heat with domestic salt caverns, ≈€100 without, at rates consistent in order with the bulk-storage literature [142, 143, 144] and applied on the cavern geology mapped by Caglayan et al. [36]), and $\text{COP}_c^{\text{cold}} = \max(0.60\,\text{SCOP}_c,\, 1.8)$ the single country-level cold-snap heat-pump coefficient of performance. The factor 0.60 is the seasonal-to-cold-snap derate calibrated to the low-temperature field measurements of Gibb et al. [55]. Across the cold-snap design window an air-source unit delivers about three-fifths of its seasonal coefficient, floored at 1.8 for the coldest, highest-flow-temperature systems. The seasonal coefficients themselves, SCOP_c of about 2.8 to 4.5 across the stock, sit at the modern variable-speed end of the field evidence, consistent with recent cold-climate and market-tracking data [55, 156]; the cold-snap derate above then keeps peak-hour dispatch from being credited with these mild-weather values.

A third coefficient sizes the electricity network. The reinforcement adder of Eq. (14) and the infrastructure bill of Eq. (10) divide by a flat $\mathrm{COP}^{\mathrm{el}} = 2.3$ applied to every country, where $\mathrm{COP}_c^{\mathrm{cold}}$ runs from 1.80 to 2.84 with a median of 2.04. The flat value therefore sits above the median and understates the reinforcement both carriers' comparison rests on, by about a tenth on the network bill, in the direction that favours the heat pump. Correcting it is the first item for the next model run. This single $\mathrm{COP}_c^{\mathrm{cold}}$ is used wherever the building or grid peaker is priced, namely the merit-order dispatch of Eq. (6), the missing-money rent of Eq. (8) and the grid-reinforcement term of Eq. (10); the deeper hour-resolved temperature derate that the 8760-hour dispatch applies within the very coldest Dunkelflaute hours (Appendix A.2, 'Explicit hourly dispatch') is a distinct, sub-peak quantity and is labelled as such where it is introduced, so that no two different cold coefficients enter the same calculation. The scarcity electricity price π_c^{peak} is derived from the power sector's own merit order. The marginal winter unit, a gas OCGT versus a hydrogen-ready turbine at wholesale fuel prices, sets it, plus a scarcity premium π^{s} spanning storage and demand-response displacement,

$$\pi_c^{\mathrm{peak}} = \min\left(\frac{p^{\mathrm{gas,w}} + \varphi_y\, e_{\mathrm{g}}}{\eta_{\mathrm{OCGT}}},\ \frac{p_c^{\mathrm{H_2,bb}}}{\eta_{\mathrm{T}}} + \sigma_c^e\right) + \pi^{\mathrm{s}}, \tag{7}$$

with turbine efficiencies $\eta_{\mathrm{OCGT}}{=}\eta_{\mathrm{T}}{=}0.40$, backbone (not retail) hydrogen prices, and $\pi^{\mathrm{s}} \in \{30, 60, 120\}$ €/MWh (low/central/high). The premium bounds the intra-seasonal flexibility cost that sits above the marginal peaker's own short-run cost, that is, the value of the storage and demand-response capacity that is displaced when the cold snap exhausts the cheaper flexibility. The low value (€30/MWh) represents a deeply interconnected system with abundant fast-responding demand response, the central value (€60/MWh) a typical continental system, and the high value (€120/MWh) an islanded system with limited battery storage in which scarcity rents approach the regulated value of lost load. The three values bracket the plausible range without estimating a point. They are stylised values we set, and no market has been observed to pay them, and we report them as such. Two consequences follow from where the premium enters.

It cancels exactly from the hydrogen-versus-gas comparison in the power arena, where both turbines buy at the same clearing price, so the power-arena win count holds at seven across the whole €30 to 120/MWh-e band. A winner's rent is the premium times its run hours, so capital recovery is exactly proportional to it. The 8.7 per cent capacity-weighted recovery computed at the central value is 4.4 per cent at the low and 17.4 at the high, none of which reaches the 100 per cent unaided financeability requires.

The building peak behaves differently, because it is precisely this scarcity price that the cold-snap heat pump is charged at, so there a higher π^{s} widens hydrogen's niche and a lower one narrows it; this is the classic scarcity-pricing margin of low-utilisation capacity [28]. The sensitivity of hydrogen's merit-order position to π^{s} is reported in full in the Results (Section 4.6, the winning-country count by scenario) and the dedicated stress test of Appendix C.7; the missing-money verdict (Eq. (8)) is invariant across the band, because it is the peaker's running hours, and not the price level, that fail the investment test. Hydrogen's heating share in each scenario is the peak slice it wins under Eq. (6), a derived ceiling that the merit order sets, and nothing in it is stipulated.

Winning the dispatch is necessary but not sufficient. An investor must also recover the peaker's capital from the hours it runs. The hydrogen peaker earns the inframarginal rent against the cheapest alternative for the $H_c \approx 500$ full-load hours of the peak slice. This 500-hour figure is not a free assumption but the duration of the slice itself. It is the slice's energy divided by its capacity, a utilisation read from the heat load-duration curve (Figure 9a). It is not the count of hours above the 2,000-hour threshold, which is 2,000 by construction. The slice covers residential space heating only; industrial and process demand are outside the model's scope and do not enter H_c.

This dwelling-level three-way contest between a gas boiler, an air-source heat pump and a hydrogen boiler is distinct from the power-sector peaker stack of Eq. (7) and the hourly dispatch of Eq. (11). The two are linked, because the heat-pump branch is priced at the endogenous scarcity electricity price π_c^{peak} that the power-sector merit order sets, but the electricity peaker that backs the grid runs far fewer full-load hours than the building heat peaker, since the electricity peak is sharper and briefer than the heat peak. The rent earns an annual margin per kW of capacity of

$$R_c = \left[\min\left(c_c^{\mathrm{gas}},\, c_c^{\mathrm{HP}}\right) - c_c^{\mathrm{H_2}}\right] \cdot \frac{H_c}{1000}, \qquad \text{build iff } R_c \geq \left[\mathrm{CRF}(r_c, n) + m\right] K, \tag{8}$$

where K is the hydrogen-boiler CAPEX (€600/kW), n=20 years, m=2% fixed O&M, and r_c the country WACC.

The power-sector peaker is a different asset and takes its own condition, which every recovery figure in Section 5.3 and Appendix A.7 uses. Where hydrogen is the cheaper firm unit it sets the clearing price at its own short-run cost plus the scarcity premium π^{s}, so its rent is that premium over the hours it runs and its capital recovery is

$$R_c^{\mathrm{pw}} = \frac{\pi^{\mathrm{s}}\,\mathrm{FLH}_c/1000}{\left[\mathrm{CRF}(r_c,\, 25) + f\right] K^{\mathrm{T}}}, \qquad f = 0.025, \tag{9}$$

with FLH_c the peaker's full-load hours, f a fixed operating charge of 2.5 per cent a year, and K^{T} the hydrogen-ready turbine's overnight capital cost of €600/kW, against

€450/kW for a conventional open-cycle machine. The turbine's 25-year life and its 2.5 per cent fixed operating charge differ from the boiler's 20 years and 2 per cent above, and the two assets share the €600/kW figure by coincidence rather than by construction, so the two conditions are not interchangeable. The denominator is the annualised charge quoted as €51 to 66/kW-yr for the five markets that build: Denmark's 36.40 + 15.00 = 51.40 and Poland's 51.49 + 15.00 = 66.49. Section 4.7 shows that the heat-side build condition fails everywhere, and the magnitude of the failure is large. Over the roughly 500 full-load hours of the slice the inframarginal rent recovers only a minority of the heating-side peaking boiler's annualised capital, with the best cases (Sweden and Denmark, under H2 Push) at approximately 40 per cent, and 12 of the 29 winning country-scenarios recovering under 20 per cent, because the inframarginal rent a peaker earns over so few hours falls far short of an annualised capital requirement that runs from €52 to 73/kW-yr across the 29 markets, and from €52 to 71 across the winning ones.

Resolved to the hour and accumulated across the milestone years to 2050, the same shortfall is about €357 bn for the hydrogen peaker fleet EU-wide under the H2 Push scenario, the central estimate carried in Section 5.3, and sweeps to roughly €379 to 563 bn across the remaining scenarios (€379 bn Net Zero, €418 bn Stated Policies, €563 bn Current Policies; against about €130 to 142 bn for the gas fleet), so the gap is structural and persists under an explicit 8760-hour treatment; the full per-country-scenario detail is in Section 4.7 and Appendix A.7.

Finally, each pathway's network capital requirement aggregates three components from the model's own technology shifts,

$$B_c = \underbrace{\Delta S_c^{\mathrm{HP}} \frac{P_c^{\mathrm{pk}}}{\mathrm{COP}^{\mathrm{el}}} \kappa\, u^{\mathrm{el}}}_{\text{grid reinforcement}} + \underbrace{\Delta S_c^{\mathrm{DH}} N_c\, u^{\mathrm{DH}}}_{\text{DH expansion}} + \underbrace{S_c^{\mathrm{H_2}} N_c \big[\gamma_c u^{\mathrm{conv}} + (1-\gamma_c) u^{\mathrm{new}}\big]}_{\mathrm{H_2}\text{ network}}, \tag{10}$$

where ΔS are the 2025→2050 share shifts, P_c^{pk} the heat peak, κ=0.5 the coincidence factor, $u^{\mathrm{el}} \in \{400, 800, 1600\}$ €/kW the reinforcement cost band, N_c the dwelling stock, $u^{\mathrm{DH}} \in \{3000, 5000, 8000\}$ € per connection, γ_c the gas-grid coverage, and $u^{\mathrm{conv}}/u^{\mathrm{new}}$ the hydrogen conversion/new-build connection cost bands of Appendix A.6.

Explicit hourly dispatch. Equations (6) to (8) treat the peak as a single load-duration slice. To make the cold-snap hours and the peaker's utilisation explicit, and to test whether the missing-money result survives an hour-resolved treatment, we add an 8760-hour dispatch of the power sector against a per-country installed-capacity stack built from national 2050 strategies and NECPs and cross-checked against the ENTSO-E TYNDP 2024 (data note in the supplementary material). Hourly electricity demand $D_{c,t}$ combines a calibrated national baseline-load shape with heat-pump electricity $Q_{c,t}^{\mathrm{heat}}/\mathrm{COP}_{c,t}$, where the hourly COP $\mathrm{COP}_{c,t}$ is *temperature-dependent*, tracking the seasonal coefficient in mild hours and falling towards a sub-peak floor of $0.55\,\overline{\mathrm{SCOP}}$ in the very coldest hours of the synthetic Dunkelflaute, so the electricity peak is sharper than the heat peak, the physical reason a cold, still, dark hour stresses the grid.

The overbar marks a study-area mean. The hourly layer blends the air-source and ground-source seasonal coefficients three to one and applies the result to every country, without the climate multiplier (0.85 in Finland to 1.35 in Cyprus) that the levelised-cost layer applies through SCOP_c. Capacity-weighted across the peaker fleet that multiplier averages 0.966, so the per-country split of the firm fleet carries a bias of a few per cent and the coldest markets are the ones understated. This hour-resolved floor ($0.55\,\overline{\mathrm{SCOP}}$) is deliberately deeper than, and distinct from, the single country-level cold-snap coefficient $\mathrm{COP}_c^{\mathrm{cold}} = \max(0.60\,\mathrm{SCOP}_c, 1.8)$ of Eq. (6). The latter is the building or grid peaker's design value over the whole cold-snap window and prices the merit order and the missing-money rent, whereas the 0.55 floor applies only to the handful of extreme hours that set the electricity peak in this dispatch layer. The two are used in different calculations and are never substituted for one another, so the merit-order position of Eq. (6) and the rent of Eq. (8) are computed at $\mathrm{COP}_c^{\mathrm{cold}}$ throughout. Table A.2 states the mapping. The three derates are distinct coefficients entering distinct equations, and none is used in another's place.

Table A.2: The three cold-weather heat-pump derates and where each enters.

Derate	Definition	Enters
Merit-order / missing-money	$\mathrm{COP}_c^{\mathrm{cold}} = \max(0.60\,\mathrm{SCOP}_c,\, 1.8)$	Eqs. (6), (8)
Network reinforcement	$\mathrm{COP}^{\mathrm{el}} = 2.3$, flat across countries	Eqs. (14), (10)
Hourly extreme-hour dispatch	$\mathrm{COP}_t \to 0.55\,\overline{\mathrm{SCOP}}$ (coldest hours)	Demand $D_{c,t}$ in Eq. (11)

Source: Authors' contribution.

Each generator g supplies $a_{g,c,t}\,P_{g,c}$, where a is a calibrated hourly availability (solar diurnal to seasonal; wind an AR(1) process with a forced cold-snap *Dunkelflaute*; hydro run-of-river seasonal; nuclear flat must-run), and a flexibility budget $f_{c,t}$ (a four-hour battery plus the hydro reservoir) shaves the daily peak. The residual served by the dispatchable fleet, and the peaker energy and capacity above the firm gas fleet P_c^{gas}, are

$$r_{c,t} = D_{c,t} - \sum_{g\in\{\mathrm{VRE,nuc,bio}\}} a_{g,c,t}\,P_{g,c} - f_{c,t}, \quad E_c^{\mathrm{pk}} = \sum_t \max\!\big(0,\, r_{c,t} - P_c^{\mathrm{gas}}\big), \quad G_c^{\mathrm{pk}} = \max_t\, r_{c,t}. \tag{11}$$

The peaker sized to G_c^{pk} runs $E_c^{\mathrm{pk}}/G_c^{\mathrm{pk}}$ full-load hours; we price it as hydrogen- versus gas-fuelled with full economics, and all inputs evolve over time, among them the *rising* carbon price on gas, a *declining* gas wholesale price (€38→28/MWh) and hydrogen price, an *improving* turbine efficiency (40% in 2025, rising to 42% for gas and 44% for the hydrogen turbine by 2050, or 48% under H2 Push, which assumes an advanced simple-cycle machine, the latter slightly faster, reflecting H_2-combustor R&D), a declining seasonal-storage cost, variable O&M, country labour, and annualised CAPEX (€600/kW H_2-ready turbine, €450/kW OCGT, 25-year life).

These turbine efficiencies sit within the open-cycle range of the Danish Energy Agency technology catalogue, in which advanced and hydrogen-ready simple-cycle units reach the mid-to-high forties per cent by 2050 [147], so the peaking economics do not rest on optimistic turbine performance. We accumulate profit and system cost across the 2025 to 2050 milestone years. The structural link to the buildings model is the endogenous peak price π_c^{peak} of Eq. (7). We bound the three assumptions to which the peaker requirement is most sensitive: *interconnection* (the islanded sum of national peakers versus an EU copper-plate fleet sized to the aggregate residual peak), the *depth* of the synthetic cold-snap *Dunkelflaute* (cold-snap wind swept from 15% to 40% of normal), and the *length* of that cold-and-still event (swept from two days to ten). The resulting ranges are reported with the result.

The third was added last and is the largest of the three. The event length was a hard-coded six days, and both the heat spike and the wind lull run on the same window in every country at once, so nothing in the earlier sweeps varied it. Sweeping it moves the fleet sized on the inflexible load from 253.7 GW at two days to 302.0 GW at ten, against 277.7 at the central six, which is a wider band than either of the other two produces. It also retires an explanation this paper previously offered. The modest reduction a partially flexible heat-pump load delivers was attributed to the event being multi-day, on the argument that intra-day pre-heating cannot move load out of a six-day window. That is a restatement of the assumption, and the sweep refutes it; the reduction is nearly invariant across the range, at 8.9 to 11.6 per cent, because what binds is the share of load that can shift, and the length of the event does not. A fourth assumption, that the heat-pump load is completely inflexible, is bounded separately at 4.3 to 10.7 per cent of the fleet across three energy-conserving smoothing operators.

Unit-commitment dispatch of the firm fleet. The reduced-form merit order of Eq. (11) sizes the dispatchable layer by a simple threshold on the residual, which treats the firm fleet as perfectly flexible. To remove that idealisation, and to address the most material structural concern about the power coupling, the dispatchable layer is now re-solved by an explicit *unit-commitment* optimisation.

The dispatchable fleet is split into a firm combined-cycle (CCGT) class and fast open-cycle gas or hydrogen-turbine peakers, and for each country and milestone year the model minimises the weighted fuel, carbon and variable operating cost plus start-up cost,

$$\min \sum_d \omega_d \sum_h \left[\sum_g \left(c_g^{\mathrm{var}}\, p_{g,d,h} + c_g^{\mathrm{su}}\, v_{g,d,h} \right) + \pi^{\mathrm{voll}}\, u_{d,h}^{\mathrm{ns}} + \pi^{\mathrm{res}}\, s_{d,h}^{\mathrm{res}} \right],$$

subject to commitment and physical constraints on each class: minimum stable output ($p_g^{\min}\, n_{g,d,h} \leq p_{g,d,h} \leq \bar{p}_g\, n_{g,d,h}$), ramp limits, minimum up and down times, a 5 per cent operating-reserve margin on the fixed hourly net load (committed capacity must exceed served load plus the margin), and the battery and reservoir state of charge carried hour to hour, with ω_d the weight of representative day d, $n_{g,d,h}$ the count of committed ≈1 GW units, $v_{g,d,h}$ start-ups, and $u^{\mathrm{ns}}, s^{\mathrm{res}}$ slacks for unserved energy (value of lost load π^{voll} = €8,000/MWh) and reserve shortfall (π^{res} = €1,000/MW-h). The optimisation reuses the reduced form's demand, renewable availability (including the synthetic *Dunkelflaute*), temperature-dependent heat-pump COP derate and causal storage dispatch verbatim, so it differs from the reduced-form merit order *only* in how the dispatchable fleet is committed.

The upgrade therefore removes the fleet-flexibility idealisation, that the firm fleet could follow the residual perfectly, but it retains the reduced form's calibrated variable-renewable availability profiles and its causal, non-co-optimised storage heuristic; both remain assumptions of this layer and are carried as such in the limitations (Appendix D).

The 8760 hours are reduced to a weighted set of representative days per country, force-including as weight-one singletons the top peaker-signal days that together hold at least 99 per cent of annual peaker energy (so the convex peaker quantity is not averaged away), and clustering the remaining low-peaker days; the per-country problems are solved in parallel. Techno-economics follow the Danish Energy Agency Technology Catalogue 2023 [147] and the ENTSO-E TYNDP 2024: a CCGT minimum stable output of 0.30 of capacity, a CCGT ramp of 2.4 per hour, four-hour minimum up and three-hour minimum down times, against faster open-cycle and hydrogen peakers (minimum stable output 0.20, ramp 4.0). The CCGT ramp and minimum-stable-output values, and the 5 per cent operating-reserve standard, are judgement-call inputs carried as assumptions and tested in the limitations (Appendix D).

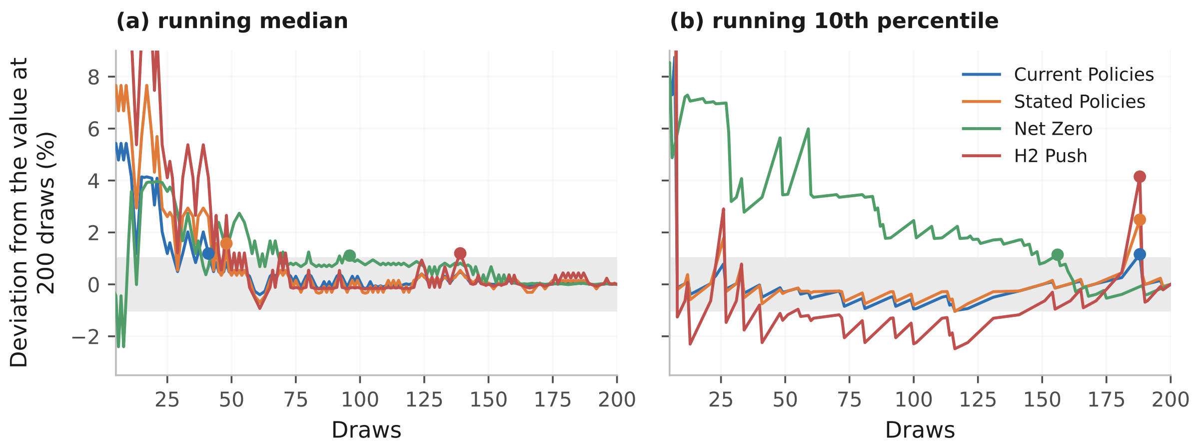
The production model is a linear relaxation of the commitment problem; against the mixed-integer formulation on spot countries the integrality gap is only 0.04 to 0.11 per cent, so commitment integrality is immaterial to the result and the relaxation is used throughout. The former reduced-form merit order of Eq. (11) is retained as a validation cross-check, run as a limit test with the commitment constraints disabled and reported with the rest of the hourly-dispatch results in Appendix A.7. The full specification, the representative-day reducer and the validation gates are documented in the unit-commitment methods note accompanying the supplementary material.

A.3 Uncertainty Quantification

Parametric uncertainty is characterised with a Monte Carlo approach using $N = 200$ samples per scenario. The following parameters are sampled stochastically: the per-country envelope-design intensity decline rate $r_{c,s} \sim \mathcal{N}(\mu_{c,s}, \sigma_{c,s})$, with $\sigma_{c,s} \approx 0.21\mu_{c,s}$ so that the published low/high bounds span the 5th to 95th percentile; the technology share targets for 2050; the heat-pump ground/air split; and the fuel-price multipliers (log-normal with $\sigma = 0.15$). Because renovation pace is driven partly by common EU policy (the EPBD and the Renovation Wave) and partly by national factors, the per-country rate shocks are correlated through a shared factor: $z_c = \sqrt{\rho}\, z_{\mathrm{EU}} + \sqrt{1-\rho}\, z_c^{\mathrm{idio}}$ with $\rho = 0.5$. This preserves each country's marginal distribution (so per-country bands and all medians are unchanged) while preventing the EU-aggregate band from being artificially narrowed by cross-country diversification; the resulting EU band is about 2.2 times wider than under independent draws. The parameter ρ is a structural knob (Appendix D). Results are reported as the 10th, 50th and 90th percentile quantiles across samples.

The choice of $N = 200$ samples is supported by a convergence test, and we report what it does and does not establish. The running median of the headline metric, 2050 EU buildings CO_2 emissions, settles inside $\pm 1\,\%$ of its final value after 41 draws under Current Policies and 48 under Stated Policies, after 96 under Net Zero, whose two-branch mixture converges slowly, and only after 139 under H2 Push, the slowest of the four. The deciles are slower still and are not converged at $N = 200$. The running 10th percentile last leaves that tube between draws 156 and 188 in all four scenarios (Figure A.2). A running quantile can re-enter the tube and leave it again, so each figure above is the last draw at which the series is still outside, which is the conservative of the two readings. Medians are therefore reliable at this sample size and decile bands should be read as indicative.

Figure A.2: Running-quantile convergence of the 2050 buildings CO_2 metric. Each series is the quantile recomputed on the first k draws, expressed as a deviation from its value at $N = 200$. The shaded band is the $\pm 1\,\%$ tube and the marker on each series is the last draw at which it sits outside. Panel (b)'s first few dozen draws overshoot to $+27\,\%$ and are clipped, so the axis resolves the settled tail the claim is about.



Source: Authors' contribution.

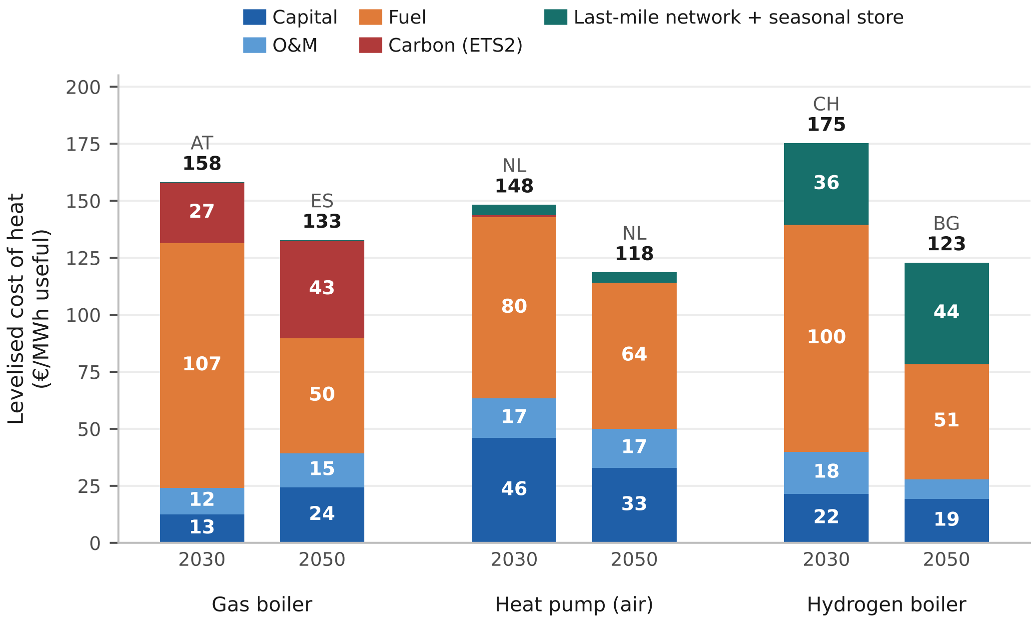
A.4 The Anatomy of the Cost Ordering

We decompose these levelised costs into their components, to show where the ordering comes from. Figure A.3 resolves the gas boiler, the air-source heat pump and the hydrogen boiler into capital, fixed and variable operation and maintenance, fuel, ETS2 carbon, and delivery and storage. On the EU median stacks, which total €129.3/MWh for the gas boiler, €111.5/MWh for the air-source heat pump and €124.8/MWh for the hydrogen boiler, the first reading is that fuel is the largest single component of every one of them, at 45.3 per cent of the gas-boiler stack, 54.2 per cent of the heat-pump stack and 43.6 per cent of the hydrogen-boiler stack, so it is the fuel- and electricity-price trajectories of Figure 6a that govern long-run competitiveness in all three cases, the heat pump included.

Three structural features then separate them. The first is carbon. The ETS2 wedge sits on the gas stack alone, at 33.2 per cent of its cost, and it is what holds gas at the top of the 2050 ordering. The second is capital, which runs the opposite way from the usual intuition. The heat pump is the *most* capital-heavy of the three, at 27.7 per cent of its levelised cost against 17.2 per cent for the hydrogen boiler and 11.8 per cent for the gas boiler, which is why the household-level barrier to adoption is an up-front financing one even though fuel remains its single largest cost component. The third, and the decisive one for the comparison this paper turns on, is what it costs to move the molecule and to hold it.

Delivery and seasonal storage together take 27.0 per cent of the hydrogen stack, just under the 27.8 per cent its capital and fixed operation and maintenance take together, against 4.4 per cent of the heat-pump stack for network reinforcement and nothing at all for gas, whose distribution grid is already built and already inside the retail price. This is the hydrogen boiler's largest structural burden. The three stacks therefore describe an efficient machine that is expensive to buy and runs on expensive taxed electricity, set against a cheap molecule that is expensive to deliver and expensive to keep from the summer in which it is made to the winter in which it is burned, and the ordering between them turns on the balance of those burdens, with no single dominant item.

Figure A.3: The anatomy of the levelised cost of heat by technology, 2030 and 2050 under Stated Policies. Each component is a median taken independently across the 29 markets and the bars are their sum, so the stacked totals of about €112, €125 and €129 differ from the €118.5, €122.8 and €132.7 per-country median costs quoted throughout the text. The figure is for the composition, not the level.



Source: Authors' contribution.

A.5 Decomposition: Demand, Efficiency and Grid Cleanness

The headline emissions fall is driven by what happens on the supply side. Table A.3 separates the Stated Policies result into three layers. Useful heat demand, what the building physically requires, falls only 9 per cent, since envelope renovation barely outpaces stock growth. *Delivered* (final) energy falls 31 per cent, because the shift to heat pumps (with a seasonal coefficient of performance of approximately 3) decouples delivered energy from useful heat. The implied EU system efficiency rises from 0.97 in 2025 to 1.29 by 2050. Emissions then fall 64 per cent, as the carrier mix moves off gas and oil and the grid behind the electrified share decarbonises. Decarbonisation is therefore a supply-side phenomenon, driven by source switching, conversion efficiency and grid cleanness, with no contribution from demand destruction.

Table A.3: The three layers of the result (Stated Policies, 2025 to 2050).

Layer	2025	2050	Δ
Useful heat demand (TWh)	3,827	3,483	−9%
Final/delivered energy (TWh)	3,928	2,695	−31%
CO_2 emissions ($MtCO_2$)	644	231	−64%

Source: Authors' contribution.

The depth of the cut nonetheless depends in part on the exogenous grid trajectory, and we test how much. This attribution runs on a single central draw in place of the 200-draw median. It shares the 644 $MtCO_2$ 2025 level with the ensemble, and reaches a Stated Policies 2050 level of about 264 $MtCO_2$ against the 231 $MtCO_2$ median reported elsewhere. The two are the same pathway read at different points of its own distribution, and only the split between the two causes is at issue here, so the percentages below are computed on the 644 to 264 fall of this draw and not on the median pair.

Recomputing that pathway with both grid and district-heat carbon intensity frozen at their 2025 levels, while keeping the same technology mix, leaves 2050 emissions under Stated Policies at approximately 407 $MtCO_2$, still a 36 per cent reduction on 2025. Technology switching alone therefore delivers roughly *three-fifths* of the scenario's emissions fall (62 per cent), achievable even on a 2025-vintage grid, with grid and district-heat decarbonisation contributing the remaining two-fifths (38 per cent). The result is primarily a technology-switching outcome, and the assumed grid trajectory is not what produces it; but the grid assumption governs the remaining two-fifths, and the deep (98 per cent) outcome of Net Zero is unattainable without it (Appendix D).

A.6 Hydrogen Distribution-Infrastructure Cost

The headline gap of Figure 8 charges each carrier for its own last-mile network. Hydrogen pays a distribution adder set at the coverage-weighted blend of conversion and new-build cost, at the central cost bound, and electric heating pays the low- and medium-voltage reinforcement that its added coincident winter peak requires (Appendix A.13). Free reuse of the existing gas grid at no incremental cost is retained only as the generous end of the sweep below, because a methane network cannot carry 100 per cent hydrogen unmodified. Polyethylene mains are largely reusable, but cast-iron and ductile-iron mains, meters, regulators and purging and leak-detection equipment require replacement or upgrade, and ungridded areas need a network built from scratch [90, 157]. We therefore sweep hydrogen's delivery cost across four per-country treatments, each annualised over a 40-year network life at the country WACC and spread over the useful heat a dwelling consumes per year (climate-scaled around an EU average of approximately 13 MWh):

1. **Free gas-grid reuse (lower bound).** Hydrogen inherits the existing distribution grid at no incremental cost. This is the assumption earlier drafts carried in the headline comparison.

2. **Retrofit conversion.** The whole existing gas grid is *converted* to 100 per cent hydrogen. This is still optimistic, since it assumes a usable grid already reaches every home and needs only conversion.

3. **Coverage-weighted blend (the model default).** Conversion cost is charged where the gas grid already reaches the dwelling and new-build cost where it does not, weighted by each country's gas-grid coverage γ_c, as in the hydrogen term of Eq. (10).

4. **New-build (upper bound).** A dedicated hydrogen distribution network is built from scratch to every dwelling.

The per-connection capital cost is itself genuinely uncertain, so we carry each treatment as a sensitivity band over a low, central and high range from the conversion literature: conversion at €1,000/1,500/3,000 per dwelling, and new-build at €3,000/5,000/8,000 per dwelling.

Two things follow from the sweep, one about direction and one about geography. The direction is monotone. Free gas-grid reuse is the most favourable treatment for hydrogen and full new-build the least, with retrofit conversion and the coverage-weighted blend in between, and the EU-mean gap between the hydrogen boiler and the best available heat pump widens against hydrogen at every step along that sequence. The blend is the treatment the headline of Section 4.5 reports, so the headline sits at the third of the four rungs.

The sweep has been recomputed on the storage-inclusive basis this paper uses, so every arm now carries the seasonal store and all of them are comparable both with one another and with the headline. The EU-mean hydrogen-minus-heat-pump gap runs from €0.0/MWh under free gas-grid reuse, through +€8.0 under retrofit conversion and +€19.7 under the coverage-weighted blend used as the default, to +€26.9 under full new build, reaching +€43.1 at the new-build high bound. The count of markets at or below parity for hydrogen falls from 12 to 9, 7, 3 and finally 2 across those treatments. Only Denmark and Germany survive the most expensive one, Denmark running from −€44.3/MWh under free reuse to −€28.0 under new build and −€18.2 at the high bound, and Germany from −€33.2 to −€16.2 and −€6.0.

The earlier reading drawn from the pre-storage version of this sweep, that hydrogen's base-load advantage survives every network treatment in a shrinking but never empty set of countries, does not survive the recomputation. The set does shrink to almost nothing, and outside Denmark and Germany hydrogen's base-load case depends on inheriting distribution assets it does not pay to build. For scale, the seasonal-storage term the earlier sweep omitted is €12.1/MWh of useful heat in a cavern market such as Germany and €28.4/MWh in a cavern-free one such as Spain (Appendix A.13), which is of the same order as the €27/MWh the four network rungs span between them.

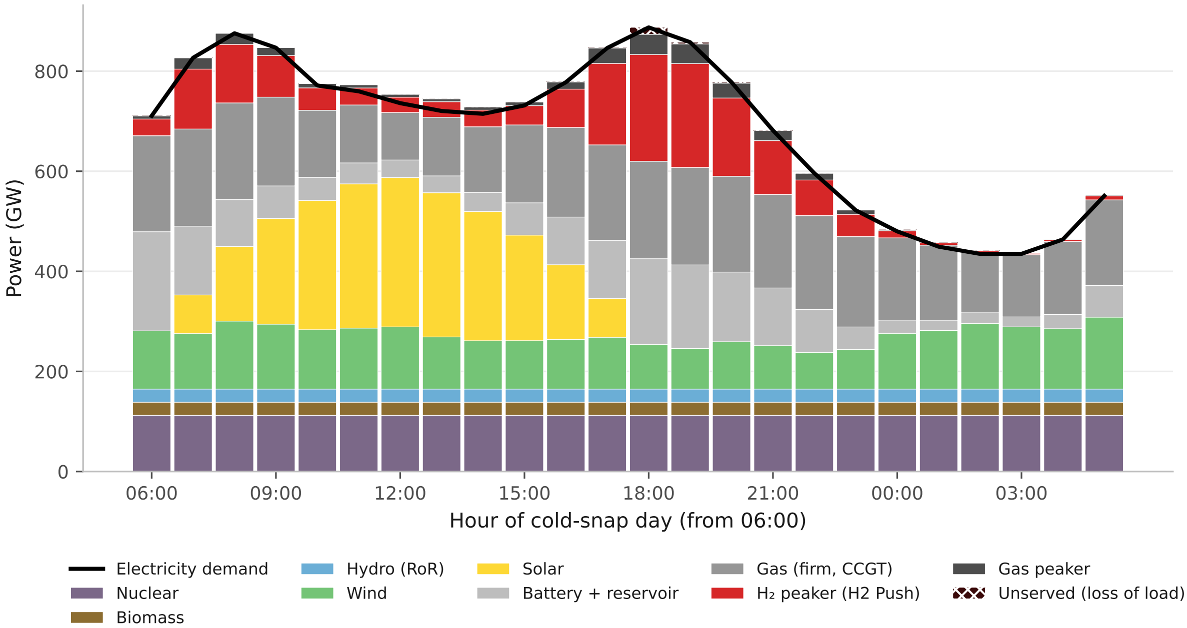
What the sweep does establish, and what does not depend on the level of the base-load cost, is that network cost redraws the map along storage geology and never across it. The countries least affected by charging hydrogen for its pipes are the cold, high-electricity-price north-west, where the cold season is long enough to spread a network charge thinly over a large annual heat demand. The warm south and the low-price east are affected most, because a small annual heat demand carries the same per-connection capital over far fewer megawatt-hours. Those are the same two groups the seasonal-storage charge separates, so the two effects reinforce one another. Two qualifications belong with this result. The gas-grid asset-utilisation co-benefit is real but only partial, since even the legacy network is not free to repurpose. And what advantage hydrogen retains is a fiscal artefact as much as a technology one, because the comparison sets an untaxed molecule priced delivered against electricity carrying the full retail levy stack; that asymmetry, and the sensitivity that would test it, is set out in Appendix D.

A.7 Hourly Dispatch, Capacity-Short Hours and the Peaker's Capital

This completes Test 3 by resolving its single load-duration slice to the hour. The missing-money result above rests on that one slice; resolved to the hour it holds and sharpens. Figure A.4 makes the point that peaker utilisation is small. It is the one-day hourly snapshot of the capacity-short hours. As a robustness check we dispatch an 8,760-hour power system against a per-country capacity stack (Eq. (11)). The model reproduces the standard pattern (Figure A.4). On a cold, still, dark winter day the residual demand after must-run and renewable generation spikes in the morning and again in the evening, exactly when electrified-heat demand peaks, solar is absent and wind has collapsed into a *Dunkelflaute*. These are the hours in which the system runs short and a firm peaker must fill the gap. Two features matter for hydrogen.

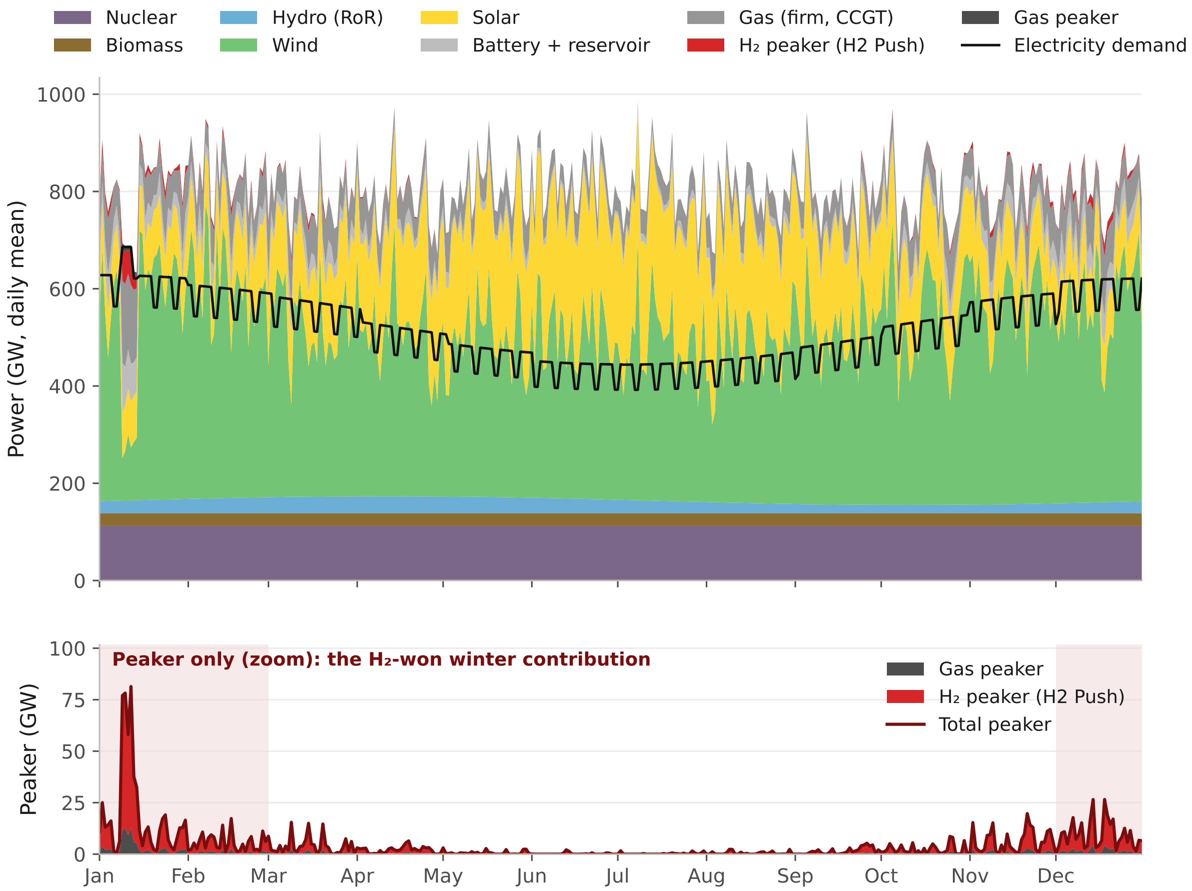
First, the scarcity is a *system-adequacy* phenomenon, and heating alone does not create it. Heat-pump electricity *alone* never exhausts the generation fleet in any country-year we model; it is the *sum* of baseline load and electrified heat, set against a low-VRE cold snap, that creates the gap. Second, the peaker's utilisation is small and falls as renewables and storage build out, from over a thousand full-load hours in the gas-dependent 2025 system to about 139 by 2050, spread across a mean of roughly 440 capacity-short hours in the markets that build a peaker, squarely in the pure-peaking regime that the TYNDP 2024 scenarios assign to the EU gas and hydrogen-turbine fleet. The same scarcity is sharply seasonal. Aggregated to daily means across the year (Figure A.5), the firm-capacity deficit concentrates in the winter months and all but vanishes in summer, so the peaker earns its keep in a narrow cold-season window.

Figure A.4: Cold-snap dispatch and the peaker gap (EU-27+UK+CH, 2050).



Source: Authors' contribution.

Figure A.5: Seasonality of the peaking need (EU-27+UK+CH, 2050, daily mean). The shaded bands are the two winter halves of the calendar year, December to February. In the upper panel the stacked supply runs above the demand line through the summer, which is surplus available capacity and not generation; the model does not curtail it explicitly. In the lower panel the total-peaker line is the sum of the two stacked areas beneath it.



Source: Authors' contribution.

Pricing that peaker with full economics confirms and sharpens the base-case result. On operating cost, hydrogen undercuts the gas turbine only where the carbon price is high or hydrogen is deliberately cheap. In 2050 the hydrogen peaker's variable cost beats gas in 0 of 29 countries under Current and Stated Policies, but, on the even-handed series holding both turbines at 0.40, in 7 of 29 under both Net Zero and H2 Push. In each the €350/tCO_2 carbon price lifts the gas turbine to about €256/MWh, and under H2 Push the rapid hydrogen decline lowers the hydrogen turbine beneath it in the same seven markets. The same carbon-driven pattern documented in the main text recurs at the building and district-heating levels.

On the investment question, however, **both fuels miss the money, cumulatively, to 2050** (Figure A.7). The cumulative loss is the milestone-year accumulation of each fleet's annual profit shortfall defined with Eq. (11), that is, the sum over the modelled years of the gas and hydrogen peaker losses. Summed across the EU-27+UK+CH and integrated over the milestone years, the gas peaker fleet loses about €142 bn and the hydrogen peaker fleet about €357 bn under the central H2 Push scenario, sweeping to €379 to 563 bn across the remaining scenarios. Those figures come from the reduced-form dispatch, which also sets the fleet and the run hours; the explicit unit-commitment layer below runs the same economics on its own dispatch and gives €358.4 bn with its constraints off and €356.0 bn with them on. The losses arise because the firm capacity that electrified heat requires for adequacy, about 278 GW EU-wide in the central islanded case and lower under interconnection (see below), runs far too few hours to repay itself from market revenue. The gas peaker recovers between about 7 and 20 per cent of its capital, the classic missing money of low-utilisation plant.

We size and dispatch this fleet against demand with the storage fleet's state of charge tracked hour by hour, so that a multi-day *Dunkelflaute* depletes the battery and the firm peaker must cover the sustained deficit; batteries serve the daily peak and cannot cover a week-long cold snap. The hydrogen peaker loses two to four times as much as gas, driven by its higher fuel-plus-storage cost, and the worst case is Current Policies, where stranded-expensive hydrogen never approaches viability. The system itself stays adequate; modelled loss of load averages about two hours per year and never exceeds the three-hour standard, so this is an economic verdict. The peaker requirement, and hence the absolute totals, turns on two assumptions that we test explicitly.

Cross-border interconnection lowers the requirement from the islanded case (no trade, with each country self-supplying its peak) towards an EU copper-plate fleet sized to the aggregate residual peak (perfect pooling), and pooling helps least in a deep, continent-wide *Dunkelflaute*, when every country is short at once, and most in milder, regional cold snaps, so trade cannot be relied upon to remove the firm-capacity need. The *depth* of the modelled cold snap moves the requirement in both directions around the central figure. Across cold-snap wind availabilities of 0.15, 0.25 and 0.40 the islanded requirement runs 291, 278 and 251 GW against a copper-plate 267, 239 and 205 GW, so the whole two-way grid spans 205 to 291 GW and the central islanded case reproduces the 278 GW headline exactly. The euro losses scale with this capacity, but the sign and ranking are invariant. In every case the hydrogen peaker loses two to four times what the gas peaker does, and neither recovers its capital from market revenue.

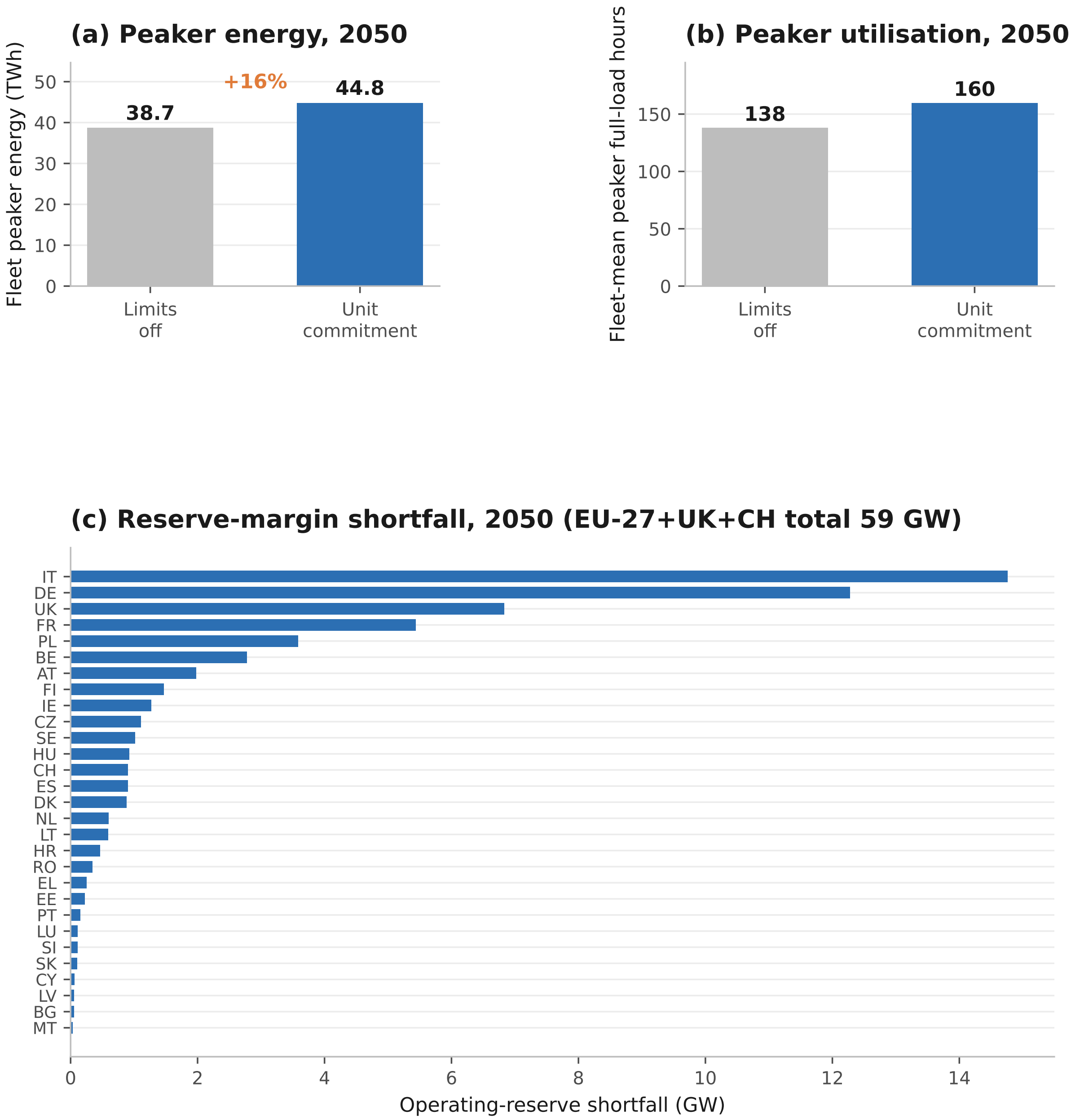
The verdict under explicit unit commitment. Resolving the dispatchable layer with the unit-commitment optimisation of Appendix A.2 confirms this result on a more realistic representation of how the firm fleet runs. The validation is exact first. With the commitment constraints disabled the optimisation reproduces the reduced-form merit order to within 0.33 per cent per country on dispatched energy and 0.40 per cent on peak capacity, and puts the fleet at 280.5 GW. The 2.8 GW difference is a Romanian peaker that the sizing pass builds and the reduced form does not. The unchanged peaker economics then feed through to a cumulative-2050 *limit case* of −€358.4 bn for hydrogen and −€143.2 bn for gas under H2 Push, within €1.1 and €0.8 bn of the reduced-form headline of Section 5.3, the residual being the Romanian peaker. The unit-commitment layer is therefore anchored to the reduced form before any constraint is switched on.

With the commitment constraints active, minimum stable output, ramp limits, minimum up and down times and start-up costs stop the cheap CCGT from following the residual perfectly, so the fast peakers pick up modestly more energy; peaker energy rises by about 16 per cent, from 38.7 to 44.8 TWh across the EU-27, the United Kingdom and Switzerland, and the fleet's full-load hours rise from 139 on the reduced form's 277.7 GW fleet to about 160 on the unit commitment's own 280.5 GW fleet (Figure A.6a,b).

The extra running cuts both ways. The hydrogen peaker burns more expensive fuel, but it also earns in more hours, and on balance the missing-money position eases marginally. In the *unit-commitment case* the hydrogen peaker loses about €356.0 bn and the gas peaker about €142.3 bn cumulatively to 2050, against €358.4 and €143.2 in the constraints-off limit case, so both positions improve, hydrogen by about €2.3 bn and gas by about €0.9 bn, and the hydrogen-minus-gas gap narrows by about €1.5 bn, from 215.2 to 213.7. The two figures are intentionally distinct, the limit case being the constraints-off reproduction of the reduced-form merit order and the unit-commitment case the constraints-on result, and the small spread between them is the effect of the commitment constraints.

What the extension does not do is rescue the capital-recovery verdict; at 160 hours the plant still recovers a small fraction of its capital. The 5 per cent operating-reserve standard, which the reduced-form merit order does not impose, implies a further about 59 GW of firm capacity the system would need to add to meet the reserve margin (Figure A.6c). This reserve adder is an adequacy signal distinct from, and additional to, the 278 GW peak-capacity headline. Read against an installed firm fleet of about 473 GW it is a roughly 13 per cent uplift, to an implied about 532 GW, the kind of capacity-market or firm-build requirement a system planner would have to procure beyond the energy-only peaker stack, and it leaves the 278 GW peaker headline itself untouched. The 278 GW peak-capacity headline, the per-country win counts and the qualitative verdict are all unchanged; the verdict now rests on a unit-commitment power model, and the hydrogen peaker still loses heavily in every scenario.

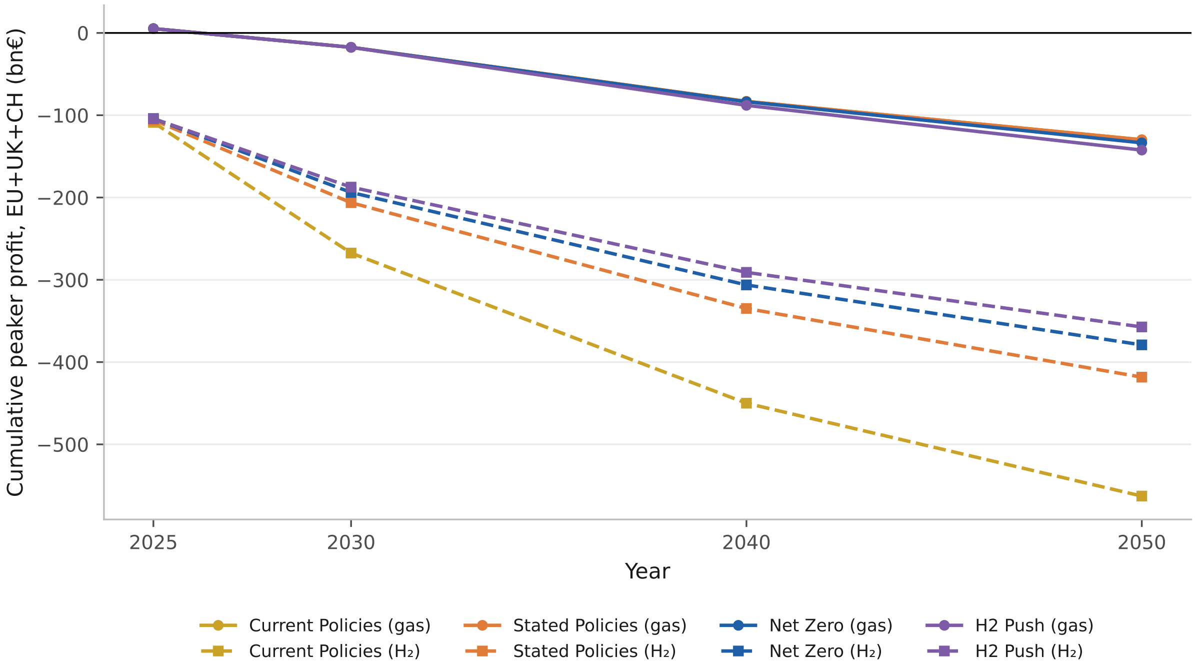
Figure A.6: What the unit-commitment upgrade changes (EU-27+UK+CH, 2050). Panels (a) and (b) compare the unit-commitment run against the same run with its commitment limits switched off, both on that layer's own 280.5 GW fleet, so the left bar's 138 full-load hours is not the 139 the reduced-form dispatch gives on its 277.7 GW fleet. Panel (c) gives the per-country operating-reserve shortfall, whose 59 GW total is the firm capacity the reserve standard would have the system add.



Source: Authors' contribution.

One mitigation deserves note. In combined-heat-and-power (CHP) mode the turbine's recovered exhaust heat (raising total efficiency to approximately 85 per cent) can be sold to district heating, and a winter peaker runs precisely when heat demand peaks. Crediting that heat, scaled by each country's district-heating share and valued at €50 to 100/MWh-heat, shrinks the loss only modestly (under H2 Push the gas fleet moves to approximately €125 to 134 bn and the hydrogen fleet to approximately €342 to 350 bn), because a *power* peaker runs too few hours for the heat by-product to be large, and most of Europe lacks the district-heating density to absorb it. The fuller cogeneration case, a *heat-led* unit running through the heating season, competes inside the district-heating merit order, where hydrogen ranks last (Appendix A.14).

Figure A.7: Cumulative peaker profit to 2050, hydrogen against gas (EU-27+UK+CH).



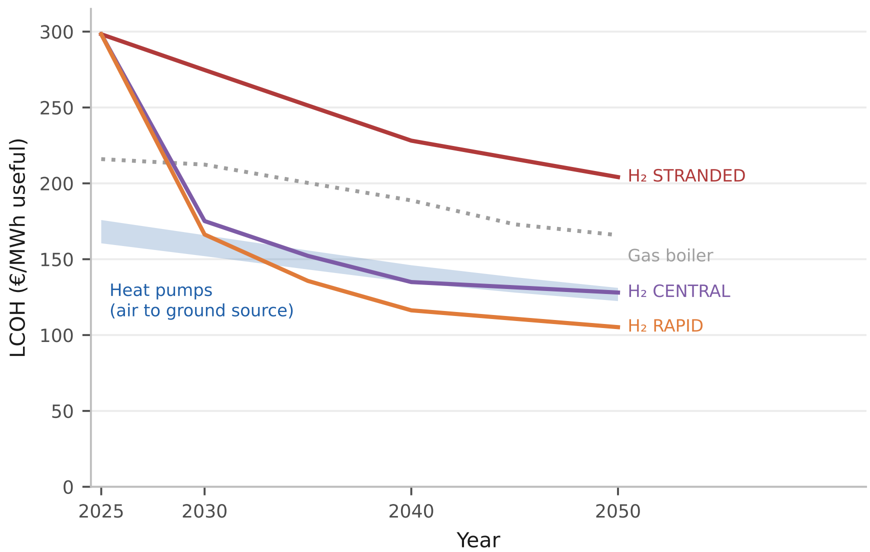
Source: Authors' contribution.

A.8 Switzerland: Hydrogen Imports and Buildings Decarbonisation (RQ2)

To address RQ2 we integrate Switzerland into the analysis *endogenously*, closing the three couplings that an earlier one-directional exogenous check left open. First, the Swiss delivered-hydrogen cost is made consistent with the EU supply side. It is the bottom-up EU supply cost plus a pipeline-transport adder for the import distance (a representative 600 km from the major European hydrogen producers), never read from a standalone OIES import-price trajectory, so the Swiss molecule price and the EU ensemble are mutually consistent. Second, a fixed-point feedback couples incremental Swiss heat-pump load to the EU winter electricity price and back. Extra Swiss demand raises the EU price, which raises the Swiss heat-pump cost, which feeds back to demand.

Because Switzerland is only about 2.27 per cent of the coupled demand and is interconnector-bound, this loop converges to a tolerance of €0.01/MWh in a single price update (the winter price moves once, from €240 to €246/MWh-e in 2050, and is then stationary; the iteration history is reported in the Swiss integration outputs in the supplementary material). Third, a seasonal-hydro credit recognises the Swiss reservoir fleet (about 8 GW of firm winter headroom) as cheap firm winter capacity. Together these make the Swiss case a member of the EU ensemble in place of a standalone sensitivity, and the result generalises the finding to a high-renewable, cavern-poor, import-reliant grid with its own seasonal hydro flexibility.

Figure A.8: Switzerland: hydrogen-boiler levelised cost against the heat-pump band. The three hydrogen lines are the stranded, central and rapid delivered-price paths, which Table 1 assigns to the four scenarios. The band spans the air-source and ground-source heat pump, and the dotted line is the gas boiler. Every series is Swiss, on the uncoupled arm that takes EU prices without feeding Swiss load back.



Source: Authors' contribution.

The Swiss case runs one way on both margins, and the reason is the same on each. On base load the heat pump wins comfortably against the incumbent fossil boiler and narrowly against hydrogen by 2050; at the winter peak hydrogen loses in every scenario once the Swiss reservoir fleet is credited as firm winter capacity, and wins under H2 Push on the uncoupled 29-country series that does not credit it.

The Swiss grid is predominantly hydro and nuclear, at 50 gCO_2/kWh in 2025 and on a trajectory to near-zero by 2050, and against gas the heat pump is already the clear 2025 choice, at €160.5/MWh (air-source) and €175.8/MWh (ground-source) against a Swiss gas-boiler cost of €216.0/MWh.

The ordering within the heat-pump category matters here as it does EU-wide, since the air-source unit, and not the ground-source one, is the cheaper of the two once equipment capital is charged against peak sizing, which annual output does not enter. By 2050 the air-source unit falls to €122.4/MWh and the ground-source unit to €131.1/MWh, and the gas boiler to €165.9/MWh, so the fossil comparison is not close. The base-load band quoted here and in the next paragraph is the uncoupled arm. We report it rather than the endogenous one because it is the narrower of the two and so the conservative choice for the conclusion the section draws. Coupling the Swiss load to the EU winter price moves the 2050 air-source unit to €120.9/MWh and the hydrogen boiler to €130.6/MWh, widening the heat pump's margin from €5.6 to €9.6/MWh. The peaking figures below are the endogenous arm's own, since that is the test the coupling was built for.

The hydrogen comparison is closer, and it turns on Swiss geology. On the EU-supply-consistent delivered cost, with the 600 km pipeline adder in place, the hydrogen-boiler LCOH in Switzerland falls from €298.3/MWh in 2025 to €128.0/MWh by 2050, which leaves the best Swiss heat pump €5.6/MWh the cheaper (Figure A.8). Switzerland has no domestic salt caverns, so a Swiss hydrogen heating load is held across the season at about €3.0/kg rather than the €1.5/kg the cavern markets pay, and that charge is what holds the ordering. An earlier draft reported the opposite, with hydrogen crossing below the Swiss heat-pump band under the central supply trajectory. That reversal was an artefact of omitting seasonal storage from the base-load comparison while charging it in the peak and dispatch tests, so that the same molecule was priced two different ways, and it does not survive the correction.

Closing the price feedback moves these levels only slightly, and in hydrogen's disfavour. The fixed point raises the Swiss winter electricity price by about €6/MWh-e, from €240 to €246/MWh-e in 2050, and crediting the reservoir fleet as firm winter capacity brings it down to about €141/MWh-e. The net effect on the levelised cost of heat is between €1 and 3/MWh. The heat-pump LCOH is about €1.5/MWh *cheaper* once the seasonal-hydro credit and the consistent EU price are applied, while the hydrogen route is about €2.5/MWh *dearer* once its cost is tied to the EU supply side in place of the more optimistic standalone trajectory. Both movements widen the Swiss heat pump's margin, so the feedback works with the ordering. It is nonetheless of the same order as the €5.6/MWh margin itself, which is why the Swiss base-load result is reported here as the weaker of the two Swiss arguments and the peaking verdict is left to carry the case.

What the endogenous treatment confirms most emphatically is the peaking verdict, which is where the Swiss argument against hydrogen in heating has its margin. Cheap firm Swiss hydro beats the imported hydrogen peaker in every one of the four scenarios. The Swiss hydrogen peaker prices at €134.8 to 235.9/MWh-heat against a heat-pump peak of €62.3 to 90.3/MWh-heat. That credit is what decides it. The uncoupled 29-country series, which prices the Swiss peak at €155/MWh-heat without the reservoir fleet, gives hydrogen the Swiss peak under H2 Push, so Switzerland sits inside the count of 16 that series produces and outside the Swiss verdict this subsection reports. Switzerland's lack of domestic salt-cavern storage leaves the seasonal-storage adder at about €100/MWh-heat, roughly double the cavern-endowed level, and that adder is what shuts the peaking niche of Section 4.6 in the Swiss winter peak in every scenario. The base-load and peaking results now share that mechanism. The same absent geology that prices the Swiss peaker out of the cold snap prices the Swiss seasonal store at twice the cavern rate on the base-load comparison, where it is spread thinly over the whole year's heat without concentrating on the peak, which is why it decides the base-load case by €5.6/MWh and the peaking case by an order of magnitude more.

The Swiss Energy Perspectives 2050+ [132] project heat pumps and direct electrification as the dominant Swiss building-decarbonisation pathway, and the model reaches that pathway on every margin it tests. Switzerland's geographic proximity to major EU hydrogen producers provides a genuine cost advantage through shorter pipeline distances [93], and on 2050 base-load cost that advantage is nearly, but not quite, sufficient to overcome the efficiency advantage of the heat pump on a near-zero-carbon grid. It comes as close as it does only because Swiss retail electricity carries a full levy stack while hydrogen is priced delivered and untaxed, which is the same fiscal asymmetry that qualifies the EU-wide result and is set out in Appendix D.

The endogenous integration therefore generalises the EU pattern to Switzerland, and the pattern it generalises is the storage one, a cold, high-price, import-connected but cavern-free system falling on the heat pump's side of both the base-load line and the peak line. The two judgement-call inputs, the 600 km representative pipeline distance and the 8 GW of firm reservoir headroom, are both conservative towards hydrogen, so relaxing either would narrow the Swiss base-load margin and never widen it, and that margin is already narrow; the peaking verdict, by contrast, has margin to spare against both (Appendix D).

Hydrogen's residual role in Switzerland lies outside the base-load comparison altogether, and two segments remain unquantified here: the hard-to-electrify historic building stock in city centres constrained by noise or space, and seasonal energy storage integrated with Swiss hydropower. Quantifying these niches requires city-level spatial modelling and sector-coupling analysis beyond the current scope.

A.9 Technology Mix

Technology shares evolve by interpolation from an observed 2025 baseline to the scenario-specific 2050 targets (Table 1), and the interpolation has three parts that we set out in full here, because the mix is described elsewhere in this paper as scenario-imposed and that is only two-thirds true.

The first part is the path. Each share moves as $s_t = s_{2025} + (s_{2050} - s_{2025})\,\tau^{p_i}$, where τ is the fraction of the 2025 to 2050 window elapsed and the shape exponent is drawn per technology each sample: $U(0.7, 2.0)$ for heat pumps, $U(0.8, 2.2)$ for district heat and

$U(1.0, 2.5)$ for hydrogen, with fossil and resistance reusing the district-heat draw and biomass the heat-pump draw. A draw can therefore front-load or back-load a transition without changing its endpoints, except for hydrogen, whose exponent is bounded below at one and so can only back-load. The second part is a cost-responsive reallocation among the non-fossil options. Their combined mass is redistributed by a softmax on levelised cost, $w_i \propto \exp(-c_i/T)$ with $T = 50$ €/MWh, so a cheaper technology takes a larger share without taking all of it. The third part is the weight given to that reallocation, which ramps linearly from zero in 2025 to 0.50 by 2050. The realised share is the weighted blend of the interpolated path and the cost-driven allocation.

A fourth part applies to hydrogen alone and belongs here because it bounds a headline. After the blend, hydrogen is capped at its own drawn 2050 target, any excess is routed to heat pumps up to their feasibility ceiling and then to district heat, and the mix is renormalised. The cap is one-directional, so cost can push hydrogen below its drawn target but never above it. It exists because hydrogen enters this model as an exogenous adoption setting for a peaking share rather than a base-load winner, and without it the softmax and the fossil-ban reallocation inflate hydrogen well past its design value: an earlier build realised about 13 per cent under Stated Policies against a 5 per cent setting. The statement that hydrogen reaches at most about 8.5 per cent of useful heat in any scenario we run is therefore a property of this cap and not an independent result of the cost mechanism.

Three further consequences follow and are load-bearing elsewhere. The mix is not purely a scenario input, since by 2050 half the non-fossil allocation is decided by cost. Nor is it purely cost-driven, since the fossil share is governed by the policy-ambition lever throughout and the cost-responsive weight is only a tenth at 2030, which is why the one-at-a-time tornado of Appendix C.1 returns such small numbers at that date. And the softmax temperature is a judgement. We chose it so that a cheaper technology wins moderately rather than absolutely, and we did not estimate it from market data, which is the largest unestimated parameter in the demand-side layer. Replacing this heuristic with a discrete-choice allocation estimated on observed switching is the natural extension.

The 2025 baseline shares are EU averages drawn from EHPA heat-pump market data and Euroheat & Power district-heating statistics: heat pumps approximately 18 %, district heat approximately 12 %, biomass approximately 10 %, and fossil fuels (gas and oil) approximately 60 %. These EU-average shares are stated on a different basis from the per-country useful-heat shares reported in Section 4.2 (heat pumps 8 per cent, gas 44 per cent, oil 14 per cent). The figures here are EU-aggregate EHPA and Euroheat & Power statistics; the results section reports the per-country useful-heat distribution actually used in the model, so the two are not contradictory. For each NUTS3 region and building type, the 2050 heat-pump target is modulated by a feasibility score:

$$\mathrm{HP}_{\mathrm{target}}(r, b) = s_{\mathrm{HP}} \left(0.4 + 0.6\, f_{\mathrm{HP}}(b)\right), \tag{12}$$

where s_{HP} is the scenario heat-pump target and $f_{\mathrm{HP}}(b) \in [0, 1]$ is a building-type feasibility factor (single-family 0.9, multi-family or high-rise 0.5, other 0.6); the intercept of 0.4 ensures that even a zero-feasibility region still reaches 40 % of the scenario target. District heat is weighted by an analogous feasibility factor (single-family 0.3, multi-family or high-rise 0.8), reflecting the economics of network heat at different building densities.

Feasibility-scoring rationale. The feasibility factors are anchored to the central tendency of the heat-pump retrofit and district-heating literature, synthesised across several sources, and they are flagged as a key modelling assumption in the Limitations section. Table A.4 sets the adopted central values against the indicative feasibility ranges reported in that literature, by building type; the values used here sit inside the cited ranges. The ordering encodes two well-documented patterns. First, air-source heat pumps retrofit most readily into detached single-family homes, which have the facade access, emitter sizing headroom and outdoor-unit space that low-temperature operation needs, and least readily into dense apartment blocks, where shared distribution, planning consent and outdoor-unit siting raise the barrier; this is the gradient reported across the heat-pump retrofit and barriers literature [55, 58], hence the descending single-family > other > multi-family ordering of f_{HP}.

Second, district heat is economic where linear heat density is high, that is in compact multi-family stock and uneconomic in dispersed single-family stock [61], which inverts the ordering for the district-heat factor. The non-zero intercept of 0.4 reflects that no region is structurally barred from heat pumps. Even the least favourable stock can reach a substantial share through ground-source units, communal and high-temperature heat pumps and new build. Because these coefficients shape the 2050 heat-pump and district-heat shares, they are treated as a structural assumption. The technology-share targets are sampled in the Monte Carlo (Appendix A.3) and the feasibility design is revisited in the Limitations section (Appendix D). The qualitative conclusions of the paper turn on the merit-order economics rather than on the precise feasibility split, since hydrogen's heating share is derived from the peak dispatch (Appendix A.2) and is not set by these factors.

Table A.4: Heat-pump and district-heat feasibility factors by building type.

Building type	Adopted f_{HP}	Indicative literature range	Basis
Single-family (SFH)	0.9	0.7 to 0.95	Gibb et al. [55], International Renewable Energy Agency [158]
Other / mixed	0.6	0.4 to 0.7	Corbett et al. [58], Joint Research Centre [159]
Multi-family / high-rise	0.5	0.3 to 0.6	International Renewable Energy Agency [158], Joint Research Centre [159]
	Adopted f_{DH}	Indicative literature range	Basis
Single-family (SFH)	0.3	0.1 to 0.4	Persson et al. [61]
Multi-family / high-rise	0.8	0.6 to 0.9	Persson et al. [61]

The ranges are indicative bands synthesised from the cited heat-pump retrofit, adoption-barrier and linear-heat-density literature; they bracket the structural feasibility of each technology in each building type. The heat-pump gradient (single-family > other > multi-family) follows the retrofit and barriers evidence [55, 58]; the district-heat gradient inverts it because network heat is economic only at high linear heat density [61]. Because the central values are structural, the technology-share targets they shape are sampled in the Monte Carlo (Appendix A.3) and the design is revisited in the Limitations section (Appendix D).

Source: Authors' contribution.

A.10 The Emissions Layer

The emissions inventory runs after the technology split, on the same country-by-year final-energy quantities the cost layer prices. Each Monte Carlo draw carries its own inventory, so the fan of Section 4.4 is the spread of the sampled technology mix, with no separate emissions uncertainty laid on top of it. The scope is direct, scope-1 combustion at the building plus the indirect content of the two delivered carriers, and it excludes embodied emissions in equipment, network construction and hydrogen production upstream of the delivered price.

Combustion fuels carry fixed factors on a lower-heating-value basis of final energy, natural gas at 0.202 and heating oil at 0.265 tCO_2/MWh, taken from Annex II of the Sixth Assessment Report's Working Group III volume and cross-checked against the UK government conversion factors. Biomass and hydrogen are entered at zero. Zero is the European accounting convention for a renewable fuel under the Renewable Energy Directive; it is not a physical claim about the molecule, and at 17 per cent of the 2025 mix the biomass convention is material to the level of every pathway. It flatters all four equally, because the biomass share is a scenario input common to them, so it moves the levels without moving the ranking or the gap between Current Policies and Net Zero that Section 4.4 reports.

Electricity and district heat carry country-by-year intensities, because the whole electrification case turns on the grid cleaning up. Both start from observed 2025 country values [101] and follow declining paths to 2050 that each scenario scales by its grid-decarbonisation lever. The electricity paths run, for example, from 600 to 30 gCO_2/kWh in Poland, 350 to 15 in Germany, 80 to 8 in France and 50 to 5 in Switzerland; district heat follows a parallel and generally lower country path [141]. Values between the milestone years are linearly interpolated. Every country in the study area carries its own series, so the implementation's fallback to the German path is unreachable here.

Two properties of that construction bound what the emissions results can carry, and both are set out again with their direction in Appendix D. The post-2025 intensity paths are scenario-consistent trajectories calibrated to published 2025 levels and to European decarbonisation targets, not published country projections, so the absolute 2050 levels inherit that calibration while the ranking across scenarios does not, since every scenario reads the same underlying path through its own lever. And the intensity is an annual average applied to all heating electricity, while the cost layer prices the heat pump's winter draw at an hourly scarcity price. Heat-pump peak electricity is both dearer and dirtier than the annual mean, so the inventory carries a systematic bias in the heat pump's favour that no hourly ledger in this model can quantify.

A.11 Techno-Economic Assumptions

Table A.5 summarises the key techno-economic parameters for each technology. All monetary values are in EUR_{2024}.

Table A.5: Techno-economic parameters by heating technology (EUR_{2024}).

Technology	CAPEX 2025 (EUR/kW)	CAPEX 2050 (EUR/kW)	Efficiency / SCOP 2025 / 2050	Fuel
Gas boiler	420	400	0.92 / 0.94	Natural gas
Oil boiler	450	430	0.90 / 0.90	Oil
Biomass boiler	950	850	0.88 / 0.90	Biomass
Resistance heater	200	180	0.98 / 0.99	Electricity
HP (air-source)	1,200	800	3.3 / 3.5	Electricity
HP (ground-source)	2,000	1,400	3.8 / 4.3	Electricity
District heat	500	450	0.95 / 0.97	District heat
H_2 boiler	600	550	0.90 / 0.92	Hydrogen

CAPEX and efficiency values are anchored to the Danish Energy Agency Technology Catalogue installed-cost figures [147], cross-checked against the JRC Technology Data repository [159, 160], the IEA *Future of Heat Pumps* [3] and IRENA [161]; the provisional first-pass values were revised down to these DEA-anchored figures in a source audit (Appendix B). The Danish Energy Agency catalogue is the primary anchor for every technology because it reports installed (turnkey) costs in place of equipment-only costs on a consistent basis; the other sources are used as cross-checks and to set the heat-pump and hydrogen values where the catalogue is less granular. The two tabulated years are the endpoints of the cost path. Intermediate-year CAPEX is obtained by linear interpolation between the 2025 and 2050 values, so the table fully specifies the trajectory, and no separate learning-curve exponent is applied. Heat-pump performance is the seasonal coefficient of performance (SCOP); the values shown are EU averages following EHPA data [162] and EU Regulation 813/2013, with the air-source SCOP set to the EHPA/Eurovent field value (3.3). The single SCOP figure is an EU-average input only. Because the seasonal coefficient of performance falls in colder climates and at higher flow temperatures, the model applies a temperature-dependent derate at the country level through the cold-snap performance COP_c^{cold} and the hourly dispatch (Appendix A.2). CAPEX is additionally scaled by a country-specific construction labour-cost multiplier derived from Eurostat `lc_lci_lev` (2024), and is country- and year-resolved in the model.

Source: Authors' contribution.

A.12 Fuel Price Assumptions

Residential end-user energy prices for the 2025 baseline are taken from Eurostat: natural gas from dataset `nrg_pc_202` and electricity from `nrg_pc_204`, both for the first half of 2025. Heating-oil, biomass and district-heat prices are per country as well, assembled from the EC Weekly Oil Bulletin and GlobalPetrolPrices, from ENplus, proPellets and national pellet series, and from Euroheat and national heat regulators; their provenance is lighter than the two Eurostat carriers and we treat it as a limitation. The green-hydrogen price follows the European Hydrogen Observatory (2024). Table A.6 gives the 2025 levels averaged over the 29 markets and the multipliers applied to reach 2050. Prices vary substantially between countries. Residential gas, for example, ranges from about €31/MWh in Hungary to €213/MWh in Sweden.

Table A.6: Energy-carrier prices: 2025 baseline and 2050 multiplier.

Carrier	2025, mean of the 29 markets (EUR/MWh)	2050 multiplier	Source
Natural gas	103	0.55	Eurostat `nrg_pc_202` (H1 2025)
Electricity	257	0.80	Eurostat `nrg_pc_204` (H1 2025)
Heating oil	154	0.55	EC Weekly Oil Bulletin, GlobalPetrolPrices
Biomass	69	1.05	ENplus, proPellets, national series
District heat	88	0.80	Euroheat, national regulators
Hydrogen	200	see text	European Hydrogen Observatory (2024)

Every carrier is priced per country, and the column above is the unweighted mean of those 29 values, so it reconciles with the per-country prices the model runs on. It is not the Eurostat headline aggregate, which is weighted by consumption and reads €114.3/MWh for gas and €287.2 for electricity; those are the figures the 2025 parity discussion of Section 4.3 quotes, and the difference is the weighting; the vintage is the same. The hydrogen price follows scenario-specific trajectories. The central trajectory falls from €200/MWh in 2025 to €90/MWh in 2030 and €50/MWh by 2050. For Switzerland, hydrogen import prices are treated as a separate key sensitivity variable (Appendix A.3). The oil, biomass and district-heat series carry lighter provenance than the two Eurostat carriers, and five district-heat entries (Spain, Portugal, Ireland, Cyprus, Malta) are neutral placeholders for markets whose district-heat share is near zero.

Source: Authors' contribution.

A.13 Hydrogen Supply Cost and the Last-Mile Delivery Networks

Hydrogen requires two country-specific treatments beyond the hub price trajectory: where the molecule comes from (the delivered commodity cost) and how it reaches the dwelling (the distribution network). Electric heating requires the symmetric treatment of the second, since its winter peak also drives incremental last-mile capital, and the two adders defined below are both active in the headline levelised costs so that each carrier is charged for its own delivery network.

Per-country delivered-cost multiplier. The hub trajectory is scaled to each country by a delivered-cost multiplier m_c, so the residential hydrogen price is $p_{c,t} = p_t^{\mathrm{hub}} \cdot m_c$. Rather than a uniform or heuristic assumption, m_c is derived from a bottom-up delivered-cost model that, for each country, levelises every supply route, namely domestic *green* electrolysis, *pink* (nuclear) electrolysis, *blue* (gas reforming with CCS), pipeline *import*, and seaborne *ammonia* import, and takes the cheapest. Each route depends on the country's cost of capital (WACC), construction labour cost, electricity or feedstock price, renewable capacity factor, water scarcity, pipeline connectivity, and nuclear or CO_2-storage availability, so the multiplier captures cost drivers beyond renewable resource quality alone.

The technology constants are calibrated to verified 2050 benchmarks: electrolyser CAPEX (approximately €300/kW) and green LCOH from IRENA [163]; blue (approximately €2.7/kg) from IEAGHG [88]; pipeline transport (approximately €0.16/kg per 1,000 km) from the European Hydrogen Backbone [91]; and delivered or import costs from OIES [93, 164]. The multiplier is the country's delivered cost divided by the EU-median delivered cost, so the median country pays the hub price ($m_c = 1.0$). The resulting m_c ranges from 0.77 (Ireland: strong wind, low capital cost) to 1.33 (Cyprus: an isolated island with no pipeline); pink and blue are not the cheapest route in 2050. Each multiplier is held as a low/central/high band, with the sensitivity of the central findings reported in Appendix C.6 and the full derivation and sources in the supplementary material.

Hydrogen distribution cost. Because a methane network cannot carry 100% hydrogen unmodified, the hydrogen-boiler LCOH carries a distribution adder a_c ($\mathrm{EUR/MWh_{useful}}$) [90, 157]. A per-dwelling capital cost K_c is annualised over a 40-year network life at the country discount rate r_c and spread over the annual useful heat per dwelling $Q \approx 13$ MWh:

$$a_c = \frac{\mathrm{CRF}(r_c,\, 40) \cdot K_c}{Q}, \tag{13}$$

where CRF is the capital-recovery factor. The per-connection cost is genuinely uncertain, so K_c is carried as a low/central/high range (conversion €1,000/1,500/3,000 per dwelling; new-build €3,000/5,000/8,000) and across three network modes, namely *retrofit* conversion of the existing grid, *new-build* of a dedicated network, and the coverage-weighted *blend* of the two. The adder is *on by default*, in blend mode and at the central cost bound, so every headline levelised cost reported in this paper already carries it; free reuse, retrofit and new-build are reported as a sweep around that default (Appendix A.6).

Seasonal storage on the base-load hydrogen load. A hydrogen heating load has to be held from the season in which the molecule is cheapest to make to the season in which it is burned, and the base-load levelised cost charges it for doing so. On the model's own monthly shape, space heating draws a median 0.748 of its annual energy in the October-to-March half of the year, within a range of 0.710 to 0.787 across the 29 countries, whereas electrolytic production is far flatter, so the difference between the two profiles is the part of the load that must be time-shifted.

We charge the seasonal store on that difference, at each country's own winter share less one half, a fraction whose median across the study area is 0.25. The store itself is priced at the same rates the peak and dispatch tests use, €1.5/kg where domestic salt caverns are available and €3.0/kg where they are not, at about one cycle a year [36]. The base-load charge divides those rates by the year-specific boiler efficiency, 0.92 at 2050, where the dispatch tests divide by the 0.90 of their own peaking convention, so per unit of useful heat this is worth €12.1/MWh in cavern-endowed Germany and €28.4/MWh in cavern-free Spain, and it is that difference, and not the delivered price of the molecule, that separates the countries in which hydrogen holds base load from those in which it does not.

Earlier drafts charged this term in the building-peak and power-dispatch tests but not in the base-load levelised cost, so the same molecule was priced two different ways and the base-load comparison omitted a cost the dispatch comparison imposed. Charging it on both is the correction, and the choice of fraction is deliberately conservative towards hydrogen in two respects. It credits hydrogen with meeting the whole non-seasonal half of its load from contemporaneous production, which no real supply chain would achieve exactly, and it ignores the within-winter swing, which also has to be stored somewhere. A fuller accounting would only charge more. The sensitivity is correspondingly one-sided, and because the count is close to linear in the charged fraction we sweep it continuously. Charging nothing, as earlier drafts did, leaves the best heat pump cheaper in 12 of 29 countries at a median gap of −€5.1/MWh.

A quarter of the load-derived fraction gives 15 countries at €0.0/MWh, half gives 19 at +€5.2, three-quarters gives 21 at +€10.3, and the fraction itself, which is the basis adopted here, gives 22 at +€15.4. Above the adopted basis the count keeps climbing, to 25 at one and a half times and 26 at twice, and charging about four times the load-derived fraction leaves the heat pump cheaper in all 29 at +€81.1/MWh, with hydrogen winning nowhere. That last arm is a multiple of each country's own fraction and not a flat full-storage charge, so it bills the southern markets slightly more than their whole load and the Nordics slightly less; charging the full adder everywhere instead gives +€87.8/MWh and the same count of 29. The adopted basis therefore sits near the low end of the range and well below what a fuller accounting would charge, and the defensible summary of the base-load verdict is 22 countries on that basis, 15 to 26 across the fractions we regard as plausible.

Electricity distribution reinforcement. Electric heating carries the symmetric charge. Mass heat-pump deployment raises the coincident winter peak that the low- and medium-voltage network must carry, and we treat that incremental reinforcement as not already recovered in today's retail tariff. We therefore add a reinforcement adder a_c^{el} (EUR/MWh$_{\mathrm{useful}}$) to the levelised cost of every electric heating technology. The added peak per dwelling P_c^{dw} is the annual useful heat divided by the country's climate-scaled full-load hours h_c and by the cold-snap coefficient of performance a cold-snap coefficient of performance $\mathrm{COP}^{\mathrm{el}} = 2.3$, and it is multiplied by a coincidence factor $\kappa = 0.5$ so that dwellings are not assumed to draw their maxima at the same instant:

$$a_c^{\mathrm{el}} = \frac{\mathrm{CRF}(r_c,\, 40) \cdot u^{\mathrm{el}} \cdot \kappa\, P_c^{\mathrm{dw}}}{Q}, \qquad P_c^{\mathrm{dw}} = \frac{Q}{h_c\, \mathrm{COP}^{\mathrm{el}}}, \tag{14}$$

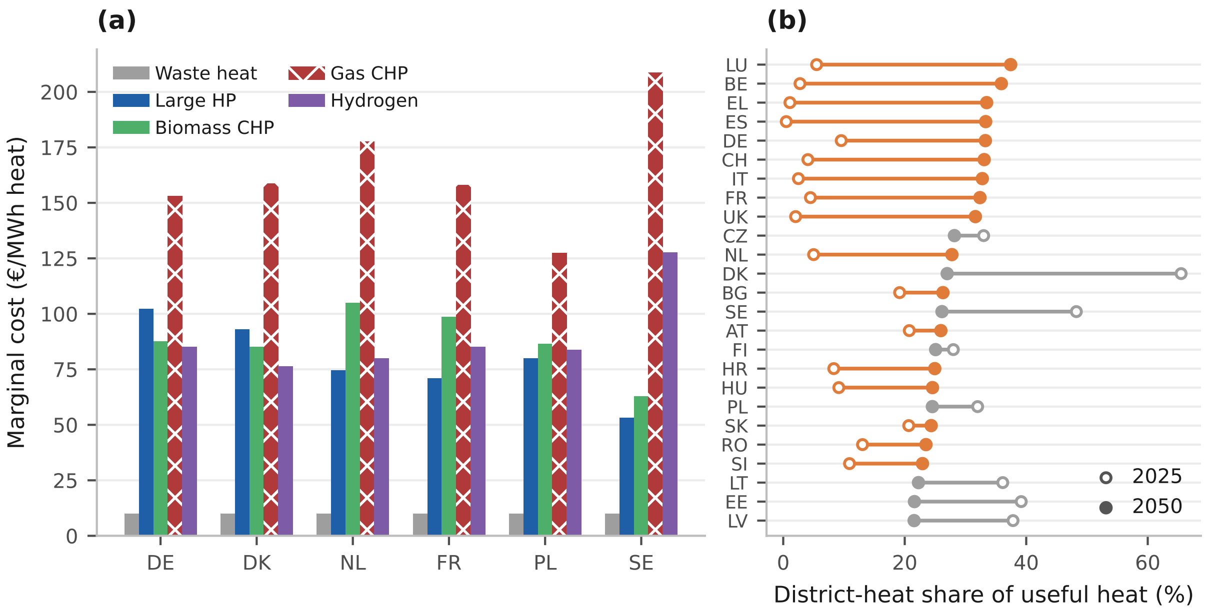
with $u^{\mathrm{el}} \in \{400, 800, 1{,}600\}$ EUR per added kW of peak (low, central and high), and with the same 40-year network life and country discount rate as the hydrogen adder, so the two carriers are annualised on identical terms. Across the band the adder is worth €1.1 to 38.2/MWh, and it is highest in peaky, mild-climate systems where a small annual heat demand is delivered through a short, sharp season. Like the hydrogen adder it is *on by default* at the central bound. Charging each carrier its own last-mile network removes the asymmetry earlier drafts carried, in which hydrogen paid for its pipes only inside a dedicated scenario while electric heating never paid for its wires. It also introduces a residual risk, that retail electricity tariffs already recover part of today's network cost so that the incremental charge may double-count, which is stated in Appendix D.

A.14 District Heating: Source-Switching Merit Order

The peaking test extends to a third scale, the district-heating peak, where the same merit-order logic governs hydrogen inside a single operator's stack of sources dispatched into the same pipes. Ranked by marginal cost, the 2050 stack runs: waste and excess heat (must-run, approximately €10/MWh), large heat pumps, biomass CHP, gas CHP (carbon-priced), and hydrogen. Hydrogen's rank in the stack depends on its delivered price. With the expensive hydrogen of the low-carbon scenarios it is the dearest source and catches only the residual peak; under the cheap derived hydrogen of H2 Push it ranks just below gas CHP. It undercuts gas CHP on operating cost in 0, 0, 7 and 7 of 29 countries across the four scenarios on the symmetric basis, and in 0, 5, 11 and 20 on the residential-tariff basis the next paragraph sets out; gas CHP nonetheless remains the marginal source that sets the district-heat peak in 20 of 29 (hydrogen in 9), since hydrogen is dispatched below it, so 20 is the arena's structural ceiling on either basis. As at the building level, it is the carbon price on the gas alternative, reinforced by cheap hydrogen, that opens the door, and the addressable share is the peak sliver above the waste/heat-pump/biomass base load, an EU-weighted 0 to 12 per cent of district heat (Figure A.9).

One price basis carries this arena and should be stated. The stack charges gas at the Eurostat residential retail tariff including taxes, a study-area median of €55/MWh in 2050, because that is the series the whole cost layer runs on; a district-heat operator buys wholesale. Repricing the gas unit at the €30/MWh hub price the power block already uses takes the four counts from 0, 5, 11 and 20 to 0, 0, 7 and 7. The test compares hydrogen against gas alone, so the electricity price the large heat pump pays does not enter it. We headline the 7, because it is what a wholesale-buying operator would see and it is the figure a district-heat investor should use, and report the 20 alongside it as what the residential basis returns.

Figure A.9: District heating. Panel (a) is the merit-order stack under H2 Push, the scenario most favourable to hydrogen; panel (b) is the Net Zero build-out. The two panels are on different scenarios deliberately, so the stack is read at hydrogen's best and the expansion at the deepest decarbonisation.

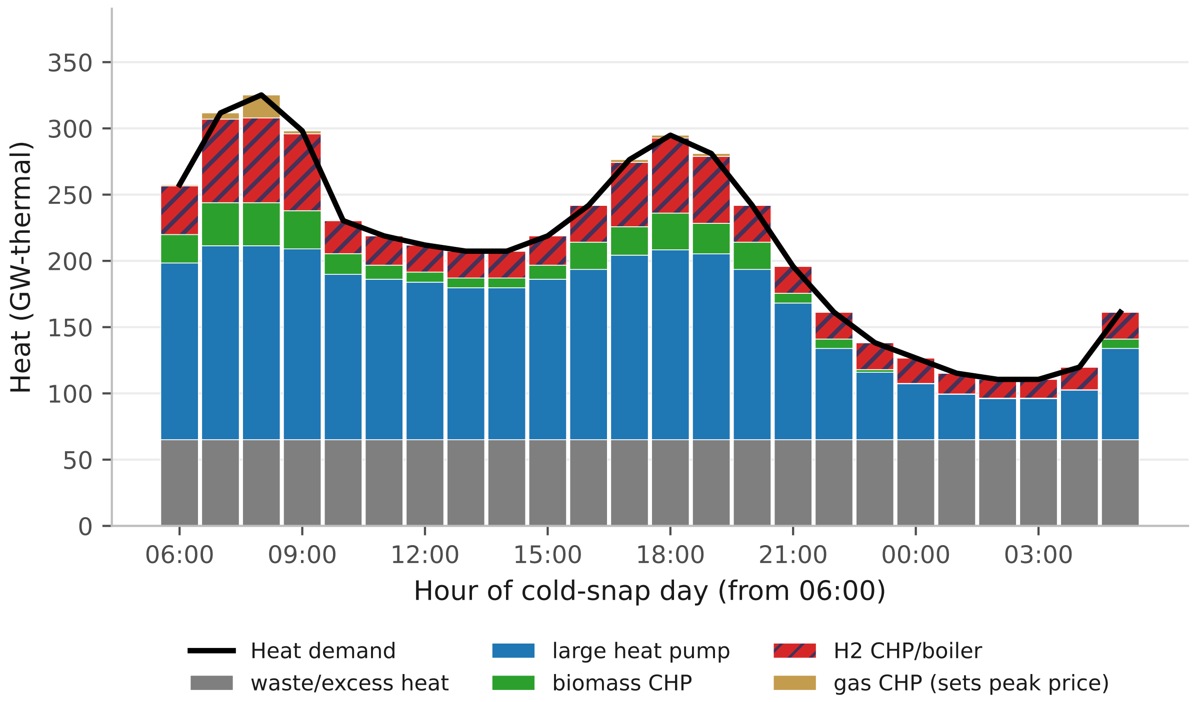


Source: Authors' contribution.

Hourly heat dispatch. Resolving the same stack to the hour (Figure A.10) confirms the picture and adds the heat-led case. On the cold-snap day, the decarbonised base of waste heat, large heat pumps and biomass carries the bulk, and hydrogen fills the morning and evening peaks, the heat-side mirror of the electricity peaker. This is hydrogen's most favourable framing; as a *heat-led* cogeneration unit it can run many winter hours, against the about 139 of the power peaker, so its capital is far better utilised.

Even so, in the H2-favourable scenario it reaches only 9.3 per cent of EU district-heat energy as an aggregate, hydrogen's total over the study area's total (14.3 per cent on the single representative cold-snap day of Figure A.10, which loads the high-share markets), approximately 29 per cent in the three salt-cavern markets where its delivered-plus-storage cost undercuts gas CHP (Germany, the United Kingdom and Denmark), but a marginal 0.33 per cent across the other 26 on a demand-weighted basis, and at most 0.5 per cent in any single one of them, including cavern-endowed Poland, where cheaper sources fill the district-heat peak. The dispatch runs on the Stated Policies district-heat footprint, in which district heating is 18.6 per cent of residential heat, so this best case is 1.7 per cent of total EU heat. Both figures are taken on that one footprint. The H2 Push district-heat share is higher, at 22.9 per cent, and combining it with a dispatch solved on the Stated Policies footprint would mix two scenarios. Cogeneration is therefore hydrogen's strongest niche in heating, but a narrow one, and one realised only where cheap hydrogen meets salt-cavern storage.

Figure A.10: Hourly district-heat dispatch on the cold-snap day (EU-27+UK+CH, 2050). The marginal costs that order the stack are H2 Push; the district-heat footprint the stack is scaled to is Stated Policies, so the two are on different scenarios and the shares in the paragraph above are read on the second. Waste and excess heat is a constant band because it is a by-product flow that does not follow the load. Gas cogeneration appears in seven hours, four across the morning peak and three across the evening one, and it sets the price in each of them.

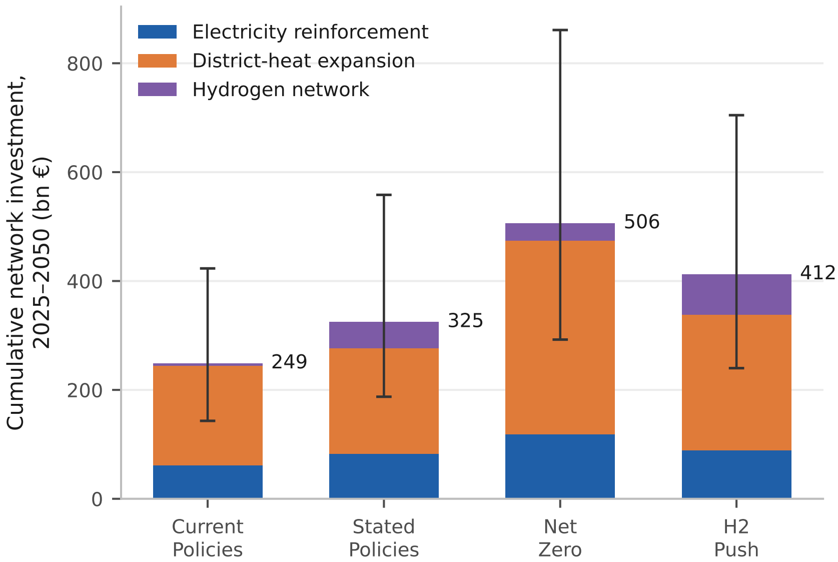


Source: Authors' contribution.

A.15 The Network-Infrastructure Bill of Each Pathway

Having tested hydrogen on the base-load, peaking and investment margins, we close with the cross-pathway comparison those per-technology tests omit, namely the network capital each whole pathway commits. Each pathway carries a different wires-and-pipes bill that per-MWh cost comparisons obscure. We aggregate three network capital requirements from the model's own technology shifts: electricity distribution reinforcement (the added coincident winter heat-pump peak, at €400/800/1,600 per added kW of peak), district-heat network expansion (new connections at €3,000/5,000/8,000 per dwelling), and the hydrogen distribution network (the coverage-weighted conversion and new-build blend of Appendix A.6).

Figure A.11: The cumulative 2025 to 2050 network-infrastructure bill of each pathway, all 29 markets. The number beside each bar is the central estimate; the whiskers are the low and high unit-cost cases. Every pathway is dominated by district-heat expansion and grid reinforcement, which a hydrogen network does not drive.



Source: Authors' contribution.

The cumulative 2025 to 2050 bills (central [low to high]), costed across all 29 markets, are €249 bn [143 to 423] under Current Policies, €325 bn [187 to 558] under Stated Policies, €506 bn [292 to 861] under Net Zero, and €412 bn [240 to 705] under H2 Push. Two findings stand out. First, *the deep pathway's bill is dominated by district heating*, at €356 bn of Net Zero's €506 bn against €119 bn of electricity reinforcement, so deep decarbonisation is a district-heating build-out as much as a heat-pump roll-out. Second, *hydrogen does not buy its way out of grid costs*. H2 Push pays €74 bn for a hydrogen network and still requires €89 bn of electricity reinforcement, because hydrogen serves only the peak sliver while heat pumps carry the bulk of the load, so the hydrogen network bill is additive to the electrification bill and substitutes for no part of it. The same pattern holds in reverse across the sweep, since even Current Policies, with the smallest hydrogen network at €4bn, still carries €62 bn of reinforcement and €183 bn of district-heat expansion.

A complementary robustness check charges the heat pump the *high* reinforcement bound in the levelised-cost comparison while charging hydrogen nothing at all for its network, which is the most conservative treatment of electric heating we can construct. Under it heat pumps are cheaper than hydrogen in 10 of 29 countries and cheaper than gas in 21 of 29, and the mean heat-pump-to-hydrogen headroom is −€6.2/MWh, meaning that hydrogen leads on average by that margin (Appendix C). Restoring hydrogen's own network charge, so that both carriers pay, returns the count to 22 of 29. This is a materially weaker robustness statement than earlier drafts carried, and it is stated as such. The base-load ranking does depend on how the two last-mile networks are charged.

B Cost-Optimal Pathway and Country-Specific Heating Mixes

This section derives the 29-country least-cost residential heating pathway through three steps: a source audit of the techno-economic inputs, a country-specific reconstruction of the 2025 residential heating mix, and a least-cost optimisation that *derives* the per-country technology mix from source-audited levelised costs instead of imposing it. The headline result is that cost-optimal decarbonisation rests on district heat and heat pumps at an approximately 53 to 45 per cent split, and that the depth of the scope-1 emissions cap barely moves the system cost, with 0.05 per cent separating the −75, −90 and −100 per cent targets (Figure B.1). The cap is nonetheless not slack everywhere, and in the handful of countries where it binds it does so at a substantial shadow price. Where this endogenous pathway is read alongside the four exogenous-share scenarios, and what the comparison implies for RQ1, is taken up in Section 4.2.5; here the task is to establish the least-cost pathway itself and to test its robustness.

B.1 Source-Audited Techno-Economics

Every capital-cost input was re-audited against three public catalogues: the Danish Energy Agency technology data catalogue [147], the International Energy Agency heat-pump cost work [3] and European Heat Pump Association market data [156]. The audit selects, for each technology, the value supported by these sources in preference to the prior model input, and adopts the Danish Energy Agency central figure as the default because it is the most complete and conservative of the three for the combustion and resistance technologies; the IEA and EHPA sources are heat-pump-specific and corroborate the heat-pump inputs, not the boiler inputs. The audit made two material corrections.

First, combustion-boiler capital costs were found to be overestimated in the prior model, at approximately €1,000 to 1,600/kW, and were revised down to Danish Energy Agency central values (€420/kW gas, €450/kW oil, €950/kW biomass and €200/kW resistance). Second, the hydrogen-boiler capital cost carried in the prior model was unsupported by any of the three catalogues, none of which reports a residential hydrogen-boiler input; we therefore adopt the Danish Energy Agency central value of €600/kW, the most conservative public figure for this nascent technology. Alongside the capital costs, the air-source heat-pump seasonal coefficient of performance was raised from the regulatory average-climate floor of 3.0 to the field value of 3.3, and residential gas and electricity prices were re-confirmed against Eurostat [131]. Table B.1 itemises the audit by technology.

Table B.1: Source-audited technology capital costs.

Technology	Prior model	DEA 2023	IEA 2022	EHPA 2024	Chosen	Basis
Gas boiler	€1,000 to 1,600/kW	€420/kW	n.r.	n.r.	€420/kW	Prior value unsupported; DEA central adopted.
Oil boiler	overestimated	€450/kW	n.r.	n.r.	€450/kW	DEA central; mature technology.
Biomass boiler	overestimated	€950/kW	n.r.	n.r.	€950/kW	DEA central.
Resistance	n.r.	€200/kW	n.r.	n.r.	€200/kW	DEA central; lowest-capital option.
Hydrogen boiler	unsupported	€600/kW	n.r.	n.r.	€600/kW	No catalogue input; DEA central, conservative for a nascent technology.
Air-source HP	SCOP 3.0 (floor)	corroborates	corrob.	corrob.	SCOP 3.3	Field value replaces regulatory floor.

n.r. = not reported by that source. Source: Authors' contribution.

The same audit benchmarks the 2050 scenario technology shares against the EU Reference Scenario 2024 [49], REPowerEU [11], Gas for Climate [165] and the hydrogen-in-buildings consensus [32, 166], and is documented in full in the supplementary material. One material reframing follows. The high hydrogen share assumed under the H2 Push scenario is retained only as a deliberate high-side stress test, since the mainstream literature places the residential hydrogen share below 1 to 2 per cent.

B.2 Country-Specific 2025 Heating Mix

Each country's 2025 residential heating mix is estimated over the eight model technologies from two independent bases and cross-checked. The first is an *energy* basis, the share of household final energy delivered by each fuel, drawn from Eurostat disaggregated household final energy by fuel [167]. The second is a *dwelling* basis, the share of dwellings equipped with each technology, drawn from national statistics, Odyssee-Mure [168] and the European Heat Pump Association. The two bases carry different units, energy delivered against dwellings equipped, and so bound the quantity the model actually needs from opposite sides.

The model allocates useful heat and not delivered energy, and the two bases bracket that quantity. The energy basis over-weights inefficient technologies, such as wood, oil and coal, which burn more delivered energy per unit of useful heat; the dwelling basis over-weights technologies that deliver more useful heat per unit of input, such as heat pumps and district heat. The per-country starting mix is therefore taken as the mean of the two. As an illustrative example, a country whose energy basis reads 65 per cent gas while its dwelling basis reads 45 per cent gas would enter the model at the 55 per cent mean; the true useful-heat share lies between the two bounds by construction.

The divergence between the two bases at country level is retained as a validation diagnostic and reported per country in the supplementary material; it is widest where efficient and inefficient technologies coexist in similar shares and narrowest in single-technology stocks. The resulting mixes are strongly heterogeneous. The United Kingdom and the Netherlands begin at approximately 80 per cent gas; Sweden and Denmark are district-heat dominated, at approximately 48 per cent and 66 per cent respectively; the Baltic states rely on a combination of biomass and district heat; and Cyprus and Malta, which have no gas grid, heat largely with reversible air-to-air units.

This per-country reconstruction supersedes the single EU-average starting mix used in earlier runs, and the change is methodologically material because the optimisation operates on each country's own prices, feasibility limits and baseline. A single EU-average mix would erase exactly the cross-country heterogeneity, from gas-dominated to district-heat-dominated stocks, that determines where each technology is least-cost. The per-country mix is used both within the least-cost program and as the 2025 base mix from which the four multi-lever scenarios (Section 3) interpolate.

B.3 The Least-Cost Optimisation

Whereas the four core scenarios impose exogenous 2050 technology shares, the least-cost program solves for the cheapest mix. It is a linear program that minimises the discounted, period-weighted system cost of the residential heating pathway from 2025 to 2050, where w_y is a period weight, $(1 + r)^{-(y-2025)}$ discounts future years at rate r, $\mathrm{LCOH}_{c,t,y}$ is the levelised cost of heat for technology t in country c and year y, $D_{c,y}$ is the useful-heat demand and $x_{c,t,y}$ is the technology share. The objective is

$$\min \sum_{c,t,y} w_y \,(1+r)^{-(y-2025)}\, \mathrm{LCOH}_{c,t,y}\, D_{c,y}\, x_{c,t,y}, \tag{15}$$

minimised over technology shares $x_{c,t,y}$ for country c, technology t and milestone year y, subject to a set of constraints. A demand balance requires the shares in each country and year to serve the full useful-heat demand,

$$\sum_t x_{c,t,y} = 1, \qquad x_{c,t,y} \geq 0, \tag{16}$$

with a 2025 anchor fixing $x_{c,t,2025}$ to the country's observed mix. Feasibility ceilings cap the deployable share of district heat, ground-source heat pumps, hydrogen and sustainable biomass,

$$x_{c,t,y} \leq \bar{x}_{c,t}, \qquad t \in \{\mathrm{DH}, \mathrm{GSHP}, \mathrm{H}_2, \mathrm{biomass}\}, \tag{17}$$

and a stock-turnover limit bounds the year-on-year change in each share by the fraction of the boiler stock that reaches end of life within the milestone interval,

$$|x_{c,t,y} - x_{c,t,y-1}| \leq \delta_{c,t}, \tag{18}$$

alongside the boiler-ban phase-out drawn from the policy layer. Finally, a per-country emissions cap limits 2050 scope-1 emissions,

$$\sum_t e_t\, D_{c,y}\, x_{c,t,y} \leq \left(1 - \kappa\, \frac{y-2025}{25}\right) E^{\text{scope-1}}_{c,2025}, \tag{19}$$

where e_t is the on-site emission factor of technology t, $E^{\text{scope-1}}_{c,2025}$ the country's 2025 scope-1 baseline, and $\kappa \in [0,1]$ the required reduction fraction relative to that baseline at 2050. The cap glides linearly from the country's own 2025 baseline to $(1-\kappa)$ of it at 2050 rather than binding at the terminal value in every year; imposing the terminal cap throughout is infeasible at $\kappa = 0.90$ given the turnover limit of Equation (18). The cap is applied to scope-1 emissions only, that is, to on-site gas and oil combustion, consistent with the zero-emission building definition of the recast Energy Performance of Buildings Directive [12], so that the exogenous decarbonisation of the grid and of district heat is not double-counted. The scope-1 boundary ensures the model does not double-count exogenous grid decarbonisation or district-heat clean-up. It reflects the fact that building-level policy levers, namely boiler choice and envelope depth, directly control on-site emissions, and grid decarbonisation is an EU-wide system outcome that the building sector does not set. Three ambition variants set κ to 0.75, 0.90 and 1.00, corresponding to scope-1 cuts of

−75 per cent, −90 per cent and −100 per cent against each country's own 2025 baseline. The objective uses the source-audited levelised cost of heat, with the ETS2 carbon price embedded, so that the cost-optimal allocation reflects both the corrected techno-economics and carbon pricing.

The shadow price of each country's cap is the implied carbon price that the country would require in order to meet its target.[1] To isolate the least-cost technology mix independent of assumptions about envelope retrofit depth, the demand $D_{c,y}$ is fixed at the Stated Policies LMDI trajectory (Section 3), using the central per-country envelope rate.

This choice reflects the modelling premise. The least-cost program answers which heating technology is cost-optimal at a given demand level, so the demand path is held at a moderate, policy-consistent trajectory and not the deeper Net Zero path, whose own demand reduction is itself an optimisation outcome that would prejudge the supply-versus-demand margin the optimisation is meant to test. The robustness test below, which co-optimises retrofit depth, confirms that supply-side switching is the least-cost margin and that this choice does not drive the result. Accordingly the 2050 EU demand of approximately 3,480 TWh tracks Stated Policies.

The 29-country least-cost 2050 mix is approximately 53 per cent district heat and 45 per cent heat pump, with hydrogen at 0.34 per cent and residual gas at 1.4 per cent. The shares are expressed on a useful-heat basis, that is, as a fraction of the building's physical heat demand, the convention used for every share in this paper. The ordering of the two leading technologies is the reverse of the one earlier drafts reported, at approximately 59 per cent heat pump and 41 per cent district heat, and that earlier figure is superseded. The reversal follows directly from charging equipment capital against peak sizing, not against annual output, which raises the heat pump's levelised cost by more than it raises any boiler's, since a heat pump is sized to the same peak whether or not it is efficient.

The composition of the mix is now governed almost entirely by the deployment ceilings and not by the cost ranking, and the two have come apart. On 2050 levelised cost the cheapest single technology is district heat in 25 countries, the hydrogen boiler in 2 and a heat pump in 2. Heat pumps nonetheless take 45 per cent of EU useful heat, because air-source units are the only major option in the program that carries no feasibility ceiling of its own (Eq. (17)), so they absorb whatever demand the ceiling-limited technologies cannot. The corollary matters for the paper's central question. Hydrogen's 0.34 per cent share is a ceiling-and-turnover outcome, not a cost outcome, and it can no longer be read as evidence that the least-cost program rejects hydrogen on price. The program does not reject hydrogen on price; the feasibility ceiling prevents it from selecting more.

[1] Four countries, Estonia, Malta, Portugal and Sweden, return a degenerate dual because their absolute scope-1 baseline is small enough that a tiny tonnage divides into a large per-tonne value that carries no meaning as a marginal abatement cost. Their shadow prices are undefined and they are excluded from the binding-country count throughout. The largest well-defined value anywhere on the path is Cyprus at about €207/tCO_2 in 2030.

That ceiling should be read for what it is, and it is stronger than a low cap. It is 3 per cent in the Netherlands and 2 per cent in the United Kingdom and *zero in the other 27 countries*, so in 27 markets the optimisation is not permitted to select any hydrogen at all. Denmark is the sharpest case, because the hydrogen boiler is one of the two markets where it is the cheapest single 2050 technology and its ceiling is zero. The ceiling encodes the mainstream literature's sub-2 per cent residential share and the physical pace at which a distribution network could be converted, which is defensible, but a near-zero hydrogen share in this optimisation is an assumption and not a finding.

Within the heat-pump category the optimisation now favours air-source units almost exclusively, at 44.4 per cent of useful heat against 1.05 per cent ground-source, so the EU mix moves from a fossil-dominated 2025 baseline to a 2050 mix led jointly by district heat and air-source heat pumps (Figure B.1). Ground-source units no longer track their feasibility ceiling, as they did before the capital charge was corrected; at 1.05 per cent they sit well below it, because their higher capital cost is now charged in full against peak sizing and is no longer offset by their better seasonal coefficient of performance. District heat takes the balance in dense stock and in areas served by an existing network.

The second result of the optimisation concerns the price of carbon, and it is weaker than earlier drafts of this paper claimed. The cost of *ambition* is negligible. The discounted present-value system cost is €7,426.3 bn EUR-yr at the −75 per cent cap, €7,428.4 bn at −90 per cent and €7,429.8 bn at −100 per cent, a spread of 0.05 per cent across the three depths, so tightening the scope-1 target all the way to net zero on-site combustion costs almost nothing at the audited costs. Co-optimising retrofit alongside the supply mix lowers the −90 per cent present value by €24.8 bn, and the −100 per cent one by €24.7 bn, 0.33 per cent in each case. That is small against the level and about seven times the 0.05 per cent the cap depth moves, so supply-side switching remains the cheap abatement margin. The renovation the least-cost solution does take, it takes at the interim milestones and lets lapse by 2050, at 2.8 per cent of EU useful heat in 2030 and 5.8 per cent in 2040 (Appendix B).

The *shadow price*, however, is not uniformly zero, and where it is not the policy reading changes. The cap binds in 4 of the 29 countries in 2050, namely Cyprus, Croatia, Poland and the United Kingdom, at an implied carbon price of up to €81/tCO_2, and it binds in seven countries at some point along the 2025 to 2050 path, the three additions being Spain, Hungary and Romania. The four degenerate cases noted in the optimisation set-up above are excluded from both counts. A cap that binds in a handful of markets is a materially different policy statement from a cap that is slack everywhere. The first says that ETS2 pricing carries most of the EU to its target but leaves an identifiable minority needing a carbon price several times higher, or an equivalent non-price instrument, to close the last increment. A cap that were slack everywhere would instead say that no supplementary instrument is needed anywhere, and that is not what the optimisation gives.

The implication of the cost-optimal result is therefore narrow, and should be stated narrowly. For most of the EU, residential heating decarbonisation is cost-effective at current ETS2 pricing, in the sense that the cap does not bind and no carbon price supplement is required in the model to reach the target. For a minority of countries it is not, and the binding shadow price there is large enough that the policy question in those markets is a live one. This finding rests on the two pillars documented above, the audited techno-economics and the embedding of ETS2 in the levelised-cost ranking, and it is conditional throughout on the optimisation's perfect-foresight least-cost framing.

Robustness of the cost-optimal mix to cap ambition. The perturbation that matters most for the policy reading is the depth of the emissions cap, and Table B.2 reports it directly. The present-value system cost moves by 0.05 per cent across the three depths, and district heat and hydrogen hold their 2050 shares across them while the adjustment falls entirely on the heat-pump against residual-gas margin, which runs 43.3 to 45.4 to 46.8 per cent against 3.5 to 1.4 to zero. The *cost* of ambition is negligible in the sense the earlier drafts claimed. What is not stable is the binding status. The cap is slack in most countries at every depth, but it binds in 4 countries in 2050 and in 7 at some point on the path. Where it binds in 2050 the implied carbon price reaches €81/tCO_2, and it is not monotone along the path. Six countries bind in 2030 at up to €207/tCO_2 in Cyprus, three in 2040 at up to €37, and four in 2050. The shadow-price column can no longer be read as a uniform zero, and the near-term figures are the larger ones.

Table B.2: The cost of ambition and the cap's binding status across the scope-1 caps.

Cap ambition	Present-value system cost (bn EUR-yr)	Spread	Binding status
−75 % scope-1	7,426.3	reference	Binds in 4 of 29 at up to €81/tCO_2.
−90 % scope-1	7,428.4	+0.03 %	Binds in the same 4 at the same shadow price.
−100 % scope-1	7,429.8	+0.05 %	Binds in the same 4 even at net zero on-site combustion.

The spread column is measured against the −75 per cent cap. In the headline program the cap binds in 4 of the 29 countries in 2050 (Cyprus, Croatia, Poland and the United Kingdom) at an implied carbon price of up to €81/tCO_2, and in seven countries at some point on the 2025 to 2050 path (adding Spain, Hungary and Romania). The four degenerate cases (Estonia, Malta, Portugal and Sweden) are excluded throughout.

Source: Authors' contribution.

Two further perturbations bear on the mix alone. The first is the feasibility ceiling on ground-source heat pumps. In earlier runs the ground-source share tracked that ceiling almost one for one, so the ceiling set the answer. It no longer does. With equipment capital charged against peak sizing, ground-source units carry their higher capital in full and are no longer compensated by their better seasonal coefficient of performance, so the program selects only 1.05 per cent of useful heat from them, well inside any ceiling in the tested range. The ground-source ceiling has accordingly stopped being a binding assumption of the result, and the per-ceiling sweep is documented in the supplementary material. The biomass ceiling is likewise not binding, since the linear program selects approximately 0 per cent biomass across a ceiling range of 0.10 to 0.30. The second is the accounting boundary of the cap, scope-1 against a scope-2-inclusive total, which is reported in the supplementary material and discussed as a limitation in the emissions-boundary subsection of Appendix D.

Finally, as a robustness test outside the central parameter set, we relax the fixed-demand assumption itself by endogenising renovation depth. A linear negawatt retrofit option avoids useful heat at a country-specific cost drawn from the medium-depth band of the European Commission's renovation-activity study [45], set at €154/m^2 for an approximately 41 per cent saving and annualised over a 30-year envelope life, which gives a cost of €117 to 1,187/MWh of useful heat avoided, cheapest in cold, high-intensity stock. Co-optimising retrofit with the supply mix leaves the 2050 mix essentially unchanged, and the least-cost solution selects 0.0 per cent retrofit in 2050 even at the net-zero cap. That endpoint is not the whole story, and reporting it alone would misstate the program's behaviour. Retrofit is selected at the interim milestones and abandoned by 2050: 2.8 per cent of EU useful heat in 2030, in six countries, and 5.8 per cent in 2040, all of Germany's. Every one of those shares sits exactly on its ramp limit, so the program wants more retrofit at those dates than the 2 percentage-point-a-year build rate allows. The ramp is one-sided, with no floor holding the stock in place, so the program builds envelope for the interim caps and lets it lapse when clean supply has caught up. The €24.8 bn present-value improvement the variant delivers is the value of that interim renovation, and not a residual.

The cheapest negawatt costs about €117/MWh of useful heat avoided, and the heat-pump levelised cost now spans about €65 to 197/MWh across the 29 countries, so the two ranges overlap and the inequality no longer holds universally. In the dearest heat-pump markets a deep retrofit is cheaper per avoided megawatt-hour than the supply switch it would displace, and the zero-retrofit result survives on the median alone. This still confirms the modelling choice to fix the least-cost program's demand at the Stated Policies trajectory, because the optimum is unchanged, but it should no longer be presented as a strict cost dominance. It is consistent in that qualified form with the Joint Research Centre finding [46] that deep envelope renovation is cost-optimal only in specific contexts, namely central-European, high-demand stock, not as a general abatement lever.

B.4 Limitations of the Cost-Optimal Model

The robustness tests above show that the headline mix and the negligible cost of ambition survive the central parameter choices, while the zero implied carbon price does not. Three structural assumptions of the linear program further qualify how its results, and in particular the zero-retrofit outcome, should be read. They are stated here in full so that the cost-optimality headline is interpreted correctly; none overturns the mix, but each bounds its reach.

First, the baseline program assumes *perfect foresight*. It minimises a discounted system cost computed against known future levelised costs, so agents in effect know the entire 2025 to 2050 price and carbon trajectory in advance, and under that assumption the cost-optimal solution selects approximately zero per cent envelope retrofit because, at the audited costs, renovation is dearer per megawatt-hour than switching supply in most of the stock. The natural concern is that the near-zero retrofit is an artefact of this foresight. It is not. Relaxing perfect foresight to a *myopic rolling-horizon* optimisation, in which each milestone year is solved on the costs visible at the time with a one- or two-period look-ahead, not the full trajectory, leaves the result unchanged. The 2050 retrofit share stays at 0.0 per cent, and the heat-pump, district-heat and hydrogen mix is identical in all 29 countries across the foresight modes at each target, reproducing the headline mix reported above with hydrogen at 0.34 per cent. Under the one- and two-period look-ahead cases the realised present-value cost matches the perfect-foresight optimum of about €7,430 bn.

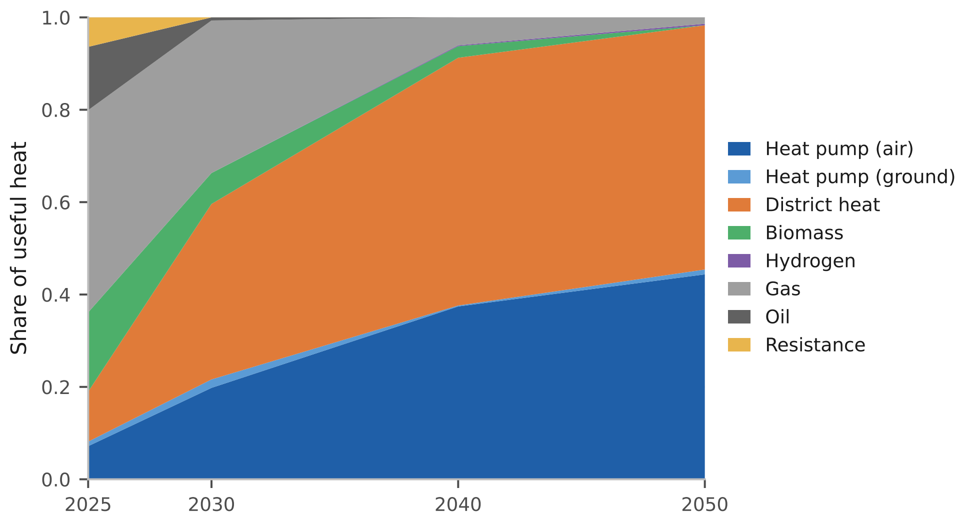
The zero-look-ahead snapshot, which re-optimises each milestone year in isolation with no inter-period stock memory, is the exception. It *diverges* on total cost, because the isolated single-year program does not carry the cumulative replacement obligation, yet still returns the same 0.0 per cent 2050 retrofit and the identical technology mix, so the divergence is in the cost accounting only and the mix verdict is intact. The result is foresight-invariant because the pathway is bound by stock turnover and the feasibility ceilings, with no room for intertemporal arbitrage to bind.

It is not, however, cost-dominated at every horizon in the way earlier drafts claimed. The cheapest negawatt retrofit costs about €117/MWh of useful heat avoided, against a heat-pump levelised cost of heat spanning about €65 to 197/MWh, so the two ranges overlap and retrofit is cheaper than a heat pump in the dearest heat-pump markets. The solved program still selects 0.0 per cent retrofit in 2050, and it does so because district heat, at a 2050 EU-median levelised cost of €96.5/MWh, has by then displaced the cheapest negawatts, while the ramp limit is what holds the interim shares down. Renovation is not dearer than every supply option everywhere, and the program takes it where the interim caps force early abatement. The zero-retrofit outcome therefore survives as a result of the optimisation. The caveat on the zero-retrofit outcome therefore moves from "perfect-foresight-conditioned" to "cost-and-constraint-conditioned", and it now carries a country caveat as well. Real building owners still face genuine uncertainty over future energy and carbon prices, which can justify precautionary envelope investment that neither optimiser values, so the modelled retrofit share remains a lower bound on real-world behaviour.

Second, the model charges *no switching cost beyond annualised capital.* It applies only the stock-turnover constraint of Equation (18), which limits the pace of replacement to the rate at which boilers reach end of life but does not penalise replacing equipment before its useful life is over. The economic cost of premature scrappage, and any transaction or disruption cost of switching technology, is not represented, so the modelled pace of transition is an upper bound on what an equivalent switching-cost-aware optimisation would deliver.

Third, the model holds the *demand path fixed* at the Stated Policies LMDI trajectory instead of co-optimising it with supply. This is a deliberate choice, made so that the least-cost program answers which heating technology is cost-optimal at a given demand level and not how much to invest in renovation; the endogenous-retrofit variant above confirms that relaxing it does not change the mix. It nonetheless means the reported pathway is the least-cost way to serve a fixed demand, and the deeper demand reductions of the Net Zero path are excluded by construction. These limitations do not invalidate the cost-optimal result, which is robust across the sweeps reported above, but they delimit it. The result is a least-cost *supply* pathway under foresight, and is best read as one bound on real-world behaviour.

Figure B.1: The EU least-cost heating pathway under the −90 per cent scope-1 cap. Each band is a technology's share of EU useful heat, so the vertical axis runs from 0 to 1 and the bands sum to it in every year. Hydrogen carries a band of 0.3 per cent, which is not visible at this scale, and that absence is the figure's point.



Source: Authors' contribution.

Country-specific outcomes under the −90 per cent cap are not uniform, and the nonuniformity is where the policy content sits. With ETS2 embedded in levelised costs, most countries reach the target with the cap slack and an implied carbon price of zero, but 4 of the 29 reach 2050 with the cap binding, at up to €81/tCO_2, and 7 have a binding cap at some point on the path. The cost-optimal district-heat and heat-pump shares vary with feasibility and climate around the EU mix of Figure B.1. The per-country shares and shadow prices are tabulated in the supplementary material.

C Sensitivity Analysis

The central findings rest on a set of uncertain parameters and structural assumptions, and it is necessary to establish which of them the conclusions actually turn on. This section examines that sensitivity through six methods, hierarchically organised and grouped by the side of the model they probe. The first four address the demand side: a one-at-a-time tornado on the three economic axes that govern the technology mix, the ETS2 carbon-price trajectory, the green-hydrogen import price and the discount rate, reported for 2030 buildings-sector emissions under Stated Policies (Appendix C.1); a parametric band on the cross-country renovation correlation ρ, which sets the width of the demand distribution (Appendix C.3); a variance-based (Sobol) global decomposition that apportions the variance of 2050 useful heat across its four demand-side drivers (Appendix C.4); and a one-sided climate-warming bound on demand (Appendix C.5).

The remaining two are targeted stress tests on the supply side: on the per-country hydrogen-supply cost (Appendix C.6) and on the electricity network reinforcement and scarcity premium (Appendix C.7). The methods are complementary. The tornado and the parametric band probe individual axes, the Sobol decomposition then apportions the demand variance across all four drivers at once, and the supply-side tests bound the cost axes the demand methods do not reach. The conclusions below rest on all four methods together. The reported bands rest on a 200-draw Monte Carlo per scenario. The convergence diagnostic of Appendix A.3 finds the medians settled well inside that sample and the deciles not converged at it, so the medians carry the results and the 80 per cent bands should be read as indicative of spread rather than as precise intervals.

Two caveats bound the applicability of these methods and should be borne in mind before the results are read. First, the per-NUTS3 building-stock feasibility constraints on heat-pump and district-heat deployment are held deterministic; probabilistic feasibility is reserved for future work. Second, the tornado is one-at-a-time and does not capture covariation, so that, for example, a simultaneous high carbon price and stranded hydrogen trajectory would reinforce and not offset one another. The Sobol screen apportions demand variance and the tornado covers the three economic axes, but a full Morris screen of the cost-side interactions is the identified next step.

C.1 Tornado on the Three Economic Axes

Each of the three principal economic axes of the model, the ETS2 carbon-price trajectory, the green-hydrogen import price and the discount rate, is moved between its low and high bound while the other two are held at their central values. The screen is deterministic. One central draw is used for every arm, with the same seed throughout, so the only thing that differs between arms is the axis under test and the result is not measuring sampling noise. The metric is buildings-sector carbon dioxide emissions in 2030 ($MtCO_2$ per year) under Stated Policies (Table C.1). The near-term horizon is deliberate, because by 2050 the cross-scenario spread is dominated by the policy levers themselves.

The discount rate is the widest axis, at −0.1 to +0.4 $MtCO_2$, the ETS2 carbon price next at −0.3 to +0.1, and the hydrogen price the narrowest at −0.0 to +0.3. All three act through the levelised-cost layer and therefore only through the cost-responsive part of the technology mix, which is what bounds them. The variance-based global sensitivity of the demand side, which the tornado does not reach, is set out in Appendix C.4 and Figure C.1a.

Table C.1: Tornado on 2030 buildings-sector emissions (Stated Policies), against a central-draw baseline of 539 $MtCO_2$. Each axis moves alone; every other lever is held and the model is re-run end to end. The screen is deterministic and is one draw, not a distribution.

Axis	Emissions-lowering bound ($MtCO_2$/yr)	Emissions-raising bound ($MtCO_2$/yr)
Discount rate	−0.1 (low)	+0.4 (high)
ETS2 carbon price	−0.3 (high price)	+0.1 (low price)
Hydrogen price	−0.0 (rapid)	+0.3 (stranded)

Source: Authors' contribution.

These magnitudes are small, and the reason is structural rather than a sign that the price levers do not matter. The share allocation blends a scenario interpolation with a cost-responsive term whose weight ramps linearly from zero in 2025 to 0.50 by 2050 (Appendix A.9), so at 2030 only a tenth of the non-fossil allocation responds to cost at all. A price axis moved alone at that date therefore reaches very little of the mix, and the 2030 spread across the four scenarios, from 512 to 548 $MtCO_2$, is set by their share and demand levers rather than by the prices those levers are paired with. An earlier version of this table reported bounds of about ±5 $MtCO_2$ on each axis; those figures were never produced by a committed run and do not reproduce. The ordering the screen returns, with the discount rate widest and hydrogen price narrowest, is itself informative. At 2030 the levelised comparison is more sensitive to the cost of capital on equipment than to either fuel price, because the capital charge is annualised over the whole equipment life while a fuel price moves only the running cost of the small cost-responsive fraction.

The cost leadership behind these magnitudes is not uniform across countries, and it does reverse in a few of them, so the tornado should be read as moving the size of a margin that mostly runs one way, not as flipping the leader. The heat pump's advantage widens in central and eastern Europe, where lower construction labour costs reduce heat-pump capital expenditure and where cheaper retail electricity lowers its running cost, and narrows in the cold, high-tariff north-west, where lower seasonal performance and dearer electricity erode it. Against the incumbent gas boiler the best available heat pump is cheaper in 24 of the 29 countries by 2050 on the central carbon path, losing only in Cyprus, Croatia, Malta, Poland and the United Kingdom, and in 16 of 29 in 2025, so cost leadership over fossil heating is broad but not universal at either date. That comparison is the carbon-sensitive one, since the gas boiler pays ETS2 and the heat pump does not, and its 2050 count runs from 19 of 29 on the low carbon path through 24 on the central path to all 29 on the high path.

Against the hydrogen boiler the margin is narrower, and the 2050 levelised-cost ordering of the heat pump, the hydrogen boiler and the gas boiler across all 29 countries is reported in the Results section (Figure 6b). Which side of that ordering a country falls on is governed by its storage geology first and by the hydrogen price path second. Heat pumps are the cheaper base-load option in 13 of 29 countries on a rapid hydrogen price decline, 22 on the central path, 26 on the slow path and all 29 if hydrogen costs stall.

C.2 Sweeping the Seasonal-Storage Adder

The same storage adder decides the power arena's seven-market concentration, and it is set by hand at €50/MWh-heat where salt caverns exist and €100 where they do not, so an explanation of that concentration which appeals to the adder is circular until the adder is itself varied. We vary it. Scaling the pair from 0.5 to 2.0 times their central values, holding the cavern-to-non-cavern ratio at 2.0, moves the power-arena win count as Table C.2 shows. The base-load count above is not swept on this axis, so the base-load concentration is bounded here by the hydrogen price paths alone.

Table C.2: Power-arena hydrogen wins under a swept seasonal-storage adder, H2 Push 2050. The cavern and non-cavern adders are scaled together at a fixed ratio of 2.0. The count is invariant to the scarcity premium across €30 to 120/MWh-e, so one column is shown. That invariance is an algebraic identity in the power arena, where the premium enters both turbines' clearing price and cancels, rather than an empirical result.

Cavern adder (€/MWh-heat)	25	38	45	**50**	55	62	75	88	100
Markets won, of 29	29	29	8	**7**	7	7	7	2	0
All winners cavern-endowed	no	no	no	yes	yes	yes	yes	yes	n/a

Source: Authors' contribution.

The seven-market result is not an artefact of one point estimate, since it holds across €50 to 75/MWh-heat, about a third of the €25 to 100 swept range. It is also confined to that band. Below €45 the count jumps to every market and the cavern confinement dissolves, and above €75 it collapses towards zero. The concentration is a real property of the model over a band; it is not a structural feature of the problem. One caveat bounds even that. The sweep scales the cavern and non-cavern adders together at a fixed ratio, so the geological *spread* that actually drives the concentration is held constant throughout and is not tested here at all.

A second caveat bears on how the seven should be read. The classification is a two-value national binary, and over the whole €50 to 75 band the winning set is that binary exactly, so the seven is conditional on the classification rather than emerging independently of it. Alternative readings of the same geological source move it. Adding Iberia, which the source also maps, gives nine wins; restricting France to its northern basins gives six. The defensible statement is therefore that, given a two-value national classification of seven markets, the power-arena winners are exactly those seven, with a range of six to nine under alternative readings. The ratio itself is corroborated independently: a European techno-economic assessment of the same stores puts levelised salt-cavern storage at about \$0.8/kg against \$1.5/kg for porous media [144], a ratio near 1.9 against the 2.0 assumed here.

The carbon-price axis, by contrast, does not enter this particular comparison at all, since neither a heat pump nor a hydrogen boiler pays a carbon price on its own fuel, so the count is identical under low, central and high carbon assumptions and the tornado's carbon axis acts only through the fossil technologies it displaces. Regional labour-cost variation moves the size of the margin without setting the ordering.

C.3 Parametric Band on the Cross-Country Renovation Correlation (ρ)

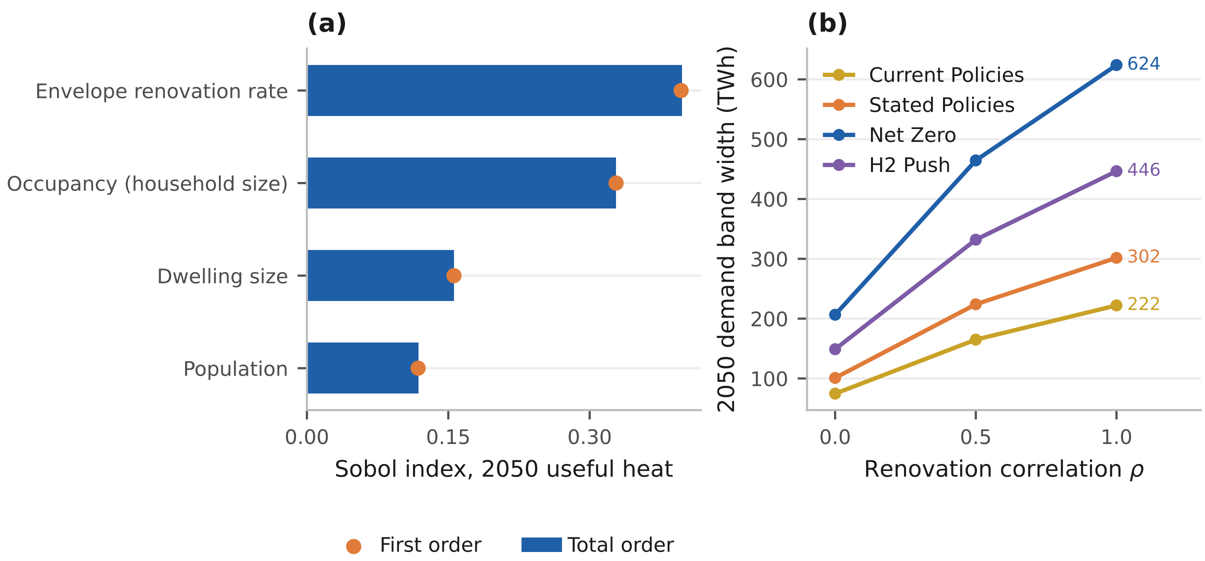
The Monte Carlo samples each country's envelope-renovation rate through a shared EU-policy factor plus idiosyncratic national noise, with correlation ρ (Appendix A.3). The central 2050 demand trajectory is invariant to ρ, because the shared factor preserves every country's marginal distribution, so only the width of the band moves. At the central value $\rho = 0.5$, the EU 2050 figure is 3,651 TWh under Current Policies, 3,483 TWh under Stated Policies, 2,660 TWh under Net Zero and 3,154 TWh under H2 Push, essentially unchanged across ρ. The uncertainty band, by contrast, widens from 1.0× (independent, $\rho = 0$) to 2.2× (shared, $\rho = 0.5$) to 3.0× (fully correlated, $\rho = 1$), by the same factor across all four scenarios (Figure C.1b; the per-scenario bands are collected in Table C.3). At $\rho = 0$, independent national renovation pace, the EU band is artificially narrow because national errors diversify away; $\rho = 0.5$, the shared central value, reflects a common policy environment under the EPBD and the Renovation Wave. The demand bands should therefore be read as conditional on $\rho = 0.5$, while the headline scenario *medians* are unaffected.

Table C.3: 2050 useful-heat demand band (TWh/yr, 10th to 90th percentile) against the cross-country renovation correlation ρ.

Scenario	$\rho = 0$ (indep.) (TWh/yr)	$\rho = 0.5$ (central) (TWh/yr)	$\rho = 1$ (full) (TWh/yr)
Current Policies	3,612 to 3,687	3,566 to 3,731	3,536 to 3,758
Stated Policies	3,432 to 3,533	3,369 to 3,593	3,328 to 3,630
Net Zero	2,563 to 2,770	2,435 to 2,900	2,353 to 2,977
H2 Push	3,080 to 3,229	2,987 to 3,319	2,928 to 3,375
Band-width vs $\rho = 0$	1.0×	2.2×	3.0×

Source: Authors' contribution.

Figure C.1: Demand-side uncertainty structure. Panel (a) gives total-order (S_T, bars) and first-order (S_1, points) Sobol indices for 2050 useful heat; panel (b) gives the demand band width against the cross-country renovation correlation ρ, labelled at $\rho = 1$.



Source: Authors' contribution.

C.4 Global (Variance-Based) Sensitivity of 2050 Demand

The tornado and the parametric band above each move one axis at a time and so, by construction, cannot apportion variance among the demand drivers or detect interactions between them. The variance-based (Sobol) global sensitivity introduced here complements them by decomposing the variance of EU useful heat in 2050 across all four uncertain demand drivers at once: the envelope-renovation rate, the scenario lever, sampled over a ±35 per cent band; population; occupancy, expressed as household size; and dwelling size. The decomposition uses a Saltelli design with $N = 2{,}048$. Because useful demand in 2050 depends only on these drivers, it is evaluated analytically per country, so a full Sobol design is both exact and immediate.

The total-order indices are an envelope rate of $S_T = 0.40$, occupancy of 0.33, dwelling size of 0.16 and population of 0.12; the first-order indices approximately equal the total-order indices, indicating a near-additive response with negligible interactions. Two readings follow. First, the scenario lever, the renovation rate, governs the largest single share of demand uncertainty, as intended. Second, occupancy, the household-size trend that sets dwellings per capita, is almost as influential, which is why the LMDI decomposition treats it as a first-class driver instead of folding it into population. The cost and emissions axes, namely the carbon price, the hydrogen trajectory, the discount rate, ETS2 pass-through and fuel-price correlation, act through the levelised-cost layer, whose global screen is compute-bound on the full NUTS3 Monte Carlo; the tornado above covers them, and a full Morris screen is the documented next step.

Cost-side propagation. The two caveats that bound this exercise, the one-at-a-time tornado and the deterministic feasibility scores, are stated at the head of the section. The cost-side axes are themselves fully propagated. The hydrogen-price scenario, the discount rate, the carbon price, ETS2 pass-through and the per-carrier log-normal fuel-price multipliers, covering gas, electricity, oil and biomass, all enter the levelised-cost routine, which is recomputed for each Monte Carlo sample, so the cost-side uncertainty is active throughout even though its global variance decomposition, which would require a screen of the full NUTS3 Monte Carlo, is left to the identified Morris next step.

C.5 Climate-Warming Demand Sensitivity

The last of the demand-side methods bounds a structural assumption, not a sampled parameter. The central demand trajectory holds heating degree days at the 1991 to 2020 climate normal, so it excludes the demand-reducing effect of a warming climate. Because space heating scales approximately linearly with heating degree days and domestic hot water does not, we overlay a warming factor on the space-heating share, approximately 85 per cent of useful heat, using EU population-weighted degree-day decline rates from Spinoni et al. [169]: a central rate of −0.35% per year, which is that study's RCP4.5 result, swept over a −0.20 to −0.55% per year band. The upper end follows its RCP8.5 result; the study runs only those two pathways, so the lower end is our own conservative bound and not a reading of a lower-emissions scenario.

This is a one-sided robustness bound, not the central case. Under the central warming rate, Stated Policies useful heat in 2050 falls a further 7 per cent, from 3,483 to 3,235 TWh, deepening the 2025 to 2050 reduction from −9% to −15.5%; relative to the frozen-climate trajectory the additional 2050 reduction spans −4.1% at our own low bound to −10.9% under RCP8.5. In the announced-policy pathway, a warming climate thus delivers roughly half of the demand reduction that deep renovation delivers under Net Zero (−31%). The frozen-climate central trajectory is therefore conservative on demand. The technology-mix and the RQ1 and RQ2 conclusions are unaffected, because warming scales demand without altering relative levelised costs.

C.6 Per-Country Hydrogen-Supply Cost and Import Connectivity

The headline hydrogen prices apply a per-country multiplier to a single north-west-European hub trajectory, which falls from €200/MWh in 2025 to €50/MWh in 2050. This multiplier was originally a crude ±10 per cent heuristic, under which a landlocked or import-dependent country was simply assumed dearer, and it was the least-grounded parameter in the cost layer. It is replaced by the bottom-up delivered-cost model of Appendix A.13, which costs out five supply routes for each country and takes the cheapest, and normalises the result so that the median country pays the hub price. This subsection reports what that model finds.

Three findings emerge. First, pink and blue hydrogen are not the cheapest route in 2050. Nuclear electrolysis, at about €116 to 117/MWh and set by the cost of firm nuclear power, and gas with carbon capture, at about €81/MWh [88], are both beaten by cheap green or imported hydrogen, so the nuclear and carbon-capture 'optionality' often invoked for central Europe does not in fact lower its delivered cost. Second, domestic green electrolysis is the cheapest route in the majority of the EU, with 23 countries self-supplying green hydrogen in 2050, while the remaining six import by pipeline. No country takes the seaborne-ammonia route at 2050, and the two non-backbone islands, Cyprus and Malta, self-supply green at about €63 to 64/MWh against €108 for shipped ammonia (Figure C.2a).

Third, capital and labour set the ranking, ahead of sunshine. High-WACC Greece, at 8 per cent in the per-country cost-of-capital assumption of the hydrogen-supply methodology (Appendix A.13), sits at a multiplier of 1.04 despite superb solar resource, whereas cheap-capital, strong-wind Ireland, Denmark and the Baltic states self-supply green hydrogen well below the median, at multipliers of 0.77 to 0.84. Pipeline import caps the six connected countries at the hub price plus 9 to 16 per cent, so the dearest supply is found in the landlocked central-European states with mediocre resource, namely Slovakia, the Czech Republic, Hungary, Slovenia and Italy at multipliers of about 1.14 to 1.16, and in the non-backbone islands, Cyprus and Malta at about 1.31 to 1.33, whose isolated-grid, small-scale solar is the costliest self-supply of all and still undercuts the shipped-ammonia alternative.

Carrying each multiplier as a low, central and high band, the 2050 hydrogen-versus-heat-pump gap is negative, meaning hydrogen is cheaper, in a small minority of countries. At central supply values, the default network treatment and the seasonal store charged on the winter-weighted fraction of the hydrogen heating load, 7 of the 29 countries sit at or below heat-pump parity (Section 4). Cheap supply is not on its own enough to put a country in that set. Finland and the Baltic states all sit below the median on the delivered multiplier and none of them is in it, because none has the salt-cavern endowment that makes seasonal storage cheap, which five of the seven that remain do have. Charging hydrogen a fuller share of its distribution cost shrinks the set further (Appendix A.6).

The colour of the molecule. A second, qualitative finding runs with the base-load result above, and it deserves more weight than its position in a sensitivity section suggests. In nearly every assessed country the national hydrogen strategy explicitly *excludes* residential heating. More consequentially, the one dedicated assessment of hydrogen for buildings concludes that any heating hydrogen would in practice be *blue* [73], and our own blue route prices that molecule at about €81/MWh [88], against the green or imported molecule at about €50/MWh that the model prices. The model therefore charges hydrogen a delivered commodity cost roughly €31/MWh below the only sourcing the sectoral literature regards as realistic for this end use. That wedge is wider than the widest of the seven surviving hydrogen leads, so a commodity price that high would move every country further towards the heat pump.

On the storage-inclusive basis this paper uses, that repricing leaves the heat pump cheaper in all 29 countries at a flat €81/MWh, and in 28 with Denmark alone still holding if the German figure is read as a green-to-blue ratio rather than a European price. The model's delivered residential-hydrogen price is thus a counterfactual that it prices, not a planned supply chain, and the counterfactual is generous to hydrogen. The base-load result should be read in that light, as a statement about what a cheap green molecule would do to the household economics of heating, not as a forecast that such a molecule will be the one delivered to households.

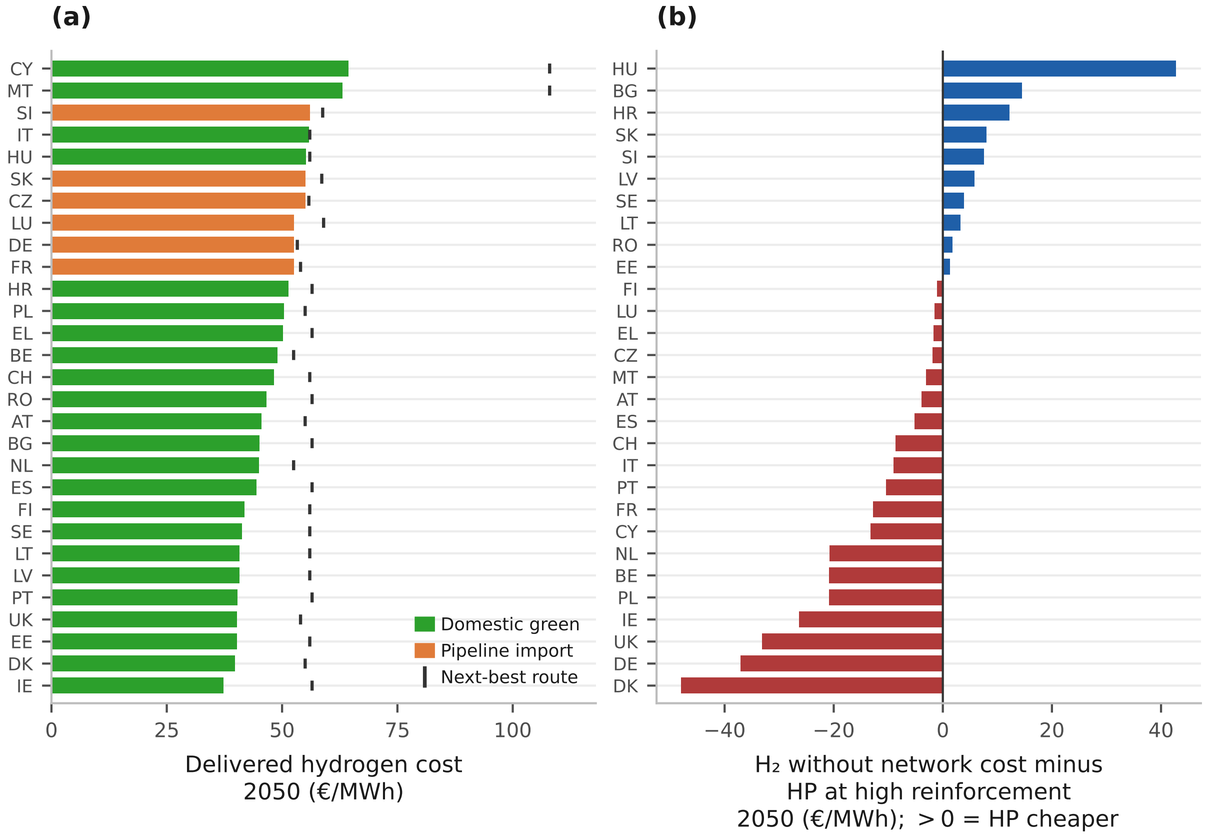
C.7 Electricity Distribution Reinforcement and the Scarcity Premium

Two network-side assumptions deserve explicit stress tests. First, earlier drafts charged hydrogen for its distribution network only inside a dedicated infrastructure scenario and never charged heat pumps for the low- and medium-voltage grid *reinforcement* that their added coincident winter peak requires. That asymmetry has been removed. Both adders are now active by default in every levelised cost this paper reports, so each carrier pays for its own last mile (Appendix A.13). The reinforcement adder is no longer opt-in, and it is no longer optional; it is part of the headline number. The added electric peak per dwelling, computed as annual heat over climate-scaled full-load hours, divided by the cold-snap coefficient of performance, with a coincidence factor of 0.5, is charged at €400, 800 and 1,600 per kW (low, central and high), annualised at the country WACC over a 40-year network life. Across that band the adder amounts to €1.1 to 38.2/MWh, and it is highest in peaky mild-climate systems where a modest annual heat demand arrives through a short, sharp season.

The stress test that follows is the conservative-for-heat-pump case, and it has two arms that must be kept apart, because they charge hydrogen differently and return different counts. Both charge the heat pump the *high* reinforcement bound. In the first arm hydrogen pays *nothing* at all for its network, and heat pumps then remain cheaper than hydrogen in 10 of 29 countries, namely Bulgaria, Estonia, Croatia, Hungary, Lithuania, Latvia, Romania, Sweden, Slovenia and Slovakia, and cheaper than gas in 21 of 29, with a mean heat-pump-to-hydrogen margin of −€6.2/MWh and a median of −€3.1/MWh, which is to say that hydrogen leads on average by those amounts. In the second arm hydrogen pays the blended distribution network it is charged by default, and heat pumps remain cheaper than hydrogen in 22 of 29 at a mean of +€13.6/MWh, which reproduces the headline count at the top of the reinforcement band (Figure C.2b).

The two arms should never be combined, since quoting the second count against the first arm's network treatment would overstate the heat-pump case. The reading is the same in either arm. The base-load ranking *does* depend on how the two last-mile networks are charged, and a reinforcement assumption at the top of its plausible band is enough to move a majority of countries from one side of the comparison to the other. This is one of the three most load-bearing assumptions in the levelised-cost layer, alongside the seasonal-storage charge and the levy treatment of the two carriers, and all three are carried explicitly in Appendix D.

Figure C.2: Supply-side robustness in the hydrogen supply route and in the network charge. Panel (a) gives each country's cheapest delivered hydrogen cost in 2050, coloured by which route wins, with the tick marking the next-best route so the margin between them is visible. Panel (b) asks the question most favourable to hydrogen: hydrogen charged no last-mile network at all against the heat pump charged its high reinforcement band, per country, where a positive value still leaves the heat pump cheaper.



Source: Authors' contribution.

Second, the merit-order results of Section 4.6 use an endogenous scarcity electricity price, the marginal-peaker short-run marginal cost plus a scarcity premium of €30, 60 and 120 per MWh (low, central and high). The premium band spans the displacement of the thermal peaker by storage and demand response across a varying number of hours. At the low premium the cold-snap heat pump becomes cheaper and hydrogen's winning-country count at the building peak falls, while at the high premium it rises. The counts are 0, 1, 8 and 15 of 29 across the four scenarios at €30/MWh-e, 0, 4, 9 and 16 at the central €60 and 0, 5, 10 and 18 at €120. Each sweep in this section moves one lever and holds the rest at their published settings, so the baseline here is the asymmetric accounting and the central column is its 0, 4, 9 and 16 rather than the symmetric 0, 2, 7 and 14 the results section headlines; the last-mile charge is swept separately. The demand-weighted derived ceiling under Stated Policies runs from 0.2 per cent at the low premium to 5.3 at the high one against 5.1 at the central. That is the widest of the three independent reasons the Stated Policies ceiling is a range rather than a point, and it is the scenario in which the ceiling actually binds. The power arena does not move at all, because there the premium enters both turbines' clearing price and cancels. The missing-money conclusion is invariant across the band, because it is the peaker's running hours, not the price level, that fail the investment test.

D Limitations

This section examines the principal classes of simplification, each of which introduces a known bias: the synthetic single weather year behind every power-sector number; the fiscal asymmetry between the tax treatments of the two carriers compared; the uniform within-country shares and static feasibility scores of the bottom-up building-stock allocation; the scope and accounting boundaries of the two modelling frameworks; the correlation, scenario-scaling and forward-extrapolation choices in the uncertainty and demand-side design; the annual-average treatment of the emissions accounting and the unit-commitment parameters of the power dispatch; and the boundary choices in the grid and network infrastructure. For each simplification we state its direction of bias, as an overstatement, an understatement or ambiguous, and, where the bias is material to the result, the sensitivity of the core hydrogen verdict to relaxing it. The first four are the most consequential and are stated first: the synthetic weather year, the three network charges the symmetry does not cover, the fiscal asymmetry between the carriers, and the energy-only market behind the recovery figures.

Three structural limitations bear on the results, the reduced-form power dispatch, the perfect-foresight cost-optimal program, and the one-directional exogenous treatment of Switzerland. Each is addressed here, by an explicit unit-commitment dispatch (Appendix A.2 and Appendix A.7), a myopic rolling-horizon robustness test (Appendix B), and an endogenous Swiss integration with EU-price feedback (the dedicated subsection below). All three strengthen the verdict they bear on, and the Swiss case, which an earlier draft reported as a base-load reversal, does so once the seasonal store is charged on the base-load comparison as well as on the peak. The residual sensitivities those upgrades introduce are documented below. The dispatch layer now preserves system adequacy through an explicit operating-reserve constraint, with a modelled loss of load of about 2 h/yr on average and at most the three-hour planning standard (Appendix A.7).

D.1 The Synthetic Weather Year Behind Every Power-Sector Number

This is the largest simplification in the paper and we place it first because the whole power-sector half rests on it. The firm fleet of about 278 GW, its roughly 139 full-load hours, the €357 bn cumulative peaker shortfall and the finding that no market recovers its capital are all computed on *one realisation of a single synthetic year*, and not on reanalysis or metered weather.

Three properties of that construction bound what the numbers can carry. Onshore wind is an AR(1) process around a winter-higher mean, forced down to a quarter of its normal level over six consecutive days in mid-January *in all 29 countries simultaneously*, so the scarcity event is perfectly correlated across the study area because we impose it, and not because the weather is. Solar availability and the non-heat baseline load are deterministic seasonal and diurnal functions, carrying no year-to-year variation at all. And the three-hour planning standard the fleet is sized to is an order statistic on that one year, not a loss-of-load expectation over a distribution of weather years, which is what a capacity-adequacy assessment reports.

The event length is a choice and the result moves with it. Sweeping the cold snap from two days to ten takes the fleet sized on the inflexible load from 253.7 to 302.0 GW against the 277.7 GW we report, a band of about ±9 per cent from that single assumption. Nothing here bounds the larger uncertainty from the wind process itself, from the correlation structure, or from the choice of year, because one realisation cannot.

Direction of bias: *ambiguous in level, conservative in use.* A perfectly correlated lull is more demanding than a real one, which overstates the fleet; a single year cannot reach the tail a thirty-year record would, which understates it. We do not benchmark any of these figures against a published European adequacy study such as the ENTSO-E resource adequacy assessment or the TYNDP, and a reader who wants an adequacy number should not take one from here. The method that would support one is established and we do not use it. Bias-corrected reanalysis gives calibrated hourly wind and solar series over three decades [170, 171], and recent work derives European renewable-drought events from 35 recorded weather years and finds that geographical balancing, which our country-autarky sizing suppresses, dominates the storage and firm-capacity answer [172]. Our copper-plate arm bounds that effect crudely, at 239 GW against 278. What the layer is for is the *comparison* between two firm technologies inside one demand envelope, and that comparison is unaffected by the level of the envelope, because both technologies face the same event.

Resolving it requires re-running the power layer on multi-year reanalysis, reporting the requirement as a distribution over weather years, and deriving the cross-country correlation from the data rather than imposing it. That is beyond this paper's scope and it is the single most valuable extension to it.

D.2 Three Network Charges That the Symmetry Does Not Cover

The rule that each carrier pays its own last mile holds at the distribution layer of the levelised comparison and is incomplete in three places. All are quantified here.

The first is that the rule does not travel into the dispatch arenas. In the base-load comparison the hydrogen boiler is charged the blended conversion-and-new-build adder, a study-area median of €13.4/MWh. In the building winter-peak arena it is not. Hydrogen enters as delivered fuel plus the seasonal store, while gas enters at the Eurostat residential retail tariff, which carries network charges and taxes inside it. The asymmetry runs in hydrogen's favour and it is not small. Charging hydrogen the model's own adder in that arena takes the four counts from 0, 4, 9 and 16 of 29 to 0, 2, 7 and 14, and the demand-weighted derived ceiling under Stated Policies from 5.1 to 0.6 per cent. The symmetric pair is what we now report, because a comparison that charges one carrier a network the other avoids is not a comparison, and the asymmetric counts are kept beside it so a reader can see what the choice is worth. With the scarcity-premium bracket, this is the second of two independent reasons the Stated Policies ceiling is a range and not a point.

The second is the electricity side of the seasonal problem. Hydrogen is charged the store its winter-weighted load requires, while electrified heat is charged only distribution reinforcement, leaving out the firm generating capacity the same winter evenings call on. This paper computes that fleet in Section 5, at about 278 GW costing €51 to 66/kW-yr, or €14.3 to 18.5 bn a year. Two attributions bracket its value on the heat side.

Charged pro rata to heating's 9.2 per cent share of power demand it is €1.0 to 1.3/MWh of heat-pump heat and the base-load count is unchanged at 22 of 29. Charged in full to the heat-pump load, which treats the whole firm fleet as built for electrified heat, it is €11.2 to 14.5/MWh over the 1,273 TWh of heat-pump heat the same year carries, and the count falls to 17 and then 15 of 29, with hydrogen picking up Austria, Switzerland, Finland, France and Luxembourg at the low bound, and the Czech Republic and Italy in addition at the high one.

Between those two sits the attribution the model computes directly, and it is the one cost causation asks for, since a peaking asset is built for a peak and not for annual energy. Re-running the dispatch with the heat-pump load removed drops the firm fleet from 277.7 to 56.6 GW and the summed national peak from 888.8 to 650.0 GW, so 221 GW, or 79.6 per cent of the fleet, exists because heat is electrified. A 9.2 per cent increment in energy drives a 36.8 per cent increment in peak, because the cold snap collapses the heat pump's coefficient of performance just as demand peaks. At that share the charge is €8.9 to 11.5/MWh and the count falls to 19 and then 17 of 29.

We therefore carry three attributions. Pro rata is the lower bound and treats the fleet as serving the whole load; charging the fleet in full is the upper bound and no one would argue it; the incremental figure is what a cost-causation regulator would use, and it sits nearer the upper bound. The headline of 22 holds on the lower bound alone, and is 19 to 17 on the incremental one. The incremental share is itself an ordering choice, since it charges electrified heat as the last load added and adding it first would attribute far less; that ambiguity is intrinsic to marginal cost allocation and we do not resolve it.

The third gap is district heat, which pays no last-mile charge while the other two carriers do. Charging it the same €5,000 per connection that this paper's own infrastructure bill already assigns it, annualised over 40 years at the country cost of capital, adds a median €21.0/MWh with a range of €7.8 to 184, and it moves district heat from cheaper than the best heat pump in 26 of 29 countries to 7 of 29. That charge leaves the hydrogen-versus-heat-pump comparison untouched, since neither carrier is district heat, but it bears directly on the least-cost program's district-heat-led mix (Appendix B), and it is the largest unpriced network asymmetry remaining in the cost layer.

D.3 Fiscal Asymmetry Between the Carriers Compared

The most important remaining asymmetry in the levelised-cost comparison is that its two sides are not taxed alike. It no longer decides the base-load result, since the heat pump now leads in 22 of 29 countries, but it sets how large that lead is and it is what keeps the seven remaining markets on hydrogen's side. Electric heating is priced at the *retail* tariff a European household actually pays, an EU median of about €209/MWh at 2050, which embeds energy taxes, value-added tax, renewables and social levies and the existing network charge. Hydrogen is priced *delivered and untaxed*, at about €50/MWh at the north-west-Europe hub in 2050 under the central trajectory, scaled by each market's own delivered-cost multiplier and carrying the distribution adder of Appendix A.13 on top, and it carries no levy stack at all, because no such stack yet exists for a residential hydrogen carrier. The comparison therefore sets a fully taxed carrier against an untaxed one.

This is not a technology result and must not be read as one. It is what sustains hydrogen's residual base-load advantage in the cold, high-tariff north-west, precisely the countries where the retail electricity tariff is highest relative to the wholesale cost of the electricity inside it. The direction of the bias is unambiguous, since the treatment favours hydrogen, and the magnitude of the favour is far larger than the leads it sustains, since the widest of the seven is €30/MWh while the levy wedge on electricity is measured in tens of euros per megawatt-hour. Two policy futures bracket the question, and the model prices neither. If a residential hydrogen carrier were to be built out at scale it would almost certainly attract a comparable levy and network-charge treatment, in which case those seven leads largely disappear. If instead electricity levies were rebalanced towards fossil carriers, as several proposals for a heating-fuel tax shift envisage, they would disappear faster still. Either way the correction widens the heat pump's margin, so the asymmetry bounds the headline result from one side only.

The sensitivity that this limitation calls for, and which we did not run, is a levy-equalised comparison in which both carriers are charged the same non-energy cost stack per unit of delivered heat, at a minimum the energy-tax and value-added-tax components. We recommend it as the first extension of this work, ahead of any refinement of the technology parameters, because no plausible movement in capital costs or coefficients of performance is worth as much to the comparison as the tax treatment is. Until it is run, the seven surviving hydrogen leads should be read as conditional on the current, asymmetric fiscal treatment of the two carriers. The paper's policy conclusion does not depend on them, since it rests on two arguments that are independent of the tax treatment, namely that the heat pump is the cheaper appliance in 22 of 29 countries once each carrier is charged the seasonal storage its own load shape requires, and that no hydrogen peaker recovers its capital from market rent in any country or scenario we model, on the assumed scarcity premium and with no capacity payment in place.

D.4 The Energy-Only Market Behind the Recovery Figures

The 9 per cent capital recovery, which the abstract, the results and the policy implications all quote, is conditional on an energy-only market in which the shortage hours the sizing standard admits are not paid at the value the model itself puts on them. It is the largest single swing of any limitation listed here, larger than the flexibility correction and larger than the event-length band, and it is a modelling choice; the plant itself has no such property. The coal-grid markets of Section 6.5 are where it bites hardest, since they build the largest peaking fleets.

The dispatch enforces a three-hour loss-of-load standard and prices unserved energy at a value of lost load of €8,000/MWh, but it pays that price to no generator. Crediting the peaker for those three hours at the model's own value lifts capacity-weighted recovery to about half, on the arithmetic set out in Section 5.3. Even then no market recovers in full, so the direction of the missing-money result survives and its magnitude does not. The 9 per cent is a statement about market rents under an administrative price cap. It does not say the plant cannot be paid for.

D.5 Uniform Building-Type Shares for the United Kingdom

We derive national building-type shares for the United Kingdom from the Office for National Statistics 2021 Census, table TS044, which covers England and Wales only [110], and we apply them uniformly across every UK NUTS3 region (Nomenclature of Territorial Units for Statistics, level 3). This abstracts from the spatial variation in the UK building stock, for example the higher proportion of tenement-style multi-family buildings in Scottish cities relative to English suburban areas. The same England-and-Wales national average stands in for Scotland's distinct housing stock, which carries a higher share of tenement flats in Glasgow and Edinburgh; the approximation therefore mislabels both the Scottish dwelling mix and its NUTS3 distribution. Future work could narrow this gap with a Local Authority to ITL3 (International Territorial Level 3) lookup that maps the census geography onto the model's regions.

D.6 Static Feasibility Scores

We assign heat-pump and district-heat feasibility scores by building type as fixed parameters, without deriving them dynamically from local grid capacity, retrofit cost or planning constraint. These scores compress a spatially heterogeneous reality into a small set of constants, which produces a within-country allocation of the two leading technologies smoother than the patchwork that local network headroom and consenting would in practice yield.

D.7 Two Modelling Frameworks, Each with a Distinct Scope

By design we run two distinct modelling frameworks whose results should not be read as substitutes, since they rest on distinct scopes and answer different questions. The four scenarios simulate technology shares under exogenous policy targets, while the least-cost program minimises system cost over the supply mix at a fixed demand path. Neither framework co-optimises demand and supply jointly, and that is the simplification at issue here. The four core scenarios (Current Policies, Stated Policies, Net Zero and H2 Push) are simulation-based. Technology shares interpolate towards stochastically sampled scenario targets, with the hydrogen target bounded above by the merit-order analysis, and the Monte Carlo characterises parametric uncertainty around those shares rather than deriving a cost-minimising allocation. The complementary least-cost run is a linear program, but it optimises only the *supply* mix at a demand path fixed to the Stated Policies LMDI trajectory; it does not co-optimise renovation depth. It should therefore be read as the cheapest way to serve a given demand, and not as a joint demand-and-supply optimum.

The two frameworks still supply a cross-check, but a narrower one than earlier drafts claimed. Hydrogen remains negligible in both, at a scenario-set share in the simulations and at 0.34 per cent of useful heat in the independently solved cost-optimal program, so the conclusion that hydrogen does not become a major residential carrier does not depend on which framework sets demand. Two qualifications now attach to that agreement. First, hydrogen's near-absence from the cost-optimal program is a feasibility-ceiling outcome and not a cost outcome, since the hydrogen boiler is the cheapest single technology in only 2 of the 29 countries in 2050 (Appendix B); the program is prevented from selecting more of it, and price does not deter it.

Second, the median cost ordering supports the agreement directly. At the 2050 EU median the hydrogen boiler is dearer than district heat, at €122.8 against €96.5/MWh, and dearer than the best available heat pump at €118.5/MWh, so the claim that hydrogen ranks below both alternatives holds against each of them. The endogenous-retrofit variant of the program (Appendix B) confirms the share result, since even when renovation depth is optimised jointly with supply, and the program takes renovation up to its ramp limit at the interim milestones, hydrogen's share does not improve. We retain the limitation here, because the reader's natural question is whether the verdict is an artefact of the framework split, and the cross-framework agreement on hydrogen's share is the direct answer to it.

D.8 Cost-Optimal Emissions Boundary

We apply the least-cost program's emissions cap to scope-1 emissions, that is on-site gas and oil combustion, alone. Electricity-grid, district-heat and hydrogen-upstream emissions lie outside the cap, and the grid carbon intensity is exogenous and near-zero by 2050. A scope-1 cap systematically favours heat pumps and electric heating, which shift emissions to the grid, over hydrogen, which internalises them on site; a -100% target therefore means net-zero on-site combustion, satisfied largely mechanically by removing fossil boilers, short of economy-wide net-zero for the heat supplied. The boundary matters less to the mix than it once appeared to, but it is no longer costless. The cap is slack in most countries, because with ETS2 embedded in the levelised costs the least-cost mix already satisfies it, and the cost of even the deepest cap is 0.05 per cent of present-value system cost. It is not slack everywhere. It binds in 4 of the 29 countries in 2050 and in seven at some point on the path, at an implied carbon price of up to €81/tCO_2 (Appendix B).

Where the cap binds, the scope-1 boundary is doing real work, since a scope-2-inclusive accounting would charge the electric technologies for their grid emissions and change which countries bind and by how much. We therefore no longer claim the boundary is immaterial. The headline verdict does not rest on it, but the reason has changed. It rests on the missing-money problem, which is a capital-recovery result independent of any emissions accounting, and on the cost of holding a winter-weighted hydrogen load in seasonal store, which is a physical result equally independent of it. The levelised-cost difference now runs the same way in 22 of 29 countries, but it is not invoked here as an accounting-invariant anchor, because it depends on the storage, network and levy treatments the rest of this section sets out. We retain the directional bias here to note that, where it acts, it favours the electric technologies, so it works against hydrogen and cannot be what produces hydrogen's remaining disadvantages.

D.9 Cross-Country Correlation of Renovation Rates

The Monte Carlo samples each country's envelope-renovation rate through a shared EU-policy factor plus an idiosyncratic country-specific component, with correlation $\rho = 0.5$, and not by drawing countries independently. The choice reflects the structure of the underlying process; renovation pace is driven partly by common EU policy, principally the recast Energy Performance of Buildings Directive and the Renovation Wave [12], and partly by national factors. Independent draws would diversify away most of the aggregate uncertainty and understate the EU band. The chosen ρ leaves each country's marginal distribution unchanged, so per-country bands and medians are unaffected, and widens the EU-aggregate band by about 2.2 times relative to independent draws, rising to 3.0 times at $\rho = 1$. Because $\rho = 0.5$ is a central assumption, with $\rho = 0$ recovering independence and $\rho = 1$ full correlation, the reported EU band is itself conditional on this knob.

The knob does not, however, touch the technology ranking. Rankings are set by levelised cost within each country, which the EU aggregate does not set, so ρ enters the width of the demand band alone. It leaves the relative ordering untouched, and the central scenario trajectories and medians are invariant across the full range from $\rho = 0$ to $\rho = 1$ (Appendix C.3). What has changed is what that invariance buys. Earlier drafts read it as confirming a hydrogen cost disadvantage that held country by country. The disadvantage holds in 22 of the 29 countries, so the correct statement is the weaker one that ρ leaves each country's own cost comparison, whichever way it falls, exactly where it was.

D.10 Scenario Rates Scale Proportionally, Without Converging on a Common Target

The Net Zero and H2 Push scenarios scale each country's envelope-renovation rate by a single EU-wide factor, without converging it on a common policy target, which preserves the existing cross-country ordering but leaves the implied national paces unequal. This affects the demand path; the technology rank is untouched. Hydrogen loses to district heat on levelised cost and loses the winter peak to the alternatives in most country-scenarios at every demand level, and its position against the heat pump is set by each country's own prices, which the level of demand does not enter, so a different renovation profile would shift absolute demand without reordering the technologies. Concretely, the Net Zero and H2 Push envelope rates are each country's Stated Policies rate multiplied by a single EU-wide factor ($\times 2.73$ and $\times 1.64$ respectively; Current Policies $\times 0.70$).

This preserves cross-country heterogeneity but does *not* represent convergence on the uniform 3%/yr Renovation-Wave target. High-baseline countries such as Switzerland reach an implied $\approx$2.7%/yr envelope-intensity decline under Net Zero, at the upper edge of physical plausibility for a sustained 25-year rate, while low-baseline countries such as Finland stay below 1%/yr. The scenarios should therefore be read as a proportional acceleration of current national paces, not as a common policy endpoint. The same applies to the technology-share ambitions. Each scenario's heat-pump and district-heat targets are EU-wide numbers, differentiated by country only through feasibility scaling and each country's own 2025 starting mix, so the realised within-scenario country spread of about 5 to 8 percentage points between the tenth and ninetieth percentiles is modest relative to the cross-scenario spread.

D.11 Forward Extrapolation of Demand Drivers

The two principal demand drivers are extrapolated with biases that act in opposite directions and are not separately quantified. Household size, which sets dwellings per capita, is linearly extrapolated from the 2015 to 2023 trend to 2050; because household-size decline typically decelerates towards saturation, this may modestly overstate the occupancy-driven rise in demand. Average dwelling floor area is held flat, and it has risen historically in most countries, which may modestly understate demand. The net effect on the demand path is therefore bounded but ambiguous in sign. It does not reach the hydrogen verdict. Occupancy and dwelling size are carried as explicit uncertain drivers in the variance-based demand sensitivity, where they sit within the ± 35 per cent sampled band (Appendix C.4), and because every technology's levelised cost is expressed per MWh of useful heat, a perturbation of either driver rescales absolute demand without reordering the technologies at any demand level. The plausible range is thus immaterial to the ranking, and we do not quantify the two biases separately because they enter only the demand level, which the ranking is invariant to.

D.12 Heat-Demand Basis and Labelling

The reported heat-demand totals, for example the $\approx$3,830 TWh EU-aggregate figure for 2025, follow the Hotmaps regional heat-demand basis [103], against which we reconcile the independent bottom-up build (Appendix A.1). The two-significant-figure precision and the approximation sign attach to this EU aggregate, where country-level deviations partially offset (validation below); we do not carry the same precision down to any single country, whose figure inherits the wider per-country band. This is a useful (that is, net) heat-demand quantity, which the model converts to final energy per technology through each technology's efficiency or coefficient of performance; the near-equality of summed final and useful energy reflects the heat-pump COP pulling the fleet-average conversion factor close to unity, so no loss term is missing. Because the Hotmaps basis sits above some narrower Eurostat residential-heating definitions, the *absolute* TWh and the derived emission tonnages should be read on the Hotmaps basis and are not directly comparable to every national statistic. The levelised-cost and hydrogen-versus-heat-pump gap results, being expressed per MWh, are invariant to this basis.

D.13 Endogenous Switzerland

Earlier drafts treated Switzerland through a one-directional exogenous sensitivity on the OIES hydrogen import-price trajectory [93, 164], with the Swiss heating market taking EU prices but not affecting them. That one-way coupling has been replaced by an *endogenous* integration, reported in full in the RQ2 results (Appendix A.8). The net effect of the feedback itself is small, about €1 to 3/MWh on the levelised cost of heat, and it moves against hydrogen. It leaves the direction of the Swiss base-load verdict unchanged, the best Swiss heat pump ending €5.6/MWh below the hydrogen boiler on the uncoupled arm and €9.6/MWh below it on the endogenous one. An earlier draft reported a Swiss base-load reversal in hydrogen's favour; that was an artefact of omitting the seasonal store from the base-load comparison while charging it in the peak and dispatch tests, and it does not survive the correction.

The margin is nonetheless narrow relative to the €1 to 3/MWh the feedback moves, so the Swiss base-load result should be read as the weaker of the two Swiss arguments, with the peaking verdict carrying the case. The residual point that belongs here is that two inputs to the endogenous treatment are judgement calls carried as assumptions: the 600 km representative pipeline distance from the major European hydrogen producers, which sets the transport adder on delivered Swiss hydrogen, and the 8 GW of firm winter reservoir headroom credited to the Swiss hydro fleet. Both are conservative towards hydrogen, so relaxing either would narrow the Swiss base-load margin and never widen it, and given how narrow that margin already is, the direction of that residual uncertainty is a further reason to rest the Swiss case on the peaking verdict.

D.14 Demand-Side Behaviour

The model does not endogenise consumer behaviour, the split incentive between landlords and tenants, or the political-economy constraints on technology mandates. These factors may materially affect the pace and the spatial distribution of technology adoption in practice, and they fall outside a cost-and-dispatch framework.

D.15 Grid Trajectory and the Annual-Average Emissions Bias

The exogenous grid-decarbonisation path, on which the EU median falls to ≈ 8 gCO_2/kWh by 2050, is load-bearing for the deepest outcomes. Freezing the grid at its 2025 intensity caps the Stated Policies reduction at $\approx 36\%$ (Appendix A.5), and the Net Zero -98% is unattainable without a near-fully decarbonised power system. The scenarios flex this dependence through the grid-speed lever ($\times 0.70$ to 1.15), but a power-sector failure scenario lies outside the modelled range. Relatedly, heat pumps exhibit seasonal degradation, with the cold-snap coefficient of performance falling to ≈ 0.6 of the seasonal value (floored at

1.8), and they draw hardest when the grid is dirtiest, in the winter thermal-peaker hour. The cost accounting incorporates this through endogenous scarcity pricing; the emissions accounting does not, instead applying *annual-average* grid carbon intensity to all heating electricity. This produces a systematic bias in favour of heat pumps and against hydrogen in the emissions accounting, because heat-pump peak electricity is both costlier and dirtier than the annual mean, and hydrogen marginal emissions scale with the peaker. We do not quantify the magnitude of this bias here, since the model carries no hourly emissions ledger.

This limitation is material. It changes the emissions ranking, and its priority should be read accordingly. The median country gap is now about €15.4/MWh in the heat pump's favour and the mean about €19.7/MWh, which restores some headroom to the cost comparison, but it does not restore it everywhere. The seven countries in which hydrogen still leads run from €2 to €30/MWh, and Poland at €2/MWh and the Netherlands, Belgium and Ireland below €10/MWh have no headroom at all with which to absorb a bias of this direction and of unquantified magnitude in the accompanying emissions ranking. An hourly emissions ledger, which would charge heat-pump electricity at the marginal grid intensity in the hours it is actually drawn, in place of the average, is accordingly promoted here from a desirable refinement to the highest-priority methodological extension of this work after the levy-equalised comparison of the first subsection. Until it exists, the heat-pump emissions rank should be read as an upper bound, and no emissions-based argument in this paper should be relied on to settle any of the country comparisons that remain close to a tie.

D.16 Power Dispatch and Unit-Commitment Parameterisation

The peaking headline is produced by the reduced-form merit order, which sets the fleet size and the running hours, and is validated against an explicit unit-commitment optimisation with minimum stable output, ramp limits, minimum up and down times, start-up costs, an operating-reserve margin and storage state of charge (Appendix A.2). With its constraints disabled the unit commitment reproduces the reduced form in 28 of the 29 countries (Appendix A.7). Romania is a structural disagreement, the reduced form finding no peaker needed under the three-hour standard where the unit commitment's sizing pass builds 2.8 GW. Switching the constraints on raises peaker energy by about 16 per cent and moves the hydrogen-minus-gas missing-money gap by about €1.5 bn, so the upgrade leaves the economic verdict intact (Appendix A.7).

Three residual sensitivities the upgrade introduces belong here. First, the unit-commitment outcome depends on the firm-fleet flexibility parameters, in particular the CCGT minimum stable output of 0.30 and ramp of 2.4 per hour, taken from the Danish Energy Agency Technology Catalogue 2023 [147] and the ENTSO-E TYNDP 2024;

these are most consequential in the southern, low-peaker markets, where a small absolute peaker energy makes the split between the firm CCGT and the fast peakers proportionally more sensitive to how tightly the CCGT can follow load, though the headline fleet and the missing-money verdict are unchanged.

Second, the 5 per cent operating-reserve standard, applied to the fixed hourly net load, is a planning convention, and we have not run a country-specific reserve study; it sets the 59 GW of additional firm capacity the reserve margin implies (an implied firm fleet of about 532 against an installed 473 GW, a roughly 13 per cent uplift), and a different standard would scale that adequacy figure without touching the peaker economics. This adder is a capacity-market or firm-build signal for the system planner, distinct from the 278 GW peaker headline, and it adds nothing to the missing-money loss. Third, the result also rests on the cold-snap COP derate (0.6 of seasonal, floored at 1.8) and the peaker's full-load hours; the missing-money conclusion is insensitive to these, since even doubling peaker full-load hours leaves every observed inframarginal rent below the capital-recovery threshold. The verdict here is accordingly an economic one.

The derate does, however, move the building-peak count, and it had not been swept until now. $\mathrm{COP}^{\mathrm{cold}} = \max(f\,\mathrm{SCOP}, \mathrm{floor})$ is published at $f = 0.60$ with a 1.8 floor, giving a study-area median of 2.04, and the low-temperature field evidence supports roughly 0.5 to 0.75. Sweeping it, the four counts run 0, 5, 10 and 18 of 29 at $f = 0.50$; 0, 4, 10 and 18 at 0.55; 0, 4, 9 and 16 at the published 0.60; 0, 1, 7 and 13 at 0.70; and 0, 1, 7 and 10 at 0.75. The demand-weighted derived ceiling under Stated Policies moves with it, from 5.3 per cent at the shallow end to 0.2 at the deep end. A better heat pump in the cold snap narrows hydrogen's peaking niche, which is the expected direction, and the size of the move makes this the largest single lever on that arena. It is the third independent reason the Stated Policies ceiling should be read as a range.

A fourth belongs here because it is an omission rather than a parameter. Non-heat national electricity demand is grown by a single smooth factor standing for road transport and industry, and it carries no explicit electrolyser load. Heat-pump electricity is added on top of that baseline and is what drives the firm-capacity requirement; the electricity that would produce the hydrogen is never added. The 889 GW summed peak, the 278 GW firm fleet, the 4,513 TWh denominator and the 0.86 per cent energy share are therefore all computed on a 2050 power system containing no hydrogen production, in a study whose hydrogen-favourable scenario puts hydrogen in 8.5 per cent of residential heat. Adding that load would raise the denominator, and so lower the energy share, while raising the peak only to the extent that electrolysers run through the cold snap rather than avoiding it, which a flexible operator would not do. We flag the coupling as unmodelled rather than bounded, since we have not run it.

D.17 Monte Carlo Aggregation of the Deepest Scenario

Net Zero carries the widest Monte Carlo spread, since its renovation and carbon levers are the most uncertain, and two presentational artefacts follow. Its per-technology medians under-sum by about 2.5 per cent before renormalisation, because a median taken technology by technology need not sum to one even though every individual draw does. Displayed mixes are therefore renormalised. Its 2025 baseline is the common 644 $MtCO_2$ figure, identical to the other three scenarios, since the base year carries no scenario lever, and every aggregation order returns that figure. The artefact affects presentation only, leaving every sample-level computation untouched.

D.18 Validation Precision

The bottom-up demand build matches the Hotmaps benchmark within ±15% in 19 of 29 countries and within ±25% in 28 of 29, with an EU aggregate deviation of −0.8%. The two scales of precision answer different questions. The country-level errors are signed and largely independent, so they offset in the sum and leave the EU aggregate tight to −0.8%, which is why we report the aggregate to two significant figures and with an approximation sign.

The wider per-country deviations concentrate where EUBUCCO footprint coverage or the TABULA proxy mapping is weakest [34, 35]; the EU-level results are insensitive, but every single-country absolute demand figure inherits the ±15 to 25% implicit band and should be read at that tolerance, and never at the precision the EU aggregate carries. The dwelling counts that feed the density proxy and the network-bill per-connection aggregation are estimates we calibrate to census multi-family shares, and their levels carry a separate and larger ±20 to 30% calibration error in some countries, well outside the demand-validation tolerance above. This error is immaterial because it operates only on the density proxy, where it cancels in the cross-scenario comparison, and the network cost-band uncertainty dominates it in any case.

D.19 Electricity Network Costs and the Risk of Double-Counting Them

The incremental low- and medium-voltage reinforcement that mass heat-pump electrification requires is now charged in the headline levelised cost of heat, for every electric heating technology and in every result this paper reports, at the central bound of a €400 to 1,600 per added kW band with a coincidence factor of 0.5 and a 40-year annualisation (Appendix A.13). The hydrogen boiler is charged its own last-mile network on symmetric terms. What was previously an omission, and was defended as conservative towards the heat pump, is therefore no longer an omission, and the defence that went with it, that the headline conclusion survived the omission with about €19/MWh of headroom, does not survive the correction.

Charging the heat pump the *high* reinforcement bound while charging hydrogen nothing at all for its network leaves heat pumps cheaper than hydrogen in 10 of 29 countries, down from 22. The mean margin becomes €6.2/MWh in hydrogen's favour, where the headline accounting gives €19.7/MWh in the heat pump's. Charging the heat pump the same high bound while hydrogen pays the blended network it is charged by default leaves heat pumps cheaper in 22 of 29 (Appendix C.7). The two arms are distinct and must not be combined, since attaching the 22 to the arm in which hydrogen pays nothing would overstate the heat-pump case. The base-load ranking depends on this assumption.

The correction introduces a new residual risk that runs the other way and that nothing else in this paper accounts for. Retail electricity prices already embed today's network tariff, which is how the existing distribution grid is recovered from households. Adding an incremental reinforcement charge on top of a retail tariff that already contains a network component may therefore *double-count* part of the network cost of electric heating, and would in that case over-charge the heat pump relative to a hydrogen boiler whose delivered price contains no comparable embedded network element at all. The size of the overlap depends on how much of each country's existing network charge is a sunk-asset recovery, which the incremental charge should not duplicate, and how much is a forward-looking reinforcement signal, which it would duplicate.

We do not decompose national network tariffs on that basis, so we cannot bound the overlap here. The direction of the bias is nonetheless clear. It acts against electric heating and points the same way as the fiscal asymmetry of the first subsection, so the two qualifications on hydrogen's modelled base-load advantage compound and never offset. A properly specified extension would charge each carrier the incremental network cost of its own peak while netting out the embedded network component already inside its retail price, and we identify it as a necessary companion to the levy-equalised comparison recommended above.

Funding

This project has received funding from the European Union's Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie grant agreement N° 945363.

Data Availability

The model and the underlying data are available from the authors on reasonable request. Enquiries should be directed to the corresponding author.

The supplementary material referred to in the text comprises the scenario parameter register of per-country envelope rates and scenario levers; the derivation of the renovation depth-band weights; the full per-country and per-scenario missing-money table; the capacity-stack data note; the Swiss fixed-point iteration history; the per-country delivered hydrogen supply costs; the source audit behind the techno-economic inputs; the per-country energy against dwelling-share diagnostic; the ground-source per-ceiling sweep; the scope-2-inclusive emissions accounting; and the per-country cost-optimal shares and shadow prices. These are available on the same basis.